

SGA 7

Exposé I

Summary of the First Lectures by A. Grothendieck

written up by P. Deligne

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0. Preliminaries

0.0. Terminology. Recall the following definitions:

(0.0.1) trait: the spectrum of a discrete valuation ring. The generic and closed points of a trait are often denoted by η and s .

(0.0.2) strictly local: for a ring, local henselian with separably closed residue field; for a scheme, the spectrum of such a ring.

(0.0.3) geometric point of S (in this exposé): a morphism $\bar{x} : \text{Spec}(\bar{k}) \rightarrow S$ with image $x \in S$, where $k(\bar{x}) \stackrel{\text{dfn}}{=} \bar{k}$ is a separable closure of $k(x)$. For a henselian trait $S = \text{Spec}(V)$, a generic geometric point of S (that is, one with image η) is often denoted by $\bar{\eta}$. The residue field of the valuation ring $\bar{V}$, the integral closure of V in $k(\bar{\eta})$, is a separably closed algebraic extension of $k(s)$ and defines a geometric point $\bar{s}$ of S centered at s .

(0.0.4) strict localization of X at the geometric point $\bar{x}$: SGA 4 VIII 4.4.

(0.0.5) essentially of finite type over S : for a local (respectively, henselian local; respectively, strictly local) S -scheme, the spectrum of a local ring (respectively, the henselization of a local ring; respectively, a strict localization) of a scheme of finite type over S .

(0.0.6) $F \mapsto F(n)$: Tate twist.

(0.0.7) An endomorphism T of a finite-dimensional vector space is quasi-unipotent if its eigenvalues are roots of unity, that is, if there exist $N \geq 1$ and $M \geq 1$ such that $(T^N - 1)^M = 0$.

0.1. Review of ℓ -adic cohomology

Let X be a scheme of finite type over an algebraically closed field k of characteristic exponent p , and let ℓ be a prime different from p .

The ℓ -adic cohomology groups of X were defined in SGA 4 and SGA 5.

a) $H^i(X, \mathbb{Z}/\ell^n)$ is the i -th cohomology group of the étale site $X_{\text{ét}}$ of X with coefficients in $\mathbb{Z}/\ell^n$;

b) by definition,

$$H^i(X, \mathbb{Z}_\ell) = \varprojlim_n H^i(X, \mathbb{Z}/\ell^n);$$

c) by definition,

$$H^i(X, \mathbb{Q}_\ell) = H^i(X, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The ℓ -adic cohomology groups of X with proper supports are defined in the same way from the groups $H_c^i(X, \mathbb{Z}/\ell^n)$ of SGA 4 XVII.

In each of the following cases, one knows how to prove that these groups $H^i(X)$ have reasonable finiteness properties:

(0.1.1) for cohomology with proper supports;

(0.1.2) when X is proper over k ;

(0.1.3) when X is smooth over k (by (0.1.1) and Poincaré duality, SGA 4 XVIII);

(0.1.4) when Hironaka's resolution of singularities is available for schemes of finite type over k of dimension at most $\dim(X)$;

(0.1.5) when $i = 0, 1$.

In all these cases, the groups $H^i(X, \mathbb{Z}/\ell^n)$ are finite, the groups $H^i(X, \mathbb{Z}_\ell)$ are finitely generated, the groups $H^i(X, \mathbb{Q}_\ell)$ are finite-dimensional, and (except perhaps in case (0.1.5)) there are split short exact sequences

$$0 \longrightarrow H^i(X, \mathbb{Z}_\ell) \otimes \mathbb{Z}/\ell^n \longrightarrow H^i(X, \mathbb{Z}/\ell^n) \longrightarrow \mathrm{Tor}_1(H^{i+1}(X, \mathbb{Z}_\ell), \mathbb{Z}/\ell^n) \longrightarrow 0.$$

Moreover, in each case the proof of the finiteness theorem also proves a generic constructibility theorem: if k is the field of fractions of an integral noetherian scheme S , and X is the generic fibre of a morphism $f_1 : X_1 \longrightarrow S$ of finite type, then there exists a nonempty open subset $U \subset S$ such that over U the sheaves $R^i f_* \mathbb{Z}/\ell^n$ are locally constant and their formation commutes with every base change.

0.2. We no longer suppose that k is algebraically closed. Let $\bar{k}$ be an algebraic closure of k (or a separable closure, which amounts to the same thing here). We consider the “geometric” cohomology groups $H^i(X_{\bar{k}})$, where $X_{\bar{k}}$ is obtained from X by extension of scalars from k to $\bar{k}$. By transport of structure, the Galois group $\mathrm{Gal}(\bar{k}/k)$ acts on these groups. In $\mathbb{Q}_\ell$ -cohomology, in any of the cases (0.1.1)–(0.1.5), this gives continuous representations of $\mathrm{Gal}(\bar{k}/k)$ in a linear group $\mathrm{GL}(n, \mathbb{Q}_\ell)$.

0.3. Review of traits (0.0.1)

For the proofs, we refer to [4]. Let S be a henselian trait, and let η , s , V , $\bar{V}$, $\bar{\eta}$, and $\bar{s}$ be as in (0.0.1) and (0.0.3). Let p denote the characteristic exponent of $k(s)$, and let $\mathrm{Gal}(\bar{\eta}/\eta)$ (respectively, $\mathrm{Gal}(\bar{s}/s)$) denote the Galois group $\mathrm{Gal}(k(\bar{\eta})/k(\eta))$ (respectively, the corresponding group for $\bar{s}/s$). By transport of structure, $\mathrm{Gal}(\bar{\eta}/\eta)$ acts on $k(\bar{s})$; hence there is a homomorphism from $\mathrm{Gal}(\bar{\eta}/\eta)$ onto $\mathrm{Gal}(\bar{s}/s)$ whose kernel is the inertia group I . The subfield of $k(\bar{\eta})$ fixed by I is a maximal unramified extension of $k(\eta)$.

The profinite group I has a largest pro- p subgroup P . The quotient I/P is the tame inertia group. Canonically,

$$I/P \simeq \varprojlim_n \mu_n(k(\bar{s})) \stackrel{\mathrm{def}}{=} \widehat{\mathbb{Z}}(1)(k(\bar{s})).$$

If ϕ_n is the canonical map from I/P to $\mu_n(\bar{k})$ and u is an element of $\bar{V}$ of valuation $1/n$, then, for $\sigma \in I$,

$$\sigma(u) \equiv \phi_n(\sigma) u \quad (\text{modulo elements of valuation greater than } 1/n).$$

In summary,

$$\begin{aligned} & \mathrm{Gal}(\bar{\eta}/\eta) \supset I \supset P, \\ & \mathrm{Gal}(\bar{\eta}/\eta)/I \simeq \mathrm{Gal}(\bar{s}/s), \\ (0.3.1) \quad & I/P \simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ & P : \text{a pro-}p \text{ group} \quad (\{e\} \text{ if } p = 1), \end{aligned}$$

and the action (by inner automorphisms in $\mathrm{Gal}(\bar{\eta}/\eta)$) of $\mathrm{Gal}(\bar{\eta}/\eta)/I$ on I/P is identified with the natural action of $\mathrm{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.4. More generally, let S be strictly local and regular, let D be a regular divisor in S , and let p' be the characteristic exponent of the generic point of D . Let $\bar{\eta}$ be a geometric generic point of S , and let $\bar{s}$ be the resulting geometric point centered at the closed point s . By Abhyankar’s lemma, the fundamental group $\pi_1(S - D, \bar{\eta})$ has a filtration analogous to (0.3.1):

$$\begin{aligned} & \pi_1(S - D, \bar{\eta}) \supset I \supset P, \\ (0.4.1) \quad & \pi_1(S - D, \bar{\eta})/I \simeq \mathrm{Gal}(\bar{s}/s), \\ & I/P \simeq \widehat{\mathbb{Z}}(1)(k(\bar{s})), \\ & P : \text{with no quotient of order prime to } p'. \end{aligned}$$

The action by inner automorphisms of $\pi_1(S - D, \bar{\eta})/I$ on I/P is identified with the natural action of $\text{Gal}(\bar{s}/s)$ on $\widehat{\mathbb{Z}}(1)(k(\bar{s}))$.

0.5. Néron desingularization

The method of Néron desingularization gives the following result.

Lemma 0.5.1. Let $S \rightarrow S_o$ be a morphism of henselian traits, with $S = \text{Spec}(V)$ and $S_o = \text{Spec}(V_o)$, and let π be a uniformizer of V_o . Suppose that π is a uniformizer of V , and that the extensions $k(s)/k(s_o)$ and $k(\eta)/k(\eta_o)$ are separable. Then V is a filtered direct limit of smooth henselian V_o -algebras B_i essentially of finite type over V_o .

To construct the desired V_o -algebras, one takes $V_o \subset B \subset V$, with B of finite type, and desingularizes $\text{Spec}(B)$ by Néron's method ([1], § 4).

Let $S = \text{Spec}(V)$ be a henselian trait.

(0.5.2) If S is of equal characteristic, with uniformizer π , one may apply (0.5.1) with $S_o = \mathbf{F}_p[\pi]_{(\pi)}$. The extension s/s_o is separable because $\mathbf{F}_p$ is perfect; the extension η/η_o is separable because π , being a uniformizer, is not a p -th power (if $p \neq 1$), and hence belongs to a p -basis. It follows that S is a limit of spectra of smooth henselian $\mathbf{F}_p[\pi]_{(\pi)}$ -algebras essentially of finite type.

(0.5.3) If S is complete and of mixed characteristic, with perfect residue field k , then V is an Eisenstein extension of the ring of Witt vectors (that is, the Cohen ring) $W(k)$:

$$V \simeq W(k)[\pi] / \left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i \right).$$

If the a_i are perturbed slightly, one obtains an isomorphic extension. We may therefore suppose that the a_i belong to $W(k')$, where k' is of finite type over $\mathbf{F}_p$. Let k_o be the perfect closure of k' in k , and set

$$V_o = W(k_o)[\pi] / \left(\pi^n + \sum_{i=0}^{n-1} a_i \pi^i \right).$$

Lemma 0.5.1 applies to V/V_o , and the residue field of V_o is a radicial extension of a field of finite type over $\mathbf{F}_p$.

1. Arithmetic proof of the monodromy theorem

The reader will find in [5], Appendix, a proof of the following result of A. Grothendieck.

Proposition 1.1. Let S be a henselian trait, let ℓ be a prime different from the characteristic exponent p of $k(s)$, and let ρ be a continuous representation of $\text{Gal}(\bar{\eta}/\eta)$ in $\text{GL}(n, \mathbb{Q}_\ell)$. Suppose that the following condition holds:

$(*_\ell)$ No finite extension of $k(s)$ contains all roots of unity of ℓ -power order.

Then there exists an open subgroup I_1 of the inertia group I such that $\rho(\sigma)$ is unipotent for every $\sigma \in I_1$.

Condition $(*_\ell)$ is automatically satisfied when $k(s)$ is of finite type over the prime field (or is radicial over such a field).

The following theorem follows immediately from 1.1.

Monodromy theorem 1.2. With the preceding notation, let X be a scheme of finite type over η , and let H be a cohomology space of $X_{\bar{\eta}}$ with coefficients in $\mathbb{Q}_\ell$, of one of the types (0.1.1)–(0.1.5). If condition $(*_\ell)$ is satisfied, the action on H of every element of I is quasi-unipotent.

The limit arguments of 0.5 allow us to dispense with hypothesis $(*_\ell)$.

Variant 1.3. Condition $(*_\ell)$ in 1.2 is superfluous.

Let H'_o be a $\mathbb{Z}_\ell$ -cohomology space of $X_{\bar{\eta}}$ giving rise to H , and put $H_o = H'_o/\text{torsion}$. By hypothesis, $H = H_o \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$, and the action of $\text{Gal}(\bar{\eta}/\eta)$ is induced by

$$\rho : \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{GL}(H_o).$$

Let $I_{\ell'}$ be the largest prime-to- ℓ subgroup of I . Its image in $\text{GL}(H_o)$ is an ℓ -adic Lie group of order prime to ℓ , and is therefore finite.

Make an extension of traits $S' \rightarrow S$. This does not change H_o (SGA 4 XVI 6.6), and $\rho(I')$ has finite index in $\rho(I)$, because I'/P' has finite index in I/P , by what precedes. It therefore suffices to prove the theorem over S' ; we may suppose that

- a) $\text{Gal}(\bar{\eta}/\eta)$ acts trivially on $H_o/\ell H_o$;
- b) if S is of mixed characteristic, then S is complete with perfect residue field (EGA O_{III} 10.3.1).

Put $S = \text{Spec}(V)$, define V_o as in (0.5.2) and (0.5.3), and apply (0.5.1). We have

$$V = \varinjlim_i B_i.$$

Here $S_i = \text{Spec}(B_i)$ is essentially of finite type and smooth over V_o . Let $D_i \subset S_i$ be defined by the equation $\pi = 0$. For sufficiently large i , H_o comes from a twisted constant $\mathbb{Z}_\ell$ -sheaf $\mathcal{H}_i$ on $S_i - D_i$ (by “generic constructibility,” (0.1)), and $\mathcal{H}_i/\ell\mathcal{H}_i$ is constant.

The filtrations (0.3.1) and (0.4.1) give rise to commutative diagrams:

$$(1.3.1) \quad \begin{array}{ccccc} \text{Gal}(\bar{\eta}/\eta) & \supset & I & \supset & P \\ \downarrow & & \downarrow & & \downarrow \\ \pi_1(S_i - D_i, \bar{\eta}) & \supset & I_i & \supset & P_i \end{array} \quad \text{and} \quad \begin{array}{ccc} I/P & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \\ \downarrow & & \parallel \\ I_i/P_i & \xrightarrow{\sim} & \widehat{\mathbb{Z}}(1) \end{array}$$

The representation on H_o of $\text{Gal}(\bar{\eta}/\eta)$ is induced by an action of $\pi_1(S_i - D_i, \bar{\eta})$ on H_o that is trivial on $H_o/\ell H_o$. The group

$$\text{Ker}(\text{GL}(H_o) \rightarrow \text{GL}(H_o/\ell H_o))$$

is a pro- ℓ group. Thus P_i acts trivially, and we apply the following variant of 1.1.

Variant 1.4. Retain the notation of 0.4, and let ρ be a continuous representation of $\text{Gal}(\bar{\eta}/\eta)$ in $\text{GL}(n, \mathbb{Q}_\ell)$. Suppose that $k(s)$ satisfies $(*_\ell)$ and that $\rho(P)$ is finite. Then the action of I is quasi-unipotent.

2. Vanishing cycles

2.1. Let S be a strictly local trait, and retain the notation of 0.3. Thus, by hypothesis, $s = \bar{s}$. Let $f : X \rightarrow S$ be a scheme of finite type over S . The formalism of vanishing cycles is a tool for studying the relation between the cohomology of X_s and that of $X_{\bar{\eta}}$, as well as the action of the inertia group on the latter, using information about f that is local on X .

A key result of this kind is the specialization theorem (SGA 4 XVI 2.2), according to which X_s and $X_{\bar{\eta}}$ have the same cohomology when f is proper and smooth.

In the transcendental theory, if $f : X \rightarrow D$ is a proper morphism from an analytic space X to the disk D , vanishing cycles arise as follows:

- a) After shrinking D if necessary, one may suppose that X is a topological fibre bundle over the punctured disk D^* (Thom’s isotopy theorem). In particular, the fibres

X_t ($t \in D^*$) are all isomorphic.

- b) It is not unreasonable to picture the special fibre X_s as obtained from “the” general fibre X_t by contracting polyhedra in X_t ; one would then have a map $\pi : X_t \rightarrow X_s$. The method of vanishing cycles consists in studying the difference between the cohomology of X_s and that of X_t by means of the Leray spectral sequence for π .

The geometric theory explained below is very close to this transcendental theory: the general point $t \in D$ must be replaced by $\bar{\eta}$, the geometric generic point of S . Conversely, the transcendental theory can be modeled on the geometric theory by replacing t or $\bar{\eta}$ with the universal cover $\tilde{D}^*$ of D^* , and X_t or $X_{\bar{\eta}}$ with $X \times_D \tilde{D}^*$. This eliminates many ε ’s and η ’s, but sometimes loses information.

2.2. Fix a torsion ring Λ in which p is invertible (for example, $\Lambda = \mathbb{Z}/\ell^n\mathbb{Z}$, with ℓ prime to p). Consider the morphisms

$$\begin{array}{ccccc}
X_s & \xrightarrow{i} & X & \xleftarrow{\bar{j}} & X_{\bar{\eta}} \\
\downarrow f_s & & \downarrow f & & \downarrow f_{\bar{\eta}} \\
s & \longrightarrow & S & \longleftarrow & \eta \longleftarrow \bar{\eta}
\end{array}$$

The sheaves of vanishing cycles ψ^i on X_s are

$$(2.2.1) \quad \psi^i = i^* R^i \bar{j}_* \Lambda.$$

If f is proper, then (SGA 4 XII 5.5)

$$(2.2.2) \quad i^* : H^p(X, R^q \bar{j}_* \Lambda) \xrightarrow{\sim} H^p(X_s, i^* R^q \bar{j}_* \Lambda),$$

and the Leray spectral sequence for $\bar{j}$ becomes

$$(2.2.3) \quad H^p(X_s, \psi^q) \implies H^{p+q}(X_{\bar{\eta}}, \Lambda).$$

When nothing happens at infinity, this spectral sequence can sometimes be obtained without a properness hypothesis. That some hypothesis at infinity is necessary is already clear from the trivial case $X_s = \emptyset$.

The definitions immediately give:

Proposition 2.3. Let $\bar{x}$ be a geometric point of X_s (for example, a closed point), let $X_{(\bar{x})}$ be the strict henselization of X at $\bar{x}$, and let $X_{(\bar{x})\bar{\eta}}$ be the geometric generic fibre of the projection $X_{(\bar{x})} \rightarrow S$. Then

$$\psi_{\bar{x}}^i = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda).$$

Corollary 2.4. Wherever f is smooth, $\psi^i = 0$ for $i > 0$ and $\psi^0 = \Lambda$.

This is SGA 4 XV 2.1.

The group $\text{Gal}(\bar{\eta}/\eta)$ (which, by hypothesis, equals the inertia group) acts on the ψ^i by transport of structure, and

Scholium 2.5. The spectral sequence (2.2.3) is $\text{Gal}(\bar{\eta}/\eta)$ -equivariant.

Variant 2.6. Put $\bar{S} = \text{Spec}(\bar{V})$ and $\bar{X} = X \times_S \bar{S}$. The scheme $X_{\bar{\eta}}$ is the complement of $X_s = X_{\bar{s}}$ in $\bar{X}$. Put

$$\phi^i = \underline{H}_{X_s}^{i+1}(\bar{X}, \Lambda)$$

(the local-cohomology sheaves). There is an exact sequence of sheaves

$$0 \longrightarrow \phi^{-1} \longrightarrow \Lambda \longrightarrow \psi^0 \longrightarrow \phi^0 \longrightarrow 0,$$

and, for $i > 0$, isomorphisms $\psi^i \xrightarrow{\sim} \phi^i$. If f is smooth, the ϕ^i vanish.

Put $\phi_{\text{gl}}^i = H_{X_s}^i(\bar{X}, \Lambda)$, the global analogue of ϕ^i . There is a spectral sequence

$$(2.6.1) \quad H^p(X_s, \phi^q) \implies \phi_{\text{gl}}^{p+q},$$

and a long exact cohomology sequence which, when f is proper, becomes

$$(2.6.2) \quad \cdots \longrightarrow H^i(X_s, \Lambda) \longrightarrow H^i(X_{\bar{\eta}}, \Lambda) \longrightarrow \phi_{\text{gl}}^i \longrightarrow \cdots.$$

Variant 2.7. Let K_t be the maximal tamely ramified extension of K in $\bar{K}$. Put $X_t = X \times_S \text{Spec}(K_t)$, and let $\bar{j}_t$ be the projection from X_t to X . Define tame variants of the ψ^i by

$$\psi_t^i = R^i \bar{j}_{t*} \Lambda.$$

Since P is a pro- p group and p is invertible in Λ , one obtains

$$(2.7.1) \quad H^*(X_t, \Lambda) = H^*(X_{\bar{\eta}}, \Lambda)^P,$$

$$(2.7.2) \quad \psi_t^i = (\psi^i)^P.$$

When f is proper, the Leray spectral sequence for $\bar{j}_t$ is obtained from (2.2.3) by passing to P -invariants:

$$(2.7.3) \quad H^p(X_s, \psi_t^q) \implies H^{p+q}(X_t, \Lambda) = H^{p+q}(X_{\bar{\eta}}, \Lambda)^P.$$

3. Geometric proof of the monodromy theorem

Let Λ be as in 2.2.

Conjecture 3.1 (purity). Let A be an excellent regular strictly local ring (0.0.2) and let D be a regular divisor in $\text{Spec}(A)$ defined by a parameter x . Put $U = \text{Spec}(A) - D = \text{Spec}(A[1/x])$. Then

$$(3.1.1) \quad H^q(U, \Lambda) = \begin{cases} \Lambda & \text{if } q = 0, \\ \Lambda(-1) & \text{if } q = 1 \quad (\text{notation (0.0.6)}), \\ 0 & \text{if } q > 1. \end{cases}$$

Here are some cases in which this conjecture is known (SGA 4 XIX): for $q = 0$ or 1 ; when A has characteristic 0 ; when A has equal characteristic p and resolution of singularities is available in characteristic p in dimensions $\leq \dim(A)$; or when $(\text{Spec}(A), D)$ is a smooth pair (the relative purity theorem, SGA 4 XVI 3.7).

When purity is available and $D = \sum_{i \in B} D_i$ is a normal-crossings divisor in $\text{Spec}(A)$ defined by part of a regular system of parameters, the cohomology of $U = \text{Spec}(A) - D$ is given by

$$(3.1.2) \quad \begin{aligned} H^0(U, \Lambda) &= \Lambda, \\ H^1(U, \Lambda) &= \Lambda(-1)^B, \\ H^q(U, \Lambda) &= \bigwedge^q H^1(U, \Lambda) \quad (\text{via the cup product}). \end{aligned}$$

These formulas are identical to those which, over $\mathbf{C}$, give the cohomology of $D^{*B} \times D^k$.

3.2. Let S be a strictly local trait, and retain the notation of (0.0.1) and (0.0.3). Let $f : X \rightarrow S$ be a morphism of finite type. Suppose that X is regular, that X_{η} is smooth, and that X_s is a normal-crossings divisor in X . We shall suppose that this divisor is, globally and not merely locally for the étale topology, a sum of regular divisors:

$$(X_s)_{\text{red}} = \sum_{i \in B} D_i$$

(the general case can be recovered by localization). Denote their multiplicities by m_i , so that $X_s = \sum_{i \in B} m_i D_i$.

Let x be a point of X_s , let $\bar{x}$ be a geometric point of X lying over x (for example, x may be closed and equal to $\bar{x}$), put $C = \{i \mid x \in D_i\}$, and let K be the complex concentrated in degrees 0 and -1

$$K : \mathbf{Z}^C \xrightarrow{d} \mathbf{Z}, \quad d((n_j)) = \sum_j m_j n_j.$$

Theorem 3.3 (uses purity). The groups of tame vanishing cycles $\psi_{t\bar{x}}^i$ are as follows:

(i) There is a canonical isomorphism of graded Λ -algebras

$$\psi_{t\bar{x}}^* \simeq \bigwedge^* (H_1(K) \otimes \Lambda(-1)) \otimes_{\Lambda} \psi_{t\bar{x}}^0.$$

(ii) $\psi_{t\bar{x}}^0$ is the Λ -algebra Λ^E , where E is a set on which the tame inertia group $I_t = I/P = \widehat{\mathbf{Z}}(1)(k(s))$ acts transitively, and whose cardinality is the largest prime-to- p divisor of $\#H_0(K)$.

Here is the method of proof.

a) Let $X_{(\bar{x})}$ be the strict localization of X at $\bar{x}$, let $U = X_{(\bar{x})} - X_{(\bar{x})s}$ (the complement of a normal-crossings divisor), let t_j be a local equation for D_j , put

$$X_n = X_{(\bar{x})}[(t_j^{1/n})] \quad ((n, p) = 1),$$

which is a regular scheme, and let U_n be the inverse image of U in X_n . By (3.1.2),

$$H^q(U_n, \Lambda) = \bigwedge^q (\Lambda(-1)^C).$$

For $m = nd$, the pullback map from $H^q(U_n, \Lambda)$ to $H^q(U_m, \Lambda)$ is multiplication by d^q . Putting $\tilde{U} = \varprojlim_{(n,p)=1} U_n$, one therefore has

$$(3.3.1) \quad \begin{cases} H^0(\tilde{U}, \Lambda) = \Lambda, \\ H^q(\tilde{U}, \Lambda) = \varinjlim_n H^q(U_n, \Lambda) = 0 \quad \text{if } q \neq 0. \end{cases}$$

b) $\tilde{U}$ is a pro-covering of U with group $\widehat{\mathbf{Z}}(1)(k(s))^C$. By definition,

$$\psi_{t\bar{x}}^* = H^*(U_{\eta_t}, \Lambda).$$

$\tilde{U}$ is a pro-covering, with group $H_1(K) \otimes \widehat{\mathbf{Z}}(1)(k(s))$, of each connected component of U_{η_t} , and these components are permuted transitively by the tame inertia group. Formula (3.3.1) and the Hochschild–Serre spectral sequence now give the formulas in 3.3.

Pictorially, one may say that $U_{\bar{\eta}}$ is the fibre of a morphism δ from the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s))^C, 1)$ represented by U to the cohomological $K(\widehat{\mathbf{Z}}(1)(k(s)), 1)$ represented by η ; the morphism δ induces

$$d : (n_j) \mapsto \sum_j m_j n_j : \widehat{\mathbf{Z}}(1)(k(s))^C \longrightarrow \widehat{\mathbf{Z}}(1)(k(s)).$$

Corollary 3.4 (uses purity). Suppose that f is proper. Let N be the least common multiple of the multiplicities m_j ($j \in B$). Let $q \geq 0$, and let q' be the minimum of q and the greatest number of divisors D_j passing through one point. For T in the tame inertia group $I_t = I/P$, one has

$$(T^N - 1)^{q'+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \Lambda)^P.$$

This follows at once from 3.3, Scholium 2.5, and Variant 2.7.

The required purity results are available for H^1 , in characteristic 0, or when S is essentially of finite type over a field.

Theorem 3.5. Let C be a smooth projective curve over the fraction field $k(\eta)$ of a discrete valuation ring V . The inertia group I acts on $H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell)$ by quasi-unipotent automorphisms of level 2: there exists N such that, for $T \in I$,

$$(T^N - 1)^2 = 0 \quad \text{on} \quad H^1(C_{\bar{\eta}}, \mathbf{Q}_\ell).$$

We reduce to the case in which V is strictly local. The assertion is that there exists an open subgroup I_1 of I such that $(T - 1)^2 = 0$ on H^1 for $T \in I_1$.

We may extend the scalars of $k(\eta)$ to a finite extension, replacing V by its normalization in that extension. Since the image of the pro- p group P in an ℓ -adic Lie group $\mathrm{GL}(n, \mathbf{Q}_\ell)$ is finite, we may in particular suppose that P acts trivially.

Let C_o be a minimal model of C over V (for the existence of C_o , which is based on Abhyankar's theorem, see the discussion in [3], § 2, p. 87; one could instead reduce to the case where V is excellent by passing to its completion and invoking SGA 4 XVI 1.6).

Let C_1 be a regular model of C , obtained from C_o by blowups, whose special fibre is a normal-crossings divisor. The arguments of 3.3 and 3.4 apply to C_1 , ψ_t^0 , ψ_t^1 , and

$$H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n)^P = H^1(C_{\bar{\eta}}, \mathbb{Z}/\ell^n),$$

with N independent of n ; Theorem 3.5 follows.

Remark 3.6. Let A be an abelian variety over $k(\eta)$. Since A is a quotient of a Jacobian, it follows again that the action of I on $H^1(A_{\bar{\eta}}, \mathbb{Q}_\ell)$, or on $T_\ell(A)$, is quasi-unipotent of level 2.

Remark 3.7. The action of I on $H^1(X_{\bar{\eta}}, \mathbb{Q}_\ell)$ is in fact quasi-unipotent of level 2 for every scheme X of finite type over η .

Theorem 3.5 (or Remark 3.6) is the basis for the proof of the stable reduction theorem for abelian varieties that will be given in Exposé IX, § 3.

3.8. It is clear that whenever purity and resolution of singularities are available (for example, in characteristic 0), the preceding method proves the monodromy theorem and gives bounds for the nilpotence exponent. Thus, in equal characteristic 0, if X is proper and smooth over η and T belongs to an open subgroup I_1 of I , one has

$$(T - 1)^{q+1} = 0 \quad \text{on} \quad H^q(X_{\bar{\eta}}, \mathbb{Z}_\ell) \quad (T \in I_1 \subset I).$$

4. Vanishing criteria for the sheaves of vanishing cycles

4.1. Let S be a strictly local trait and let $f : X \rightarrow S$ be a scheme of finite type over S . Put $n = \dim(X_\eta)$.

If X' is the scheme-theoretic closure of X_η in X , then:

- a) On $X_s - X'_s$, one has $\phi^{-1} = \Lambda$, and the other ϕ^i vanish;
- b) on X'_s , the ϕ^i and ψ^i relative to X/S or to X'/S coincide, and $\phi^{-1} = 0$.

Since X' is flat over S , this often permits a reduction to the case in which X is flat over S .

Theorem 4.2.

(i) One has $\psi^i = 0$ for $i > n$.

(ii) More precisely, let x be a point of X_s whose closure has dimension k ($k = \mathrm{trdeg}_{k(s)} k(x)$), and let $\bar{x}$ be a geometric point lying over x . Then

$$\psi_{\bar{x}}^i = 0 \quad \text{for} \quad i > n - k, \quad \text{i.e.} \quad d(\psi^i) \leq n - i \quad (\text{notation of SGA 4 XIV 2.1}).$$

Indeed,

$$\psi_{\bar{x}}^i = H^i(X_{(\bar{x})\bar{\eta}}, \Lambda),$$

and $X_{(\bar{x})\bar{\eta}}$ is an inverse limit of affine schemes of dimension at most n over $k(\bar{\eta})$. Assertion (i) therefore follows from the affine Lefschetz theorem (SGA 4 XIV 3.2) by passage to the limit.

To prove (ii), we may suppose that X is flat over S (4.1) and that there is a quasi-finite flat S -morphism

$$g : X \rightarrow \mathbb{A}_S^k$$

such that $g(x)$ is the generic point of $\mathbb{A}_S^k$. Let S' be the strict localization of $\mathbb{A}_S^k$ at $\bar{x}$. Then:

- a) S'_η is the spectrum of a field, and $\text{Gal}(\bar{\eta}'/s'_\eta)$ is a p -group Q ;
- b) by (i), applied to g ,

$$H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda) = 0 \quad \text{for } i > n - k;$$

- c) one has

$$H^i(X_{(\bar{x})\bar{\eta}}, \Lambda) = H^i(X_{(\bar{x})\bar{\eta}'}, \Lambda)^Q,$$

which proves the theorem.

Corollary 4.3. If f is proper and flat and the non-smooth locus of f in X_s has dimension at most d , the specialization morphism

$$H^i(X_s) \longrightarrow H^i(X_{\bar{\eta}})$$

is an isomorphism for $i > n + d + 1$ and an epimorphism for $i = n + d + 1$.

4.4. To prove, in certain cases, that the ϕ^i vanish for small i , we shall use local duality, either directly or through the local Lefschetz theorems of SGA 2 XIV. This tool is available only in characteristic 0, or in equal characteristic p when resolution of singularities à la Hironaka is available in the dimensions under consideration.

Theorem 4.5. Suppose that S has characteristic 0. Let $f : X \longrightarrow S$ be a morphism of finite type and let x be a closed point of X_s . Suppose that:

- (a) x is an isolated non-smooth point in its fibre;

- (b) X has relative geometric depth at least n at x : in a neighbourhood of x , the scheme X is identified with a subscheme defined by k equations in a smooth scheme of relative dimension N at x , with $n \leq N - k$.

Then $\phi_x^i = 0$ for $i < n$.

Let X_x be the strict localization of X at x , and put $V = X_x - \{x\}$. For every trait S' finite over S , put similarly

$$X'_x = X_x \times_S S', \quad V' = V \times_S S'.$$

Let $\bar{S}$ be the normalization of S in $\bar{\eta}$, and put

$$\bar{X}_x = X_x \times_S \bar{S}, \quad \bar{V} = V \times_S \bar{S}.$$

We have:

- (1) If $\phi^i = 0$ on $(X_x)_s$ (i.e. at every generization of x in X_s) for $i \leq m$, then

$$H_{\{x\}}^{i-1}(\bar{X}_x) = \tilde{H}^{\cdot, i}(\bar{V}) \xrightarrow{\sim} \tilde{H}^{\cdot, i}(\bar{V}_{\bar{\eta}}) = \phi^i \quad \text{for } i \leq m.$$

The ϕ^i are indeed the cohomology sheaves of $\bar{V}$ with support in V_s ; under the hypotheses of 4.5, condition (a) makes (1) applicable with $m = \infty$ (SGA 4 XV 2.1).

- (2) Hypothesis (b) of 4.5 is stable under base change. By SGA 2 XIV 5.6, it implies

$$\tilde{H}^{\cdot, i}(V') = H_{\{x\}}^{i-1}(X'_x) = 0 \quad \text{for } i < n.$$

We conclude by observing that $\tilde{H}^{\cdot, i}(\bar{V})$ is the filtered colimit of the $\tilde{H}^{\cdot, i}(V')$.

Combining 4.2 and 4.5 gives:

Corollary 4.6. Let S be a henselian trait of characteristic 0, let X/S be a relative complete intersection of relative dimension n , and let $x \in X_s$ be an isolated non-smooth point in its fibre. Then $\phi_x^i = 0$ for $i \neq n$.

Corollary 4.7. Under the hypotheses of 4.6, ϕ_x^n is a flat Λ -module.

Indeed, if Λ' is a finite Λ -algebra, a general nonsense (SGA 4 XVII 5.2.11) gives, for the $\phi_{\Lambda'}^i$,

$$\phi_{\Lambda'}^{n-i} = \text{Tor}_i^{\Lambda}(\Lambda', \phi_{\Lambda}^n),$$

and we apply 4.6 to Λ' .

Variant 4.8. In 4.5, omit condition (a), and let Z be the non-smooth locus of f . One may still conclude that

$$\phi_x^i = 0 \quad \text{for } i < n - \dim Z_s.$$

We proceed by descending induction on $\dim \overline{\{x\}}$. The induction hypothesis permits us to apply 4.5 (1) with $m = n - \dim Z_s$, and the proof is completed as before.

4.9. It is possible to obtain a vanishing criterion using only information about the geometric fibres of f . For complete intersections, however, the result is one degree weaker than 4.5.

Proposition 4.10. Under the general hypotheses of 4.5, suppose that:

- (a) x is an isolated non-smooth point in its fibre;
- (b) for every point y of $X_{\bar{\eta}}$ whose closure in $X_{\bar{\eta}}$ has dimension $d(y)$ and which is a generization of x , the étale depth satisfies

$$\text{depth}_{\text{ét},y}(X_{\bar{\eta}}) \geq n - d(y), \quad \text{i.e.} \quad H_{\{y\}}^i(X_{\bar{\eta}}) = 0 \quad \text{for } i \leq n - d(y);$$

- (c) $\text{depth}_{\text{ét},x}(X_s) \geq n$.

Then $\phi_x^i = 0$ for $i < n - 1$.

Retain the notation of 4.5. We may apply 4.5 (1) with $m = \infty$. By the local Lefschetz theorem (SGA 2 XIV), hypothesis (b) gives

$$H^i(V') \xrightarrow{\sim} H^i(V_s) \quad (i < n-1)$$

$$H^i(V') \hookrightarrow H^i(V_s) \quad (i = n-1)$$

Condition (c) then gives

$$H^i(\bar{V}) = \varinjlim H^i(V') = 0 \quad \text{for } i \leq n - 1,$$

which proves 4.10.

4.11. As in 4.8, one can give a variant of 4.10 in which (a) is not assumed.

5. Action of monodromy on π_1

In this section (Grothendieck's exposé of 19 November 1968), we shall use the semistable reduction theorem for curves. Artin and Winters [2] gave a direct proof of this theorem; it can also be deduced [3] from the analogous theorem for abelian varieties (IX § 3, using 3.6). We shall also use Schlessinger's theory (Rim's Exposé VI).

5.1. Let k be an algebraically closed field, let C be a projective curve over k , and let $x_1, \dots, x_n$ be a finite set of closed points of C . The pointed curve

$$X = (C; x_1, \dots, x_n)$$

is called stable if:

- (a) C is reduced, and its only singularities are double points with distinct tangents;
- (b) the x_i are distinct smooth points of C ;
- (c) if C_1 is an irreducible component of C , and E is the set of points of the normalization C'_1 of C_1 lying above a singular point of C or above one of the x_i , then, if C'_1 has genus 0 (respectively 1), the set E has at least 3 (respectively 1) elements.

Condition (c) is equivalent to each of the following:

- (c') X has no nontrivial infinitesimal automorphisms.
- (c'') If ω is the invertible dualizing sheaf, the invertible sheaf

$$\omega' = \omega \left(\sum_i x_i \right)$$

is ample.

An S -scheme C equipped with sections $x_1, \dots, x_n$ is a stable curve over S if its geometric fibres are stable curves. In [3], §§ 1–2, only the case $n = 0$ is considered. The extension of the results of loc. cit. to the general case is immediate:

- (1) If X/k is stable, then $H^1(C, \omega'^{\otimes n}) = 0$ for $n \geq 2$, and $\omega'^{\otimes n}$ is very ample for $n \geq 3$ (cf. [3] 1.2).
- (2) The formal moduli scheme M of X is smooth, and C remains singular over a normal-crossings divisor of M .
- (3) If X and Y are stable over S , then $\text{Isom}_S(X, Y)$ is finite and unramified over S (isomorphisms are required to respect the marked points).
- (4) If X/k is stable, the map

$$\text{Aut}_k(X) \longrightarrow \text{Aut}_k \text{Pic}^\circ(C)$$

is injective.

Finally, we have:

Proposition 5.1.1. Let $X_\eta = (C_\eta; x_1, \dots, x_n)$ be stable over the generic point η of a trait S , with C_η smooth. There exist a finite separable extension η' of η and a stable curve X' over the normalization S' of S in η' such that

$$X' \otimes_{S'} \eta' = X_\eta \otimes_\eta \eta'.$$

This follows from the usual semistable reduction theorem, which permits one to choose η' so that the geometric special fibre of the minimal model C'' of $C_{\eta'}$ over S' satisfies (a). One then blows up, iteratively if necessary, smooth points of C'' so that (b) is satisfied. Next one contracts the chains of smooth rational components of self-intersection -2 which contain none of the $\bar{x}_i$ in the special fibre; the result is C' (cf. [3], proof of 2.3, p. 88).

As in loc. cit., one can check that C' exists as soon as the Jacobian of C_η has semistable reduction. The smoothness hypothesis on C_η is moreover unnecessary.

5.2. Let X_η be a geometrically connected scheme of finite type over the fraction field of a strictly local trait S . For a geometric point ξ of $X_{\bar{\eta}}$, there is an exact sequence

$$(5.2.1) \quad 0 \longrightarrow \pi_1(X_{\bar{\eta}}, \xi) \longrightarrow \pi_1(X, \xi) \longrightarrow \text{Gal}(\bar{\eta}/\eta) \longrightarrow 0.$$

It induces a homomorphism from $\text{Gal}(\bar{\eta}/\eta)$ to the group of outer automorphisms of $\pi_1(X_{\bar{\eta}})$. Passing to the largest prime-to- p quotient $\pi_1^{(p)}(X_{\bar{\eta}})$ of $\pi_1(X_{\bar{\eta}})$, one obtains

$$(5.2.2) \quad \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Aut}_{\text{ext}}(\pi_1^{(p)}(X_{\bar{\eta}})).$$

Suppose that X_η has a rational point x , and let $\bar{x}$ be the geometric point

$$\bar{\eta} \longrightarrow \eta \xrightarrow{x} X_\eta.$$

For $\xi = \bar{x}$, the exact sequence (5.2.1) splits canonically, giving

$$(5.2.3) \quad [x] : \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{Aut}(\pi_1(X_{\bar{\eta}}, \bar{x})) \longrightarrow \text{Aut}(\pi_1^{(p)}(X_{\bar{\eta}}, \bar{x})).$$

Theorem 5.3. The image of P under (5.2.2) (or under (5.2.3), which amounts to the same thing) is finite.

5.3.1. Reduction to the case in which X_η is geometrically normal and integral.

Let X'_η be the normalization of X_η , let $(X'_{i\eta})_i$ be the family of connected components of X'_η , put

$$X''_\eta = X'_\eta \times_{X_\eta} X'_\eta,$$

and let $(X''_{j\eta})_j$ be the family of connected components of X''_η . After a finite extension of $k(\eta)$, we may suppose that the $X'_{i\eta}$ are geometrically normal and integral and have rational points x_i , and that the $X''_{j\eta}$ are geometrically connected and have rational points y_j .

For $k = 1$ or 2 , and whenever $p_k(y_j) \in X'_{i\eta}$, choose a class of paths ℓ_{kij} from $\text{pr}_k(\bar{y}_j)$ to $\bar{x}_i$. It is an element of a torsor under $\pi_1(X'_{i\eta}, \bar{x}_i)$. Let $\bar{\ell}_{kij}$ be the corresponding prime-to- p path (the same path modulo $\text{Ker}(\pi \rightarrow \pi^{(p)})$). We may choose $\bar{\ell}_{kij}$ to be invariant under P : the set of prime-to- p paths is an inverse limit of finite sets of cardinality prime to p on which the pro- p group P acts.

Consequently, if a finite-index subgroup P' of P acts trivially on the groups $\pi_1^{(p)}(X'_{i\eta}, \bar{x}_i)$, the Van Kampen formulas—equivalently, the fact that the normalization map $X'_\eta \rightarrow X_\eta$ is effective descent for étale coverings—show that P' acts trivially on $\pi_1^{(p)}(X_\eta, \bar{x})$.

5.3.2. If X_η is normal and U is an open subset, then $\pi_1(U)$ maps onto $\pi_1(X_\eta)$; we may therefore suppose that X_η is smooth and affine. By cutting with generic hyperplane sections (which replaces S by its strict localization at the generic point of the special fibre of an $\mathbb{A}_S^n$) and applying Bertini, we reduce to the case in which X_η is smooth, geometrically connected, and of dimension one.

After a finite extension of $k(\eta)$, we may suppose that X_η is the complement in a smooth projective curve $\bar{X}_\eta$ of a finite set of rational points x_i , whose number we are free to increase. By the semistable reduction theorem (5.1.1), we reduce to the case in which there is a stable curve $X = (C; x_1, \dots, x_n)$ over S such that

$$X_\eta = C_\eta - \{x_1, \dots, x_n\}.$$

We then apply the following result.

Theorem 5.4. Let S be a strictly local scheme, let $f : \bar{X} \rightarrow S$ be a proper flat morphism whose special fibre is a curve with ordinary double points as its only non-smooth points, and let Y be a subscheme of $\bar{X}$ which is the union of a finite number of disjoint sections contained in the smooth locus. Let x be a rational point of $X = \bar{X} - Y$. Let U be the open subset of S over which X is smooth, and let ξ be a geometric point of U . Then the image of $\pi_1(U, \xi)$ in $\text{Aut}(\pi_1^{(p)}(X_\xi, x\xi))$ is abelian and prime to p .

Observe that $\pi_1(U, \xi)$ can act on $\pi_1^{(p)}(X_\xi, x\xi)$ only because the groups $\pi_1^{(p)}$ satisfy the specialization theorem.

We reduce first to the case in which S is complete, and then to the universal case in which S is a formal moduli scheme for the special fibre equipped with its sections. Then S is the spectrum of a formal power-series ring over a Cohen ring:

$$S = \text{Spec}(W[[t_1, \dots, t_n]]),$$

and U is the complement of a normal-crossings divisor with equation $t_1 \cdots t_m = 0$. In this universal case, Abhyankar's lemma shows that $\pi_1(U, \xi)$ is abelian and prime to p , which proves the theorem.

Remark 5.5. One can prove a result analogous to 5.4 in arbitrary relative dimension by taking a generic plane section of dimension one and applying Bertini.

6. Appendix: arithmetic proof of the stable reduction theorem (by P. Deligne)

Let S be a trait and let A_η be an abelian variety over $k(\eta)$, of dimension $d \neq 0$. For a local morphism of traits $S' \rightarrow S$, denote by $A_{\eta'}$ the abelian variety over $k(\eta')$ obtained from A_η by extension of scalars. Let $\mathcal{A}_{S'}$ be the Néron model of $A_{\eta'}$ over S' , let $\mathcal{A}_{S', s'}$ be its special fibre, and let $\mathcal{A}_{S', s'}^\circ$ be the identity component of $\mathcal{A}_{S', s'}$. The stable reduction theorem is as follows.

Theorem 6.1. For a suitable S' , $A_{\eta'}$ has stable reduction, i.e. $\mathcal{A}_{S', s'}^\circ$ is an extension of an abelian variety by a torus.

It follows fairly readily from 6.1 that one may choose S' such that $k(\eta')$ is a finite separable extension of $k(\eta)$. Suppose first that the residue field of S is of finite type over the prime field (the same argument would work if $k(s)$ were radicial over such a field).

Let ℓ be a prime different from the residual characteristic exponent p , let $T_\ell(A)$ be the Tate module, let $V_\ell(A) = T_\ell(A) \otimes \mathbf{Q}_\ell$, and let

$$\rho : \text{Gal}(\bar{\eta}/\eta) \longrightarrow \text{GL}(T_\ell(A))$$

be the natural representation. After a preliminary extension of scalars, we are reduced to the case in which the inertia group I acts unipotently (1.2), and hence through its largest pro- ℓ quotient $\mathbf{Z}_\ell(1)$ (0.3). Let T be the image in $\text{GL}(T_\ell(A))$ of a generator of $\mathbf{Z}_\ell(1)$, and let r be the largest integer ≥ 0 such that $(T - 1)^r \neq 0$. We have:

Lemma 6.2. The map

$$(\sigma, v) \in I \times V_\ell(A) \longmapsto (\rho(\sigma) - 1)^r(v)$$

induces the morphism and the isomorphism in the commutative diagram

$$(6.2.1) \quad \begin{array}{ccc} V_\ell(A)_I \otimes \mathbf{Q}_\ell(r) & \xrightarrow{M} & V_\ell(A)^I \\ \downarrow & & \uparrow \\ (V_\ell(A)/\text{Ker}(T - 1)^r) \otimes \mathbf{Q}_\ell(r) & \xrightarrow{\sim} & \text{Im}(T - 1)^r \end{array}$$

$V_\ell(A)_I$ and $V_\ell(A)^I$ are $\text{Gal}(\bar{s}/s)$ -modules, and M is Galois-equivariant.

6.3. An ℓ -adic representation

$$\tau : \text{Gal}(\bar{s}/s) \longrightarrow \text{GL}(V)$$

is said to be of weight n ($n \in \mathbf{Z}$) if the following condition is satisfied:

(*) For a suitable model X of $k(s)$ (that is, an irreducible variety of finite type over the prime field such that $k(s) = k(X)$), τ factors through $\pi_1(X, \bar{s})$ and, for every closed point x of X , the eigenvalues of $\tau(F_x)$ are algebraic numbers all of whose complex conjugates have absolute value $|k(x)|^{n/2}$ (here F_x denotes the geometric Frobenius ϕ_x^{-1}).

If V (respectively W) has weight n (respectively m) and $n \neq m$, then $\text{Hom}_{\text{Gal}}(V, W) = 0$.

6.4. It follows from the universal property of the Néron model that $V_\ell(A)^I = V_\ell(\mathcal{A}_{S,s}^\circ)$. If a is the dimension of the largest abelian quotient of $\mathcal{A}_{S,s}^\circ$ and m is the dimension of its multiplicative part, then, by Weil, $V_\ell(A)^I$ is an extension of a representation of dimension $2a$ and weight -1 by a representation of dimension m and weight -2 .

6.5. Let A^* be the abelian variety dual to A . Recall that $V_\ell(A^*)$ and $V_\ell(A)$ (and hence $V_\ell(A^*)^I$ and $V_\ell(A)_I$) are in duality with values in $\mathbf{Q}_\ell(1)$. If a^* and m^* are defined for A^* as a and m are for A , it follows from 6.4 that $V_\ell(A)_I$ is an extension of a representation of dimension m^* and weight 0 by a representation of dimension $2a^*$ and weight -1 .

6.6. The Galois module $\mathbf{Q}_\ell(r)$ has weight $-2r$. Since $\text{Im}(T - 1)^r \neq 0$, it follows from 6.4, 6.5, and the isomorphism in 6.2 that $r \leq 1$. If $r = 0$, then

$$\begin{cases} a + m \leq d, \\ 2a + m = 2d, \end{cases}$$

so $m = 0$, $a = d$, and A_η has good reduction. Suppose that $r = 1$. It then follows from 6.2 that $\text{Im}(T - 1)$ has weight -2 and that $V_\ell(A)/V_\ell(A)^I$ has weight 0 ; these spaces have the same dimension $m_1 \leq m$.

We then have

$$\dim V_\ell(A)^I = 2d - m_1 = 2a + m,$$

and

$$2 \dim \mathcal{A}_{S,s} \geq 2(a + m) \geq 2a + m + m_1 = 2d = 2 \dim \mathcal{A}_{S,s},$$

so A_η has stable reduction ($\dim \mathcal{A}_{S,s} = a + m$). At the same time, we have obtained $m_1 = m$; that is,

$$\mathrm{Im}(T - 1) \subset V_\ell(A)^I \simeq V_\ell(\mathcal{A}_{S,s}^\circ)$$

is the V_ℓ of the multiplicative part of $\mathcal{A}_{S,s}^\circ$.

6.7. To treat the general case, we shall use (0.5) (cf. 1.3). We reduce to the case in which:

- (a) the conditions of (0.5.2) or (0.5.3) are satisfied;
- (b) $\mathrm{Gal}(\bar{\eta}/\eta)$ acts trivially on A_ℓ ;
- (c) the action of I on $T_\ell(A)$ is unipotent.

Under these hypotheses, we shall prove that A_η has stable reduction. Hypothesis (a) allows us to apply (0.5.1) and write $S = \varprojlim S_i$ as in 1.3, whose notation we retain. Let s_i be the closed point of S_i .

For sufficiently large i , the identity component $\mathcal{A}^\circ$ of the Néron model $\mathcal{A}_S$ descends to a smooth group scheme $\mathcal{A}_i$ with connected fibres over S_i , and $(\mathcal{A}_i)_\ell$ is constant over $S_i - D_i$. The group $\pi_1(S_i - D_i, \bar{\eta})$ acts on $T_\ell(A)$ and acts trivially on $T_\ell(A)/\ell T_\ell(A) \simeq A_\ell$. As in 1.3, one sees that P_i acts trivially. We then have, by (1.3.1),

$$T_\ell(A)_I = T_\ell(A)_{I_i}, \quad T_\ell(\mathcal{A}_{i,s_i}) = T_\ell(\mathcal{A}_s^\circ) = T_\ell(A)^I = T_\ell(A)^{I_i},$$

and (6.2.1) is a diagram of $\mathrm{Gal}(\bar{s}_i/s_i)$ -modules to which the arguments of 6.4–6.6 may be applied.

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Exposé II

Finiteness properties of the fundamental group

by Michèle Raynaud

Let k be a separably closed field of characteristic $p \geq 0$, let X be a connected scheme of finite type over k , and let $\Pi_1^{(p)}(X)$ be the largest “prime-to- p ” quotient of $\Pi_1(X)$. In this exposé we prove that, when X is a surface, $\Pi_1^{(p)}(X)$ is topologically finitely presented, and that the same holds for arbitrary X if resolution of singularities is assumed.

For a surface, the proof method, due to J.-P. Murre, consists in reducing to a fibration over P^1 having properties (i), (ii), and (iii) of the following proposition.

Proposition 2.1. Let k be an algebraically closed field and let S be a projective, irreducible, smooth surface over k . Then there exists a blow-up

$$g : S' \longrightarrow S,$$

where S' is a regular surface equipped with a fibration

$$f : S' \longrightarrow P^1,$$

such that f has a section σ and the following conditions hold:

- (i) f is smooth at the points of $\sigma(P^1)$.
- (ii) The fibres of f are geometrically integral curves having only ordinary singularities.
- (iii) There exists a nonempty open subset U of P^1 such that the fibres of f over the points of U are smooth.

The proposition follows from the existence of a projective embedding $S \longrightarrow P^r$ for which a Lefschetz pencil exists (XVII 2.5). Indeed, let $D = \{H_t\}_{t \in P^1}$ be such a pencil of hyperplanes. Recall that it has the following properties (loc. cit. 2.2):

(a) The axis Δ of D meets S transversely; the closed subscheme $\Delta \cap S$ is therefore reduced and consists of finitely many closed points $x_1, \dots, x_n$ of S .

(b) There exists a nonempty open subset U of P^1 such that, for every $t \in U$, H_t meets S transversely.

(c) For every $t \in P^1 - U$, H_t meets S transversely except at one point, which is an ordinary quadratic singularity.

Through every point x of S not belonging to Δ there passes a unique hyperplane H_t . The map assigning $t \in P^1$ to x is a rational map $a : S \dashrightarrow P^1$ with domain of definition $S - (\Delta \cap S)$.

Consider the blow-up $g : S' \longrightarrow S$ of the closed subscheme $S \cap \Delta$. The scheme S' is regular, and the fibre S'_{x_i} over x_i is the projective bundle defined by the tangent space of S at x_i , hence is isomorphic to P^1 (EGA IV 19.4.2, 19.4.4). Moreover, the rational map $f = ag$ is everywhere defined (XVIII 3.1). Finally, to each x_i one can associate the section of f that sends $t \in P^1$ to the tangent vector at x_i of $H_t \cap S$.

Let us verify conditions (i), (ii), and (iii). For each $t \in P^1$, the curves S'_t and $S_t = H_t \cap S$ are isomorphic. Indeed, the morphism $S'_t \longrightarrow S_t$ is an isomorphism except possibly at the points x_i , where S_t is regular; moreover, since S'_t is a complete intersection in S' , it is integral, and is therefore isomorphic to S_t .

The asserted properties of the fibres follow from (a)–(c). Finally, f is smooth at the points of σ , since f is flat and the fibres S'_t are geometrically regular at $\sigma(t)$.

Corollary 2.2. With the notation of 2.1, let Y be a normal-crossings divisor in S , and let $Y' = Y \times_S S'$. Then f , g , and U may be chosen so that $Y' \times_{P^1} U$ is a divisor with normal crossings relative to U .

It is known that, if a Lefschetz pencil exists for an embedding $S \longrightarrow P^r$, the Lefschetz pencils form an open subset of the variety of lines in P^r (XVII 3.2.1). The same is true of the set of

Lefschetz pencils whose axis does not meet Y and for which there is a nonempty open subset U_1 of P^1 such that $Y \cap H_t$ is smooth for $t \in U_1$. It is then enough to construct f and g from a pencil satisfying these two conditions and replace U by $U \cap U_1$.

2.3.0. Let G be a profinite group. Recall that G is said to be topologically finitely generated if there is an integer n such that G is isomorphic to a quotient of the free pro-group $F(n)$ on n generators. Put $K = \text{Ker}(F(n) \rightarrow G)$. The group G is said to be topologically finitely presented if, in addition, there are finitely many elements $k_1, \dots, k_r$ of K such that the smallest closed normal subgroup of K containing $k_1, \dots, k_r$ is K .

Theorem 2.3.1. Let k be a separably closed field of characteristic $p \geq 0$, and let X be a connected scheme of finite type over k . Suppose that schemes of finite type and dimension at most $\dim X$ over an algebraic closure of k admit strong resolution of singularities (SGA 5 I 3.1.5). This condition is satisfied when $\dim X = 2$, by [1], or when $\text{char } k = 0$, by [2]. Then the fundamental group $\Pi_1^{(p)}(X)$ is topologically finitely presented.

By SGA 1 IX 4.10, we may suppose that k is algebraically closed.

1) Reduction to the case in which X is a regular surface.

If $f : X' \rightarrow X$ is a morphism of finite type, put $X'' = X' \times_X X'$, and let $(X'_a)_{a \in A}$ and $(X''_b)_{b \in B}$ be the families of connected components of X' and X'' , respectively. When f is an effective descent morphism for the category of finite étale coverings, the fundamental group $\Pi_1(X)$ can be computed explicitly

in terms of the fundamental groups $\Pi_1(X'_a)$ and $\Pi_1(X''_b)$. It follows (SGA 1 IX 5.2, 5.3) that if the $\Pi_1(X'_a)$ are topologically finitely generated, then so is $\Pi_1(X)$; and if the $\Pi_1(X'_a)$ are topologically finitely presented while the $\Pi_1(X''_b)$ are topologically finitely generated, then $\Pi_1(X)$ is topologically finitely presented.

Since X admits resolution of singularities, there is a regular scheme X' and a proper birational morphism $f : X' \rightarrow X$. Since f is an effective descent morphism for finite étale coverings, it suffices to prove the theorem under the additional assumption that X is regular. One first deduces that $\Pi_1(X)$ is topologically finitely generated; knowing that $\Pi_1(X')$ is topologically finitely presented and that the $\Pi_1(X''_b)$ are topologically finitely generated then shows that $\Pi_1(X)$ is topologically finitely presented.

Now let $(U_i)_{i \in I}$ be a finite affine open covering of X , and let $f : \bigsqcup U_i \rightarrow X$ be the canonical morphism. Since f is an effective descent morphism for finite étale coverings, it is enough to prove the theorem for the U_i . We are thus reduced to the case in which X is affine and smooth.

Assume henceforth that X is affine and smooth over k , with $\dim X > 2$. The Lefschetz theorem for quasi-projective schemes ([3] V 7.4) applies: if Y is a sufficiently general hyperplane section of X , then

$$\Pi_1(Y) \simeq \Pi_1(X).$$

It therefore suffices to prove the theorem for Y , reducing us to the case $\dim X = 2$.

2) The case in which X is a smooth surface over k .

By the strong resolution theorem for surfaces, there exist a smooth, integral, projective surface S and an open immersion $i : X \rightarrow S$ such that $Y = S - X$ is a normal-crossings divisor. We apply Corollary 2.2 and retain its notation.

We show that it is enough to prove that the fundamental group of $X' = X \times_S S'$ is topologically finitely presented. Since X and X' are regular and

$$\text{codim} \left(\bigcup_{1 \leq i \leq n} \{x_i\}, X \right) \geq 2,$$

every étale covering of X' induces on $X - \bigcup_{1 \leq i \leq n} \{x_i\}$ an étale covering which extends uniquely to an étale covering of X (SGA 2 1.11). This gives an isomorphism

$$\Pi_1(X') \simeq \Pi_1(X).$$

Consider the cartesian diagram

$$\begin{array}{ccc} X' & \longleftarrow & X'_U \\ \downarrow h & & \downarrow h_U \\ P^1 & \longleftarrow & U \end{array}$$

where $h = f|_{X'}$. Let L be the set of primes different from p , and let $\bar{\eta}$ be a geometric point over the generic point η of P^1 . We verify the hypotheses of SGA 1 XIII 4.8. The morphism h has a section σ , because f does. To prove condition (a) of SGA 1 XIII 4.7, note first that, since the fibres of h are geometrically integral, h is 0-acyclic and locally 0-acyclic (SGA 4 XV 4.1, 1.16). It remains to show that, for every locally constant sheaf of sets F on X' , (F, h) is cohomologically proper in dimensions ≤ 0 . If ξ is a closed point of P^1 and Z denotes the integral closure of $\text{Spec } \mathcal{O}_{P^1, \xi}$ in $\bar{\eta}$, then $X'_Z = X' \times_{P^1} Z$ satisfies condition (S_2) at its closed points and is regular at every other point, hence is normal. Consequently, an étale covering of X'_Z whose restriction to $X'_{\bar{\eta}}$ is connected is necessarily connected. This proves the assertion, and hence condition (a).

Since X'_U is the complement in S'_U of a divisor with normal crossings relative to U , h_U is locally 1-aspherical for L (SGA 4 XV 2.1); and, for every finite constant sheaf of L -groups F on X'_U , (F, h_U) is cohomologically proper in dimensions ≤ 1 (SGA 1 XIII 2.4), which gives condition (b). Since P^1 is normal and $T = P^1 - U$ has codimension 1, we have

$$\text{prof ét}_T(P^1) \geq 2,$$

which gives condition (c). The condition

$$\text{prof hop}_x(X') \geq 3$$

holds at every closed point of X' by the purity theorem, which gives condition (e). Finally, for every $t \in P^1$, h is smooth at $\sigma(t)$, and hence is locally 1-aspherical for L at $\sigma(t)$, which gives condition (d).

By SGA 1 XIII 4.7, if a is a geometric point of $X'_{\bar{\eta}}$, there is an exact sequence

$$1 \longrightarrow \Pi_1^{(p)}(X'_{\bar{\eta}}, a) \longrightarrow \Pi_1'(X'_U, a) \longrightarrow \Pi_1(U, a) \longrightarrow 1,$$

and the choice of the section σ defines a splitting of this exact sequence. Let K be the image of $\Pi_1(U, a)$ in $\text{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$. The group of coinvariants under K ,

$$\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K,$$

is the quotient of $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ by the closed normal subgroup generated by the elements $x^{-1}q(x)$, where $x \in \Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ and $q \in K$. By loc. cit. (4.7.4), there is an isomorphism

$$\Pi_1^{(p)}(X', a) \simeq \Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K.$$

Write $P^1 - U = \coprod_{1 \leq i \leq n} \{t_i\}$. For each i , let $\bar{T}_i$ be the strict localization of P^1 at a geometric point $\bar{t}_i$ over t_i , and let s_i be the generic point of $\bar{T}_i$. An étale covering of U whose restriction to every s_i is trivial extends to an étale covering of P^1 , and is therefore trivial. For each i , let $\bar{k}(s_i)$ be an algebraic closure of $k(s_i)$. The preceding assertion says that $\Pi_1(U, a)$ is the smallest closed normal subgroup containing the images of the Galois groups $\text{Gal}(\bar{k}(s_i)/k(s_i))$ in $\Pi_1(U, a)$. By I 5.3, the image K_i of $\text{Gal}(\bar{k}(s_i)/k(s_i))$ in $\text{Aut}(\Pi_1^{(p)}(X'_{\bar{\eta}}, a))$ is finite. Since K is the smallest closed normal subgroup containing all the K_i , and since $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)$ is topologically finitely presented (SGA 1 XIII 3.2), this proves that $\Pi_1^{(p)}(X'_{\bar{\eta}}, a)_K$, and hence also the isomorphic group $\Pi_1^{(p)}(X)$, is topologically finitely presented.

Remark 2.3.2. Resolution of singularities is not needed to prove Theorem 2.3.1 when X is the open complement of a normal-crossings divisor in a projective scheme smooth over k .

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Exposé VI

FORMAL DEFORMATION THEORY

by D. S. Rim

1. Formal existence theorem

We recall the notion of a cofibered category (S.G.A. 1 IV). Let $\underline{C}$ be a fixed category. A cofibered category $\underline{A}$ over $\underline{C}$ is, by definition, a category $\underline{A}$ together with a functor $P : \underline{A} \rightarrow \underline{C}$ such that:

Ax. 1. For every map f in $\underline{C}$ and every object a in $\underline{A}$ for which $P(a)$ is the source of f , there is a cocartesian f -morphism $\alpha : a \rightarrow b$. Thus, for every f -morphism $\alpha' : a \rightarrow b'$, there is a unique T -morphism $\tau : b \rightarrow b'$ in $\underline{A}$ such that $\alpha' = \tau\alpha$, where T is the target of f .

Ax. 2. A composite of cocartesian morphisms in $\underline{A}$ is again cocartesian.

The following properties of a cofibered category $P : \underline{A} \rightarrow \underline{C}$ follow immediately from the definitions.

(Cancellation) Let $a \xrightarrow[\beta]{\alpha} a'$ be two cocartesian maps such that $P(\alpha) = P(\beta)$. If there is a cocartesian map $b \xrightarrow{\gamma} a$ such that $\alpha\gamma = \beta\gamma$, then $\alpha = \beta$.

(Factorization) Given cocartesian maps $a \xrightarrow{\alpha} a'$ and $a \xrightarrow{\beta} a''$, there is a cocartesian map $a' \xrightarrow{\gamma} a''$ such that $\gamma\alpha = \beta$ if and only if there is a map $f : P(a') \rightarrow P(a'')$ in $\underline{C}$ such that $fP(\alpha) = P(\beta)$. Moreover, γ is unique.

Let $\underline{A}$ be a cofibered category over $\underline{C}$. For each object S of $\underline{C}$, denote by $\underline{A}(S)$ the subcategory of $\underline{A}$ whose objects are the objects a for which $P(a) = S$, and in which

$$\mathrm{Hom}_{\underline{A}(S)}(a, a') = \{\alpha \in \mathrm{Hom}_{\underline{A}}(a, a') \mid P(\alpha) = \mathrm{id}_S\}.$$

A cofibered groupoid over $\underline{C}$ is a cofibered category $\underline{A}$ over $\underline{C}$ such that $\underline{A}(S)$

is a groupoid for every object S of $\underline{C}$. Recall that a groupoid is a category in which every morphism is an isomorphism. Thus a cofibered groupoid over $\underline{C}$ is a cofibered category $P : \underline{A} \rightarrow \underline{C}$ in which every map of $\underline{A}$ is cocartesian. Furthermore, for every $\alpha \in \mathrm{Fl}(\underline{A})$, α is an isomorphism in $\underline{A}$ if and only if $P(\alpha)$ is an isomorphism.

Throughout the rest of this section, fix a complete noetherian local ring Λ with residue field k , and let $\underline{C}_\Lambda$ be the category of artinian local Λ -algebras with residue field k , whose maps are local homomorphisms over Λ . From 1.2 onward, a cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is always assumed to satisfy $\underline{A}(k) = \{e\}$, where $\{e\}$ denotes the category in which $\mathrm{Hom}_{\{e\}}(a, b)$ consists of exactly one element for every two objects a, b of $\{e\}$. A map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will simply be called a $\underline{C}_\Lambda$ -map if $P(a') = P(a)$ and $P(\alpha) = 1_{P(a)}$. By abuse of language, a map $a' \xrightarrow{\alpha} a$ in $\underline{A}$ will be called surjective if $P(\alpha) : P(a') \rightarrow P(a)$ is surjective in $\underline{C}_\Lambda$.

Ex. 1.1(a). Let $F : \underline{C} \rightarrow (\text{categories})$ be a functor. Define the category $\underline{F}$ as follows. An object of $\underline{F}$ is a pair (S, a) with $S \in \mathrm{ob}(\underline{C})$ and $a \in \mathrm{ob}F(S)$; a map $(S, a) \rightarrow (S', a')$ is a pair (f, u') , where $f : S \rightarrow S'$ is a map in $\underline{C}$ and $u' : F(f)a \rightarrow a'$ is a map in $F(S')$. The composite of

$$(S, a) \xrightarrow{(f, u')} (S', a') \xrightarrow{(g, u'')} (S'', a'')$$

is

$$(gf, u'' \cdot F(g)u') : (S, a) \rightarrow (S'', a'').$$

With respect to the functor $P : \underline{F} \rightarrow \underline{C}$ given by $(S, a) \mapsto S$, $\underline{F}$ becomes a cofibered category over $\underline{C}$. If $F(S)$ is a groupoid for every $S \in \mathrm{ob}(\underline{C})$, then $\underline{F}$ is a cofibered groupoid over $\underline{C}$. In particular, a functor $F : \underline{C} \rightarrow (\text{groups})$ yields a cofibered group, and a functor $F : \underline{C} \rightarrow (\text{Ens})$ yields a cofibered discrete groupoid. A natural transformation $\varphi : F \rightarrow G$, where $F, G : \underline{C} \rightarrow (\text{categories})$, induces a cocartesian functor $\varphi : \underline{F} \rightarrow \underline{G}$ over $\underline{C}$.

Ex. 1.1(b). Let X and Y be schemes over Λ together with a fixed

isomorphism

$$X \otimes_{\Lambda} k \xleftarrow{\tau} Y \otimes_{\Lambda} k.$$

Then the functor $\text{Isom}_{X,Y} : \underline{C}_{\Lambda} \longrightarrow (\text{groupoids})$ given by

$$\text{Isom}_{X,Y}(S) = \left\{ X \otimes_{\Lambda} S \xleftarrow{\sigma} Y \otimes_{\Lambda} S \mid \sigma \otimes_{\Lambda} k = \tau \right\}$$

yields a cofibered groupoid $P : \underline{\text{Isom}}_{X,Y} \longrightarrow \underline{C}_{\Lambda}$.

Ex. 1.1(c). Let X_0 be a fixed scheme over k , and let $\underline{M}_{X_0}$ be the category of deformations of X_0 over $\underline{C}_{\Lambda}$ (see §4). Thus an object of $\underline{M}_{X_0}$ is a pair (R, X) , where $R \in \underline{C}_{\Lambda}$ and X is a deformation of X_0 to R . Then the functor $P : \underline{M}_{X_0} \longrightarrow \underline{C}_{\Lambda}$ given by $(R, X) \longmapsto R$ defines a cofibered groupoid over $\underline{C}_{\Lambda}$.

Ex. 1.1(d). For any groupoid X , denote by $[X]$ the discrete groupoid consisting of the isomorphism classes of objects of X . If $P : \underline{F} \longrightarrow \underline{C}_{\Lambda}$ is a cofibered groupoid, we obtain a functor $[F] : \underline{C}_{\Lambda} \longrightarrow (\text{Ens})$ given by $[F](S) = [F(S)]$, and hence a cofibered discrete groupoid which, by abuse of notation, is again denoted by $[F]$. Definition 1.2. A cofibered groupoid $\underline{A}$ over $\underline{C}_{\Lambda}$ is called semi-homogeneous if the following properties hold:

(1) Every diagram

$$\begin{array}{ccc} a'' & \text{-----} & a' \\ \vdots & & \downarrow \\ a_1 & \text{-----} & a \end{array}$$

of solid arrows in $\underline{A}$, where $a' \longrightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(2) Let

$$\begin{array}{ccc} & & R \\ & \nearrow \pi_1 & \\ R \times_k k[\varepsilon] & & \\ & \searrow \pi_2 & \\ & & k[\varepsilon] \end{array}$$

be the projection maps, where $k[\varepsilon]$ is the ring of dual numbers over k , and let

$$\begin{array}{ccc} a' & & \\ & \searrow \alpha & \\ & & a \\ & \nearrow \beta & \\ b' & & \end{array}$$

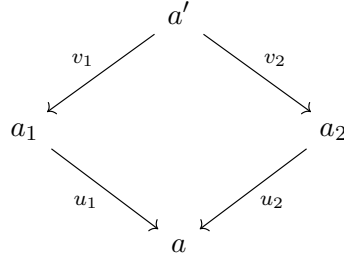
be π_1 -maps in $\underline{A}$. Then there is a $\underline{C}_{\Lambda}$ -map $\gamma : a' \longrightarrow b'$ such that $\alpha = \beta\gamma$ if and only if $(\pi_2)_*(a') \simeq (\pi_2)_*(b')$.

Remark 1.3(a). A reformulation of the notion in terms of 2-categories will be given later in 2.6(b). If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. In particular, 1.2(2) implies that the canonical map

$$[\underline{A}(R \times_k k[\varepsilon])] \longrightarrow [\underline{A}(R)] \times [\underline{A}(k[\varepsilon])]$$

is bijective.

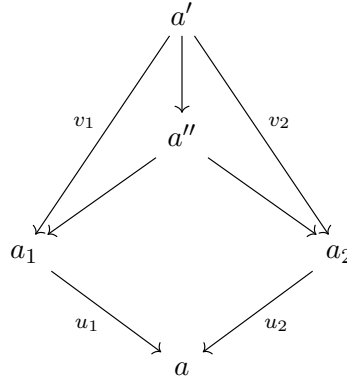
Remark 1.3(b). Consider a commutative diagram



in a cofibered category $\underline{A}$ over $\underline{C}_\Lambda$. Put $R = P(a)$, $R' = P(a')$, and $R_i = P(a_i)$ for $i = 1, 2$. If a'' is the direct image under the map

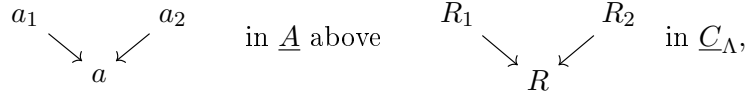
$$(P(v_1), P(v_2)) : R' \longrightarrow R_1 \times_R R_2,$$

we obtain a commutative diagram

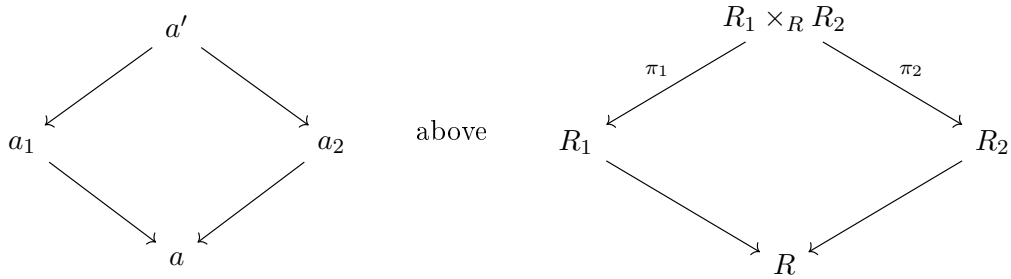


Thus 1.2(1) may be replaced by:

(1)' Every diagram



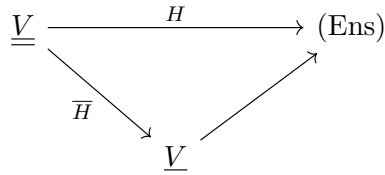
where $R_2 \longrightarrow R$ is surjective, can be completed to a commutative diagram



where π_i ($i = 1, 2$) are the projections.

Remark 1.3(c). If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, then $[\underline{A}(k[\varepsilon])]$ is canonically endowed with a vector space structure over k , where $k[\varepsilon]$ is the ring of dual numbers over k . Indeed, let $\underline{\underline{V}}$ be the category of

finite-dimensional vector spaces over k . Every product-preserving functor $H : \underline{\underline{V}} \longrightarrow (\text{Ens})$ factors canonically as



where $\underline{\underline{V}} \rightarrow (\text{Ens})$ is the forgetful functor. On the other hand, the functor

$$G : \underline{\underline{V}} \rightarrow \underline{\underline{C}}_\Lambda, \quad W \mapsto k[W] = k \oplus W, \quad W^2 = 0,$$

takes products in $\underline{\underline{V}}$ to fibre products over k in $\underline{\underline{C}}_\Lambda$. If $\underline{A}$ is a semi-homogeneous cofibered groupoid over $\underline{\underline{C}}_\Lambda$, then

$$[\underline{A}(k[V] \times_k k[W])] \rightarrow [\underline{A}(k[V])] \times [\underline{A}(k[W])]$$

is bijective by 1.3(a). Hence the functor $V \mapsto [\underline{A}(k[V])]$ commutes with products, proving the assertion.

Definition 1.4. (1) Let R be an object of $\underline{\underline{C}}_\Lambda$. A small extension of R in $\underline{\underline{C}}_\Lambda$ is a surjective map $R' \xrightarrow{f} R$ in $\underline{\underline{C}}_\Lambda$ such that $(\ker f)^2 = 0$ and $\ker f \simeq k$. In other words, $R' \xrightarrow{f} R$ is a small extension of R if

$$0 \rightarrow k \xrightarrow{t} R' \xrightarrow{f} R \rightarrow 0$$

is exact for some $t \in R'$ with $t^2 = 0$.

(2) Let $\underline{A}$ be a cofibered groupoid over $\underline{\underline{C}}_\Lambda$, and let a be an object of $\underline{A}$. A small prolongation of a is an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ such that $P(\alpha)$ is a small extension in $\underline{\underline{C}}_\Lambda$. An arrow $a' \xrightarrow{\alpha} a$ will be called versal for small prolongations of a if, for every small prolongation $a'_1 \rightarrow a$, there exists a commutative diagram

$$\begin{array}{ccc} a' & \xrightarrow{\quad} & a'_1 \\ & \searrow \alpha & \swarrow \\ & a & \end{array}$$

Proposition 1.5. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{\underline{C}}_\Lambda$.

(1) Let

$$\begin{array}{ccc} & & R \\ & \nearrow \pi_1 & \\ R \times_k k[\varepsilon] & & \\ & \searrow \pi_2 & \\ & & k[\varepsilon] \end{array}$$

be the projection maps in $\underline{\underline{C}}_\Lambda$, and let $a' \xrightarrow{\alpha} a$ be a π_1 -map. Then there exists a map $a \xrightarrow{\beta} a'$ such that $\alpha\beta = 1_a$ if and only if there exists a map $a \rightarrow (\pi_2)_* a'$.

(2) Let $R' \xrightarrow{f} R$ be a small extension in $\underline{\underline{C}}_\Lambda$, and consider a commutative diagram

$$\begin{array}{ccc} & a'' & \\ \alpha' \swarrow & & \searrow \alpha'_1 \\ a' & & a' \\ \alpha \searrow & & \swarrow \alpha_1 \\ & a & \end{array} \quad \text{above} \quad \begin{array}{ccc} & R' \times_R R' & \\ \swarrow & & \searrow \\ R' & & R' \\ \searrow & & \swarrow \\ & R & \end{array}$$

Define the canonical map $\tilde{f} : R' \times_R R' \rightarrow k[\ker f]$ by

$$\tilde{f}(r'_1, r'_2) = r'_1(0) + (r'_1 - r'_2).$$

If there exists a map $a' \rightarrow (\tilde{f})_*(a'')$, then there exists a map $\gamma : a' \rightarrow a'$ such that $\alpha = \alpha_1\gamma$.

Proof. (1) The “only if” part is clear. Assume conversely that there exists a map $\varphi : a \longrightarrow (\pi_2)_* a'$. Set

$$h : R \longrightarrow R \times_k k[\varepsilon], \quad h = 1 \times P(\varphi).$$

The composite $\pi_1 h$ is the identity. We therefore obtain maps

$$a \xrightleftharpoons[\alpha_1]{\gamma} h_* a, \quad \alpha_1 \gamma = 1_a, \quad P(\gamma) = h, \quad P(\alpha_1) = \pi_1.$$

On the other hand, $\pi_2 h = P(\varphi)$, so there exists a π_2 -map

$$h_* a \longrightarrow (\pi_2)_* a'.$$

Thus $(\pi_2)_*(h_* a) = (\pi_2)_* a'$, and 1.2(2) gives a $\underline{C}_\Lambda$ -map $u : h_* a \longrightarrow a'$ such that $\alpha u = \alpha_1$. Define $\beta = u\gamma$. Then

$$\alpha\beta = \alpha u\gamma = \alpha_1 \gamma = 1_a.$$

(2) The map

$$1 \times \tilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\ker f]$$

is an isomorphism compatible with the projections onto the first factor, and $\pi_2(1 \times \tilde{f}) = \tilde{f}$, where $\pi_2 : R' \times_k k[\ker f] \longrightarrow k[\ker f]$ is the second projection. Since $R' \xrightarrow{f} R$ is a small extension, $k[\ker f] \simeq k[\varepsilon]$. Hence, if there exists a map

$$a' \longrightarrow (\tilde{f})_* a'' = (\pi_2)_*((1 \times \tilde{f})_* a''),$$

part (1) yields a map $\gamma' : a' \longrightarrow a''$ such that $\alpha'\gamma' = 1_{a'}$. Define $\gamma = \alpha'_1 \gamma' : a' \longrightarrow a'$. Then $\alpha_1 \gamma = \alpha$.

Corollary 1.6. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$\begin{array}{ccc} a' & & a_1 \\ & \searrow \alpha & \swarrow \alpha_1 \\ & a & \end{array}$$

in $\underline{A}$, where α_1 is a small prolongation. Assume that a is versal for small prolongations of e . Then there exists a map $\rho : a' \longrightarrow a_1$ such that $\alpha = \alpha_1 \rho$ if and only if there exists a map $f : P(a') \longrightarrow P(a_1)$ such that $P(\alpha) = P(\alpha_1)f$ in $\underline{C}_\Lambda$.

Proof. Assume that $P(\alpha) = P(\alpha_1)f$, where $f : P(a') \longrightarrow P(a_1)$. Replacing a' by its direct image under f , we may and shall assume that $P(\alpha) = P(\alpha_1)$. Put $R' = P(a_1)$ and $R = P(a)$. Since $\underline{A}$ is semi-homogeneous, we obtain a commutative diagram

$$\begin{array}{ccc} & a'' & \\ \beta \swarrow & & \searrow \beta_1 \\ a' & & a_1 \\ \alpha \swarrow & & \searrow \alpha_1 \\ & a & \end{array} \quad \text{above} \quad \begin{array}{ccc} & R' \times_R R' & \\ \swarrow & & \searrow \\ R' & & R' \\ \swarrow & & \searrow \\ & R & \end{array}$$

Since a is versal for small prolongations of e , so is a' ; the assertion therefore follows from 1.5(2).

Definition 1.7. For each object R of $\underline{C}_\Lambda$, put

$$\bar{\Omega}_R = M/(m_\Lambda R + M^2),$$

where m_Λ and M are the maximal ideals of Λ and R , respectively. This is a finite-dimensional vector space over k , and $R \mapsto \overline{\Omega}_R$ defines a functor

$$\overline{\Omega} : \underline{C}_\Lambda \longrightarrow \underline{V},$$

where $\underline{V}$ is the category of finite-dimensional vector spaces over k . If $\underline{A}$ is a cofibered category over $\underline{C}_\Lambda$, the composite functor $\underline{A} \xrightarrow{P} \underline{C}_\Lambda \longrightarrow \underline{V}$ is again denoted $\overline{\Omega}$, by abuse of notation.

Proposition 1.8. Let $\underline{A}$ be a semi-homogeneous cofibered groupoid over $\underline{C}_\Lambda$. Given an object a of $\underline{A}$ that is versal for small prolongations of e , there exists an arrow $a' \longrightarrow a$ such that:

- (i) a' is versal for small prolongations of a ;
- (ii) $P(a') \longrightarrow P(a)$ is surjective, and its kernel is annihilated by the maximal ideal of $P(a)$;
- (iii) $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

Proof. Put $R = P(a)$, and consider an exact sequence

$$0 \longrightarrow J \longrightarrow S \longrightarrow R \longrightarrow 0,$$

where S is a formal power-series ring over Λ in finitely many variables and $J \subset M^2$, with M the maximal ideal of S . Let Σ be the set of ideals J_1 of S such that

- (i) $J \supset J_1 \supset MJ$, and
- (ii) there exists a π_1 -morphism $a_1 \longrightarrow a$ in $\underline{A}$, where $\pi_1 : S/J_1 \longrightarrow R$ is the canonical map.

Property 1.2(1) of $\underline{A}$, together with the fact that J/MJ is a finite-dimensional vector space over k , implies that Σ has a smallest ideal. Denote it by J' , and choose a π' -morphism $a' \xrightarrow{g} a$, where $\pi' : S/J' \longrightarrow R$ is the canonical map. Since $J' \subset J \subset M^2$, the map $\overline{\Omega}_{a'} \longrightarrow \overline{\Omega}_a$ is an isomorphism.

We claim that $a' \xrightarrow{g} a$ is versal for small prolongations of a . Indeed, let $a_1 \longrightarrow a$ be a small prolongation, and put $R_1 = P(a_1)$. We obtain an exact commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & J/MJ & \longrightarrow & S/MJ & \longrightarrow & R \longrightarrow 0 \\ & & \downarrow h & & \downarrow h_1 & & \parallel \\ 0 & \longrightarrow & k & \longrightarrow & R_1 & \longrightarrow & R \longrightarrow 0. \end{array}$$

If $h = 0$, then $\pi_1 : R_1 \longrightarrow R$ admits a cross-section, so h_1 induces a map $f : S/J' \longrightarrow R_1$. If $h \neq 0$, then $h_1 : S/MJ \longrightarrow R_1$ is surjective and hence, by the definition of J' , again induces $f : S/J' \longrightarrow R_1$. In either case we obtain a commutative diagram

$$\begin{array}{ccc} S/J' & \xrightarrow{f} & R_1 \\ & \searrow \pi' & \swarrow \pi_1 \\ & R & \end{array}$$

Since a is versal for small prolongations of e , Corollary 1.6 gives a commutative diagram

$$\begin{array}{ccc} a' & \xrightarrow{\rho} & a_1 \\ & \searrow & \swarrow \\ & a & \end{array}$$

as required.

Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category in groupoids. It then gives a cofibered category in groupoids

$$\text{pro}(P) : \text{pro}(\underline{A}) \longrightarrow \text{pro}(\underline{C}_\Lambda)$$

(for the definition of the category of pro-objects $\text{pro}(\underline{A})$, see A. Grothendieck, Séminaire Bourbaki 195), fitting into the commutative diagram

$$\begin{array}{ccc} \underline{A} & \longrightarrow & \text{pro}(\underline{A}) \\ P \downarrow & & \downarrow \text{pro}(P) \\ \underline{C}_\Lambda & \longrightarrow & \text{pro}(\underline{C}_\Lambda). \end{array}$$

Let $\widehat{\underline{C}}_\Lambda$ be the full subcategory of $\text{pro}(\underline{C}_\Lambda)$ whose objects are systems $(R_i)_{i \in \mathbb{N}}$ with $R_i \in \underline{C}_\Lambda$ such that each map $\varphi_i : R_i \rightarrow R_{i-1}$ is surjective and, for all sufficiently large i , induces an isomorphism

$$\frac{m_i}{m_\Lambda R_i + m_i^2} \xrightarrow{\sim} \frac{m_{i-1}}{m_\Lambda R_{i-1} + m_{i-1}^2},$$

where $\mathbb{N}$ denotes the ordered category of natural numbers and m_i is the maximal ideal of R_i .

Let $\widehat{\underline{A}}$ be the full subcategory of $\text{pro}(\underline{A})$ defined by

$$\text{ob}(\widehat{\underline{A}}) = \left\{ a \in \text{ob}(\text{pro}(\underline{A})) \mid \text{pro}(P)(a) \in \text{ob}(\widehat{\underline{C}}_\Lambda) \right\}.$$

We obtain a cofibered category in groupoids $\widehat{P} : \widehat{\underline{A}} \rightarrow \widehat{\underline{C}}_\Lambda$ and a commutative diagram

$$\begin{array}{ccc} \underline{A} & \hookrightarrow & \widehat{\underline{A}} \\ P \downarrow & & \downarrow \widehat{P} \\ \underline{C}_\Lambda & \hookrightarrow & \widehat{\underline{C}}_\Lambda. \end{array}$$

The category $\widehat{\underline{C}}_\Lambda$ is canonically equivalent to the category of complete local Λ -algebras of formally finite type over Λ with fixed residue field k ; we henceforth make this identification.

Definition 1.9. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. An object ξ of $\widehat{\underline{A}}$ is called quasi-initial if, for every object a of $\widehat{\underline{A}}$, there exists a map $\xi \rightarrow a$ in $\widehat{\underline{A}}$. A quasi-initial object ξ is called minimal if the morphism $\xi : h_R \rightarrow [\underline{A}]$ induces a bijection

$$\xi(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\sim} [\underline{A}(k[\varepsilon])], \quad R = P(\xi).$$

An object ξ of $\widehat{\underline{A}}$ is called a versal formal object of $\underline{A}$ if every surjective map $\alpha : a' \rightarrow a$ in $\underline{A}$ induces a surjective map

$$\text{Hom}_{\widehat{\underline{A}}}(\xi, a') \rightarrow \text{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

A versal formal object ξ is called minimal if $\xi : h_R \rightarrow [\underline{A}]$ induces the above bijection on $k[\varepsilon]$, where $R = P(\xi)$.

Proposition 1.10. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$.

- (i) Every (minimally) versal formal object of $\underline{A}$ is a quasi-initial (minimal) object of $\widehat{\underline{A}}$.
- (ii) Suppose that ξ is an object of $\widehat{\underline{A}}$ for which

$$\widetilde{\xi}(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \rightarrow [\underline{A}(k[\varepsilon])]$$

is injective, where $P(\xi) = R$. Then every endomorphism of ξ in $\widehat{\underline{A}}$ is an automorphism. In particular, a quasi-initial minimal object of $\widehat{\underline{A}}$, if it exists, is unique up to noncanonical isomorphism.

Proof. (i) Let ξ be a versal formal object of $\underline{A}$. For an object a of $\widehat{\underline{A}}$, choose a presentation by surjective maps in $\underline{A}$,

$$\cdots \rightarrow a_n \xrightarrow{\alpha_n} a_{n-1} \rightarrow \cdots \rightarrow a_1 \xrightarrow{\alpha_1} a_0, \quad a = \varprojlim_n a_n.$$

Suppose maps $\varphi_i : \xi \rightarrow a_i$ have been chosen with $\alpha_i \varphi_i = \varphi_{i-1}$ for all $i < n$. Since ξ is versal and α_n is surjective, there is a map $\varphi_n : \xi \rightarrow a_n$ with $\alpha_n \varphi_n = \varphi_{n-1}$. Thus $\varphi = \varprojlim_n \varphi_n$ defines a map $\xi \rightarrow a$.

(ii) Let $\alpha : \xi \rightarrow \xi$ be an endomorphism in $\widehat{\underline{A}}$. For every $\lambda \in \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon])$, we have

$$\lambda_* \xi \simeq (\lambda \circ P(\alpha))_* \xi.$$

Injectivity therefore gives $\lambda = \lambda \circ P(\alpha)$ for every λ . It follows that $P(\alpha) : R \rightarrow R$ is surjective. Every surjective endomorphism of a noetherian ring is an automorphism, so $P(\alpha)$, and hence α , is an automorphism.

Theorem 1.11 (Schlessinger). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then $\underline{A}$ admits a minimally versal formal object if and only if:

(i) $\underline{A}$ is semi-homogeneous;

(ii) $[\underline{A}(k[\varepsilon])]$ is a finite-dimensional vector space over k (cf. 1.3(c)).

Proof. Necessity. Let $\xi \in \widehat{\underline{A}}$ be a minimally versal formal object of $\underline{A}$, and consider a diagram

$$\begin{array}{ccc} a_1 & & a_2 \\ & \searrow & \swarrow \alpha \\ & a & \end{array}$$

in $\underline{A}$, where α is surjective. Since ξ is quasi-initial, there is a map $\xi \rightarrow a_1$. Since α is surjective, the composite $\xi \rightarrow a_1 \rightarrow a$ lifts to a_2 ; hence we obtain a commutative diagram

$$\begin{array}{ccc} & \xi & \\ \swarrow & & \searrow \\ a_1 & & a_2 \\ & \searrow & \swarrow \alpha \\ & a & \end{array}$$

which proves condition 1.2(1).

We now prove condition 1.2(2). Let

$$\begin{array}{ccc} & & S \\ & \nearrow \pi_1 & \\ S \times_k k[\varepsilon] & & \\ & \searrow \pi_2 & \\ & & k[\varepsilon] \end{array}$$

be the projections, and let

$$\begin{array}{ccc} a' & & b' \\ & \searrow \alpha & \swarrow \beta \\ & a & \end{array}$$

be π_1 -maps in $\underline{A}$. By the definition of ξ , there is a commutative diagram

$$\begin{array}{ccc} & \xi & \\ \swarrow u & & \searrow v \\ a' & & b' \\ & \searrow \alpha & \swarrow \beta \\ & a & \end{array}$$

and therefore $\pi_1 P(u) = \pi_1 P(v)$. Suppose now that $(\pi_2)_* a' \simeq (\pi_2)_* b'$. Minimality of ξ gives $\pi_2 P(u) = \pi_2 P(v)$, whence

$$P(u) = \pi_1 P(u) \times \pi_2 P(u) = \pi_1 P(v) \times \pi_2 P(v) = P(v).$$

Since $\underline{A}$ is a cofibered groupoid, there is a unique isomorphism $\sigma : a' \xrightarrow{\sim} b'$ in $\underline{A}(S \times_k k[\varepsilon])$ such that $\sigma u = v$. The equality $\beta \sigma u = \alpha u$ then implies $\beta \sigma = \alpha$.

Sufficiency. Let $c_1, \dots, c_n$ be objects of $\underline{A}(k[\varepsilon])$ whose classes form a k -basis of $[\underline{A}(k[\varepsilon])]$. Choose

$$a_1 \in \underline{A}(k[\varepsilon] \times_k \cdots \times_k k[\varepsilon])$$

such that $(\pi_j)_* a_1 \simeq c_j$. Then

$$\tilde{a}_1(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \longrightarrow [\underline{A}(k[\varepsilon])]$$

is bijective, where $R_1 = P(a_1)$. In particular, $a_1 \longrightarrow e$ is versal for small prolongations of e . Choose $a_2 \longrightarrow a_1$ versal for small prolongations of a_1 such that $\overline{\Omega}_{a_2} \longrightarrow \overline{\Omega}_{a_1}$ is bijective. Continue inductively, choosing $a_{n+1} \longrightarrow a_n$ versal for small prolongations and inducing a bijection on $\overline{\Omega}$, and set

$$\xi = \varprojlim_n a_n.$$

Then $\overline{\Omega}_\xi \longrightarrow \overline{\Omega}_{a_1}$ is bijective, so

$$\xi(k[\varepsilon]) : \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \xrightarrow{\sim} [\underline{A}(k[\varepsilon])]$$

is bijective.

It remains to prove versality. Consider a diagram

$$\begin{array}{ccc} & a' & \\ & \downarrow \alpha & \\ \xi & \xrightarrow{h} & a \end{array}$$

where α is surjective. We must lift h to a' . By induction we may assume that a' is a small prolongation of a . Since $P(a)$ is artinian, h factors through some a_q ,

$$\xi \longrightarrow a_q \longrightarrow a.$$

Semi-homogeneity gives a commutative diagram

$$\begin{array}{ccccc} & a'' & \longrightarrow & a' & \\ & \downarrow \alpha' & & \downarrow \alpha & \\ \xi & \longrightarrow & a_q & \longrightarrow & a, \end{array}$$

where α' is also a small prolongation. Since $a_{q+1} \longrightarrow a_q$ is versal for small prolongations of a_q , there is a commutative diagram

$$\begin{array}{ccccc} & & & a'' & \\ & & \nearrow & \downarrow \alpha' & \\ \xi & \longrightarrow & a_{q+1} & \longrightarrow & a_q, \end{array}$$

which supplies the required lift.

Remark 1.12. Let

$$\cdots \longrightarrow a_n \longrightarrow a_{n-1} \longrightarrow \cdots \longrightarrow a_2 \longrightarrow a_1 \longrightarrow a_0 = e$$

be an infinite sequence of arrows in $\underline{A}$ such that

$$\begin{cases} a_i \text{ is versal for small prolongations of } a_{i-1}, \\ P(a_i) \longrightarrow P(a_{i-1}) \text{ is surjective,} \\ \overline{\Omega}_{a_i} \longrightarrow \overline{\Omega}_{a_{i-1}} \text{ is bijective for } i \gg 0. \end{cases}$$

The preceding proof shows that $\varprojlim_i a_i$ is a versal formal object of $\underline{A}$.

Remark 1.13. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. If $\underline{A}$ satisfies 1.11(i) and (ii), then so does $[\underline{A}]$; moreover, $\xi \in \text{ob}(\widehat{A})$ is a minimally versal formal object of $\underline{A}$ if and only if it is so for $[\underline{A}]$, by 1.10(ii).

Remark 1.14. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ . Any invariant of $R = P(\xi)$, a complete local Λ -algebra of formally finite type over Λ , is automatically an invariant of $\underline{A}$ over $\underline{C}_\Lambda$. Thus, for example, we say that $\underline{A}$ is (formally) smooth over $\underline{C}_\Lambda$ if R is (formally) smooth over Λ , that is, a formal power-series algebra over Λ ; and we say that $\underline{A}$ is irreducible over $\underline{C}_\Lambda$ if $\text{Spec}(R)$ is irreducible.

In the next section we introduce smoothness for a functor $\underline{A} \longrightarrow \underline{B}$ over $\underline{C}_\Lambda$, where $\underline{A}$ and $\underline{B}$ are cofibered categories in groupoids over $\underline{C}_\Lambda$. It follows from 2.4 that $\underline{A}$ being (formally) smooth over $\underline{C}_\Lambda$ is equivalent to the functor $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ being smooth, so no ambiguity arises. We define

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim(k \otimes_\Lambda R).$$

For this, let $\underline{C}_k$ be the category of artinian local k -algebras with residue field k . The functor $\underline{C}_k \longrightarrow \underline{C}_\Lambda$ is fully faithful, and its left adjoint is

$$\underline{C}_\Lambda \longrightarrow \underline{C}_k, \quad R \longmapsto k \otimes_\Lambda R.$$

Let $\underline{A}|_{\underline{C}_k}$ be the pullback of $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ along $\underline{C}_k \longrightarrow \underline{C}_\Lambda$. If ξ is a minimally versal formal object for $\underline{A}$ over $\underline{C}_\Lambda$, then $k \otimes_\Lambda \xi$ is minimally versal for $\underline{A}|_{\underline{C}_k}$ over $\underline{C}_k$; the arrow $\xi \longrightarrow k \otimes_\Lambda \xi$ is cocartesian over $R \longrightarrow k \otimes_\Lambda R$. Consequently,

$$\dim_{\underline{C}_\Lambda} \underline{A} = \dim_{\underline{C}_k} (\underline{A}|_{\underline{C}_k}).$$

1.15. In the remainder of this paragraph, we briefly explain how to modify the preceding theory to allow residue-field extensions.

Let Λ and K be complete noetherian local rings, with K a Λ -algebra. Assume that:

- (a) the natural map $\Lambda \longrightarrow K$ is a local homomorphism;
- (b) the residue field K' of K is an extension of finite type of the residue field k of Λ .

The most interesting case is $K = K'$ with K a finite inseparable extension of k .

Denote by $\widehat{C}_{\Lambda, K}$ the category whose objects are complete noetherian local rings R equipped with a Λ -algebra structure and a Λ -epimorphism $s : R \longrightarrow K$. Let $C_{\Lambda, K}$ be its full subcategory consisting of pairs (R, s) for which $\ker(s)$ has finite length as an R -module. The category $\widehat{C}_{\Lambda, K}$ identifies with a full subcategory of $\text{Pro}(C_{\Lambda, K})$. If K is artinian, then every object of $C_{\Lambda, K}$ is artinian.

For a finite-dimensional vector space V over K' , write $D_V(K)$ for the algebra of dual numbers over K defined by V . Its elements have the form $x + v\varepsilon$, where $x \in K$, $v \in V$, and $\varepsilon^2 = 0$, with multiplication

$$(x + v\varepsilon)(x' + v'\varepsilon) = xx' + (xv' + x'v)\varepsilon.$$

The map

$$s : D_V(K) \longrightarrow K, \quad s(x + v\varepsilon) = x,$$

makes $D_V(K)$ an object of $C_{\Lambda, K}$. The role previously played by $k[\varepsilon]$ is now played by $D_{K'}(K)$.

As in 1.8, every cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ extends naturally to a cofibered category $\widehat{\underline{A}}$ over $\widehat{C}_{\Lambda,K}$. A map in $\underline{A}$ or $\widehat{\underline{A}}$ is called surjective when the corresponding algebra homomorphism in $C_{\Lambda,K}$ or $\widehat{C}_{\Lambda,K}$ is surjective.

We assume that $\underline{A}(K) = \{e\}$.

Definition 1.16. A cofibered groupoid $\underline{A}$ over $C_{\Lambda,K}$ is called semi-homogeneous if the following properties hold:

(I) Every diagram of solid arrows

$$\begin{array}{ccc} a'' & \xrightarrow{\quad\quad\quad} & a' \\ \downarrow & & \downarrow \\ a_1 & \xrightarrow{\quad\quad\quad} & a \end{array}$$

in $\underline{A}$, where $a' \rightarrow a$ is surjective, can be completed to a commutative diagram by including the dotted arrows.

(II) For every R , the natural map

$$[\underline{A}(R \times_K D_{K'}(K))] \longrightarrow [\underline{A}(R)] \times [\underline{A}(D_{K'}(K))]$$

is bijective.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$.

1.17. First restrict $\underline{A}$ to the full subcategory of $C_{\Lambda,K}$ whose objects are the $D_V(K)$. Put

$$\Omega = \Omega_{K/\Lambda}^1 \otimes_K K', \quad d : K \longrightarrow \Omega.$$

A map $f : D_V(K) \longrightarrow D_W(K)$ can be written

$$f(x + v\varepsilon) = x + (u(v) + D(dx))\varepsilon,$$

with $u \in \text{Hom}_{K'}(V, W)$ and $D \in \text{Hom}_{K'}(\Omega, W)$. Identifying f with (u, D) , composition is given by

$$(u, D) \circ (u', D') = (uu', D + uD').$$

Condition (II) implies:

(II') the functor from K' -vector spaces to sets

$$V \longmapsto [\underline{A}(D_V(K))]$$

commutes with products.

Lemma 1.18. Assume that condition (II') is satisfied by $\underline{A}$. Then:

(1) $[\underline{A}(D_{K'}(K))]$ carries a unique vector-space structure such that, functorially in V ,

$$[\underline{A}(D_V(K))] \simeq V \otimes_{K'} [\underline{A}(D_{K'}(K))].$$

(2) There is a linear map

$$\gamma : \text{Hom}(\Omega, K') \longrightarrow [\underline{A}(D_{K'}(K))]$$

such that, for every $D \in \text{Hom}(\Omega, K')$, the map

$$[\underline{A}((1, D))] : [\underline{A}(D_{K'}(K))] \longrightarrow [\underline{A}(D_{K'}(K))]$$

is $x \longmapsto x + \gamma(D)$.

(3) For every $(u, D) : D_V(K) \longrightarrow D_W(K)$, the induced map

$$(u, D) : [\underline{A}(D_V(K))] \longrightarrow [\underline{A}(D_W(K))],$$

that is,

$$(u, D) : V \otimes [\underline{A}(D_{K'}(K))] \longrightarrow W \otimes [\underline{A}(D_{K'}(K))],$$

is given by

$$x \longmapsto u(x) + \gamma(D).$$

Introduce the notation

$$\begin{aligned} F(V) &= [\underline{A}(D_V(K))], \\ A &= F(K'), \\ F(u, D) : F(V) &\longrightarrow F(W) \quad \text{for the map induced by } (u, D) : D_V(K) \longrightarrow D_W(K), \\ F(D) : F(V) &\longrightarrow F(V), \quad F(D) = F(1, D). \end{aligned}$$

By (II'), and because K' is a K' -vector-space object in the category of K' -vector spaces, $F(K')$ is a K' -vector-space object in the category of sets, hence a K' -vector space. This proves (1). We henceforth identify $F(V)$ with $V \otimes A$.

The identity

$$(u, D) = (1, D) \circ (u, 0)$$

gives

$$F(u, D) = F(D) \circ (u \otimes 1_A).$$

The functoriality of F amounts to the identities

$$\begin{aligned} F(0) &= \text{Id}, \\ F(D) \circ F(D') &= F(D + D'), \\ (u \otimes 1) \circ F(D) &= F(uD) \circ (u \otimes 1_A). \end{aligned}$$

For any $D : \Omega \longrightarrow V$ and any $n \in \mathbb{N}$, let

$$i : V \hookrightarrow V \oplus K'^n, \quad p : V \oplus K'^n \longrightarrow K'^n$$

be the canonical inclusion and projection. Then $piD = 0$, and the following diagram is commutative:

$$\begin{array}{ccccc} V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A \\ \downarrow F(D) & & \downarrow F(i, D) & & \parallel \\ V \otimes A & \longrightarrow & (V \oplus K'^n) \otimes A & \longrightarrow & K'^n \otimes A. \end{array}$$

Let $x : K'^n \longrightarrow V$ be a map, put $x_i = x(e_i)$, and let (a_i) be a family of elements of A . We have

$$(1_V + x) \circ i \circ D = D.$$

Consequently, if

$$F(i, D) \left(\sum_i e_i a_i \right) = \sum_i e_i a_i + a, \quad a \in V \otimes A,$$

then

$$F(D) \left(\sum_i x_i a_i \right) = \sum_i x_i a_i + a,$$

and a is independent of the x_i . Hence

$$\begin{aligned} F(D)(y) &= y + G(D), \\ G(D + D') &= G(D) + G(D'), \\ uG(D) &= G(uD) \quad \text{for every } u : V \longrightarrow W. \end{aligned}$$

Every such G is given, for some $\gamma \in \Omega \otimes A$, by

$$G(D) = (D \otimes 1_A)(\gamma).$$

This proves (2) and (3).

Definition 1.19. (1) An object $\xi \in \widehat{\underline{A}}$ is called a versal formal object if every surjective map $\alpha : a' \rightarrow a$ in $\underline{A}$ induces a surjective map

$$\mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a') \rightarrow \mathrm{Hom}_{\widehat{\underline{A}}}(\xi, a).$$

(2) A versal formal object $\xi \in \widehat{\underline{A}}$, with image (R, s) in $\widehat{C}_{\Lambda, K}$, is called minimal if the following condition holds. For any two maps

$$\varphi_1, \varphi_2 : R \rightrightarrows D_{K'}(K),$$

if $\varphi_1(\xi)$ is isomorphic to $\varphi_2(\xi)$, then there is a derivation $D : K \rightarrow K'$ such that

$$(1, D)(\varphi_1) = \varphi_2, \quad \text{that is,} \quad \varphi_2 - \varphi_1 = Ds.$$

Theorem 1.20. If $\underline{A}$ is semi-homogeneous and

$$\dim_{K'}[\underline{A}(D_{K'}(K))] < \infty,$$

then $\underline{A}$ admits a minimally versal formal object, and any two such objects are isomorphic.

With the notation of 1.18, let H^* be a subspace of $[\underline{A}(D_{K'}(K))]$ complementary to the image of γ , and let H be the dual of H^* . Set

$$R_0 = D_H(K).$$

Choose $\xi_0 \in \underline{A}(R_0)$ whose class, under the identifications

$$[\underline{A}(R_0)] \simeq H \otimes [\underline{A}(D_{K'}(K))] \simeq \mathrm{Hom}(H^*, [\underline{A}(D_{K'}(K))]),$$

is the inclusion $H^* \hookrightarrow [\underline{A}(D_{K'}(K))]$. Then:

(1) for every V , the maps

$$\mathrm{Hom}_{C_{\Lambda, K}}(D_H(K), D_V(K)) \rightarrow [\underline{A}(D_V(K))]$$

are surjective;

(2) the minimality condition of 1.19(2) holds for (R_0, ξ_0) .

Choose $S \in \widehat{C}_{\Lambda, K}$ formally smooth over Λ , together with an epimorphism $t : S \rightarrow R_0$, such that

$$\Omega_{R_0/\Lambda} \otimes K' \xrightarrow{\sim} \Omega_{S/\Lambda} \otimes K'.$$

If M is the maximal ideal of S , define inductively, for $n \geq 1$,

$$R_n = S/I_n, \quad \xi_n \in \mathrm{Ob}(\underline{A}(R_n)),$$

by the following conditions:

(a) I_{n+1} is the intersection of the ideals I such that

$$I_n M \subset I \subset I_n$$

and ξ_n extends to an object $\xi \in \underline{A}(S/I)$;

(b) ξ_{n+1} is an extension of ξ_n .

If

$$R = \varprojlim_n R_n, \quad \xi = \varprojlim_n \xi_n,$$

then (R, ξ) is a minimally versal formal object, and it is unique up to isomorphism.

2. Prorepresentable cofibered groupoids

Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a minimally versal formal object ξ . In general, ξ alone does not determine $\underline{A}$ up to $\underline{C}_\Lambda$ -equivalence. One must also know when two maps

$$R \xrightarrow[g]{f} S$$

give rise to a $\underline{C}_\Lambda$ -isomorphism $\xi(f) \rightarrow \xi(g)$, where $R = P(\xi)$. As an application of 1.11 and 1.12, we now consider the problem of recovering $\underline{A}$, up to $\underline{C}_\Lambda$ -equivalence. We obtain a satisfactory answer for a broad class of cofibered groupoids.

Many cofibered groupoids that arise in practice satisfy a property slightly stronger than semi-homogeneity; it will be defined explicitly in 2.5. To investigate them, it is convenient to recast the preceding discussion in terms of “smooth functors.” All the notation and terminology of the preceding section remain in force throughout this section.

Definition 2.1. Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. We say that φ is smooth if every surjective map $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram

$$(*) \quad \begin{array}{ccc} \varphi(a') & \xrightarrow{\quad} & b' \\ & \searrow \varphi(\alpha) & \downarrow \beta \\ & & \varphi(a) \end{array}$$

for some $\alpha : a' \rightarrow a$ in $\underline{A}$. We say that φ is minimally smooth if it is smooth and

$$\varphi(k[\varepsilon]) : [\underline{A}(k[\varepsilon])] \rightarrow [\underline{B}(k[\varepsilon])]$$

is bijective.

Remark 2.2(a). Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. The following conditions are equivalent:

- (i) φ is smooth;
- (ii) every surjective map $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed to a diagram $(*)$ in which $\varphi(a') \rightarrow b'$ is a $\underline{C}_\Lambda$ -isomorphism;
- (iii) every small prolongation $\beta : b' \rightarrow \varphi(a)$ in $\underline{B}$ can be completed to a commutative diagram $(*)$.

Remark 2.2(b). For every cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$, the natural functor

$$\underline{A} \rightarrow [\underline{A}]$$

is minimally smooth.

Remark 2.2(c). For each $R \in \widehat{\underline{C}}_\Lambda$, let

$$h_R : \underline{C}_\Lambda \rightarrow (\text{Ens})$$

be the functor defined by

$$h_R(S) = \text{Hom}_{\Lambda\text{-alg}}(R, S) \quad (S \in \underline{C}_\Lambda).$$

It determines a cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$. A map $R \rightarrow S$ in $\widehat{\underline{C}}_\Lambda$ induces a functor

$$\underline{h}_S \rightarrow \underline{h}_R.$$

This functor is smooth if and only if S is a formal power-series ring over R .

Remark 2.2(d). Given $R \in \widehat{\underline{C}}_\Lambda$, denote by $(R, \underline{C}_\Lambda)$ the category of arrows in $\widehat{\underline{C}}_\Lambda$ with source R and target in $\underline{C}_\Lambda$. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Given $\xi \in \text{Ob } \widehat{\underline{A}}$ with $P(\xi) = R$, let

$$\underline{A}' \rightarrow (R, \underline{C}_\Lambda)$$

be the pullback of $\underline{A} \rightarrow \underline{C}_\Lambda$ along the canonical functor $(R, \underline{C}_\Lambda) \rightarrow \underline{C}_\Lambda$. Choose a section $\xi^\#$ such that $\xi^\#(1_R) = \xi$. Thus, for every map $f : R \rightarrow S$, we choose an f -map $\xi \rightarrow \xi(f)$.

We obtain a functor

$$\xi^\# : \underline{h}_R \rightarrow \underline{A}$$

over $\underline{C}_\Lambda$ by sending $(R \xrightarrow{f} S)$ to $\xi(f)$. Up to natural equivalence, this functor does not depend on the choice of $\xi^\#$.

The assertion that $\xi^\# : \underline{h}_R \rightarrow \underline{A}$ is (minimally) smooth means that every surjective map $a' \xrightarrow{\alpha} \xi(f)$ in $\underline{A}$ can be completed to a commutative diagram

$$\begin{array}{ccc} \xi(f') & \xrightarrow{\quad} & a' \\ & \searrow \xi(\quad) & \downarrow \alpha \\ & & \xi(f) \end{array}$$

(and, in the minimal case, that $\underline{h}_R(k[\varepsilon]) \rightarrow \underline{A}(k[\varepsilon])$ is bijective). This means precisely that ξ is a (minimally) versal formal object of $\underline{A}$ over $\underline{C}_\Lambda$.

Thus the existence of a (minimally) versal formal object of $\underline{A}$ is equivalent to the existence of a (minimally) smooth functor

$$\underline{h}_R \rightarrow \underline{A}$$

over $\underline{C}_\Lambda$ for some $R \in \text{Ob}(\widehat{\underline{C}}_\Lambda)$.

The following general properties of smoothness are immediate.

Proposition 2.3. Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$.

(a) If φ and ψ are both smooth, then so is $\psi \circ \varphi$.

(b) If $\psi \circ \varphi$ is smooth and φ is surjective on isomorphism classes of objects, then ψ is smooth.

Remark 2.4. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$ admitting a versal formal object ξ . By 2.2(d), this means that

$$\xi : \underline{h}_R \rightarrow \underline{A}$$

is smooth, where $R = P(\xi)$. Therefore the following assertions are equivalent:

- $\underline{A}$ is formally smooth over $\underline{C}_\Lambda$ (cf. 1.13), that is, R is a formal power-series algebra over Λ ;
- the composite functor

$$\underline{h}_R \xrightarrow{\xi} \underline{A} \xrightarrow{P} \underline{C}_\Lambda (= \underline{h}_\Lambda)$$

is smooth;

- $P : \underline{A} \rightarrow \underline{C}_\Lambda$ is smooth.

The last equivalence follows from Proposition 2.3(b).

We now consider a stronger property than semi-homogeneity.

Definition 2.5. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called homogeneous if every diagram

$$\begin{array}{ccc} & & a' \\ & & \downarrow \\ b' & \longrightarrow & a \end{array}$$

in which $a' \rightarrow a$ is surjective admits a fibre product $a' \times_a b'$ in $\underline{A}$ such that

$$P(a' \times_a b') = P(a') \times_{P(a)} P(b').$$

Remark 2.6(a). Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and consider a diagram

$$(*) \quad \begin{array}{ccc} & a' & \\ \swarrow & & \searrow \\ a_1 & & a_2 \\ \searrow & & \swarrow \\ & a & \end{array} \quad \text{in } \underline{A} \text{ above the diagram} \quad \begin{array}{ccc} & R_1 \times_R R_2 & \\ \swarrow \pi_1 & & \searrow \pi_2 \\ R_1 & & R_2 \\ \searrow & & \swarrow \\ & R & \end{array}$$

in $\underline{C}_\Lambda$, where the π_i are the projections. If a fibre product $a_1 \times_a a_2$ exists in $\underline{A}$, there is a unique map

$$\varphi : a' \longrightarrow a_1 \times_a a_2$$

compatible with the projections. If

$$P(a_1 \times_a a_2) = R_1 \times_R R_2,$$

then $P(\varphi)$ is the identity map of $R_1 \times_R R_2$; hence φ is a $\underline{C}_\Lambda$ -isomorphism, that is, $a' = a_1 \times_a a_2$. Thus, if $\underline{A}$ is homogeneous, every diagram $(*)$ defines a fibre product $a_1 \times_a a_2$ whenever $R_1 \longrightarrow R$ is surjective. It follows at once

that every homogeneous cofibered groupoid is, a fortiori, semi-homogeneous.

Remark 2.6(b). Consider a diagram of categories and functors

$$\begin{array}{ccc} X & & Y \\ & \searrow \varphi & \swarrow \psi \\ & Z & \end{array}$$

The ordinary fibre product $X \times_Z Y$ is the category whose objects are pairs (x, y) satisfying $\varphi(x) = \psi(y)$ and whose maps $(x, y) \longrightarrow (x', y')$ are pairs

$$\alpha : x \longrightarrow x', \quad \beta : y \longrightarrow y'$$

such that $\varphi(\alpha) = \psi(\beta)$. The fibre product $X \times_Z^2 Y$ in the sense of 2-categories is the category whose objects are triples $(x, y; \gamma)$, where

$$\varphi(x) \xrightarrow{\gamma} \psi(y)$$

is an isomorphism. A map $(x, y; \gamma) \longrightarrow (x', y'; \gamma')$ is a pair $\alpha : x \longrightarrow x'$ and $\beta : y \longrightarrow y'$ for which the diagram

$$\begin{array}{ccc} \varphi(x) & \xrightarrow{\varphi(\alpha)} & \varphi(x') \\ \gamma \downarrow & & \downarrow \gamma' \\ \psi(y) & \xrightarrow{\psi(\beta)} & \psi(y') \end{array}$$

commutes.

The notions of a (semi-)homogeneous cofibered groupoid and of a smooth functor can be rephrased in terms of fibre products of categories in this 2-categorical sense. Let $P : \underline{A} \longrightarrow \underline{C}_\Lambda$ be a cofibered category, and consider $R_1 \times_R R_2$ in $\underline{C}_\Lambda$. Choose direct-image functors

$$(\pi_i)_* : \underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_i), \quad i = 1, 2.$$

They induce a functor

$$\underline{A}(R_1 \times_R R_2) \longrightarrow \underline{A}(R_1) \times_{\underline{A}(R)}^2 \underline{A}(R_2).$$

Up to canonical natural equivalence, it is independent of the chosen direct-image functors. The groupoid $\underline{A}$ is homogeneous if and only if this functor is an equivalence whenever $R_1 \rightarrow R$ is surjective; it is semi-homogeneous if and only if

$$[\underline{A}(R_1 \times_R R_2)] \rightarrow [\underline{A}(R_1)] \times_{[\underline{A}(R)]} [\underline{A}(R_2)]$$

is surjective whenever $R_1 \rightarrow R$ is surjective, and is bijective whenever $R = k$ and $R_1 = k[\varepsilon]$.

Now let $\varphi : \underline{A} \rightarrow \underline{B}$ be a functor of cofibered groupoids over $\underline{C}_\Lambda$. For every map $R' \rightarrow R$ in $\underline{C}_\Lambda$, φ induces, up to canonical natural equivalence, a functor

$$\underline{A}(R') \rightarrow \underline{A}(R) \times_{\underline{B}(R)}^2 \underline{B}(R').$$

The functor φ is smooth if and only if this functor is surjective on isomorphism classes of objects whenever $R' \rightarrow R$ is surjective.

Remark 2.6(c). If $\underline{A}$ is homogeneous, then for any finite-dimensional k -vector spaces V and W the functor

$$\underline{A}(k[V \times W]) \rightarrow \underline{A}(k[V]) \times \underline{A}(k[W])$$

is an equivalence of categories. Recall our convention $\underline{A}(k) = \{e\}$. In particular,

$$\text{Aut}_{\underline{A}(k[V \times W])}(1_{V \times W}) \xrightarrow{\sim} \text{Aut}_{\underline{A}(k[V])}(1_V) \times \text{Aut}_{\underline{A}(k[W])}(1_W),$$

where 1_V is the direct image of e under the unique map $k \rightarrow k[V]$ in $\underline{C}_\Lambda$. Consequently, if $\underline{A}$ is homogeneous, both $\text{Aut}(1_\varepsilon)$ and $[\underline{A}(k[\varepsilon])]$ carry canonical k -vector-space structures.

Remark 2.6(d). For every $R \in \widehat{\underline{C}}_\Lambda$, the cofibered discrete groupoid $\underline{h}_R$ over $\underline{C}_\Lambda$ is homogeneous.

If $\underline{A}$ is semi-homogeneous, then so is $[\underline{A}]$. Homogeneity of $\underline{A}$, however, does not in general imply homogeneity of $[\underline{A}]$.

Proposition 2.7. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$. The following conditions are equivalent.

(i) $[\underline{A}]$ is homogeneous; that is,

$$[\underline{A}(R' \times_R S)] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is bijective for every small extension $R' \rightarrow R$.

(ii) For every small prolongation $a' \rightarrow a$ in $\underline{A}$, the map

$$\text{Aut}_{\underline{A}(R')}(a') \rightarrow \text{Aut}_{\underline{A}(R)}(a)$$

is surjective, where $R' = P(a')$ and $R = P(a)$.

Proof. (i) $\Rightarrow$ (ii). Let $f : R' \rightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $a' \rightarrow a$ be an f -map in $\underline{A}$. Given $\tau \in \text{Aut}_{\underline{A}(R)}(a)$, consider $a' \times_a a'$ and $a' \times_a a'_\tau$, where $a'_\tau \rightarrow a$ is the composite

$$a' \rightarrow a \xrightarrow{\tau} a.$$

Since

$$[\underline{A}(R' \times_R R')] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(R')]$$

is bijective, there is an isomorphism

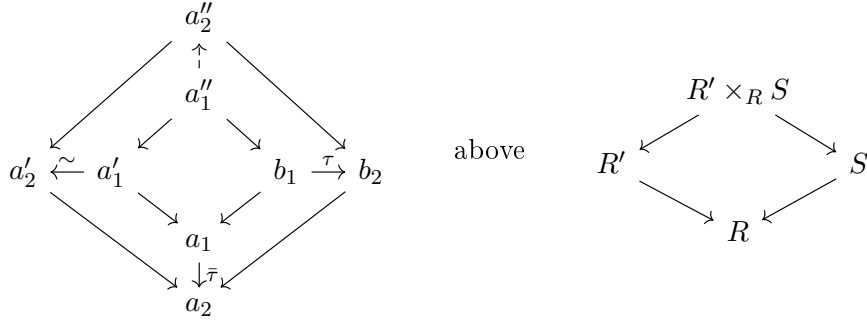
$$a' \times_a a' \xrightarrow{u} a' \times_a a'_\tau.$$

Projection onto the second factor then gives an isomorphism $\tau' : a' \rightarrow a'$ lifting τ .

(ii) $\Rightarrow$ (i). Let a''_i ($i = 1, 2$) be objects of $\underline{A}(R' \times_R S)$ with the same image under

$$[\underline{A}(R' \times_R S)] \rightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)].$$

We obtain a diagram of solid arrows



where $\bar{\tau} : a_1 \longrightarrow a_2$ is the isomorphism induced by τ , so the lower-right square commutes. Under condition (ii), the given isomorphism $a'_1 \xrightarrow{\sim} a'_2$ can be replaced so that the left square also commutes. Since

$$a''_i = a'_i \times_{a_i} b_i \quad (i = 1, 2)$$

by Remark 2.6(a), we obtain an isomorphism $a''_1 \xrightarrow{\sim} a''_2$. Hence

$$[\underline{A}(R' \times_R S)] \longrightarrow [\underline{A}(R')] \times_{[\underline{A}(R)]} [\underline{A}(S)]$$

is injective, and therefore bijective because $\underline{A}$ is homogeneous by hypothesis. $\square$

Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let $f : R' \longrightarrow R$ be a small extension in $\underline{C}_\Lambda$. There is a canonical map

$$\tilde{f} : R' \times_R R' \longrightarrow k[\text{Ker } f], \quad (r'_1, r'_2) \longmapsto r'_1(0) + (r'_1 - r'_2),$$

and

$$1 \times \tilde{f} : R' \times_R R' \longrightarrow R' \times_k k[\text{Ker } f]$$

is an isomorphism. Given two f -maps $a_i \longrightarrow a$ ($i = 1, 2$) in $\underline{A}$, the fibre product $a_1 \times_a a_2$ is an object over $R' \times_R R'$. We may therefore form

$$\tau(a_1, a_2) = (\tilde{f})_*(a_1 \times_a a_2) \in \underline{A}(k[\text{Ker } f]).$$

Since $1 \times \tilde{f}$ is an isomorphism, there is an isomorphism

$$a_1 \times_a a_2 \simeq a_1 \times \tau(a_1, a_2)$$

compatible with projection onto the first factor. Statement 1.5(2) now takes the following more definite form.

Lemma 2.8. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let

$$a_i \xrightarrow{\alpha_i} a \quad (i = 1, 2)$$

be f -maps in $\underline{A}$.

(1) There exists $\rho : a_1 \longrightarrow a_2$ in $\underline{A}$ with $\alpha_2 \circ \rho = \alpha_1$ if and only if there is an arrow $a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$.

(2) There exists such a ρ with $P(\rho) = \text{id}_{R'}$ if and only if

$$\tau(a_1, a_2) = 0 \quad \text{in} \quad [\underline{A}(k[\text{Ker } f])].$$

Proof. We prove both assertions simultaneously. The existence of $\rho : a_1 \longrightarrow a_2$ satisfying $\alpha_2 \circ \rho = \alpha_1$ (and $P(\rho) = \text{id}_{R'}$) is equivalent successively to the existence of

$$h : a_1 \longrightarrow a_1 \times_a a_2, \quad \text{pr}_1 \circ h = \text{id}_{a_1},$$

with $P(h)$ the diagonal map; to the existence of

$$g : a_1 \longrightarrow a_1 \times \tau(a_1, a_2), \quad \text{pr}_1 \circ g = \text{id}_{a_1},$$

with

$$P(g) : R' \longrightarrow R' \times_k k[\text{Ker } f], \quad r' \longmapsto (r', r'(0));$$

and to the existence of an arrow $\gamma : a_1 \longrightarrow \tau(a_1, a_2)$ in $\underline{A}$, with

$$P(\gamma) : R' \longrightarrow k[\text{Ker } f], \quad r' \longmapsto r'(0).$$

Such an arrow factors through $R' \longrightarrow k \longrightarrow k[\text{Ker } f]$ if and only if $\tau(a_1, a_2) = 0$. □

Proposition 2.9. (a) Let

$$\underline{A} \xrightarrow{\varphi} \underline{B} \xrightarrow{\psi} \underline{C}$$

be functors of cofibered groupoids over $\underline{C}_\Lambda$. Assume that

- (i) $\underline{B}$ is semi-homogeneous and $\underline{C}$ is homogeneous;
- (ii) the map

$$\psi(k[\varepsilon]) : [\underline{B}(k[\varepsilon])] \longrightarrow [\underline{C}(k[\varepsilon])]$$

is injective.

Then $\psi \circ \varphi$ is smooth if and only if φ and ψ are both smooth.

(b) Let $\varphi : \underline{A} \longrightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. If $\underline{A}$ is semi-homogeneous and $\underline{B}$ is homogeneous, then

$$\widehat{\varphi} : \widehat{\underline{A}} \longrightarrow \widehat{\underline{B}}$$

is smooth. Explicitly, for every surjective map $b' \xrightarrow{\beta} \widehat{\varphi}(a)$ in $\widehat{\underline{B}}$, there are a map $\alpha : a' \longrightarrow a$ in $\widehat{\underline{A}}$ and a $\widehat{\underline{C}}_\Lambda$ -isomorphism

$$\rho : \widehat{\varphi}(a') \xrightarrow{\sim} b'$$

such that $\beta \circ \rho = \varphi(\alpha)$.

Proof. (a) Suppose $\psi \circ \varphi$ is smooth. If φ is smooth, then ψ is smooth by Proposition 2.3(b), so it remains to prove that φ is smooth. Let $b' \xrightarrow{\beta} \varphi(a)$ be a small prolongation in $\underline{B}$. Since $\psi \circ \varphi$ is smooth, there are an arrow $a' \xrightarrow{\alpha} a$ in $\underline{A}$ and a $\underline{C}_\Lambda$ -isomorphism

$$\rho : \psi(b') \longrightarrow \psi\varphi(a')$$

for which

$$\begin{array}{ccc} \psi(b') & \xrightarrow{\rho} & \psi\varphi(a') \\ & \searrow \psi(\beta) & \swarrow \psi\varphi(\alpha) \\ & \psi\varphi(a) & \end{array}$$

commutes. Since $\underline{B}$ is semi-homogeneous, we can choose a commutative diagram

$$\begin{array}{ccc} & b'' & \\ b' \swarrow & & \searrow \varphi(a') \\ & \varphi(a) & \swarrow \varphi(\alpha) \end{array} \quad \text{above} \quad \begin{array}{ccc} & R' \times_R R' & \\ R' \swarrow & & \searrow R' \\ & R & \end{array}$$

where $R = P(a)$ and $R' = P(b') = P(a')$. Homogeneity of $\underline{C}$ gives

$$\psi(b'') = \psi(b') \times_{\psi\varphi(a)} \psi\varphi(a')$$

$$0 = \tau(\psi\varphi(a'), \psi(b')) = (\widetilde{f})_*\psi(b'').$$

(b) Let a surjective map $\beta : b' \longrightarrow \widehat{\varphi}(a)$ in $\underline{B}$ be given. It is represented by a sequence of commutative diagrams

in $\underline{B}$, where every α_v , β_v , and β'_v is surjective and $\beta = \varprojlim_v \beta_v$. It is enough to show inductively that every diagram of solid arrows below, with α_n , β_n , β_{n+1} , and β'_n surjective, can be completed by the dotted arrows:

$$\begin{array}{ccccc}
& a' & & R_{n+1} \times_{R_n} R'_n & \\
\gamma \swarrow & & \searrow \gamma' & \swarrow & \searrow \\
a_{n+1} & & a'_n & R_{n+1} & R'_n \\
\alpha_n \searrow & & \swarrow \alpha'_n & \searrow & \swarrow \\
& a_n & & R_n &
\end{array}
\quad \text{in } \underline{A} \text{ over } \quad \quad \quad \text{in } \underline{C}_\Lambda$$

$$\begin{array}{ccc}
 & \varphi(a') & \\
 \swarrow \varphi(\gamma) & & \searrow \varphi(\gamma') \\
 \varphi(a_{n+1}) & & \varphi(a'_n) \\
 \searrow \varphi(\alpha_n) & & \swarrow \varphi(\alpha'_n) \\
 & \varphi(a_n) &
 \end{array}$$
$$\rho : \varphi(a'_{n+1}) \xrightarrow{\sim} b'_{n+1}$$

such that $\delta \circ \rho = \varphi(\alpha')$. We obtain the commutative diagram

$$\begin{array}{ccccc}
\varphi(a'_{n+1}) & \xrightarrow{\varphi(\gamma' \circ \alpha')} & & \xrightarrow{\quad} & \varphi(a'_n) \\
& \searrow \rho & & & \nearrow \sim \\
& & b'_{n+1} & \xrightarrow{\quad} & b'_n \\
& \searrow \varphi(\gamma \circ \alpha') & \downarrow & & \downarrow \\
& & \varphi(a_{n+1}) & \xrightarrow{\quad} & \varphi(a_n)
\end{array}$$

$\varphi(\alpha'_n)$

which completes the induction and proves (b). $\square$

Corollary 2.10. (a) Let $\varphi : \underline{A} \rightarrow \underline{B}$ be a smooth functor of cofibered groupoids over $\underline{C}_\Lambda$. Let ξ be a versal formal object of $\underline{A}$, and let η be a minimally versal formal object of $\underline{B}$. If $\underline{B}$ is homogeneous, then for every map $\beta : \eta \rightarrow \varphi(\xi)$ in $\widehat{\underline{B}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\beta) : P(\eta) \rightarrow P(\varphi(\xi)) = P(\xi).$$

(b) Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, and let ξ and η be versal formal objects of $\underline{A}$. If η is minimal, then for every map $\alpha : \eta \rightarrow \xi$ in $\widehat{\underline{A}}$, the ring $P(\xi)$ is a formal power-series ring over $P(\eta)$ via

$$P(\alpha) : P(\eta) \rightarrow P(\xi).$$

Proof. (a) A map $\beta : \eta \rightarrow \varphi(\xi)$ induces a commutative diagram

$$\begin{array}{ccc}
\underline{h}_{P(\xi)} & \xrightarrow{\tilde{\beta}} & \underline{h}_{P(\eta)} \\
\downarrow \tilde{\xi} & & \downarrow \tilde{\eta} \\
\underline{A} & \xrightarrow{\varphi} & \underline{B}.
\end{array}$$

Since $\tilde{\xi}$ and φ are smooth, the composite $\tilde{\eta} \circ \tilde{\beta} = \varphi \circ \tilde{\xi}$ is smooth. On the other hand,

$$\tilde{\eta}(k[\varepsilon]) : \underline{h}_{P(\eta)}(k[\varepsilon]) \rightarrow [\underline{B}(k[\varepsilon])]$$

is an isomorphism. Proposition 2.9(a) therefore shows that $\tilde{\beta}$ is smooth, which means precisely that $P(\xi)$ is a formal power-series ring over $P(\eta)$.

(b) Apply (a) with $\underline{A} = \underline{B}$. $\square$ We conclude by considering the “representability” of a cofibered groupoid. Let $\underline{C}$ be a fixed category. Every object R of $\underline{C}$ defines a functor

$$h_R : \underline{C} \rightarrow (\text{Sets}), \quad h_R(S) = \text{Hom}_{\underline{C}}(R, S),$$

and hence a fully faithful functor

$$h : \underline{C}^{\text{op}} \rightarrow \underline{\text{Hom}}(\underline{C}, (\text{Sets})).$$

A *cocategory* in $\underline{C}$ is a diagram

$$h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1} \begin{array}{c} \xrightarrow{s_1} \\ \xleftarrow{\epsilon} \\ \xrightarrow{s_2} \end{array} h_{R_0} \quad \text{with } s_i \circ \epsilon = \text{id} \quad (i = 1, 2),$$

where the first arrow is denoted by c and

$$h_{R_1} \times_{h_{R_0}} h_{R_1} = (h_{R_1})_{s_1} \times_{h_{R_0} s_2} (h_{R_1}),$$

such that

- (i) c is associative and compatible with s_i for $i = 1, 2$;
- (ii) both composites

$$\begin{aligned} h_{R_1} &\xrightarrow{\sim} h_{R_0} \times_{h_{R_0}} h_{R_1} \xrightarrow{\epsilon \times 1} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1}, \\ h_{R_1} &\xrightarrow{\sim} h_{R_1} \times_{h_{R_0}} h_{R_0} \xrightarrow{1 \times \epsilon} h_{R_1} \times_{h_{R_0}} h_{R_1} \xrightarrow{c} h_{R_1} \end{aligned}$$

are identities.

If $(R_0, R_1, s_1, s_2, \epsilon, c)$ is a cocategory in $\underline{C}$, then for every $S \in \underline{C}$ we define a category $h_{(R_0, R_1, \dots)}(S)$ by

$$\text{ob } h_{(R_0, R_1, \dots)}(S) = \text{Hom}_{\underline{C}}(R_0, S), \quad \text{Fl } h_{(R_0, R_1, \dots)}(S) = \text{Hom}_{\underline{C}}(R_1, S),$$

with composition supplied by

$$c : h_{R_1} \times_{h_{R_0}} h_{R_1} \longrightarrow h_{R_1}.$$

The categories so obtained depend functorially on S and define a cofibered category $\underline{h}_{(R_0, R_1, \dots)}$ over $\underline{C}$. For example, every object R of $\underline{C}$ defines a discrete cocategory

$$\underline{h}_R = \underline{h}_{(R, R, s_1, s_2, \epsilon, c)}$$

by taking s_1, s_2, ϵ , and c all to be identity maps. A homomorphism

$$(R_0, R_1, \dots) \longrightarrow (R'_0, R'_1, \dots)$$

of cocategories is a pair of maps of represented functors $h_{R_0} \longrightarrow h_{R'_0}$ and $h_{R_1} \longrightarrow h_{R'_1}$ whose induced map

$$h_{(R_0, R_1, \dots)}(S) \longrightarrow h_{(R'_0, R'_1, \dots)}(S)$$

is a functor for every $S \in \underline{C}$. Every cocategory $(R_0, R_1, \dots)$ has, in particular, a homomorphism

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}.$$

A *cogroupoid* in $\underline{C}$ is a cocategory $(R_0, R_1, \dots)$ for which $h_{(R_0, R_1, \dots)}(S)$ is a groupoid for every $S \in \underline{C}$; equivalently, $\underline{h}_{(R_0, R_1, \dots)}$ is a category cofibered in groupoids over $\underline{C}$.

In practice, cocategories and cogroupoids arise as follows. Let $P : \underline{F} \longrightarrow \underline{C}$ be a cofibered category, and fix an object ξ of $\underline{F}$. Put $R_0 = P(\xi)$, and let $(R_0, \underline{C})$ denote the category of arrows in $\underline{C}$ with source R_0 . Let $P : \underline{F}' \longrightarrow (R_0, \underline{C})$ be the pullback of $\underline{F}$ along the canonical functor $(R_0, \underline{C}) \longrightarrow \underline{C}$. Choose a cocartesian section $\xi^\#$ of $\underline{F}'$ over $(R_0, \underline{C})$ such that $\xi^\#(\text{id}_{R_0}) = \xi$. Thus, for every arrow $R_0 \xrightarrow{f} S$ in $\underline{C}$, we choose a cocartesian f -map $\xi \longrightarrow \xi^\#(f)$.

Define

$$\text{Hom}_{\xi^\#} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\text{Sets})$$

by

$$X = (f_1, f_2) \longmapsto \text{Hom}_{\underline{F}(S)}(\xi^\#(f_1), \xi^\#(f_2)),$$

where S is the common target of f_1 and f_2 . Up to canonical natural equivalence, this functor is independent of the choice of $\xi^\#$; henceforth we write simply ξ for $\xi^\#$.

Suppose the functor is represented by an object

$$X_0 = \left(R_0 \begin{matrix} \xrightarrow{s_1} \\ \xrightarrow{s_2} \end{matrix} R_1 \right)$$

of $(R_0 \amalg R_0, \underline{C})$. Thus there is an R_0 -map $\varphi : \xi(s_1) \longrightarrow \xi(s_2)$ in $\underline{F}$ such that

$$\tilde{\varphi} : \text{Hom}_{(R_0 \amalg R_0, \underline{C})}(X_0, X) \longrightarrow \text{Hom}_{\xi}(X)$$

is bijective for every X in $(R_0 \amalg R_0, \underline{C})$. Then X_0 naturally defines a cocategory $(R_0, R_1, \dots)$ in $\underline{C}$. Indeed, a triple of arrows

$$R_0 \begin{array}{c} \xrightarrow{u} \\ \xrightarrow{v} \\ \xrightarrow{w} \end{array} S$$

in $\underline{C}$ defines a composition

$$\text{Hom}_{\underline{F}(S)}(\xi(v), \xi(w)) \times \text{Hom}_{\underline{F}(S)}(\xi(u), \xi(v)) \longrightarrow \text{Hom}_{\underline{F}(S)}(\xi(u), \xi(w)),$$

and hence a composition rule

$$h_{R_1}(S) \times_{h_{R_0}(S)} h_{R_1}(S) \longrightarrow h_{R_1}(S),$$

functorial in S . Consequently, Hom_{ξ} defines the cocategory $(R_0, R_1, \dots)$ in $\underline{C}$ and the cofibered category $\underline{h}_{(R_0, R_1, \dots)}$ over $\underline{C}$. Explicitly,

$$\text{ob } \underline{h}_{(R_0, R_1, \dots)} = \coprod_{S \in \text{ob}(\underline{C})} \text{Hom}_{\underline{C}}(R_0, S).$$

An arrow

$$(R_0 \xrightarrow{f} S) \longrightarrow (R_0 \xrightarrow{f'} S')$$

is a pair of arrows $S \xrightarrow{h} S'$ and $R_1 \xrightarrow{u} S'$ in $\underline{C}$ such that

$$u \circ s_1 = h \circ f, \quad u \circ s_2 = f'.$$

The structural functor $P : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{C}$ sends $(R_0 \xrightarrow{f} S)$ to S .

The pair (ξ, φ) now defines a functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{F} \quad \text{over } \underline{C}$$

by

$$(R_0 \xrightarrow{f} S) \longmapsto \xi(f)$$

on objects and by

$$\left[(R_0 \xrightarrow{f} S) \xrightarrow{(h, u)} (R_0 \xrightarrow{f'} S') \right] \longmapsto \left[\xi(f) \xrightarrow{h_*} \xi(hf) \xrightarrow{u(\varphi)} \xi(f') \right]$$

on arrows. For every $S \in \underline{C}$, the functor

$$(\xi, \varphi)(S) : \underline{h}_{(R_0, R_1, \dots)}(S) \longrightarrow \underline{F}(S)$$

is fully faithful by the representability of Hom_{ξ} . It follows that

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{F}$$

is a $\underline{C}$ -equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$, that is, if every object a of $\underline{F}$ admits a cocartesian arrow $\xi \longrightarrow a$.

One may instead consider the functor

$$\text{Isom}_{\xi} : (R_0 \amalg R_0, \underline{C}) \longrightarrow (\text{Sets}),$$

given by

$$X = (f_1, f_2) \mapsto \text{Isom}_{\underline{F}(S)}(\xi(f_1), \xi(f_2)),$$

where S is the common target of f_1 and f_2 . If this functor is represented by an object $(R_0 \rightrightarrows R_1)$ of $(R_0 \amalg R_0, \underline{C})$, then it defines a cogroupoid $(R_0, R_1, \dots)$ in $\underline{C}$, together with a functor

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{F}.$$

When $\underline{F}$ is a cofibered groupoid, this functor is a $\underline{C}$ -equivalence if and only if ξ is quasi-initial in $\underline{F}$ over $\underline{C}$. Definition and Proposition 2.11. A cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$ is called smooth if either, and hence both, of the following equivalent conditions holds:

- (i) R_1 is a formal power-series ring over R_0 via s_1 (and via s_2);
- (ii) the functor

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}$$

is smooth over $\underline{C}_\Lambda$.

A minimally smooth cogroupoid, also called a normalized smooth cogroupoid, in $\widehat{\underline{C}}_\Lambda$ is a smooth cogroupoid $(R_0, R_1, \dots)$ for which $\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])$ is totally disconnected; equivalently,

$$s_1(k[\varepsilon]) = s_2(k[\varepsilon]).$$

Proof. Smoothness of $\epsilon : \underline{h}_{R_0} \rightarrow \underline{h}_{(R_0, R_1, \dots)}$ means that, for every surjective map

$$(S', f') \xrightarrow{(h, u)} (S, f)$$

in $\underline{h}_{(R_0, R_1, \dots)}$, there are a lift f'' of f to S' and an isomorphism

$$(S', f'') \xrightarrow{(1, u')} (S', f').$$

Equivalently, given $u \in h_{R_1}(S)$ and a lift f' of $s_1(u)$ to S' , there is a lift u' of u to S' such that $s_1(u') = f'$, for every surjection $S' \rightarrow S$ in $\underline{C}_\Lambda$. This is precisely the smoothness of

$$s_1 : \underline{h}_{R_1} \longrightarrow \underline{h}_{R_0}$$

over $\underline{C}_\Lambda$, and that in turn says that R_1 is a formal power-series ring over R_0 via s_1 . □

Proposition 2.12. Let $(R_0, R_1, \dots)$ be a cogroupoid in $\widehat{\underline{C}}_\Lambda$.

(a) If $(R_0, R_1, \dots)$ is smooth, then $\underline{h}_{(R_0, R_1, \dots)}$ is homogeneous.

(b) If $s_1(k[\varepsilon]) = s_2(k[\varepsilon])$, then the following conditions are equivalent:

- (i) $(R_0, R_1, \dots)$ is smooth;
- (ii) $\underline{h}_{(R_0, R_1, \dots)}$ is homogeneous;
- (iii) $\underline{h}_{(R_0, R_1, \dots)}$ is semi-homogeneous.

(c) If $\underline{h}_{(R_0, R_1, \dots)}$ is semi-homogeneous, then it is equivalent to $\underline{h}_{(S_0, S_1, \dots)}$ for some minimally smooth cogroupoid S_* .

Proof. (a) Consider a diagram

$$\begin{array}{ccc} (S_1, f_1) & & (S_2, f_2) \\ & \searrow (h_1, u_1) & \swarrow (h_2, u_2) \\ & (S, f) & \end{array}$$

in $\underline{h}_{(R_0, R_1, \dots)}$, where $h_2 : S_2 \rightarrow S$ is surjective in $\underline{C}_\Lambda$. Since $s_1 : \underline{h}_{R_1} \rightarrow \underline{h}_{R_0}$ is smooth, $u_1^{-1}u_2$ lifts to an element $v \in h_{R_1}(S_2)$ with $s_1(v) = f_2$. We have

$$h_2 \circ s_2(v) = h_1 \circ f_1,$$

so $f_1 \times s_2(v)$ is an element of $h_{R_0}(S_1 \times_S S_2)$ and gives a commutative diagram

$$\begin{array}{ccc}
 & (S_1 \times_S S_2, f_1 \times s_2(v)) & \\
 \pi_1 \swarrow & & \searrow (\pi_2, v^{-1}) \\
 (S_1, f_1) & & (S_2, f_2) \\
 \searrow (h_1, u_1) & & \swarrow (h_2, u_2) \\
 & (S, f) &
 \end{array}$$

We claim that its top object is the fibre product

$$(S_1, f_1) \times_{(S, f)} (S_2, f_2)$$

in $\underline{h}_{(R_0, R_1, \dots)}$. Indeed, for every object (T, g) , a map to the top object is equivalently a tuple $(h'_1 \times h'_2, w_1, w_2)$ satisfying

$$\begin{aligned}
 h_1 w_1 &= h_2 w_2, & s_1(w_1) &= h'_1 g, & s_2(w_1) &= f_1, \\
 s_1(w_2) &= h'_2 g, & s_2(w_2) &= s_2(v).
 \end{aligned}$$

Putting $w'_2 = v^{-1} w_2$ gives a bijection with the tuples $(h'_1 \times h'_2, w_1, w'_2)$ satisfying

$$\begin{aligned}
 u_2 h_2 w'_2 &= u_1 h_1 w_1, & s_1(w_1) &= h'_1 g, & s_2(w_1) &= f_1, \\
 s_1(w'_2) &= h'_2 g, & s_2(w'_2) &= f_2.
 \end{aligned}$$

These are exactly the pairs of arrows from (T, g) to (S_1, f_1) and (S_2, f_2) whose composites to (S, f) coincide. Thus

$$\begin{aligned}
 \text{Hom}((T, g), (S_1 \times_S S_2, f_1 \times s_2(v))) \\
 \simeq \text{Hom}((T, g), (S_1, f_1)) \times_{\text{Hom}((T, g), (S, f))} \text{Hom}((T, g), (S_2, f_2)),
 \end{aligned}$$

which proves (a).

(b) It remains only to prove (iii) $\Rightarrow$ (i). If $\underline{h}_{(R_0, R_1, \dots)}$ is semi-homogeneous, then

$$\dim[\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])] = \dim h_{R_0}(k[\varepsilon]) < \infty,$$

so it admits a minimally versal formal object. On the other hand, $(R_0, 1_{R_0})$ is a quasi-initial minimal object of $\underline{h}_{(R_0, R_1, \dots)}$, because

$$h_{R_0}(k[\varepsilon]) \longrightarrow [\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])]$$

is bijective. Proposition 1.10(ii) therefore makes $(R_0, 1_{R_0})$ a minimally versal formal object of $\underline{h}_{(R_0, R_1, \dots)}$. Equivalently,

$$\epsilon : \underline{h}_{R_0} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}$$

is minimally smooth. This proves (i). (c) Let (S_0, ξ_0) be a minimally versal formal object of $\underline{h}_{(R_0, R_1, \dots)}$ (Theorem 1.11). For any object

$$\xi : R_0 \longrightarrow T \quad \text{of } \underline{h}_{(R_0, R_1, \dots)}(T),$$

consider the functor

$$\text{Isom}(\xi, \xi) : \underline{C}_\Lambda / (T \hat{\otimes}_\Lambda T) \longrightarrow (\text{Sets}).$$

It assigns to R the set of isomorphisms between the two images of ξ on R obtained from

$$R_0 \xrightarrow{\xi} T \rightrightarrows T \hat{\otimes}_\Lambda T \longrightarrow R.$$

This functor is represented by

$$R_1 \otimes_{R_0 \widehat{\otimes}_\Lambda R_0} (T \widehat{\otimes}_\Lambda T).$$

Passing to the inverse limit shows that $\text{Isom}(\xi_0, \xi_0)$ is prorepresented by an object S_1 , and that

$$\underline{h}_{(R_0, R_1, \dots)} \simeq \underline{h}_{(S_0, S_1, \dots)}.$$

By (b), the cogroupoid S_* is minimally smooth. $\square$

Definition 2.13. A cofibered groupoid $\underline{A}$ over $\underline{C}_\Lambda$ is called (minimally) prorepresentable if it is $\underline{C}_\Lambda$ -equivalent to $\underline{h}_{(R_0, R_1, \dots)}$ for some (normalized) cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$.

Proposition 2.14. If $(R_0, R_1, \dots)$ is a normalized cogroupoid in $\widehat{\underline{C}}_\Lambda$, then every homomorphism

$$(\alpha_0, \alpha_1) : (R_0, R_1, \dots) \longrightarrow (R_0, R_1, \dots)$$

which induces a $\underline{C}_\Lambda$ -equivalence

$$\underline{h}_{(\alpha_0, \alpha_1)} : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{h}_{(R_0, R_1, \dots)}$$

is an isomorphism. In

particular, a normalized prorepresentation of a cofibered groupoid over $\underline{C}_\Lambda$, if it exists, is unique up to isomorphism.

Proof. Consider the commutative diagram

$$\begin{array}{ccc} \underline{h}_{(R_0, R_1, \dots)} & \xrightarrow{\underline{h}_{(\alpha_0, \alpha_1)}} & \underline{h}_{(R_0, R_1, \dots)} \\ \uparrow \epsilon & & \uparrow \epsilon \\ \underline{h}_{R_0} & \xrightarrow{\underline{h}_{\alpha_0}} & \underline{h}_{R_0} \end{array}$$

in which $\underline{h}_{(\alpha_0, \alpha_1)}$ is a $\underline{C}_\Lambda$ -equivalence. Since $\underline{h}_{(R_0, R_1, \dots)}(k[\varepsilon])$ is totally disconnected, both

$$[\underline{h}_{(\alpha_0, \alpha_1)}(k[\varepsilon])] \quad \text{and} \quad [\epsilon(k[\varepsilon])]$$

are bijections, and hence so is $\underline{h}_{\alpha_0}(k[\varepsilon])$. Thus $\alpha_0 : R_0 \longrightarrow R_0$ is a surjective endomorphism and therefore an automorphism. It follows that α_1 is also an automorphism of R_1 . $\square$ Lemma 2.15. Let A and B be discrete groupoids over $\underline{C}_\Lambda$, and let $\varphi : A \longrightarrow B$ be a minimally smooth functor. If A and B are homogeneous, then φ is a $\underline{C}_\Lambda$ -equivalence.

Proof. We must show that $\varphi(S) : A(S) \longrightarrow B(S)$ is bijective for every $S \in \text{ob}(\underline{C}_\Lambda)$. It is surjective for every S and bijective for $S = k[\varepsilon]$. We argue by induction on the length of S . Let $S' \xrightarrow{f} S$ be a small extension in $\underline{C}_\Lambda$. Since A and B are homogeneous discrete groupoids, the canonical isomorphism

$$1 \times \tilde{f} : S' \times_S S' \xrightarrow{\sim} S' \times_k k[\text{Ker } f]$$

induces a commutative diagram

$$\begin{array}{ccc} A(S') \times_{A(S)} A(S') & \xrightarrow{\sim} & A(S') \times A(k[\text{Ker } f]) \\ \downarrow & & \downarrow \\ B(S') \times_{B(S)} B(S') & \xrightarrow{\sim} & B(S') \times B(k[\text{Ker } f]). \end{array}$$

Thus $A(S')$ and $B(S')$ are principal homogeneous sets over $A(S)$ and $B(S)$, with structural groups identified by

$$A(k[\text{Ker } f]) \xrightarrow{\sim} B(k[\text{Ker } f]).$$

If $\varphi(S)$ is bijective, it follows that $\varphi(S')$ is injective, and hence bijective. This completes the induction. $\square$

Corollary 2.16. Let $\underline{A}$ be a homogeneous groupoid over $\underline{C}_\Lambda$ such that

$$\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$$

is finite-dimensional. For every $\xi \in \mathrm{ob}(\widehat{\underline{A}})$, the functor

$$\mathrm{Isom}_\xi : \widehat{\underline{C}}_{R \widehat{\otimes}_\Lambda R} \longrightarrow (\mathrm{Sets})$$

is prorepresentable, where $R = P(\xi)$.

Proof. If Isom_ξ is homogeneous, Theorem 1.11 supplies a minimally smooth functor

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}}_\xi.$$

Lemma 2.15 makes Φ a $\widehat{\underline{C}}_{R \widehat{\otimes}_\Lambda R}$ -equivalence, and hence Isom_ξ is prorepresentable. It remains to prove homogeneity. Consider a diagram

$$\begin{array}{ccc} T_1 & & T_2 \\ & \searrow & \swarrow \\ & T & \end{array}$$

in $\widehat{\underline{C}}_{R \widehat{\otimes}_\Lambda R}$, with $T_2 \longrightarrow T$ surjective. Since $\underline{A}$ is homogeneous, the fibre products

$$\xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2)$$

exist in $\underline{A}$, and there are canonical isomorphisms

$$\xi_i(T_1 \times_T T_2) \xrightarrow{\sim} \xi_i(T_1) \times_{\xi_i(T)} \xi_i(T_2) \quad (i = 1, 2)$$

(cf. 2.6(a)). Therefore

$$\mathrm{Isom}_\xi(T_1 \times_T T_2) \longrightarrow \mathrm{Isom}_\xi(T_1) \times_{\mathrm{Isom}_\xi(T)} \mathrm{Isom}_\xi(T_2)$$

is an isomorphism. $\square$

Theorem 2.17. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$. Then it is prorepresentable by a (normalized) smooth cogroupoid if and only if

- (1) $\underline{A}$ is homogeneous;
- (2) $[\underline{A}(k[\varepsilon])]$ and $\mathrm{Aut}_{\underline{A}(k[\varepsilon])}(1_\varepsilon)$ are finite-dimensional vector spaces over k .

Proof. Suppose first that $\underline{A}$ is prorepresented by a smooth cogroupoid. Thus $\underline{A}$ is $\underline{C}_\Lambda$ -equivalent to

$$\underline{h} = \underline{h}_{(R_0, R_1, \dots)}$$

for a smooth cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$. Proposition 2.12(a) shows that $\underline{h}$ is homogeneous. Moreover,

$$\begin{aligned} [\underline{h}(k[\varepsilon])] &= \mathrm{Coker} \left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow[s_2]{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \right), \\ \mathrm{Aut}_{\underline{h}}(1_\varepsilon) &= \mathrm{Ker} \left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_1} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \right) \\ &= \mathrm{Ker} \left(\mathrm{Hom}_{\Lambda\text{-alg}}(R_1, k[\varepsilon]) \xrightarrow{s_2} \mathrm{Hom}_{\Lambda\text{-alg}}(R_0, k[\varepsilon]) \right). \end{aligned}$$

Both are finite-dimensional vector spaces over k , so the same is true for $\underline{A}$.

Conversely, assume (1) and (2). Since $\underline{A}$ is homogeneous and $[\underline{A}(k[\varepsilon])]$ is finite-dimensional, it has a minimally versal formal object ξ . Put $P(\xi) = R_0$ and consider

$$\mathrm{Isom}_\xi : \widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0} \longrightarrow (\mathrm{Sets}),$$

given by

$$T \longmapsto \mathrm{Isom}_{\underline{A}(T)}(\xi_1(T), \xi_2(T)),$$

where $\xi_i(T) = \xi(f_i)$ and $f_i : R_0 \longrightarrow T$ are the composite maps in

$$R_0 \begin{array}{c} \xrightarrow{s_1} \\ \xrightarrow{s_2} \end{array} R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow T.$$

Corollary 2.16 prorepresents this functor by an object

$$R_1 \in \mathrm{ob}(\widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0})$$

and an isomorphism $\xi_1(R_1) \xrightarrow{\varphi} \xi_2(R_1)$ for which

$$\Phi : \underline{h}_{R_1} \longrightarrow \underline{\mathrm{Isom}}_\xi$$

is an isomorphism over $\widehat{\underline{C}}_{R_0 \widehat{\otimes}_\Lambda R_0}$. The pair (R_0, R_1) therefore defines a cogroupoid, and

$$(\xi, \varphi) : \underline{h}_{(R_0, R_1, \dots)} \longrightarrow \underline{A}$$

is a $\underline{C}_\Lambda$ -equivalence. Because ξ is minimally versal, the composite

$$\underline{h}_{(R_0, R_0, \dots)} \xrightarrow{\epsilon} \underline{h}_{(R_0, R_1, \dots)} \xrightarrow{(\xi, \varphi)} \underline{A}$$

is minimally smooth. Since (ξ, φ) is an equivalence, ϵ is minimally smooth as well. Thus $(R_0, R_1, \dots)$ is a normalized smooth cogroupoid in $\widehat{\underline{C}}_\Lambda$. $\square$

Corollary 2.18. A functor $F : \underline{C}_\Lambda \longrightarrow (\mathrm{Sets})$ is prorepresentable if and only if it is homogeneous and $\dim F(k[\varepsilon]) < \infty$.

Corollary 2.19. Every smooth cogroupoid in $\widehat{\underline{C}}_\Lambda$ can be normalized up to isomorphism.

Given a cogroupoid $(R_0, R_1, \dots)$ in $\widehat{\underline{C}}_\Lambda$, set $\bar{R}_1 = R_1/J$, where J is the ideal of R_1 generated by

$$\{s_1(r) - s_2(r) \mid r \in R_0\}.$$

Then $(R_0, \bar{R}_1, \dots)$ defines a formal group scheme over R_0 with the induced composition law.

Now let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \dots)$ via (ξ, φ) . Let

$$\underline{C}_k \hookrightarrow \underline{C}_{R_0} \hookrightarrow \underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$$

be the embeddings corresponding to the surjections

$$R_0 \widehat{\otimes}_\Lambda R_0 \longrightarrow R_0 \longrightarrow k,$$

and let Aut_ξ and Aut^o be the restrictions of Isom_ξ to $\underline{C}_{R_0}$ and $\underline{C}_k$, respectively. Thus $\mathrm{Aut}_\xi(S)$ is the automorphism group of $\xi(S)$ for $S \in \mathrm{ob}(\underline{C}_{R_0})$, while $\mathrm{Aut}^o(S)$ is the automorphism group of 1_S for $S \in \mathrm{ob}(\underline{C}_k)$, where 1_S is the direct image of $1 \in \underline{A}(k)$ under $k \longrightarrow S$.

Since R_1 prorepresents Isom_ξ on $\underline{C}_{R_0 \widehat{\otimes}_\Lambda R_0}$, it follows that $\bar{R}_1$ prorepresents Aut_ξ on $\underline{C}_{R_0}$, and that $k \otimes_{R_0} \bar{R}_1$ prorepresents Aut^o on $\underline{C}_k$. Although $\underline{A}$ is prorepresented by the smooth cogroupoid $(R_0, R_1, \dots)$, neither $(R_0, \bar{R}_1, \dots)$ nor $(k, k \otimes_{R_0} \bar{R}_1, \dots)$ is smooth in general. More precisely:

Corollary 2.20. Let $\underline{A}$ be a homogeneous cofibered groupoid over $\underline{C}_\Lambda$, prorepresented by a normalized smooth cogroupoid $(R_0, R_1, \dots)$ via (ξ, φ) .

- (a) $(R_0, \bar{R}_1, \dots)$ and $(k, k \otimes_{R_0} \bar{R}_1, \dots)$ prorepresent Aut_ξ and Aut^o , respectively.

(b) The following statements are equivalent:

- (i) the formal group scheme $\underline{h}_{(R_0, \bar{R}_1, \dots)}$ is formally smooth over R_0 ;
- (ii) for every surjective arrow $a' \rightarrow a$ in $\underline{A}$, the map

$$\mathrm{Aut}_{P(a')}(a') \rightarrow \mathrm{Aut}_{P(a)}(a)$$

is surjective;

- (iii) R_0 prorepresents the functor $[\underline{A}]$ over $\underline{C}_\Lambda$.

Proof. We have already noted (a). For (b), the equivalence (i) $\Leftrightarrow$ (ii) is clear, whereas (ii) $\Leftrightarrow$ (iii) follows from 2.7 and 2.15, since

$$\tilde{\xi} : \underline{h}_{R_0} \rightarrow [\underline{A}]$$

is minimally smooth. $\square$

Remark 2.21. Let $\underline{A}$ be a cofibered groupoid over $\underline{C}_\Lambda$, and let $\underline{A}|_{\underline{C}_k}$ be the pullback of $p : \underline{A} \rightarrow \underline{C}_\Lambda$ along $\underline{C}_k \hookrightarrow \underline{C}_\Lambda$. If $\underline{A}$ is prorepresented by $(R_0, R_1, \dots)$ via (ξ, φ) , then $\underline{A}|_{\underline{C}_k}$ is prorepresented by

$$(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \dots)$$

via $(k \otimes_\Lambda \xi, k \otimes_\Lambda \varphi)$. Here $k \otimes_\Lambda \xi$ is the direct image of ξ under $R_0 \rightarrow k \otimes_\Lambda R_0$, and $k \otimes_\Lambda \varphi$ is the image of φ under

$$\mathrm{Isom}_{\underline{A}(R_0)}(\xi(s_1), \xi(s_2)) \rightarrow \mathrm{Isom}_{\underline{A}(k \otimes_\Lambda R_1)}(k \otimes_\Lambda \xi(s_1), k \otimes_\Lambda \xi(s_2)).$$

If $(R_0, R_1, \dots)$ is normalized, then so is $(k \otimes_\Lambda R_0, k \otimes_\Lambda R_1, \dots)$; moreover, the relative dimension of R_1 over R_0 equals that of $k \otimes_\Lambda R_1$ over $k \otimes_\Lambda R_0$, since R_1 is smooth over R_0 .

Remark 2.22. Suppose that $\underline{A}$ is prorepresented by smooth cogroupoids $(R_0, R_1, \dots)$ and $(R'_0, R'_1, \dots)$, with $(R_0, R_1, \dots)$ normalized. We obtain a homomorphism

$$(i_0, i_1) : (R_0, R_1, \dots) \rightarrow (R'_0, R'_1, \dots).$$

By 2.10(b), R'_0 is a formal power-series ring over R_0 , and (i_0, i_1) induces an isomorphism

$$R'_0 \otimes_{R_0} (R_0, R_1, \dots) \xrightarrow{\sim} (R'_0, R'_1, \dots).$$

3. The coherent sheaves $\underline{D}_X^i$

One important functor which is usually not prorepresentable, but nevertheless admits a formal versal object, is the deformation functor. With this in view, we recall some basic facts about extensions of commutative rings and their obstruction theory. This section contains nothing new, except perhaps for its systematic use of Grothendieck cohomology on a site. Its purpose is to make this exposé as self-contained as possible; in fact, we prove somewhat more than will be needed later. The basic results are stated in Theorem 3.12. Throughout, all rings are commutative and have a unit.

Given a ring R , let $\underline{C}_R$ denote the category of R -algebras. For every $A \in \mathrm{Ob}(\underline{C}_R)$, put

$$\mathrm{Cov}(A) = \{\text{surjective arrows } \varphi : A' \rightarrow A \text{ in } \underline{C}_R\}.$$

This defines a pretopology, and hence a site, on $\underline{C}_R$. For every $A \in \mathrm{Ob}(\underline{C}_R)$, let $\underline{C}_{A|R}$ be the category of arrows in $\underline{C}_R$ with target A . We endow $\underline{C}_{A|R}$ with the topology induced by the forgetful functor

$$j_A : \underline{C}_{A|R} \rightarrow \underline{C}_R.$$

The category $\underline{C}_{A|R}$ has a final object $(1_A : A \rightarrow A)$ and an initial object $(i_A : R \rightarrow A)$, and it admits both fibre products and coproducts. When there is no risk of confusion, we write simply B for an object $(B \rightarrow A)$ of $\underline{C}_{A|R}$.

Let $\mathcal{C}$ be an abelian category. A $\mathcal{C}$ -valued sheaf on $\underline{C}_{A|R}$ is, by definition, a functor

$$F : \underline{C}_{A|R}^{\text{op}} \longrightarrow \mathcal{C}$$

such that, for every surjective arrow $B' \longrightarrow B$ in $\underline{C}_{A|R}$, the sequence

$$0 \longrightarrow F(B) \longrightarrow F(B') \rightrightarrows F(B' \times_B B')$$

is exact. The additive presheaves are those which preserve coproducts; thus

$$F(B_1 \otimes_R B_2) = F(B_1) \oplus F(B_2).$$

For example, an A -module M defines the functor

$$\text{Der}_R(-, M) : \underline{C}_{A|R}^{\text{op}} \longrightarrow (A\text{-mod}),$$

which sends B to the R -linear derivations $\text{Der}_R(B, M)$, with M regarded as a B -module via the structural map $B \longrightarrow A$. This is plainly an additive sheaf on $\underline{C}_{A|R}$. Here $(A\text{-mod})$ denotes the category of A -modules.

A presheaf F on $\underline{C}_{A|R}$ with values in $(A\text{-mod})$ is said to be of finite type if $F(B)$ is an A -module of finite type whenever B is an R -algebra of finite type. For example, if A is noetherian and M is an A -module of finite type, then $\text{Der}_R(-, M)$ is an additive sheaf of finite type on $\underline{C}_{A|R}$.

We record a little more notation and a few basic facts from homological algebra.

3.1(a). Throughout the remainder of this section, $\tilde{\underline{C}}_{A|R}$ and $\check{\underline{C}}_{A|R}$ denote, respectively, the categories of sheaves and presheaves with values in $(A\text{-mod})$. There is a left-exact inclusion functor

$$i_{A|R} : \tilde{\underline{C}}_{A|R} \longrightarrow \check{\underline{C}}_{A|R}$$

and an exact associated-sheaf functor

$$G_{A|R} : \check{\underline{C}}_{A|R} \longrightarrow \tilde{\underline{C}}_{A|R},$$

left adjoint to $i_{A|R}$.

An object $B \in \text{Ob}(\underline{C}_{A|R})$ induces a continuous functor $j : \underline{C}_{B|R} \longrightarrow \underline{C}_{A|R}$ and hence a commutative diagram

$$(*) \quad \begin{array}{ccc} \tilde{\underline{C}}_{A|R} & \xrightarrow{\tilde{j}} & \tilde{\underline{C}}_{B|R} \\ i_{A|R} \downarrow & & \downarrow \\ \check{\underline{C}}_{A|R} & \xrightarrow{j^\vee} & \check{\underline{C}}_{B|R}. \end{array}$$

The restriction functor $\tilde{j}$, as well as $j^\vee$, is exact, because the site $\underline{C}_{B|R}$ is simply the category of arrows in $\underline{C}_{A|R}$ with target B , endowed with the induced topology.

A ring homomorphism $S \longrightarrow R$ induces a continuous functor $\beta : \underline{C}_{A|S} \longrightarrow \underline{C}_{A|R}$ and hence a commutative diagram

$$(**) \quad \begin{array}{ccc} \tilde{\underline{C}}_{A|S} & \xrightarrow{\tilde{\beta}} & \tilde{\underline{C}}_{A|R} \\ i_{A|S} \downarrow & & \downarrow \\ \check{\underline{C}}_{A|S} & \xrightarrow{\beta^\vee} & \check{\underline{C}}_{A|R}. \end{array}$$

Notice that $\widetilde{\beta}$ need not be exact.

3.1(b). The right-derived functor of the inclusion

$$i_{A|R} : \widetilde{\underline{C}}_{A|R} \longrightarrow \bigvee \underline{C}_{A|R}$$

will be denoted by $\underline{H}_{A|R}^\bullet$. For every sheaf F on $\underline{C}_{A|R}$, put

$$H^\bullet(A|R, F) = [\underline{H}_{A|R}^\bullet(F)](A).$$

For $B' \longrightarrow B$ in $\text{Cov}(B)$, define

$$H^0(\{B' \longrightarrow B\}, F) = \ker(F(B') \rightrightarrows F(B' \times_B B')).$$

We denote the right-derived functors of $H^0(\{B' \longrightarrow B\}, -)$ by $H^\bullet(\{B' \longrightarrow B\}, -)$. These are the cohomology groups of the Čech semisimplicial module

$$F(B') \begin{array}{c} \longrightarrow \\ \longrightarrow \end{array} F(B' \times_B B') \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} F(B' \times_B B' \times_B B') \begin{array}{c} \longrightarrow \\ \longrightarrow \\ \longrightarrow \\ \longrightarrow \end{array} \cdots$$

The right-derived functor of

$$\check{H}_{A|R}^0 : \bigvee \underline{C}_{A|R} \longrightarrow \bigvee \underline{C}_{A|R}$$

will be denoted by $\check{\underline{H}}_{A|R}^\bullet$, where

$$[\check{\underline{H}}_{A|R}^0(F)](B) = \varinjlim_{\{B' \rightarrow B\} \in \text{Cov}(B)} H^0(\{B' \longrightarrow B\}, F).$$

For every presheaf F on $\underline{C}_{A|R}$, put

$$\check{H}^\bullet(A|R, F) = [\check{\underline{H}}_{A|R}^\bullet(F)](A) = \varinjlim_{\{A' \rightarrow A\} \in \text{Cov}(A)} H^\bullet(\{A' \longrightarrow A\}, F).$$

For every sheaf F on $\underline{C}_{A|R}$ there is the usual spectral sequence

$$\check{H}^p(A|R, \underline{H}_{A|R}^q(F)) \Longrightarrow H^{p+q}(A|R, F).$$

Here $\check{H}^0(A|R, \underline{H}_{A|R}^q(F)) = 0$ for every $q > 0$. Consequently,

$$\check{H}^1(A|R, F) = H^1(A|R, F),$$

and the sequence

$$0 \longrightarrow \check{H}^2(A|R, F) \longrightarrow H^2(A|R, F) \longrightarrow \check{H}^1(A|R, \underline{H}_{A|R}^1(F)) \longrightarrow \check{H}^3(A|R, F)$$

is exact. 3.1(c). If B is an object of $\underline{C}_{A|R}$, diagram $(*)$ in 3.1(a), together with the exactness of $\widetilde{j}$ and $j^\vee$, gives a canonical isomorphism

$$H^\bullet(B|R, \widetilde{j}F) = [\underline{H}_{A|R}^\bullet(F)](B).$$

On the other hand, a ring homomorphism $S \longrightarrow R$ induces diagram $(**)$ in 3.1(a). The functor $\widetilde{\beta}$ need not be exact; however, β preserves fibre products, and we obtain a spectral sequence

$$\underline{H}_{A|R}^p(R^q \widetilde{\beta}F) \Longrightarrow \beta^\vee \underline{H}_{A|S}^{p+q}(F),$$

or, on evaluating at A ,

$$H^p(A|R, R^q \widetilde{\beta}F) \Longrightarrow H^{p+q}(A|S, F).$$

Since there is a canonical isomorphism

$$\tilde{\beta} = G_{A|R} \circ \beta^\vee \circ i_{A|S}$$

and the functor $\beta^\# = G_{A|R} \circ \beta^\vee$ is exact, we have

$$R^\bullet \tilde{\beta} F = \beta^\# \underline{H}_{A|S}^\bullet(F).$$

Thus the preceding spectral sequence may be written

$$H^p(A|R, \beta^\# \underline{H}_{A|S}^q(F)) \implies H^{p+q}(A|S, F).$$

Lemma 3.2. Let $B \in \text{Ob}(\underline{C}_{A|R})$, and choose a surjective arrow $P \longrightarrow B$ in $\underline{C}_{B|R}$, where P is a polynomial algebra over R . Then, for every $F \in \text{Ob}(\underline{C}_{A|S}^\vee)$, the natural map

$$H^\bullet(\{P \longrightarrow B\}, F) \longrightarrow [\check{H}^\bullet(F)](B)$$

is an isomorphism.

Proof. It suffices to prove that the natural map

$$H^0(\{P \longrightarrow B\}, F) \longrightarrow [\check{H}^0(F)](B) = \varinjlim_{\{B' \rightarrow B\} \in \text{Cov}(B)} H^0(\{B' \longrightarrow B\}, F)$$

is an isomorphism. This is clear because, for every $\{B' \longrightarrow B\} \in \text{Cov}(B)$, there is a map of covers

$$\{P \longrightarrow B\} \longrightarrow \{B' \longrightarrow B\}.$$

□

Corollary 3.3.

- (a) Suppose that A is noetherian. If $F \in \text{Ob}(\tilde{\underline{C}}_{A|R})$ is of finite type, then $\underline{H}^n(F)$ is of finite type for every n .
- (b) If P is a polynomial algebra over R , then

$$H^n(P|R, F) = 0$$

for every $n > 0$ and every $F \in \text{Ob}(\tilde{\underline{C}}_{P|R})$.

- (c) A sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

in $\tilde{\underline{C}}_{A|R}$ is exact if and only if

$$0 \longrightarrow F'(P) \longrightarrow F(P) \longrightarrow F''(P) \longrightarrow 0$$

is exact for every polynomial R -algebra $P \in \text{Ob}(\underline{C}_{A|R})$.

Proof.

- (a) In view of the spectral sequence

$$\check{H}^p(B|R, \underline{H}^q(F)) \implies H^{p+q}(B|R, F)$$

and the equality $\check{H}^0(\underline{H}^q(F)) = 0$ for $q > 0$, it is enough to show that $\check{H}^n(G)$ is of finite type for every n and every presheaf G of finite type. By Lemma 3.2,

$$\check{H}^\bullet(B|R, G) = H^\bullet(\{P \longrightarrow B\}, G),$$

where P is a polynomial algebra over R and $\{P \longrightarrow B\} \in \text{Cov}(B)$. The latter is the cohomology of the complex

$$\cdots G(P \times_B P \times_B P) \xleftarrow{\quad} G(P \times_B P) \xleftarrow{\quad} G(P).$$

If B is an R -algebra of finite type, P may be chosen of finite type over R . All the iterated fibre products $P \times_B \cdots \times_B P$ are then R -algebras of finite type. If G is of finite type, their images under G are A -modules of finite type. Hence $H^\bullet(\{P \rightarrow B\}, G)$ is an A -module of finite type because A is noetherian.

(b) Let P be a polynomial algebra over R . Every surjective R -algebra map $B \rightarrow P$ admits a section, so

$$H^i(\{B \rightarrow P\}, G) = 0 \quad (i > 0).$$

It follows that $\check{H}^i(P|R, G) = 0$ for every $i > 0$ and every presheaf G on $\underline{C}_{P|R}$. Consequently the spectral sequence

$$\check{H}^p(P|R, \underline{H}_{P|R}^q(F)) \Rightarrow H^{p+q}(P|R, F)$$

degenerates, and

$$\check{H}^0(P|R, \underline{H}^n(F)) \simeq H^n(P|R, F)$$

for every n . For a sheaf F , however, $\check{H}^0(P|R, \underline{H}^n(F)) = 0$ whenever $n > 0$. Thus $H^n(P|R, F) = 0$ for every $n > 0$.

(c) If $0 \rightarrow F' \rightarrow F \rightarrow F'' \rightarrow 0$ is exact in $\tilde{\underline{C}}_{A|R}$, then the sequence

$$0 \rightarrow F'(B) \rightarrow F(B) \rightarrow F''(B) \rightarrow H^1(B|R, F') \rightarrow H^1(B|R, F) \rightarrow \cdots$$

is exact. If $P \in \text{Ob}(\underline{C}_{A|R})$ is a polynomial algebra over R , then $H^1(P|R, F') = 0$ by (b), and hence

$$0 \rightarrow F'(P) \rightarrow F(P) \rightarrow F''(P) \rightarrow 0$$

is exact.

Conversely, let $\underline{P}_{A|R}$ be the full subcategory of $\underline{C}_{A|R}$ with

$$\text{Ob}(\underline{P}_{A|R}) = \{P \in \text{Ob}(\underline{C}_{A|R}) \mid P \text{ is a polynomial algebra over } R\},$$

endowed with the induced topology, and let $p : \underline{P}_{A|R} \rightarrow \underline{C}_{A|R}$ be the inclusion. Every object of $\underline{C}_{A|R}$ is covered by an object of $\underline{P}_{A|R}$; the comparison lemma therefore shows that

$$\tilde{p} : \tilde{\underline{C}}_{A|R} \rightarrow \tilde{\underline{P}}_{A|R}$$

is an equivalence of categories. If the sequence obtained by evaluating $F' \rightarrow F \rightarrow F''$ at every $P \in \text{Ob}(\underline{P}_{A|R})$ is exact, then

$$0 \rightarrow \tilde{p}F' \rightarrow \tilde{p}F \rightarrow \tilde{p}F'' \rightarrow 0$$

is exact in $\tilde{\underline{P}}_{A|R}$, and therefore the original sequence is exact in $\tilde{\underline{C}}_{A|R}$. $\square$

Remark 3.4. Every surjective arrow in $\underline{P}_{A|R}$ admits a section. It follows at once that every presheaf on $\underline{P}_{A|R}$ is a sheaf; that is,

$$\tilde{\underline{P}}_{A|R} = \check{\underline{P}}_{A|R}.$$

Consequently, the restriction functor

$$\tilde{\underline{C}}_{A|R} \rightarrow \check{\underline{P}}_{A|R}$$

is an equivalence of categories. Lemma 3.5. Consider a diagram of R -algebras

$$\begin{array}{ccc} & C & \\ & \downarrow & \\ B' & \xrightarrow{\text{surjective}} & B \end{array}$$

and let S be an R -algebra.

(i) If $\text{Tor}_1^R(B, S) = 0$, then the canonical map

$$(B' \times_B C) \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S)$$

is an isomorphism.

(ii) Suppose that B' is flat over R . If

$$\text{Tor}_i^R(C, S) = 0 \quad (0 < i < n) \quad \text{and} \quad \text{Tor}_i^R(B, S) = 0 \quad (0 < i \leq n),$$

then

$$\text{Tor}_i^R(B' \times_B C, S) = 0 \quad (0 < i < n).$$

Proof. Let $0 \longrightarrow J \longrightarrow B' \longrightarrow B \longrightarrow 0$ be exact.

(i) If $\text{Tor}_1^R(B, S) = 0$, then

$$0 \longrightarrow J \otimes_R S \longrightarrow B' \otimes_R S \longrightarrow B \otimes_R S \longrightarrow 0$$

is exact, and hence so is

$$0 \longrightarrow J \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) \longrightarrow C \otimes_R S \longrightarrow 0.$$

We therefore obtain a commutative diagram with exact rows:

$$\begin{array}{ccccccc} J \otimes_R S & \longrightarrow & (B' \times_B C) \otimes_R S & \longrightarrow & C \otimes_R S & \longrightarrow & 0 \\ \parallel & & \downarrow & & \parallel & & \\ 0 \longrightarrow J \otimes_R S & \longrightarrow & (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S) & \longrightarrow & C \otimes_R S & \longrightarrow & 0. \end{array}$$

It follows that

$$(B' \times_B C) \otimes_R S \longrightarrow (B' \otimes_R S) \times_{B \otimes_R S} (C \otimes_R S)$$

is an isomorphism.

(ii) Since B' is flat over R and $\text{Tor}_i^R(B, S) = 0$ for $0 < i \leq n$, the long exact Tor sequence gives

$$\text{Tor}_i^R(J, S) = 0 \quad (0 < i < n).$$

Applying the same sequence to

$$0 \longrightarrow J \longrightarrow B' \times_B C \longrightarrow C \longrightarrow 0$$

and using the hypothesis on C gives

$$\text{Tor}_i^R(B' \times_B C, S) = 0 \quad (0 < i < n).$$

□

Corollary 3.6.

(a) Let A and S be R -algebras. If $\text{Tor}_1^R(A, S) = 0$, then for every $G \in \text{Ob}(\underline{C}_{A \otimes_R S|S}^\vee)$ the canonical map

$$\check{H}^\bullet(A \otimes_R S|S, G) \longrightarrow \check{H}^\bullet(A|R, \check{S}G)$$

is an isomorphism. Here

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \quad X \longmapsto X \otimes_R S.$$

In particular, if S is flat over R , then

$$\check{S} \check{H}^\bullet(G) \longrightarrow \check{H}^\bullet(\check{S}G)$$

is an isomorphism for every such G .

(b) Let $P \longrightarrow B$ be a surjective map of R -algebras, where P is a polynomial algebra over R . If

$$\mathrm{Tor}_i^R(B, S) = 0 \quad (0 < i \leq n),$$

then, for every iterated fibre product of copies of P over B ,

$$\mathrm{Tor}_i^R(P \times_B P \times_B \cdots \times_B P, S) = 0 \quad (0 < i < n).$$

Proof.

(a) Given $B \in \mathrm{Ob}(\underline{C}_{A|R})$, choose a cover $\{P \longrightarrow B\} \in \mathrm{Cov}(B)$ with P a polynomial algebra over R . If $\mathrm{Tor}_1^R(B, S) = 0$, Lemma 3.5(i) shows that the natural map

$$\left(\overbrace{P \times_B \cdots \times_B P}^n \right) \otimes_R S \longrightarrow \overbrace{(P \otimes_R S) \times_{B \otimes_R S} \cdots \times_{B \otimes_R S} (P \otimes_R S)}^n$$

is an isomorphism for every n . Consequently,

$$\begin{aligned} \check{H}^\bullet(B|R, \check{S}G) &= H^\bullet(\{P \longrightarrow B\}, \check{S}G) \\ &= H^\bullet(\{P \otimes_R S \longrightarrow B \otimes_R S\}, G) \\ &= \check{H}^\bullet(B \otimes_R S|S, G). \end{aligned}$$

If S is flat over R , then $\mathrm{Tor}_1^R(X, S) = 0$ for every $X \in \mathrm{Ob}(\underline{C}_{A|R})$. The preceding isomorphism is therefore natural in X , and gives

$$\check{S} \check{H}^\bullet(G) \xrightarrow{\sim} \check{H}^\bullet(\check{S}G).$$

(b) Put

$$P^{(m)} = \overbrace{P \times_B \cdots \times_B P}^{m+1}.$$

For $m = 0$ there is nothing to prove, since P is flat over R . Suppose inductively that $\mathrm{Tor}_i^R(P^{(m-1)}, S) = 0$ for $0 < i < n$. Since $P \longrightarrow B$ is surjective and $P^{(m)} = P \times_B P^{(m-1)}$, Lemma 3.5(ii) gives

$$\mathrm{Tor}_i^R(P^{(m)}, S) = 0 \quad (0 < i < n).$$

□

Proposition 3.7.

(a) Let A and S be R -algebras. If

$$\mathrm{Tor}_i^R(A, S) = 0 \quad (0 < i \leq n),$$

then, for every sheaf F on $\underline{C}_{A \otimes_R S|S}$, the canonical map

$$H^i(A \otimes_R S|S, F) \longrightarrow H^i(A|R, \tilde{S}F)$$

is an isomorphism for every $i \leq n$. Here

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \quad X \longmapsto X \otimes_R S.$$

(b) Let P be a polynomial algebra over R . Then, for every additive sheaf F on $\underline{C}_{A \otimes_R P|R}$, the canonical map

$$H^n(A \otimes_R P|R, F) \longrightarrow H^n(A|R, F)$$

is an isomorphism for every $n > 0$.

(c) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every additive sheaf F on $\underline{C}_{A|S}$, there is an exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(A|R, F_R) \longrightarrow H^0(A|S, F) \longrightarrow H^0(R|S, F) \longrightarrow H^1(A|R, F_R) \\ \longrightarrow H^1(A|S, F) \longrightarrow H^1(R|S, F) \longrightarrow \cdots \longrightarrow H^n(A|R, F_R) \\ \longrightarrow H^n(A|S, F) \longrightarrow H^n(R|S, F) \longrightarrow H^{n+1}(A|R, F_R) \longrightarrow \cdots, \end{aligned}$$

where F_R is the sheaf on $\underline{C}_{A|R}$ defined by

$$F_R(X) = \ker(F(X) \longrightarrow F(R))$$

for every $X \in \text{Ob}(\underline{C}_{A|R})$, via the forgetful functor $\underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$.

Proof.

(a) Consider the continuous functor

$$S : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R S|S}, \quad S(X) = X \otimes_R S.$$

Since S sends polynomial algebras over R to polynomial algebras over S , the functor

$$\tilde{S} : \underline{\tilde{C}}_{A \otimes_R S|S} \longrightarrow \underline{\tilde{C}}_{A|R}$$

is exact by 3.3(c). We therefore obtain the canonical map

$$\check{S} \underline{H}_{A \otimes_R S|S}^\bullet(F) \longrightarrow \underline{H}_{A|R}^\bullet(\tilde{S}F)$$

and the following commutative diagram of spectral sequences:

$$\begin{array}{ccc} E_S^{p,q}(S(X)) = \check{H}^p(X \otimes_R S|S, \underline{H}_{A \otimes_R S|S}^q(F)) & \xRightarrow{\quad} & H^{p+q}(X \otimes_R S|S, F) \\ \downarrow & & \downarrow \\ \check{H}^p(X|R, \check{S} \underline{H}_{A \otimes_R S|S}^q(F)) & & \\ \downarrow & & \\ E_R^{p,q}(X) = \check{H}^p(X|R, \underline{H}_{A|R}^q(\tilde{S}F)) & \xRightarrow{\quad} & H^{p+q}(X|R, \tilde{S}F). \end{array}$$

By 3.6(a), the map

$$E_S^{p,q}(S(X)) \longrightarrow \check{H}^p(X|R, \check{S} \underline{H}_{A \otimes_R S|S}^q(F))$$

is an isomorphism whenever $\text{Tor}_1^R(X, S) = 0$. Suppose inductively that

$$[\check{S} \underline{H}_{A \otimes_R S|S}^q(F)](X) \longrightarrow [\underline{H}_{A|R}^q(\tilde{S}F)](X)$$

is an isomorphism for every $q < n$ and every $X \in \text{Ob}(\underline{C}_{A|R})$ satisfying

$$\text{Tor}_i^R(X, S) = 0 \quad (0 < i \leq n).$$

It follows from 3.6(b) that $E_S^{p,q}(S(X)) \longrightarrow E_R^{p,q}(X)$ is an isomorphism for every $q < n$ and every p , under the same Tor hypothesis. Moreover, $E_S^{0,n}(S(X)) \longrightarrow E_R^{0,n}(X)$ is an isomorphism, since both sides are zero. Hence

$$H^n(X \otimes_R S|S, F) \longrightarrow H^n(X|R, \tilde{S}F)$$

is an isomorphism whenever $\text{Tor}_i^R(X, S) = 0$ for $0 < i \leq n$. Equivalently,

$$[\check{S} \underline{H}_{A \otimes_R S|S}^i(F)](X) \longrightarrow [\underline{H}_{A|R}^i(\tilde{S}F)](X)$$

is an isomorphism for every $i \leq n$ under that hypothesis.

(b) Let

$$f : \underline{C}_{A|R} \longrightarrow \underline{C}_{A \otimes_R P|R}, \quad f(X) = X \otimes_R P.$$

Choose a surjective arrow $P' \longrightarrow X$ in $\underline{C}_{A|R}$, with P' polynomial over R . Then

$$P' \otimes_R P \longrightarrow X \otimes_R P$$

is a surjective arrow in $\underline{C}_{A \otimes_R P|R}$, and $P' \otimes_R P$ is again polynomial over R . Since P is flat over R , 3.5(i) gives

$$\begin{aligned} \check{H}^\bullet(X|R, \check{f}G) &= H^\bullet(\{P' \longrightarrow X\}, \check{f}G) \\ &= H^\bullet(\{P' \otimes_R P \longrightarrow X \otimes_R P\}, G) \\ &= \check{H}^\bullet(X \otimes_R P|R, G) \end{aligned}$$

for every $G \in \text{Ob}(\check{\underline{C}}_{A \otimes_R P|R})$. Thus $\check{f}$ sends acyclic sheaves to acyclic sheaves. On the other hand, f sends polynomial algebras over R to polynomial algebras over R , so

$$\tilde{f} : \tilde{\underline{C}}_{A \otimes_R P|R} \longrightarrow \tilde{\underline{C}}_{A|R}$$

is exact by 3.3(c). It follows that

$$H^\bullet(A \otimes_R P|R, F) \longrightarrow H^\bullet(A|R, \tilde{f}F)$$

is an isomorphism for every $F \in \text{Ob}(\tilde{\underline{C}}_{A \otimes_R P|R})$.

For each $X \in \text{Ob}(\underline{C}_{A|R})$ there is a canonical map

$$(\tilde{f}F)(X) = F(X \otimes_R P) \longrightarrow F(X) \oplus F(P),$$

and hence a canonical map $\tilde{f}F \longrightarrow F \oplus F(P)$, where $F(P)$ denotes the constant sheaf and F on the right denotes the restriction of F to $\underline{C}_{A|R}$. If F is additive, this map is an isomorphism. Therefore, for every $n > 0$,

$$H^n(A|R, \tilde{f}F) \xrightarrow{\sim} H^n(A|R, F \oplus F(P)) = H^n(A|R, F),$$

because the constant sheaf $F(P)$ is flasque. This proves (b).

(c) This argument was communicated by Rinehart. Consider the forgetful functor $\beta : \underline{C}_{A|R} \longrightarrow \underline{C}_{A|S}$. There is a spectral sequence

$$E^{p,q} = H^p(A|R, \beta^\# \underline{H}_{A|S}^q(F)) \implies H^{p+q}(A|S, F),$$

where

$$\beta^\# = G_{A|R} \circ \beta^\vee, \quad G_{A|R} : \check{\underline{C}}_{A|R} \longrightarrow \tilde{\underline{C}}_{A|R}$$

is the associated-sheaf functor.

Suppose now that F is additive. We claim that $\beta^\# \underline{H}_{A|S}^q(F)$ is a constant sheaf for every $q > 0$. Indeed, if P is a polynomial algebra over R , then $P = R \otimes_S P'$ for a polynomial algebra P' over S , and

$$[\beta^\# \underline{H}_{A|S}^q(F)](P) = H^q(P|S, F) = H^q(R \otimes_S P'|S, F) \longrightarrow H^q(R|S, F)$$

is an isomorphism for $q > 0$ by (b), and is surjective for $q = 0$. By 3.3(c), the canonical map

$$\beta^\# \underline{H}_{A|S}^q(F) \longrightarrow H^q(R|S, F)$$

to the corresponding constant sheaf is therefore an isomorphism for $q > 0$ and a surjection for $q = 0$. Hence $E^{p,q} = 0$ whenever $p, q > 0$, and the spectral sequence yields the exact sequence

$$(*) \quad 0 \longrightarrow E^{1,0} \longrightarrow H^1 \longrightarrow E^{0,1} \longrightarrow E^{2,0} \longrightarrow H^2 \longrightarrow E^{0,2} \longrightarrow E^{3,0} \longrightarrow H^3 \longrightarrow \dots$$

Here

$$E^{0,q} = [\beta^\# \underline{H}_{A|S}^q(F)](A) \xrightarrow{\sim} H^q(R|S, F) \quad (q > 0), \quad E^{p,0} = H^p(A|R, \tilde{\beta}F).$$

On the other hand, the exact sequence

$$0 \longrightarrow F_R \longrightarrow \tilde{\beta}F \longrightarrow F(R) \longrightarrow 0$$

gives the exact diagram

$$\begin{array}{ccccccccc} 0 & \rightarrow & H^0(A|R, F_R) & \rightarrow & H^0(A|R, \tilde{\beta}F) & \rightarrow & H^0(A|R, F(R)) & \rightarrow & H^1(A|R, F_R) & \rightarrow & H^1(A|R, \tilde{\beta}F) & \rightarrow & 0 \\ & & \parallel & & \parallel & & & & & & & & \\ & & H^0(A|S, F) & & H^0(R|S, F) & & & & & & & & \end{array}$$

labelled (**) in the source. Combining (*) and (**) gives the desired exact sequence. $\square$

Corollary 3.8.

(a) Let A and B be R -algebras such that

$$\mathrm{Tor}_i^R(A, B) = 0 \quad (0 < i \leq n).$$

Then, for every additive sheaf $F \in \mathrm{Ob}(\tilde{\mathcal{C}}_{A \otimes_R B|R})$, the canonical map

$$H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism for every $i \leq n$.

(b) Let A be an R -algebra such that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$. If the map $A \otimes_R A \longrightarrow A$ is an isomorphism, then, for every additive sheaf $F \in \mathrm{Ob}(\tilde{\mathcal{C}}_{A|R})$, one has

$$H^i(A|R, F) = 0$$

for every i .

Proof.

(a) Let $F \in \mathrm{Ob}(\tilde{\mathcal{C}}_{A \otimes_R B|R})$ be additive. Applied to $R \longrightarrow A \longrightarrow A \otimes_R B$, Proposition 3.7(c) gives the exact sequence

$$(*) \quad \cdots \longrightarrow H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \longrightarrow \cdots.$$

Consider the commutative diagram

$$\begin{array}{ccc} H^i(A \otimes_R B|A, F_A) & \longrightarrow & H^i(A \otimes_R B|R, F) \\ \downarrow & & \downarrow \\ H^i(B|R, \tilde{A}F_A) & \xrightarrow{\approx} & H^i(B|R, F). \end{array}$$

Here

$$A : \underline{\mathcal{C}}_{B|R} \longrightarrow \underline{\mathcal{C}}_{A \otimes_R B|A}, \quad X \longmapsto A \otimes_R X,$$

and the bottom map is induced by $\tilde{A}F_A \longrightarrow F$, which is an isomorphism because F is additive. By functoriality, the commutativity of the diagram reduces to the case $i = 0$.

If $\mathrm{Tor}_i^R(A, B) = 0$ for $0 < i \leq n$, then Proposition 3.7(a) shows that the left vertical map is an isomorphism for $i \leq n$. Consequently, both composites

$$H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(B|R, F)$$

and, by symmetry,

$$H^i(A \otimes_R B|B, F_B) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F)$$

are isomorphisms for $i \leq n$. It follows that $(*)$ breaks into split short exact sequences

$$0 \longrightarrow H^i(A \otimes_R B|A, F_A) \longrightarrow H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \longrightarrow 0$$

for $i \leq n$, and hence that

$$H^i(A \otimes_R B|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(B|R, F)$$

is an isomorphism in that range.

(b) By (a), the map

$$H^i(A \otimes_R A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F)$$

is an isomorphism for every i . On the other hand, the assumed isomorphism $A \otimes_R A \longrightarrow A$ induces an isomorphism

$$H^i(A|R, F) \longrightarrow H^i(A \otimes_R A|R, F).$$

Their composite is the diagonal map

$$H^i(A|R, F) \longrightarrow H^i(A|R, F) \oplus H^i(A|R, F).$$

Since it is an isomorphism, $H^i(A|R, F) = 0$ for every i . □

Corollary 3.9. Let S be a multiplicative subset of an R -algebra A . Suppose that

$$F \in \text{Ob}(\tilde{\mathcal{C}}_{S^{-1}A|R})$$

is additive and that F_A is also additive. Then the canonical map

$$H^i(S^{-1}A|R, F) \longrightarrow H^i(A|R, F)$$

is an isomorphism for every i . Here $F_A \in \text{Ob}(\tilde{\mathcal{C}}_{S^{-1}A|A})$ is defined by

$$F_A(X) = \ker(F(X) \longrightarrow F(A))$$

for every $X \in \text{Ob}(\mathcal{C}_{S^{-1}A|A})$.

Proof. Applied to $R \longrightarrow A \longrightarrow S^{-1}A$, Proposition 3.7(c) gives the exact sequence

$$\cdots \longrightarrow H^n(S^{-1}A|A, F_A) \longrightarrow H^n(S^{-1}A|R, F) \longrightarrow H^n(A|R, F) \longrightarrow \cdots$$

The map

$$S^{-1}A \otimes_A S^{-1}A \longrightarrow S^{-1}A$$

is an isomorphism, and F_A is additive by hypothesis. Corollary 3.8(b) therefore gives $H^n(S^{-1}A|A, F_A) = 0$ for every n . The middle map in the exact sequence is consequently an isomorphism for every n . □

Let A be an R -algebra. An A -module M defines an additive sheaf

$$\text{Der}_R(-, M) \in \text{Ob}(\tilde{\mathcal{C}}_{A|R}), \quad X \longmapsto \text{Der}_R(X, M).$$

We make the following definition.

Definition 3.10. Let A be an R -algebra. For every A -module M , set

$$D^i(A|R, M) = H^i(A|R, \text{Der}_R(-, M)).$$

We now compute the groups of low degree, namely $D^i(A|R, M)$ for $i = 1, 2$. Choose a surjective arrow $P \rightarrow A$ in $\underline{C}_{A|R}$ with P a polynomial algebra over R , and write

$$0 \rightarrow I \rightarrow P \rightarrow A \rightarrow 0.$$

Lemma 3.10.

$$\begin{aligned} D^1(A|R, M) &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_P(I, M)), \\ D^2(A|R, M) &= H^1(A|R, \underline{H}_{A|R}^1(\text{Der}_R(-, M))). \end{aligned}$$

Proof. Consider the simplicial algebra

$$(*) \quad \cdots P_A^{(2)} \rightrightarrows P_A^{(1)} \rightrightarrows P_A^{(0)},$$

where

$$P_A^{(n)} = \overbrace{P \times_A \cdots \times_A P}^{n+1}.$$

For every n there is an exact sequence

$$0 \rightarrow I^{(n)} \rightarrow P_A^{(n)} \rightarrow A \rightarrow 0, \quad I^{(n)} = \overbrace{I \times \cdots \times I}^{n+1}.$$

Put $\bar{I} = I/I^2$, and let Δ denote the diagonal map. It is straightforward to verify that, for every n , the sequence

$$(a) \quad 0 \rightarrow \text{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \rightarrow \text{Der}_R(P_A^{(n)}, M) \xrightarrow{\text{Der}(\Delta, M)} \text{Der}_R(P, M) \rightarrow 0$$

is exact. We also have the exact sequence

$$(b) \quad 0 \rightarrow \text{Hom}_A(\bar{I}^{(n)}/\Delta(\bar{I}), M) \rightarrow \text{Hom}_A(\bar{I}^{(n)}, M) \xrightarrow{\text{Hom}(\Delta, M)} \text{Hom}_A(\bar{I}, M) \rightarrow 0.$$

The sequences (a) and (b) yield the corresponding exact sequences of simplicial A -modules. Since $\{I \rightarrow 0\}$ and $\{I \rightarrow I\}$ admit sections,

$$H^i(\{I \rightarrow 0\}, \text{Hom}_A(-, M)) = H^i(\{I \rightarrow I\}, \text{Hom}_A(-, M)) = 0 \quad (i > 0).$$

It follows that

$$\left\{ \begin{array}{l} 0 \rightarrow \text{Der}_R(A, M) \rightarrow \text{Der}_R(P, M) \rightarrow \text{Hom}_A(\bar{I}, M) \\ \quad \rightarrow H^1(\{P \rightarrow A\}, \text{Der}_R(-, M)) \rightarrow 0 \quad \text{is exact,} \\ H^i(\{P \rightarrow A\}, \text{Der}_R(-, M)) = 0 \quad (i > 1). \end{array} \right.$$

By 3.1(b), therefore,

$$\begin{aligned} D^1(A|R, M) &= \check{H}^1(A|R, \text{Der}_R(-, M)) \\ &= H^1(\{P \rightarrow A\}, \text{Der}_R(-, M)) \\ &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_A(\bar{I}, M)) \\ &= \text{Coker}(\text{Der}_R(P, M) \rightarrow \text{Hom}_P(I, M)), \\ D^2(A|R, M) &= H^1(A|R, \underline{H}_{A|R}^1(\text{Der}_R(-, M))). \end{aligned}$$

□

Choose a free P -module F and a P -linear map $\gamma : F \rightarrow P$ such that $\gamma(F) = I$. Associated with γ is a Koszul complex, denoted by K_I by abuse of notation. Lemma 3.11. There is an isomorphism

$$D^2(A|R, M) \simeq \text{Coker}(H^1(K_I, M) \rightarrow \text{Hom}_A(H_1(K_I), M)).$$

Proof. We have

$$\begin{aligned} D^2(A|R, M) &= H^1(A|R, \underline{H}_{A|R}^1(\text{Der}_R(-, M))) \\ &= \ker \left(H^1(P_A^{(1)}|R, M) \rightrightarrows H^1(P_A^{(2)}|R, M) \right), \end{aligned}$$

because

$$[\underline{H}_{A|R}^1(\text{Der}_R(-, M))](P_A^{(0)}) = [\underline{H}_{A|R}^1(\text{Der}_R(-, M))](P) = 0.$$

Let

$$0 \longrightarrow K \longrightarrow F \xrightarrow{\gamma} I \longrightarrow 0$$

be exact. Write $P[F]$ for the symmetric algebra over P generated by F . Then $P[F]$ and $P[F] \otimes_P P[F]$ are polynomial algebras over R , and there are surjective maps

$$\begin{aligned} P[F] &\longrightarrow P_A^{(1)} = P \times_A P, & X &\longmapsto (0, \gamma(X)), \\ P[F] \otimes_P P[F] &\longrightarrow P_A^{(2)} = P \times_A P \times_A P, & X \otimes 1 &\longmapsto (0, \gamma(X), 0), \\ & & 1 \otimes X &\longmapsto (0, 0, \gamma(X)), \end{aligned}$$

for $X \in F$. Put

$$J = \ker(P[F] \longrightarrow P_A^{(1)}), \quad J' = \ker(P[F] \otimes_P P[F] \longrightarrow P_A^{(2)}).$$

If F_γ denotes the ideal generated by $\{X - \gamma(X) \mid X \in F\}$, then

$$\begin{aligned} J &= (F) \cap (F_\gamma) = (K) + (F \cdot F_\gamma), \\ J' &= (F \otimes 1 + 1 \otimes F) \cap (F_\gamma \otimes 1 + 1 \otimes F) \cap (F \otimes 1 + 1 \otimes F_\gamma) \\ &= (K \otimes 1 + 1 \otimes K) + (F \cdot F_\gamma \otimes 1 + 1 \otimes F \cdot F_\gamma) + (F \otimes F). \end{aligned}$$

The three maps $\pi_i : P_A^{(2)} \longrightarrow P_A^{(1)}$ admit lifts $\widehat{\pi}_i : P[F] \otimes_P P[F] \longrightarrow P[F]$ given, for $X \in F$, by

$$\widehat{\pi}_0 : \begin{cases} X \otimes 1 \longmapsto \gamma(X) - X, \\ 1 \otimes X \longmapsto X, \end{cases} \quad \widehat{\pi}_1 : \begin{cases} X \otimes 1 \longmapsto 0, \\ 1 \otimes X \longmapsto X, \end{cases} \quad \widehat{\pi}_2 : \begin{cases} X \otimes 1 \longmapsto X, \\ 1 \otimes X \longmapsto 0. \end{cases}$$

A direct computation shows that $\widehat{\pi}_0 - \widehat{\pi}_1 + \widehat{\pi}_2$ annihilates

$$K \otimes 1 + 1 \otimes K + 1 \otimes F \cdot F_\gamma$$

and maps $F \cdot F_\gamma \otimes 1 + F \otimes F$ onto $F \cdot F_\gamma$. For every

$$D \in \text{Der}_R(P[F] \otimes_P P[F], M)$$

one has $D(F \cdot F_\gamma \otimes 1 + F \otimes F) = 0$, because $I \cdot M = 0$. Consequently,

$$\begin{aligned} D^2(A|R, M) &= \ker \left(D^1(P_A^{(1)}|R, M) \rightrightarrows D^1(P_A^{(2)}|R, M) \right) \\ &= \{[f] \in D^1(P_A^{(1)}|R, M) \mid f(F \cdot F_\gamma) = 0\} \\ &= \text{Coker} \left(\text{Der}_R(P[F], M) \longrightarrow \text{Hom}_{P[F]} \left(\frac{(F) \cap (F_\gamma)}{F \cdot F_\gamma}, M \right) \right) \\ &= \text{Coker} \left(\text{Der}_P(P[F], M) \longrightarrow \text{Hom}_{P[F]} \left(\frac{(F) \cap (F_\gamma)}{(F)(F_\gamma)}, M \right) \right). \end{aligned}$$

The sequences

$$0 \longrightarrow (F) \longrightarrow P[F] \xrightarrow{P[0]} P \longrightarrow 0, \quad 0 \longrightarrow (F_\gamma) \longrightarrow P[F] \xrightarrow{P[\gamma]} P_\gamma \longrightarrow 0$$

are exact. Hence

$$\frac{(F) \cap (F_\gamma)}{(F)(F_\gamma)} = \text{Tor}_1^{P[F]}(P, P_\gamma) = H_1(K_I),$$

where K_I is the Koszul complex associated with $F \xrightarrow{\gamma} I \subset P$. We therefore obtain

$$\begin{aligned} D^2(A|R, M) &= \text{Coker}(\text{Hom}_P(F, M) \longrightarrow \text{Hom}_P(H_1(K_I), M)) \\ &= \text{Coker}(H^1(K_I, M) \longrightarrow \text{Hom}_A(H_1(K_I), M)). \end{aligned}$$

□

We can now state the basic facts needed below, summarizing the results already established.
Theorem 3.12.

(1) If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A -modules, then the sequence

$$\begin{aligned} 0 \longrightarrow D^0(A|R, M') \longrightarrow D^0(A|R, M) \longrightarrow D^0(A|R, M'') \longrightarrow D^1(A|R, M') \\ \longrightarrow D^1(A|R, M) \longrightarrow D^1(A|R, M'') \longrightarrow \cdots \longrightarrow D^n(A|R, M') \\ \longrightarrow D^n(A|R, M) \longrightarrow D^n(A|R, M'') \longrightarrow \cdots \end{aligned}$$

is exact.

(2) Let $S \longrightarrow R \longrightarrow A$ be ring homomorphisms. Then, for every A -module M , the sequence

$$\begin{aligned} 0 \longrightarrow D^0(A|R, M) \longrightarrow D^0(A|S, M) \longrightarrow D^0(R|S, M) \longrightarrow D^1(A|R, M) \\ \longrightarrow D^1(A|S, M) \longrightarrow D^1(R|S, M) \longrightarrow \cdots \longrightarrow D^n(A|R, M) \\ \longrightarrow D^n(A|S, M) \longrightarrow D^n(R|S, M) \longrightarrow \cdots \end{aligned}$$

is exact.

(3) Let A and S be R -algebras such that $\text{Tor}_i^R(A, S) = 0$ for every $i > 0$. Then, for every $A \otimes_R S$ -module M , the canonical map

$$D^i(A \otimes_R S|S, M) \longrightarrow D^i(A|R, M)$$

is an isomorphism for every i . In particular, if A is flat over R , then

$$D^i(A \otimes_R S|S, M) \xrightarrow{\sim} D^i(A|R, M)$$

for every i , every R -algebra S , and every $A \otimes_R S$ -module M .

(4) Let S be a multiplicative subset of A . For every $S^{-1}A$ -module N , the canonical map

$$D^i(S^{-1}A|R, N) \longrightarrow D^i(A|R, N)$$

is an isomorphism for every i . Consequently, for every A -module M , there is a canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M)$$

for every i .

(5) Let A and B be R -algebras such that $\text{Tor}_i^R(A, B) = 0$ for $0 < i \leq n$. Then, for every $A \otimes_R B$ -module M , the canonical map

$$D^i(A \otimes_R B|R, M) \longrightarrow D^i(A|R, M) \oplus D^i(B|R, M)$$

is an isomorphism for $i \leq n$. In particular, suppose that $\mathrm{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that the diagonal

$$\Delta : \mathrm{Spec}(A) \longrightarrow \mathrm{Spec}(A) \times_{\mathrm{Spec}(R)} \mathrm{Spec}(A)$$

is an open immersion—for example, suppose that A is *étale* over R . Then

$$D^i(A|R, M) = 0$$

for every $i > 0$ and every A -module M .

- (6) Let R be noetherian, let A be an R -algebra essentially of finite type, and let E be a flat A -module. Then the canonical map

$$E \otimes_A D^i(A|R, M) \xrightarrow{\sim} D^i(A|R, E \otimes_A M)$$

is an isomorphism for every $i \geq 0$ and every A -module M .

- (7) If R is noetherian, A is an R -algebra essentially of finite type, and M is an A -module of finite type, then $D^i(A|R, M)$ is an A -module of finite type for every i .
(8) Let P be a polynomial algebra over R , and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. Then

$$D^0(A|R, M) = \mathrm{Der}_R(A, M),$$

$$\begin{aligned} D^1(A|R, M) &= \mathrm{Coker}(\mathrm{Der}_R(P, M) \longrightarrow \mathrm{Hom}_A(I/I^2, M)) \\ &= \{ \text{isomorphism classes of commutative } R\text{-algebra extensions of } A \text{ by } M \}, \\ D^2(A|R, M) &= \mathrm{Coker}(H^1(K_I, M) \longrightarrow \mathrm{Hom}_A(H_1(K_I), M)). \end{aligned}$$

Proof.

- (1) If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A -modules, then, for every polynomial algebra P over R , the sequence

$$0 \longrightarrow \mathrm{Der}_R(P, M') \longrightarrow \mathrm{Der}_R(P, M) \longrightarrow \mathrm{Der}_R(P, M'') \longrightarrow 0$$

is exact. Hence, by 3.3(c),

$$0 \longrightarrow \mathrm{Der}_R(-, M') \longrightarrow \mathrm{Der}_R(-, M) \longrightarrow \mathrm{Der}_R(-, M'') \longrightarrow 0$$

is an exact sequence of sheaves.

- (2) The functor $\mathrm{Der}_S(-, M)$ is an additive sheaf on $\underline{\mathcal{C}}_{A|S}$, and

$$\mathrm{Der}_R(B, M) = \ker(\mathrm{Der}_S(B, M) \longrightarrow \mathrm{Der}_S(R, M))$$

for every $B \in \mathrm{Ob}(\underline{\mathcal{C}}_{A|R})$. The assertion therefore follows from 3.7(c).

- (3) Let

$$S : \underline{\mathcal{C}}_{A|R} \longrightarrow \underline{\mathcal{C}}_{A \otimes_R S|S}, \quad X \longmapsto X \otimes_R S.$$

Then $\tilde{S} \mathrm{Der}_S(-, M) = \mathrm{Der}_R(-, M)$, so the assertion follows from 3.7(a).

- (4) This follows from 3.9. Indeed, $F = \text{Der}_R(-, N)$ is an additive sheaf on $\underline{C}_{S^{-1}A|R}$, and $F_A \in \text{Ob}(\tilde{\underline{C}}_{S^{-1}A|A})$ is also additive, since

$$\begin{aligned} F_A(X) &= \ker(F(X) \longrightarrow F(A)) \\ &= \ker(\text{Der}_R(X, N) \longrightarrow \text{Der}_R(A, N)) \\ &= \text{Der}_A(X, N). \end{aligned}$$

Thus $F_A = \text{Der}_A(-, N)$.

- (5) The first assertion follows from 3.8(a). Now suppose that $\text{Tor}_i^R(A, A) = 0$ for every $i > 0$ and that

$$\Delta : \text{Spec}(A) \longrightarrow \text{Spec}(A) \times_{\text{Spec}(R)} \text{Spec}(A)$$

is an open immersion. Let $\varphi : A \otimes_R A \longrightarrow A$ be the multiplication map. For every maximal ideal $\mathfrak{m}$ of A , the induced map

$$(A \otimes_R A)_{\varphi^{-1}(\mathfrak{m})} \longrightarrow A_{\mathfrak{m}}$$

is an isomorphism. The first assertion, together with 3.12(4), therefore implies that the diagonal map

$$\Delta_{\mathfrak{m}} : D^i(A|R, M)_{\mathfrak{m}} \longrightarrow D^i(A|R, M)_{\mathfrak{m}} \oplus D^i(A|R, M)_{\mathfrak{m}}$$

is an isomorphism for every maximal ideal $\mathfrak{m}$ of A . Consequently,

$$\Delta : D^i(A|R, M) \longrightarrow D^i(A|R, M) \oplus D^i(A|R, M)$$

is an isomorphism, and hence $D^i(A|R, M) = 0$ for every $i > 0$.

- (6) In view of 3.12(4), we may assume that A is an R -algebra of finite type. Consider the morphism of sheaves

$$E \otimes_A \text{Der}_R(-, M) \longrightarrow \text{Der}_R(-, E \otimes_A M)$$

on $\tilde{\underline{C}}_{A|R}$. Since E is flat over A , the map

$$E \otimes_A \text{Der}_R(B, M) \xrightarrow{\sim} \text{Der}_R(B, E \otimes_A M)$$

is an isomorphism whenever $B \in \text{Ob}(\underline{C}_{A|R})$ is of finite type over R . By 3.3(b), it follows that

$$D^i(A|R, E \otimes_A \text{Der}_R(-, M)) \longrightarrow D^i(A|R, \text{Der}_R(-, E \otimes_A M))$$

is an isomorphism. On the other hand, $E \otimes_A -$ is exact, and therefore

$$E \otimes_A D^i(A|R, M) = D^i(A|R, E \otimes_A \text{Der}_R(-, M)).$$

This proves the assertion.

- (7) This follows from 3.3(a).
 (8) In view of 3.10 and 3.11, it suffices to show that

$$\text{Coker}(\text{Der}_R(P, M) \longrightarrow \text{Hom}_A(I/I^2, M)) = \text{Exalcomm}(A|R, M),$$

where $\text{Exalcomm}(A|R, M)$ denotes the set of isomorphism classes of R -algebra extensions of A by M . Let

$$0 \longrightarrow M \longrightarrow A' \longrightarrow A \longrightarrow 0$$

be such an extension. Since P is a polynomial algebra over R , we obtain a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A \longrightarrow 0. \end{array}$$

The map $I \longrightarrow M$ is determined by the extension A' , up to the action of $\text{Der}_R(P, M)$. We thus obtain a map

$$\Phi : \text{Exalcomm}(A|R, M) \longrightarrow \text{Coker}(\text{Der}_R(P, M) \longrightarrow \text{Hom}_A(I/I^2, M)).$$

Conversely, a P -linear map $I \longrightarrow M$ yields, by pushout, an extension fitting into a diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & I & \longrightarrow & P & \longrightarrow & A \longrightarrow 0 \\ & & \downarrow & & \downarrow & & \parallel \\ 0 & \longrightarrow & M & \longrightarrow & A' & \longrightarrow & A \longrightarrow 0. \end{array}$$

Two P -linear maps $I \rightrightarrows M$ that differ by an element of $\text{Der}_R(P, M)$ yield isomorphic extensions. Hence there is a map

$$\Psi : \text{Coker}(\text{Der}_R(P, M) \longrightarrow \text{Hom}_A(I/I^2, M)) \longrightarrow \text{Exalcomm}(A|R, M).$$

It is straightforward to verify that Φ and Ψ are inverse to each other. □

Corollary 3.13.

- (a) The R -algebra A is formally smooth over R if and only if $D^1(A|R, M) = 0$ for every A -module M .
- (b) Let R be noetherian and let A be an R -algebra of finite type. Then A is a relative complete intersection over R if and only if $D^2(A|R, M) = 0$ for every A -module M .
- (c) Let R be noetherian and let A be an R -algebra of finite type. For every multiplicative subset S of A , there is a canonical isomorphism

$$D^i(S^{-1}A|R, S^{-1}M) \xrightarrow{\sim} S^{-1}D^i(A|R, M) \quad (i \geq 0).$$

- (d) Let R be noetherian and let A be a local R -algebra essentially of finite type. Write $\hat{A}$ for the $\mathfrak{m}$ -adic completion of A , where $\mathfrak{m}$ is the maximal ideal of A . Then, for every A -module E of finite type, there is a canonical isomorphism

$$D^1(\hat{A}|R, \hat{A} \otimes_A E) \xrightarrow{\sim} \hat{A} \otimes_A D^1(A|R, E).$$

Proof.

Part (a) follows immediately from $D^1(A|R, M) = \text{Exalcomm}(A|R, M)$.

For (b), let P be a polynomial algebra over R in finitely many variables, and let

$$0 \longrightarrow I \longrightarrow P \longrightarrow A \longrightarrow 0$$

be exact. After localization, we may suppose that I is generated by a P -regular sequence. Then $H_1(K_I) = 0$, and hence

$$D^2(A|R, M) = \text{Coker}(H^1(K_I, M) \longrightarrow \text{Hom}_A(H_1(K_I), M)) = 0.$$

Conversely, suppose that $D^2(A|R, M) = 0$ for every A -module M . Choose an exact sequence

$$0 \longrightarrow K \longrightarrow F \longrightarrow I \longrightarrow 0,$$

where F is a free P -module of finite type. If

$$K_0 = \text{Im}(\bigwedge^2 F \longrightarrow \bigwedge^1 F),$$

then $H_1(K_I) = K/K_0$. Since $D^2(A|R, H_1(K_I)) = 0$, the map

$$H^1(K_I, H_1(K_I)) \longrightarrow \text{Hom}_A(H_1(K_I), H_1(K_I))$$

is surjective. Consequently, the exact sequence

$$0 \longrightarrow K/K_0 \longrightarrow F/IF \longrightarrow I/I^2 \longrightarrow 0$$

splits as a sequence of A -modules. In particular, I/I^2 is projective. After localizing A once more, we may suppose that I/I^2 is free and that $F/IF \longrightarrow I/I^2$ is an isomorphism. The split exact sequence then gives $H_1(K_I) = K/K_0 = 0$. Thus I is generated by a P -regular sequence, so A is a relative complete intersection over R .

Part (c) follows immediately from 3.12(4) and (6).

For (d), 3.12(6) first gives canonical isomorphisms

$$D^i(A|R, \widehat{A} \otimes_A E) \xrightarrow{\sim} \widehat{A} \otimes_A D^i(A|R, E)$$

for every i . It therefore remains to prove that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is an isomorphism. We use the description of D^1 in terms of algebra extensions.

Let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow A' \xrightarrow{\varphi} \widehat{A} \longrightarrow 0$$

be an R -algebra extension whose restriction to A splits. Since $\widehat{A} \otimes_A E$ is an $\widehat{A}$ -module of finite type, A' is a local R -algebra complete for its $\mathfrak{m}'$ -adic topology, where $\mathfrak{m}'$ is the maximal ideal of A' . If $\psi : A \longrightarrow A'$ is an R -algebra map lifting the canonical map $A \longrightarrow \widehat{A}$, then ψ extends uniquely to a map $\widehat{\psi} : \widehat{A} \longrightarrow A'$ satisfying $\varphi \circ \widehat{\psi} = \text{id}_{\widehat{A}}$. This proves that

$$D^1(\widehat{A}|R, \widehat{A} \otimes_A E) \longrightarrow D^1(A|R, \widehat{A} \otimes_A E)$$

is injective.

Now let

$$0 \longrightarrow \widehat{A} \otimes_A E \longrightarrow B \xrightarrow{\varphi} A \longrightarrow 0$$

be an R -algebra extension, and put $\widehat{E} = \widehat{A} \otimes_A E$. We first claim that the topology on $\widehat{E}$ induced by the $\mathfrak{m}_B$ -adic topology on B coincides with its topology as an $\widehat{A}$ -module. Indeed, since A is essentially of finite type over R , there is a subalgebra $B_0 \subset B$ such that B_0 is a local R -algebra essentially of finite type and $\varphi(B_0) = A$. Put $E_0 = \widehat{E} \cap B_0$, and denote the maximal ideals of B , B_0 , and A by $\mathfrak{m}_B$, $\mathfrak{m}_0$, and $\mathfrak{m}$, respectively. Then

$$\mathfrak{m}_B = \mathfrak{m}_0 + \widehat{E}, \quad \mathfrak{m}_B^{n+1} = \mathfrak{m}_0^{n+1} + \mathfrak{m}^n \widehat{E}$$

for every n . Since B_0 is noetherian, the Artin–Rees theorem gives an integer $r > 0$ such that, for every $n \geq r$,

$$\begin{aligned} \widehat{E} \cap \mathfrak{m}_B^{n+1} &= \mathfrak{m}^n \widehat{E} + \widehat{E} \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + E_0 \cap \mathfrak{m}_0^{n+1} \\ &= \mathfrak{m}^n \widehat{E} + \mathfrak{m}^{n+1-r} (E_0 \cap \mathfrak{m}_0^r) \subset \mathfrak{m}^{n+1-r} \widehat{E}. \end{aligned}$$

This proves the assertion about the two topologies. In particular, $\widehat{E}$ is complete for the topology induced by the $\mathfrak{m}_B$ -adic topology on B , and hence completion gives an exact commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \widehat{E} & \longrightarrow & B & \longrightarrow & A \longrightarrow 0 \\ & & \parallel & & \downarrow & & \downarrow \\ 0 & \longrightarrow & \widehat{E} & \longrightarrow & \widehat{B} & \longrightarrow & \widehat{A} \longrightarrow 0. \end{array}$$

Here $\widehat{B}$ denotes the $\mathfrak{m}_B$ -adic completion of B . Thus

$$D^1(\widehat{A}|R, \widehat{E}) \longrightarrow D^1(A|R, \widehat{E})$$

is surjective, and the proof is complete. $\square$

Corollary 3.14. Let R be noetherian, and let A be an R -algebra of finite type that is smooth everywhere except at one closed point $x \in \text{Spec}(A)$. Then there are isomorphisms

$$D^1(A|R, A) \simeq D^1(A_x|R, A_x) \simeq D^1(\widehat{A}_x|R, \widehat{A}_x).$$

Proof. Since A is smooth over R away from x , $D^1(A|R, A)$ is an A -module of finite type supported on $\{x\}$. Hence

$$D^1(A|R, A) = D^1(A|R, A)_x = D^1(\widehat{A}|R, \widehat{A})_x.$$

The assertion now follows from 3.13. $\square$

Remark 3.15. Let R be a topological ring and A a topological R -algebra. For every A -module E , define

$$D_{\text{top}}^i(A|R, E) = \varinjlim_{\alpha} D^i(A_{\alpha}|R_{\alpha}, A_{\alpha} \otimes_A E),$$

where $A_{\alpha} = A/J_{\alpha}$, $R_{\alpha} = R/I_{\alpha}$, and J_{α} and I_{α} are open in A and R , respectively. Thus A is formally smooth as a topological R -algebra precisely when $D_{\text{top}}^1(A|R, E) = 0$ for every A -module E .

Now let X be a scheme over a ring R , and let $\underline{E}$ be a quasi-coherent $\mathcal{O}_X$ -module. We define sheaves

$$\underline{D}_{X|R}^i(\underline{E}) = \underline{D}^i(X|R, \underline{E})$$

on X from the presheaves on affine open subsets $U \subset X$ given by

$$[\underline{D}_{X|R}^i(\underline{E})](U) = D^i(\Gamma(U, \mathcal{O}_X)|R, \Gamma(U, \underline{E})).$$

If R is noetherian and X is of finite type over R , the localization property 3.13(c) shows that these presheaves canonically form quasi-coherent $\mathcal{O}_X$ -modules.

When no confusion is possible, we abbreviate

$$\underline{D}_{X|R}^i = \underline{D}_{X|R}^i(\mathcal{O}_X).$$

All the assertions of 3.13 have corresponding formulations for the quasi-coherent sheaves $\underline{D}_{X|R}^i(\underline{E})$. We leave those formulations to the reader and record below the properties most relevant to our purpose.

Corollary 3.15. Let R be noetherian and let X be an R -scheme of finite type.

(a) For every coherent $\mathcal{O}_X$ -module $\underline{E}$, $\underline{D}_{X|R}^i(\underline{E})$ is again coherent for every i . Moreover,

$$\text{Supp } \underline{D}_{X|R}^i(\underline{E}) \subset X^{\text{Sing}} \quad (i > 0),$$

where

$$X^{\text{Sing}} = \{x \in X \mid X \text{ is not smooth over } R \text{ at } x\}.$$

(b) If X is flat over R , then for every R -algebra S there is a canonical isomorphism

$$\underline{D}_{X \otimes_R S|R}^i(\underline{E}) \xrightarrow{\sim} \underline{D}_{X|R}^i(\underline{E})$$

for every i .

- (c) Suppose that X is a relative complete intersection over R . For every surjective ring homomorphism $S \rightarrow R$, there is an exact sequence

$$0 \rightarrow \underline{D}_{X|R}^1(\underline{E}) \rightarrow \underline{D}_{X|S}^1(\underline{E}) \rightarrow \underline{\mathrm{Hom}}_{\mathcal{O}_X}((I/I^2) \otimes_R \mathcal{O}_X, \underline{E}) \rightarrow 0,$$

where $I = \ker(S \rightarrow R)$. If, moreover, R is local with residue field k and X is flat over R , then

$$0 \rightarrow \underline{D}_{X_0|k}^1(\mathcal{O}_{X_0}) \rightarrow \underline{D}_{X|S}^1(\mathcal{O}_{X_0}) \rightarrow D^1(R|S, k) \otimes_k \mathcal{O}_{X_0} \rightarrow 0,$$

where $X_0 = X \otimes_R k$.

Proof. Part (a) follows from the finiteness and low-degree descriptions in 3.12, while (b) follows from 3.12(3).

For (c), because X is a relative complete intersection over R , we have $\underline{D}_{X|R}^i(\underline{E}) = 0$ for every $i \geq 2$. The transitivity sequence 3.12(2) therefore gives an exact sequence

$$0 \rightarrow \underline{D}_{X|R}^0(\underline{E}) \rightarrow \underline{D}_{X|S}^0(\underline{E}) \rightarrow \underline{D}_{R|S}^0(\underline{E}) \rightarrow \underline{D}_{X|R}^1(\underline{E}) \rightarrow \underline{D}_{X|S}^1(\underline{E}) \rightarrow \underline{D}_{R|S}^1(\underline{E}) \rightarrow 0,$$

where the quasi-coherent sheaves $\underline{D}_{R|S}^i(\underline{E})$ on X are defined on every affine open subset $U \subset X$ by

$$[\underline{D}_{R|S}^i(\underline{E})](U) = D^i(R|S, \underline{E}(U)).$$

Since $S \rightarrow R$ is surjective, $\underline{D}_{R|S}^0(\underline{E}) = 0$; and the low-degree description in 3.12(8) gives

$$\underline{D}_{R|S}^1(\underline{E}) = \underline{\mathrm{Hom}}_{\mathcal{O}_X}((I/I^2) \otimes_R \mathcal{O}_X, \underline{E}).$$

This proves the first exact sequence.

If X is flat over R , then, for every quasi-coherent $\mathcal{O}_{X_0}$ -module $\underline{E}$, there are canonical isomorphisms

$$\underline{D}_{X_0|k}^i(\underline{E}) \xrightarrow{\sim} \underline{D}_{X|R}^i(\underline{E})$$

for every i . In particular,

$$\underline{D}_{X_0|k}^1(\mathcal{O}_{X_0}) \xrightarrow{\sim} \underline{D}_{X|R}^1(\mathcal{O}_{X_0}),$$

from which the second exact sequence follows. $\square$

Remark 3.16. The quasi-coherent sheaves $\underline{D}_{X|R}^i(\underline{E})$ defined above should occur as the terms $E_2^{0,i}$ of a spectral sequence relating the local theory to the global theory. L. Illusie has obtained such a theory in *Lecture Notes in Mathematics* 239. His method uses “homotopical algebra,” generalizing the methods of M. André and D. Quillen, rather than “cohomological algebra.” It seems that the approach taken here can be globalized by means of Grothendieck cohomology on the category of schemes, equipped with an appropriate topology generalizing the one on the affine category.

4. Formal moduli of deformations

Among the natural cofibered categories whose isomorphism classes of objects are generally not prorepresentable but nevertheless admit a formal versal object is the cofibered category of deformations. We return to the situation and notation of section 1. Thus, throughout this section, Λ is a fixed complete noetherian local ring with residue field k , and $\underline{\mathcal{C}}_\Lambda$ is the category of artinian local Λ -algebras with residue field k .

Let X_0 be a fixed scheme over k . By an (infinitesimal) deformation of X_0 we mean a pair (R, X) , where $R \in \text{Ob}(\underline{\mathcal{C}}_\Lambda)$ and X is a flat R -scheme, together with a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R) \end{array}$$

such that the induced morphism

$$X_0 \longrightarrow X \times_{\text{Spec}(R)} \text{Spec}(k)$$

is an isomorphism. Since R is artinian local, $i_X : X_0 \longrightarrow X$ is necessarily a closed immersion and a homeomorphism on the underlying topological spaces.

If (R, X) and (R', X') are deformations of X_0 over R and R' , respectively, an arrow

$$(R', X') \longrightarrow (R, X)$$

consists of a map $\varphi : R' \longrightarrow R$ in $\underline{\mathcal{C}}_\Lambda$ and a morphism of schemes $\Phi : X \longrightarrow X'$ fitting into a commutative diagram

$$\begin{array}{ccccc} X & & \xleftarrow{\Phi} & & X' \\ & \swarrow i_X & & \searrow i_{X'} & \\ & X_0 & & & \\ & \downarrow & & & \\ & \text{Spec}(k) & & & \\ & \swarrow & & \searrow & \\ \text{Spec}(R) & & \xrightarrow{\text{Spec}(\varphi)} & & \text{Spec}(R'). \end{array}$$

This defines the category $\underline{M}_{X_0}$ of infinitesimal deformations of X_0 . It is equipped with the functor

$$p : \underline{M}_{X_0} \longrightarrow \underline{\mathcal{C}}_\Lambda, \quad (R, X) \longmapsto R.$$

Let $(R, X) \in \text{Ob}(\underline{M}_{X_0})$ and let $\varphi : R \longrightarrow S$ be a map in $\underline{\mathcal{C}}_\Lambda$. There is a commutative diagram

$$\begin{array}{ccc} X \times_{\text{Spec}(R)} \text{Spec}(S) & \xrightarrow{p_1} & X \\ \text{flat} \downarrow & & \downarrow \text{flat} \\ \text{Spec}(S) & \xrightarrow{\text{Spec}(\varphi)} & \text{Spec}(R), \end{array}$$

and the φ -arrow

$$(\varphi, p_1) : (R, X) \longrightarrow (S, X \times_{\text{Spec}(R)} \text{Spec}(S))$$

is cocartesian in $\underline{M}_{X_0}$. Hence $\underline{M}_{X_0}$ is a cofibered category over $\underline{\mathcal{C}}_\Lambda$. We record the following facts.

4.1(a). Let

$$(1_R, \Phi) : (R, X_1) \longrightarrow (R, X_2)$$

be an arrow in $\underline{M}_{X_0}$. Thus $\Phi : X_2 \rightarrow X_1$ is an R -morphism whose reduction

$$X_2 \otimes_R k \rightarrow X_1 \otimes_R k$$

is an isomorphism. Since R is artinian and the X_i are flat over R , Φ is an isomorphism. Consequently, $\underline{M}_{X_0}$ is a cofibered category in groupoids over $\underline{C}_\Lambda$.

4.1(b). Suppose that X_0 is affine, say $X_0 = \text{Spec}(A_0)$. Then every $(R, X) \in \text{Ob}(\underline{M}_{X_0})$ has X affine as well. Consequently, for every $R \in \text{Ob}(\underline{C}_\Lambda)$, taking global sections identifies $\underline{M}_{X_0}(R)$ with the following category. An object is a flat R -algebra A together with an R -algebra map $j_A : A \rightarrow A_0$ inducing an isomorphism

$$A \otimes_R k \xrightarrow{\sim} A_0.$$

A morphism $\Phi : A_1 \rightarrow A_2$ is an R -algebra map for which the diagram

$$\begin{array}{ccc} A_1 & \xrightarrow{\Phi} & A_2 \\ j_{A_1} \downarrow & & \downarrow j_{A_2} \\ A_0 & \xlongequal{\quad} & A_0 \end{array}$$

commutes.

4.1(c). Consider the cofibered category $\widehat{\underline{M}}_{X_0}$ in groupoids over $\widehat{\underline{C}}_\Lambda$. By definition, for every $R \in \text{Ob}(\widehat{\underline{C}}_\Lambda)$, the category $\widehat{\underline{M}}_{X_0}(R)$ is canonically equivalent to the category of formal schemes $\mathfrak{X}$, adic over $\text{Spf}(R)$, such that

$$\mathfrak{X}_n = \mathfrak{X} \otimes_R R/\mathfrak{m}^{n+1}$$

is flat over $R/\mathfrak{m}^{n+1}$ for every $n \geq 0$. If X_0 is noetherian, this is equivalent to requiring $\mathfrak{X}$ to be flat over $\text{Spf}(R)$. Thus, in that case, $\widehat{\underline{M}}_{X_0}$ is canonically equivalent over $\widehat{\underline{C}}_\Lambda$ to the category of deformations of X_0 over the objects of $\widehat{\underline{C}}_\Lambda$, in the sense of formal schemes.

4.1(d). Let $(R, X) \in \text{Ob}(\underline{M}_{X_0})$, so that there is a commutative diagram

$$\begin{array}{ccc} X_0 & \xrightarrow{i_X} & X \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R) \end{array}$$

whose induced map $X_0 \rightarrow X \times_{\text{Spec}(R)} \text{Spec}(k)$ is an isomorphism.

Let $U_0 \subset X_0$ be an open subscheme. Since $i_X : X_0 \rightarrow X$ is a closed immersion and a homeomorphism on underlying spaces, restriction gives a commutative diagram

$$\begin{array}{ccc} U_0 & \xrightarrow{i_X|_{U_0}} & X|_{U_0} \\ \downarrow & & \downarrow \text{flat} \\ \text{Spec}(k) & \longrightarrow & \text{Spec}(R). \end{array}$$

The induced map

$$U_0 \rightarrow (X|_{U_0}) \times_{\text{Spec}(R)} \text{Spec}(k) = (X \otimes_R k)|_{U_0}$$

is an isomorphism, so $(R, X|_{U_0})$ is a deformation of U_0 over R . Moreover, every arrow $(\varphi, \Phi) : (R, X) \longrightarrow (R', X')$ in $\underline{M}_{X_0}$ restricts to an arrow

$$(\varphi, \Phi|_{U_0}) : (R, X|_{U_0}) \longrightarrow (R', X'|_{U_0})$$

in $\underline{M}_{U_0}$. We therefore have a restriction functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0}$$

over $\underline{C}_\Lambda$.

One may likewise localize and complete at a point. Let $x_0 \in X_0$. If $(R, X) \in \text{Ob}(\underline{M}_{X_0})$, then $\text{Spec}(\mathcal{O}_{X, x_0})$ is a deformation of $\text{Spec}(\mathcal{O}_{X_0, x_0})$ over R , while $\text{Spec}(\widehat{\mathcal{O}}_{X, x_0})$ is a deformation of $\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})$ over R ; the hats denote completion for the topology defined by powers of the maximal ideal. Thus there are functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x_0})} \longrightarrow \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})}.$$

We first prove that $\underline{M}_{X_0}$ is a homogeneous cofibered category over $\underline{C}_\Lambda$.

Lemma 4.2. Consider a diagram

$$\begin{array}{ccc} & & R' \\ & & \downarrow \\ R & \xrightarrow{\text{surjective}} & \bar{R} \end{array}$$

in $\underline{C}_\Lambda$, and let E and E' be flat modules over R and R' , respectively. Then, for every R' -homomorphism

$$E' \longrightarrow \bar{E} = E \otimes_R \bar{R},$$

the fiber product $E \times_{\bar{E}} E'$ is flat as a module over $R \times_{\bar{R}} R'$.

Proof. Let

$$0 \longrightarrow I \longrightarrow R \longrightarrow \bar{R} \longrightarrow 0$$

be exact, so that

$$0 \longrightarrow I \longrightarrow R \times_{\bar{R}} R' \longrightarrow R' \longrightarrow 0$$

is exact. Since E is flat over R ,

$$0 \longrightarrow I \otimes_R E \longrightarrow E \longrightarrow \bar{E} \longrightarrow 0$$

is exact, and hence we obtain the exact sequence

$$(*) \quad 0 \longrightarrow I \otimes_R E \longrightarrow E \times_{\bar{E}} E' \longrightarrow E' \longrightarrow 0.$$

Set $S = R \times_{\bar{R}} R'$. Since I is nilpotent, $E \times_{\bar{E}} E'$ is flat as an S -module if and only if $(E \times_{\bar{E}} E') \otimes_S R'$ is flat as an R' -module and

$$\text{Tor}_1^S(E \times_{\bar{E}} E', R') = 0.$$

But

$$I \otimes_R E = I \otimes_S (E \times_{\bar{E}} E'), \quad E' = R' \otimes_S (E \times_{\bar{E}} E'),$$

and therefore the exact sequence $(*)$ may be rewritten as

$$0 \longrightarrow I \otimes_S (E \times_{\bar{E}} E') \longrightarrow E \times_{\bar{E}} E' \longrightarrow R' \otimes_S (E \times_{\bar{E}} E') \longrightarrow 0.$$

Thus

$$\text{Tor}_1^S(R', E \times_{\bar{E}} E') = 0, \quad R' \otimes_S (E \times_{\bar{E}} E') = E',$$

and E' is flat over R' . It follows that $E \times_{\bar{E}} E'$ is flat over S . $\square$

Corollary 4.3. Given a diagram

$$\begin{array}{ccc} (R, X) & & (R', X') \\ & \searrow (\varphi, \Phi) & \swarrow (\varphi', \Phi') \\ & (\bar{R}, \bar{X}) & \end{array}$$

in $\underline{M}_{X_0}$, the amalgamated sum $X \amalg_{\bar{X}} X'$ (in the category of schemes) is a flat scheme over $R \times_{\bar{R}} R'$, provided that $\varphi : R \rightarrow \bar{R}$ is surjective. Consequently, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Proof. For every affine open $U_0 \subset X_0$, by definition we have

$$(X \amalg_{\bar{X}} X')|_{U_0} \simeq \text{Spec} \left(\Gamma(U_0, X) \times_{\Gamma(U_0, \bar{X})} \Gamma(U_0, X') \right).$$

Consequently, if $\varphi : R \rightarrow \bar{R}$ is surjective, $X \amalg_{\bar{X}} X'$ is flat over $R \times_{\bar{R}} R'$ by Lemma 4.2. Moreover, the morphism

$$i_X \amalg_{i_{\bar{X}}} i_{X'} : X_0 \rightarrow X \amalg_{\bar{X}} X'$$

induces an isomorphism

$$X_0 \rightarrow (X \amalg_{\bar{X}} X') \otimes_S k, \quad S = R \times_{\bar{R}} R'.$$

Therefore

$$(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X') \in \text{Ob}(\underline{M}_{X_0}),$$

and, in $\underline{M}_{X_0}$, we have

$$(R \times_{\bar{R}} R', X \amalg_{\bar{X}} X') = (R, X) \times_{(\bar{R}, \bar{X})} (R', X').$$

Thus $\underline{M}_{X_0}$ is homogeneous over $\underline{C}_\Lambda$. $\square$

Lemma 4.4. For every scheme X_0 over k , there is a canonical exact sequence

$$0 \rightarrow H^1(X_0, \underline{D}_{X_0}^0) \rightarrow [\underline{M}_{X_0}(k[\varepsilon])] \rightarrow H^0(X_0, \underline{D}_{X_0}^1),$$

where the $\underline{D}_{X_0}^i$ are as defined in section 3.

Proof. First suppose that X_0 is affine, and put $\Gamma(X_0, \mathcal{O}_{X_0}) = A_0$. As noted in 4.1(b), the category $\underline{M}_{X_0}(k[\varepsilon])$ is canonically equivalent to a category $\underline{E}$ whose objects are flat $k[\varepsilon]$ -algebras A equipped with a $k[\varepsilon]$ -algebra map $j_A : A \rightarrow A_0$ inducing an isomorphism

$$k \otimes_{k[\varepsilon]} A \xrightarrow{\sim} A_0.$$

Equivalently, an object of $\underline{E}$ is a k -algebra A equipped with an exact sequence

$$0 \rightarrow A_0 \xrightarrow{\varepsilon} A \xrightarrow{j_A} A_0 \rightarrow 0$$

in which $\varepsilon(A_0)^2 = 0$ as an ideal of A .

An arrow $A_1 \rightarrow A_2$ in $\underline{E}$, in the first description, is a $k[\varepsilon]$ -algebra map $\Phi : A_1 \rightarrow A_2$ such that $j_{A_1} = j_{A_2} \circ \Phi$. In the second description it is a k -algebra map $\Phi : A_1 \rightarrow A_2$ yielding a commutative diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & A_0 & \xrightarrow{\varepsilon} & A_1 & \xrightarrow{j_{A_1}} & A_0 \longrightarrow 0 \\ & & \parallel & & \downarrow \Phi & & \parallel \\ 0 & \longrightarrow & A_0 & \longrightarrow & A_2 & \xrightarrow{j_{A_2}} & A_0 \longrightarrow 0. \end{array}$$

It follows that, via $\Gamma(X_0, -)$, $[\underline{M}_{X_0}(k[\varepsilon])]$ is canonically isomorphic to the set of isomorphism classes of k -algebra extensions of A_0 by A_0 . In other words, there is a canonical bijection

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(X_0, \underline{D}_{X_0}^1)$$

which plainly respects the vector-space structures on both sides.

In general, for every affine open $U_0 \subset X_0$, restriction and the affine case give the composite map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{U_0}(k[\varepsilon])] \xrightarrow{\sim} H^0(U_0, \underline{D}_{X_0}^1).$$

These maps give a canonical map

$$[\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}_{X_0}^1).$$

Its kernel consists of the isomorphism classes of locally trivial deformations of X_0 over $k[\varepsilon]$, that is, of deformations locally isomorphic to $X_0 \otimes_k k[\varepsilon]$. The sheaf of automorphisms of $X_0 \otimes_k k[\varepsilon]$ inducing the identity on X_0 is

$$\underline{D}_{X_0}^0 = \underline{\mathrm{Der}}_k(\mathcal{O}_{X_0}, \mathcal{O}_{X_0}).$$

The isomorphism classes of such locally trivial deformations are therefore canonically identified with $H^1(X_0, \underline{D}_{X_0}^0)$, which gives the asserted exact sequence. $\square$

Theorem 4.5. Let X_0 be a scheme of finite type over k for which either

- (i) X_0 is proper over k , or
- (ii) X_0 is affine and

$$X_0^{\mathrm{sing}} = \{x \in X_0 \mid X_0 \text{ is not smooth over } k \text{ at } x\}$$

is finite.

(a) The category $\underline{M}_{X_0}$ admits a formal versal object over $\underline{C}_\Lambda$, called a formal versal deformation of X_0 . If $(R, \mathcal{X})$ is a formal versal deformation of X_0 , then for every $(S, \mathcal{Y}) \in \mathrm{Ob}(\widehat{\underline{M}}_{X_0})$ there is a map

$$(\varphi, \Phi) : (R, \mathcal{X}) \longrightarrow (S, \mathcal{Y})$$

in $\widehat{\underline{M}}_{X_0}$ such that the following diagram is cartesian:

$$\begin{array}{ccc} \mathcal{Y} & \xrightarrow{\Phi} & \mathcal{X} \\ \downarrow & & \downarrow \\ \mathrm{Spf}(S) & \xrightarrow{\mathrm{Spf}(\varphi)} & \mathrm{Spf}(R). \end{array}$$

Furthermore, a formal versal deformation $(R, \mathcal{X})$ of X_0 prorepresents the functor $[\underline{M}_{X_0}]$ if and only if, for every surjective map $R' \rightarrow R$ in $\underline{C}_\Lambda$ and every deformation (R', X') of X_0 over R' , the homomorphism

$$\mathrm{Aut}_{R'}(X' \mid X_0) \longrightarrow \mathrm{Aut}_R(X' \otimes_{R'} R \mid X_0)$$

is surjective.

(b) If X_0 is proper over k , then $\underline{M}_{X_0}$ is smoothly pseudo-representable over $\underline{C}_\Lambda$.

Proof. (a) By Corollary 4.3, $\underline{M}_{X_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$. By 1.11 and 2.9(b), it therefore suffices to prove that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k .

Consider the exact sequence

$$(*) \quad 0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}_{X_0}^1).$$

If X_0 is proper over k , both $H^1(X_0, \underline{D}_{X_0}^0)$ and $H^0(X_0, \underline{D}_{X_0}^1)$ are finite-dimensional, since the $\underline{D}_{X_0}^i$ are coherent for all $i \geq 0$ by 2.13(a).

Now suppose that X_0 is affine and X_0^{sing} is finite. The sheaf $\underline{D}_{X_0}^1$ is coherent and

$$\text{Supp } \underline{D}_{X_0}^1 \subset X_0^{\text{sing}}$$

by 2.13(a). Hence $H^0(X_0, \underline{D}_{X_0}^1)$ is finite-dimensional over k , whereas $H^1(X_0, \underline{D}_{X_0}^0) = 0$ because X_0 is affine. In either case, the exact sequence (*) shows that $[\underline{M}_{X_0}(k[\varepsilon])]$ is finite-dimensional over k . The second assertion of part (a) follows from 2.7.

(b) If X_0 is proper, then

$$\text{Aut}(X_0 \otimes_k k[\varepsilon] \mid X_0) = H^0(X_0, \underline{D}_{X_0}^0)$$

is finite-dimensional, and the assertion follows from 2.13. $\square$

Remark 4.6(a). Let X_0 be affine. A formal versal deformation of X_0 exists if and only if the support of $\underline{D}_{X_0}^1$ is a finite set. This does not imply, however, that all nonsmooth points of X_0 are isolated: there are affine schemes X_0 over k having a nonisolated nonsmooth point for which $\text{Supp } \underline{D}_{X_0}^1$ is finite.

Remark 4.6(b). Henceforth, a deformation of X_0 means an object of $\underline{M}_{X_0}$. Where confusion is possible, such an object will be called an infinitesimal deformation of X_0 .

Remark 4.6(c). Let X_0 be a scheme over k . We have fixed a complete noetherian local ring Λ with residue field k and considered deformations of X_0 over $\underline{C}_\Lambda$.

In practice one usually takes $\Lambda = k$, or $\Lambda = W(k)$ when k is a perfect field of positive characteristic. The inclusion $\underline{C}_k \subset \underline{C}_\Lambda$ is fully faithful, and $\underline{M}_{X_0}|_{\underline{C}_k}$ is simply the category of deformations of X_0 over the objects of $\underline{C}_k$. Consequently, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 over $\underline{C}_\Lambda$, then

$$(k \otimes_\Lambda R, k \otimes_\Lambda \mathfrak{X})$$

is a formal versal deformation of X_0 over $\underline{C}_k$. It follows, for example, that

$$\dim_{\underline{C}_\Lambda} \underline{M}_{X_0} = \dim(k \otimes_\Lambda R)$$

does not depend on the choice of Λ , but only on X_0 .

The preceding theorem admits many generalizations. For example, consider a pair $Y_0 \rightarrow X_0$, where X_0 is a scheme of finite type over k and Y_0 is a fixed closed subscheme of X_0 . For $R \in \text{Ob}(\underline{C}_\Lambda)$, a deformation of (X_0, Y_0) over R is a commutative diagram

$$\begin{array}{ccccc} Y_0 & \xrightarrow{\quad\quad\quad} & & & Y \\ & \searrow & & \swarrow & \\ & & X_0 & \xrightarrow{\quad\quad\quad} & X \\ & \searrow & \downarrow & \downarrow & \swarrow \\ & & \text{Spec}(k) & \xrightarrow{\quad\quad\quad} & \text{Spec}(R) \end{array} \quad \begin{array}{l} \\ \\ \text{flat} \\ \text{flat} \\ \text{flat} \end{array}$$

such that the induced map

$$(X_0, Y_0) \longrightarrow (X \times_{\text{Spec}(R)} \text{Spec}(k), Y \times_{\text{Spec}(R)} \text{Spec}(k))$$

is an isomorphism. Defining morphisms

$$(R'; X', Y') \longrightarrow (R; X, Y)$$

of deformations of (X_0, Y_0) in the evident way gives a cofibered category $\underline{M}_{X_0, Y_0}$ in groupoids over $\underline{C}_\Lambda$. This construction is frequently used to rigidify automorphisms in deformation-moduli

problems. The argument of Corollary 4.3, based on Lemma 4.2, immediately shows that $\underline{M}_{X_0, Y_0}$ is a homogeneous cofibered category in groupoids over $\underline{C}_\Lambda$.

Lemma 4.7. The sequence

$$0 \longrightarrow H^0\left(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\right) \longrightarrow [\underline{M}_{X_0, Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]$$

is exact, where $\underline{I}$ is the ideal sheaf defining Y_0 in X_0 ; thus

$$0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$$

is exact.

Proof. Let $\underline{B}$ be the full subcategory of $\underline{M}_{X_0, Y_0}(k[\varepsilon])$ whose objects have the form

$$(k[\varepsilon]; X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]), Y).$$

Then

$$[\underline{B}] = \ker([\underline{M}_{X_0, Y_0}(k[\varepsilon])] \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])]).$$

The category $\underline{B}$ is the groupoid of deformations of Y_0 over $k[\varepsilon]$ inside $X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon])$. Thus an object is a commutative diagram

$$\begin{array}{ccc} Y & \xrightarrow{i_Y} & X_0 \times_{\mathrm{Spec}(k)} \mathrm{Spec}(k[\varepsilon]) \\ & \searrow \text{flat} & \swarrow \\ & \mathrm{Spec}(k[\varepsilon]) & \end{array}$$

whose special fiber is the fixed closed immersion i_{Y_0} . An arrow $(Y_1, i_{Y_1}) \longrightarrow (Y_2, i_{Y_2})$ is a $k[\varepsilon]$ -morphism $\Phi : Y_1 \longrightarrow Y_2$ (necessarily an isomorphism) whose special fiber is id_{Y_0} and for which $i_{Y_2} \circ \Phi = i_{Y_1}$.

Equivalently, an object of $\underline{B}$ is a pair (Y, j_Y) , where Y is a deformation of Y_0 over $k[\varepsilon]$ and $j_Y : Y \longrightarrow X_0$ is a k -morphism whose composite $Y_0 \longrightarrow Y \xrightarrow{j_Y} X_0$ is i_{Y_0} . An arrow $(Y_1, j_{Y_1}) \longrightarrow (Y_2, j_{Y_2})$ is a morphism $\Phi : Y_1 \longrightarrow Y_2$ whose special fiber is id_{Y_0} and for which $j_{Y_2} \circ \Phi = j_{Y_1}$.

The assignment sending (Y, j_Y) to the diagram

$$\begin{array}{ccccccc} 0 & \longrightarrow & \underline{I} & \longrightarrow & \mathcal{O}_{X_0} & \longrightarrow & \mathcal{O}_{Y_0} \longrightarrow 0 \\ & & \downarrow & & \downarrow j_Y & & \parallel \\ 0 & \longrightarrow & \mathcal{O}_{Y_0} & \xrightarrow{\varepsilon} & \mathcal{O}_Y & \longrightarrow & \mathcal{O}_{Y_0} \longrightarrow 0 \end{array}$$

gives an isomorphism

$$\begin{aligned} [\underline{B}] & \xrightarrow{j_Y|_{\underline{I}}} \mathrm{Hom}_{\mathcal{O}_{X_0}}(\underline{I}, \mathcal{O}_{Y_0}) \\ & = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0}) \\ & = H^0\left(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\right). \end{aligned}$$

This proves exactness. $\square$

Remark 4.8. For a more general formula implying Lemma 4.7, see SGA 3, III, 4.4. One may also note the following. For every S -scheme X and every quasi-coherent $\mathcal{O}_X$ -module E , one can define quasi-coherent $\mathcal{O}_X$ -modules $\underline{D}_{X|S}^i(E)$ for all $i \geq 0$, generalizing the case in which S is affine. As usual, we write

$$\underline{D}_{X|S}^i(\mathcal{O}_X) = \underline{D}_{X|S}^i.$$

For the closed immersion $Y_0 \hookrightarrow X_0$, we then have

$$H^0\left(Y_0, \underline{\mathrm{Hom}}_{\mathcal{O}_{Y_0}}(\underline{I}/\underline{I}^2, \mathcal{O}_{Y_0})\right) = H^0(Y_0, \underline{D}_{Y_0|X_0}^1),$$

where $0 \longrightarrow \underline{I} \longrightarrow \mathcal{O}_{X_0} \longrightarrow \mathcal{O}_{Y_0} \longrightarrow 0$ is exact.

Theorem 4.9. Let X_0 be a k -scheme of finite type and Y_0 a closed subscheme of X_0 . Suppose either

- (i) X_0 is proper over k , or
 - (ii) X_0 is affine and smooth over k outside finitely many points, and Y_0 is proper over k .
- (a) The category $\underline{M}_{X_0, Y_0}$ admits a formal versal deformation over $\underline{C}_\Lambda$. It prorepresents the functor $[\underline{M}_{X_0, Y_0}]$ if and only if, for every surjective map $R' \longrightarrow R$ in $\underline{C}_\Lambda$ and every deformation $(R'; X', Y')$, the map

$$\mathrm{Aut}_{R'}((X', Y') | X_0) \longrightarrow \mathrm{Aut}_R((X' \otimes_{R'} R, Y' \otimes_{R'} R) | X_0)$$

is surjective.

- (b) If, in addition,

$$H^0\left(X_0, \underline{D}_{Y_0|k}^0 \times_{\underline{D}_{X_0|k}^0(\mathcal{O}_{Y_0})} \underline{D}_{X_0|k}^0\right)$$

is finite-dimensional, then $\underline{M}_{X_0, Y_0}$ is smoothly pseudo-representable.

Proof. This follows immediately from Lemmas 4.4 and 4.7 together with 1.11, 2.7, and 2.13. $\square$

We now consider localization and completion in the deformation theory of schemes. We first fix notation. Let X_0 be a k -scheme of finite type and let $x_0 \in X_0$ be a fixed point. There are natural functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0, x_0})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})},$$

and hence functors

$$\widehat{\underline{M}}_{X_0} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\mathcal{O}_{X_0, x_0})} \longrightarrow \widehat{\underline{M}}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x_0})}.$$

As observed in 4.1(c), $\widehat{\underline{M}}_{X_0}$ is canonically equivalent to the category whose objects are pairs $(R, \mathfrak{X})$, where $R \in \mathrm{Ob}(\widehat{\underline{C}}_\Lambda)$ and $\mathfrak{X}$ is a flat R -adic formal scheme equipped with a fixed isomorphism $\mathfrak{X}_0 \xrightarrow{\sim} X_0$.

When there is no danger of confusion, we denote the images of $(R, \mathfrak{X})$ under localization and completion by $(R, \mathfrak{X}_{(x_0)})$ and $(R, \mathfrak{X}_{(\widehat{x}_0)})$, respectively. Thus, if

$$\mathfrak{X} = \varinjlim_{\nu} \mathfrak{X}_{\nu}, \quad \mathfrak{X}_{\nu} = \mathfrak{X} \times_{\mathrm{Spf}(R)} \mathrm{Spec}(R/\mathfrak{m}^{\nu+1}),$$

then

$$\mathfrak{X}_{(x_0)} = \varinjlim_{\nu} \mathrm{Spec}(\mathcal{O}_{\mathfrak{X}_{\nu}, x_0})$$

and

$$\mathfrak{X}_{(\widehat{x}_0)} = \varinjlim_{\nu} \mathrm{Spec}(\widehat{\mathcal{O}}_{\mathfrak{X}_{\nu}, x_0}).$$

Theorem 4.10. Let X_0 be an affine scheme of finite type over k .

- (a) If X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth over $\underline{C}_\Lambda$.
- (b) Suppose that, for a fixed closed point $x \in X_0$, the open $X_0 - \{x\}$ is smooth over k . Then both functors

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\mathrm{Spec}(\mathcal{O}_{X_0, x})} \longrightarrow \underline{M}_{\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x})}$$

are minimally smooth. In particular, if $(R, \mathfrak{X})$ is a formal versal deformation of X_0 , then $(R, \mathfrak{X}_{(x)})$ and $(R, \mathfrak{X}_{(\widehat{x})})$ are formal versal deformations of $\mathrm{Spec}(\mathcal{O}_{X_0, x})$ and $\mathrm{Spec}(\widehat{\mathcal{O}}_{X_0, x})$, respectively.

Proof. We use the notation for the cohomology of commutative algebras from section 3, and write $X_0 = \text{Spec}(A_0)$.

(a) It suffices to show that $P : [\underline{M}_{X_0}] \rightarrow \underline{C}_\Lambda$ is smooth. Let $(R, \text{Spec}(A))$ be a deformation of $\text{Spec}(A_0)$ over $R \in \text{Ob}(\underline{C}_\Lambda)$. Given a small extension $R' \rightarrow R$ in $\underline{C}_\Lambda$, so that

$$0 \rightarrow k \rightarrow R' \rightarrow R \rightarrow 0$$

is exact, we must lift $(R, \text{Spec}(A))$ to R' . The transitivity sequence for D^i with coefficients in A_0 , applied to $R' \rightarrow R \rightarrow A$, gives

$$\cdots \rightarrow D^1(A|R, A_0) \rightarrow D^1(A|R', A_0) \rightarrow D^1(R|R', k) \otimes_k A_0 \rightarrow D^2(A|R, A_0) \rightarrow \cdots$$

Because $R' \rightarrow R$ is a small extension, $D^1(R|R', k)$ is one-dimensional over k , generated by the class $[R']$ represented by $0 \rightarrow k \rightarrow R' \rightarrow R \rightarrow 0$. Since A is flat over R ,

$$D^i(A|R, A_0) = D^i(A_0|k, A_0) \quad (i > 0).$$

In particular, $D^2(A|R, A_0) = 0$ because A_0 is locally a complete intersection, by the low-degree description in 3.12(8). We therefore obtain an exact sequence

$$\rightarrow D^1(A_0|k, A_0) \rightarrow D^1(A|R', A_0) \rightarrow D^1(R|R', k) \otimes_k A_0 \rightarrow 0.$$

Hence there is an R' -algebra extension A' of A by A_0 whose class maps to $[R'] \otimes 1$. Equivalently, $(R, \text{Spec}(A))$ lifts to the deformation $(R', \text{Spec}(A'))$.

(b) By 3.14 there are isomorphisms

$$D^1(A_0|k, A_0) \xrightarrow{\sim} D^1((A_0)_x|k, (A_0)_x) \xrightarrow{\sim} D^1(\widehat{(A_0)_x}|k, \widehat{(A_0)_x}),$$

and hence

$$[\underline{M}_{X_0}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x})}(k[\varepsilon])] \xrightarrow{\sim} [\underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x})}(k[\varepsilon])].$$

By 2.7(a), it remains to show that

$$[\underline{M}_{X_0}] \rightarrow [\underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x})}] \rightarrow [\underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x})}]$$

are smooth functors.

Let $(R, \text{Spec}(A))$ be a deformation of $X_0 = \text{Spec}(A_0)$, and let $0 \rightarrow k \rightarrow R' \rightarrow R \rightarrow 0$ be a small extension. As above, the homomorphisms $R' \rightarrow R \rightarrow A$ give an exact commutative diagram

$$\begin{array}{ccccccc} \rightarrow D^1(A_0|k, A_0) & \longrightarrow & D^1(A|R', A_0) & \xrightarrow{p} & D^1(R|R', k) \otimes_k A_0 & \longrightarrow & D^2(A_0|k, A_0) \rightarrow \cdots \\ & \sim \downarrow & \downarrow & & \downarrow & & \downarrow \sim \\ \rightarrow D^1(A_0|k, (A_0)_x) & \longrightarrow & D^1(A|R', (A_0)_x) & \rightarrow & D^1(R|R', k) \otimes_k (A_0)_x & \rightarrow & D^2(A_0|k, (A_0)_x) \rightarrow \cdots \\ & \uparrow & \uparrow & & \uparrow & & \uparrow \\ \rightarrow D^1((A_0)_x|k, (A_0)_x) & \rightarrow & D^1(A_x|R', (A_0)_x) & \xrightarrow{p_x} & D^1(R|R', k) \otimes_k (A_0)_x & \rightarrow & D^2((A_0)_x|k, (A_0)_x) \rightarrow \cdots \end{array}$$

The indicated isomorphisms follow from the fact that A_0 is smooth over k away from the single closed point x , so that

$$D^i(A_0|k, (A_0)_x) = D^i(A_0|k, A_0)_x = D^i(A_0|k, A_0) \quad (i > 0).$$

Suppose we are given an arrow

$$(R', \text{Spec}(B)) \rightarrow (R, \text{Spec}(A_x))$$

in $[\underline{M}_{\text{Spec}((A_0)_x)}]$. This amounts to a class $[B] \in D^1(A_x|R', (A_0)_x)$ satisfying $p_x([B]) = [R']$. A diagram chase gives a class $[A'] \in D^1(A|R', A_0)$ such that $p([A']) = [R']$ and $A'_x = B$. Thus $(R', \text{Spec}(A'))$ is a deformation of $\text{Spec}(A_0)$ over R' whose localization at x is $\text{Spec}(B)$. This proves that

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{\text{Spec}(\mathcal{O}_{X_0, x})}$$

is smooth.

The smoothness of

$$\underline{M}_{\text{Spec}((A_0)_x)} \longrightarrow \underline{M}_{\text{Spec}(\widehat{(A_0)_x})}$$

follows from the analogous diagram and the identities

$$D^i((A_0)_x|k, \widehat{(A_0)_x}) = D^i(\widehat{(A_0)_x|k}, (A_0)_x) = D^i((A_0)_x|k, (A_0)_x) \quad (i > 0),$$

since x is the only possible nonsmooth point. $\square$

Theorem 4.11. Let X_0 be a separated scheme of finite type over k , smooth away from a finite closed subset

$$X_0^{\text{Sing}} = \{x_1, \dots, x_r\}.$$

Let $U_0^1, \dots, U_0^r$ be affine open neighbourhoods of $x_1, \dots, x_r$ such that $U_0^i \cap X_0^{\text{Sing}} = \{x_i\}$. If $H^2(X_0, \underline{D}_{X_0}^0) = 0$, then the natural functor

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{U_0^1} \times \dots \times \underline{M}_{U_0^r}$$

is smooth.

Proof. Complete $U_0^1, \dots, U_0^r$ to an affine open covering of X_0 by choosing $U_0^{r+1}, \dots, U_0^N$ such that $U_0^j \cap X_0^{\text{Sing}} = \emptyset$ for every $j > r$. Let $R' \rightarrow R$ be a small extension in $\underline{C}_\Lambda$, and let $(R', U^{j1}), \dots, (R', U^{jr})$ be deformations of $U_0^1, \dots, U_0^r$ over R' such that

$$U^{ri} \otimes_{R'} R = X|_{U_0^i}, \quad 1 \leq i \leq r,$$

for some deformation (R, X) of X_0 over R . We must find a deformation (R', X') of X_0 over R' such that

$$X' \otimes_{R'} R = X, \quad X'|_{U_0^i} = U^{ri} \quad (1 \leq i \leq r).$$

For $j > r$, the scheme U_0^j is smooth over k ; hence, by 4.9(a), $X|_{U_0^j}$ lifts to a deformation (R', U^{rj}) of U_0^j over R' . Thus

$$U^{rj} \otimes_{R'} R = X|_{U_0^j} \quad (j > r).$$

For every pair $i \neq j$, the intersection $U_0^i \cap U_0^j$ is affine and smooth over k . We therefore obtain an isomorphism

$$\tau_{ij} : U^{ri}|_{U_0^i \cap U_0^j} \xrightarrow{\sim} U^{rj}|_{U_0^i \cap U_0^j}$$

such that $\tau_{ij} \otimes_{R'} R = \text{id}_{X|_{U_0^i \cap U_0^j}}$. On triple intersections, the automorphisms

$$\tau_{ij}\tau_{jk}\tau_{ik}^{-1}$$

define a cohomology class in $H^2(X_0, \underline{D}_{X_0}^0)$. This class is the obstruction to gluing the U^{ri} to a global deformation. It vanishes by hypothesis, so the U^{ri} glue to a deformation (R', X') with $X' \otimes_{R'} R = X$ and $X'|_{U_0^i} \simeq U^{ri}$. $\square$

Remark 4.12. The preceding proof assumes that X_0 is separated, but this assumption is unnecessary. Indeed, fix a deformation (R, X) of X_0 and deformations (R', U^{ri}) of the members U_0^i of

an open covering. The deformations (R', V') of open subsets V of X over R' that are compatible with the already-given deformations (R', U^i) form a “gerbe” (in Giraud’s terminology) whose structure sheaf is $\underline{D}_{X_0}^0$. The obstruction to the existence of a section of this gerbe—that is, to a global deformation (R', X') of X compatible with the given data (R', U^i) —lies in $H^2(X_0, \underline{D}_{X_0}^0)$, and hence vanishes under the hypothesis of the theorem.

Corollary 4.13. Let X_0 be a scheme of finite type over k , smooth away from a finite subset

$$X_0^{\text{Sing}} = \{x_1, \dots, x_r\},$$

and suppose that $H^2(X_0, \underline{D}_{X_0}^0) = 0$. Then:

(a) The functor

$$\underline{M}_{X_0} \longrightarrow \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}$$

is smooth. In particular,

$$\dim \underline{M}_{X_0} = \dim_k H^1(X_0, \underline{D}_{X_0}^0) + \sum_{i=1}^r \dim \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}.$$

(b) If, in addition, X_0 is locally a complete intersection, then $\underline{M}_{X_0}$ is smooth.

Proof. (a) The first assertion follows from 4.10 and 4.9(b). Let $(R, \mathfrak{X})$ and $(R_1, \mathfrak{X}_1), \dots, (R_r, \mathfrak{X}_r)$ be formal versal deformations of X_0 and of

$$\text{Spec}(\mathcal{O}_{X_0, x_1}), \dots, \text{Spec}(\mathcal{O}_{X_0, x_r}),$$

respectively. We obtain a commutative diagram

$$\begin{array}{ccc} \underline{h}_R & \longrightarrow & \underline{h}_{R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r} \\ \downarrow & & \downarrow \\ \underline{M}_{X_0} & \longrightarrow & \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}. \end{array}$$

The composite functor

$$\underline{h}_R \longrightarrow \underline{M}_{X_0} \longrightarrow \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}$$

is smooth, whereas

$$\underline{h}_{R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r} \longrightarrow \prod_{i=1}^r \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}$$

is minimally smooth. It follows from 2.7(a) that

$$\underline{h}_R \longrightarrow \underline{h}_{R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r}$$

is smooth; equivalently, R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r$.

By 4.4, the sequence

$$0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow \prod_{i=1}^r [\underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}(k[\varepsilon])]$$

is exact. Equivalently,

$$0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0) \longrightarrow \text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow \text{Hom}_{\Lambda\text{-alg}}(R_1 \widehat{\otimes}_\Lambda \dots \widehat{\otimes}_\Lambda R_r, k[\varepsilon])$$

is exact. Consequently, R is a formal power-series ring over $R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r$ in d variables, where

$$d = \dim_k H^1(X_0, \underline{D}_{X_0}^0).$$

Therefore

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim(k \otimes_\Lambda R) \\ &= d + \dim(k \otimes_\Lambda (R_1 \widehat{\otimes}_\Lambda \cdots \widehat{\otimes}_\Lambda R_r)) \\ &= d + \dim((k \otimes_\Lambda R_1) \widehat{\otimes}_k \cdots \widehat{\otimes}_k (k \otimes_\Lambda R_r)) \\ &= d + \sum_{i=1}^r \dim(k \otimes_\Lambda R_i) \\ &= d + \sum_{i=1}^r \dim \underline{M}_{\text{Spec}(\widehat{\mathcal{O}}_{X_0, x_i})}. \end{aligned}$$

(b) This follows from 4.10 and 4.9(a). $\square$

4.14. We close this section by constructing explicitly a formal versal deformation in a simple case. When X_0 is affine and a complete intersection, such a deformation is immediate to write down. Let

$$A_0 = k[x_1, \dots, x_n]/(f_1, \dots, f_r) = S/J,$$

where $S = k[x_1, \dots, x_n]$, $J = (f_1, \dots, f_r)$, and $f_1, \dots, f_r$ is an S -regular sequence. We have an exact sequence

$$\cdots \longrightarrow \text{Der}_k(S, A_0) \longrightarrow \text{Hom}_{A_0}(J/J^2, A_0) \longrightarrow D^1(A_0|k, A_0) \longrightarrow 0.$$

The A_0 -module J/J^2 is free of rank r , generated by the classes of $f_1, \dots, f_r$, because these elements form an S -regular sequence.

Assume now that $D^1(A_0|k, A_0)$ is finite-dimensional over k ; for example, assume that A_0 has only isolated nonsmooth points. Put

$$d = \dim_k D^1(A_0|k, A_0),$$

and choose maps

$$p_i : J/J^2 \longrightarrow A_0 \quad (1 \leq i \leq d)$$

whose classes form a k -basis of $D^1(A_0|k, A_0)$. For $1 \leq i \leq d$ and $1 \leq j \leq r$, choose $P_{ij} \in \Lambda[x_1, \dots, x_n]$ such that

$$P_{ij} \equiv p_i(f_j) \pmod{\mathfrak{m}_\Lambda + J},$$

where $\mathfrak{m}_\Lambda$ is the maximal ideal of Λ . Set

$$R = \Lambda[[t_1, \dots, t_d]], \quad B = R[x_1, \dots, x_n]/(G_1, \dots, G_r),$$

where

$$G_j = F_j - \sum_{i=1}^d t_i P_{ij},$$

and $F_j \in \Lambda[x_1, \dots, x_n]$ is obtained from f_j by lifting its coefficients from k to Λ .

The algebra $R[x_1, \dots, x_n]$ is flat over R , and the images of $G_1, \dots, G_r$ modulo $\mathfrak{m}_R$ form the regular sequence $f_1, \dots, f_r$ in $k \otimes_R R[x_1, \dots, x_n]$. Hence B is R -flat and $k \otimes_R B = A_0$. By construction, the map

$$\text{Hom}_{\Lambda\text{-alg}}(R, k[\varepsilon]) \longrightarrow D^1(A_0|k, A_0) = [\underline{M}_{\text{Spec}(A_0)}(k[\varepsilon])]$$

is bijective. Thus, if

$$\mathrm{Spf}(B) = \varinjlim_{\nu} \mathrm{Spec}((R/\mathfrak{m}_R^{\nu+1}) \otimes_R B),$$

then $(R, \mathrm{Spf}(B))$ is an object of $\widehat{\underline{M}}_{\mathrm{Spec}(A_0)}$. The functor

$$\underline{h}_R \longrightarrow \underline{M}_{\mathrm{Spec}(A_0)}$$

that it induces is smooth by 1.6, since R is a formal power-series ring over Λ . In other words, $(R, \mathrm{Spf}(B))$ is a formal versal object of $\underline{M}_{\mathrm{Spec}(A_0)}$.

By 4.10(b), if instead we put

$$B = R[[x_1, \dots, x_n]]/(G_1, \dots, G_r),$$

then $(R, \mathrm{Spf}(B))$ is a formal versal deformation of $\mathrm{Spec}(A_0)$ for

$$A_0 = k[[x_1, \dots, x_n]]/(f_1, \dots, f_r).$$

Remark 4.15. Let X_0 be a scheme of finite type over k with only isolated nonsmooth points, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . The preceding construction shows that, when X_0 is affine and locally a complete intersection, $\mathfrak{X}$ is the formalization of an R -scheme of finite type. The same fact extends readily to the nonaffine case provided that $H^2(X_0, \underline{D}_{X_0}^0) = 0$.

Remark 4.16. The global theory of the relative cotangent complex—here reduced to the truncated complex of length one, which nevertheless suffices for many purposes in deformation theory—generalizes and simplifies results such as 4.10. Let S be a scheme, and let $S_0 \subset S' \subset S$ be subschemes defined by ideals $J \supset I$ with $JI = 0$. Let X' be flat over S' , and write

$$L_{\bullet} = L^{X_0/S_0}$$

for the relative cotangent complex of X_0 over S_0 , regarded as an object of $D(X_0, \mathcal{O}_{X_0})$. Let $\underline{F}$ be the category of flat S -schemes X extending X'/S' ; thus an object is equipped with an S' -isomorphism

$$X \times_S S' \xrightarrow{\sim} X'.$$

Then:

(a) For an object X of $\underline{F}$, its group of automorphisms inducing the identity on X' is canonically isomorphic to

$$\mathrm{Ext}^0(X_0; L_{\bullet}, I_{X_0}),$$

where I_{X_0} is the inverse image on X_0 of the $\mathcal{O}_{S_0}$ -module I .

(b) The set of isomorphism classes of objects of $\underline{F}$ is either empty or a torsor (a principal homogeneous space) under $\mathrm{Ext}^1(X_0; L_{\bullet}, I_{X_0})$.

(c) There is a canonical element

$$c \in \mathrm{Ext}^2(X_0; L_{\bullet}, I_{X_0})$$

whose vanishing is necessary and sufficient for $\underline{F}$ to be nonempty.

To study these groups, it is often convenient to write

$$\mathrm{Ext}^i(X_0; L_{\bullet}, I_{X_0}) = H^i(X_0, R\mathrm{Hom}(L_{\bullet}, I_{X_0})).$$

The cohomology sheaves of $R\mathrm{Hom}(L_{\bullet}, I_{X_0})$ are precisely the sheaves $\underline{D}_{X_0}^i(I_{X_0})$ introduced in section 3. For $i = 0, 1, 2$, they may be viewed respectively as the sheaf of automorphisms of an object X of $\underline{F}$ (automorphisms that induce the identity on X'), the sheaf measuring the indeterminacy of isomorphism classes of local solutions to the flat-deformation problem, and the sheaf of obstructions to the existence of local solutions.

The standard spectral sequence is

$$E_2^{p,q} = H^p(X_0, \underline{D}_{X_0}^q(I_{X_0})) \implies \text{Ext}^{p+q}(X_0; L_\bullet, I_{X_0}).$$

It gives the familiar isomorphisms

$$\begin{aligned} \text{Aut}(X/X') &\simeq H^0(X_0, \underline{D}_{X_0}^0(I_{X_0})) \\ &\simeq H^0(X_0, \underline{\text{Hom}}(\Omega_{X_0/S_0}^1, I_{X_0})) \\ &= \text{Hom}(\Omega_{X_0/S_0}^1, I_{X_0}), \end{aligned}$$

and the low-degree exact sequence

$$0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0(I_{X_0})) \longrightarrow \text{Ext}^1 \longrightarrow H^0(X_0, \underline{D}_{X_0}^1(I_{X_0})) \longrightarrow H^2(X_0, \underline{D}_{X_0}^0(I_{X_0})) \longrightarrow \text{Ext}^2.$$

In particular, if $H^2(X_0, \underline{D}_{X_0}^0(I_{X_0})) = 0$ and local flat deformations of X'/S' exist, then compatible local deformations glue to a global flat deformation. More generally, for a prescribed section of the sheaf of isomorphism classes of local flat deformations, the argument of 4.10 places the obstruction to a compatible global deformation in $H^2(X_0, \underline{D}_{X_0}^0(I_{X_0}))$.

The same spectral sequence shows that Ext^2 vanishes whenever

$$(*) \quad H^0(X_0, \underline{D}_{X_0}^2) = 0, \quad H^1(X_0, \underline{D}_{X_0}^1) = 0, \quad H^2(X_0, \underline{D}_{X_0}^0) = 0.$$

The first equality always holds when X_0 is a relative complete intersection over S_0 , since then $\underline{D}_{X_0}^2(M) = 0$ for every $\mathcal{O}_{X_0}$ -module M . The second holds when the support of $\underline{D}_{X_0}^1$ is discrete, as happens when X_0 is smooth over S_0 away from a discrete subset. If all three equalities in $(*)$ hold, then $\underline{F}$ is nonempty by (c), which generalizes 4.13(b).

There is also a geometric interpretation, requiring only the globalized functors $\underline{D}_{X_0}^i$ rather than the global theory of the relative cotangent complex. The three successive obstructions to extending X'/S' flatly over S lie in the three groups displayed in $(*)$. The local theory defines the first obstruction

$$c_1 \in H^0(X_0, \underline{D}_{X_0}^2)$$

as the obstruction to the existence of local solutions. If $c_1 = 0$, the sheaf $\underline{F}$ of isomorphism classes of local solutions is canonically a torsor under $\underline{D}_{X_0}^1$. Its obstruction to having a section is

$$c_2 \in H^1(X_0, \underline{D}_{X_0}^1);$$

such a section is a coherent system of isomorphism classes of local solutions. If $c_2 = 0$, choose a section of $\underline{F}$. The obstruction to a global solution realizing the chosen local classes is then

$$c_3 \in H^2(X_0, \underline{D}_{X_0}^0).$$

Changing the section of $\underline{F}$ by a section of $\underline{D}_{X_0}^1$ gives a map

$$d : H^0(X_0, \underline{D}_{X_0}^1) \longrightarrow H^2(X_0, \underline{D}_{X_0}^0).$$

A solution exists if and only if $c_1 = 0$, $c_2 = 0$, and the class of c_3 in $\text{Coker } d$ is zero (each c_i being defined only after the preceding obstructions vanish).

The three obstruction groups

$$H^0(X_0, \underline{D}_{X_0}^2), \quad H^1(X_0, \underline{D}_{X_0}^1), \quad H^2(X_0, \underline{D}_{X_0}^0)/\text{Im } d$$

are the initial terms $E_2^{0,2}$, $E_2^{1,1}$, and $E_3^{2,0}$ of the preceding spectral sequence; up to sign, d is its differential

$$d_2 : E_2^{0,1} \longrightarrow E_2^{2,0}.$$

5. Formal Jacobian subschemes

Let X_0 be a scheme of finite type over a field k , and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . The space $\mathfrak{X}$ is not an ordinary R -scheme, but an R -adic formal scheme. The purpose of this section is to clarify the meaning of the “nonsmooth locus of $\mathfrak{X}$ over R .”

We first recall the Fitting ideals of a finite-type module. Let A be a commutative ring and let E be a finite-type A -module. Choose a presentation

$$K \longrightarrow F \longrightarrow E \longrightarrow 0,$$

where F is locally free of constant rank n . For $p = 0, 1, 2, \dots$, define

$$I_A^p(E) = \text{Im} \left(\bigwedge^{n-p} K \otimes_A \bigwedge^{n-p} F^* \longrightarrow A \right).$$

These ideals are independent of the chosen presentation. They are invariants of the A -module E , called its *Fitting ideals*. The following properties follow directly from the definition.

5.1. (a) For every ring homomorphism $A \longrightarrow B$, the canonical map

$$B \otimes_A A/I_A^p(E) \xrightarrow{\sim} B/I_B^p(B \otimes_A E)$$

is an isomorphism for every p . In particular,

$$I_{S^{-1}A}^p(S^{-1}E) = S^{-1}I_A^p(E)$$

for every multiplicative subset $S \subset A$.

(b) There are inclusions

$$I_A^0(E) \subset I_A^1(E) \subset I_A^2(E) \subset \dots$$

For every $x \in \text{Spec}(A)$,

$$I_A^p(E)_x = A_x \quad \text{for all } p \geq \dim_{\kappa(x)} E(x),$$

where $E(x) = \kappa(x) \otimes_A E$.

(c) A point $x \in \text{Spec}(A) - \text{Supp}(A/I_A^p(E))$ if and only if, for any (equivalently, for some) presentation

$$K \longrightarrow F \longrightarrow E \longrightarrow 0, \quad F \simeq A^n,$$

the module K_x contains a copy of A_x^{n-p} that is a direct summand of F_x . In particular,

$$\text{Supp}(A/I_A^0(E)) = \text{Supp}(E).$$

(d) For any two finite-type A -modules,

$$I_A^r(E_1 \oplus E_2) = \sum_{p+q=r} I_A^p(E_1) I_A^q(E_2).$$

(e) If A is a noetherian adic ring, then

$$A/I_A^p(E) = \varprojlim_{\nu} A_{\nu}/I_{A_{\nu}}^p(E_{\nu}),$$

where $A = \varprojlim_{\nu} A_{\nu}$ and $E_{\nu} = E \otimes_A A_{\nu}$.

Property 5.1(a) globalizes the definition. If X is a scheme and $\underline{E}$ is a quasi-coherent $\mathcal{O}_X$ -module of finite presentation, there are quasi-coherent ideal sheaves $\underline{I}_X^p(\underline{E})$, $p \geq 0$, characterized on every affine open $U \subset X$ by

$$\Gamma(U, \underline{I}_X^p(\underline{E})) \xrightarrow{\sim} I_{\Gamma(U, \mathcal{O}_X)}^p(\Gamma(U, \underline{E})).$$

For a morphism of schemes $X \longrightarrow Y$, the relative module of Kähler differentials $\Omega_{X|Y} = \Omega_{X|Y}^1$ is quasi-coherent. If the morphism is essentially of finite presentation, then $\Omega_{X|Y}$ is of finite presentation, so the ideal sheaves $\underline{I}_X^p(\Omega_{X|Y})$ are defined.

Definition 5.2. Let $X \longrightarrow Y$ be a morphism essentially of finite presentation. For every integer $p \geq 0$, define $J_X^p(X|Y)$ to be the closed subscheme of X defined by the ideal sheaf

$$\underline{I}_X^p(X|Y) = \underline{I}_X^p(\Omega_{X|Y}).$$

It is called the p th *Jacobian subscheme* of X over Y . For every $x \in X$, put

$$X_x = \text{Spec}(\mathcal{O}_{X,x}).$$

Theorem 5.3. Let $X \longrightarrow Y$ be a morphism essentially of finite presentation. Then:

(1) For every base change $Y' \longrightarrow Y$, the canonical map

$$J_{X'}^p(X'|Y') \xrightarrow{\sim} J_X^p(X|Y) \times_Y Y'$$

is an isomorphism for every p , where $X' = X \times_Y Y'$.

(2) For every $x \in X$, there is a canonical isomorphism

$$J_{X_x}^p(X_x|Y) \xrightarrow{\sim} J_X^p(X|Y)_x.$$

(3) There are inclusions

$$J_X^0(X|Y) \supset J_X^1(X|Y) \supset J_X^2(X|Y) \supset \cdots,$$

and

$$x \notin J_X^p(X|Y) \quad \text{for every } p \geq \dim_{\kappa(x)} \Omega_{X|Y}(x).$$

(4) A point x does not belong to $J_X^p(X|Y)$ if and only if it has an open neighbourhood U that admits a closed immersion $i : U \hookrightarrow U'$ into a Y -scheme U' such that U' is smooth over Y at $x' = i(x)$ and

$$\dim_{x'} U'_{f(x)} = p.$$

(5) For the projections $\pi_i : X_1 \times_Y X_2 \longrightarrow X_i$,

$$\underline{I}_{X_1 \times_Y X_2}^r(X_1 \times_Y X_2|Y) = \sum_{p+q=r} \pi_1^* \underline{I}_{X_1}^p(X_1|Y) \pi_2^* \underline{I}_{X_2}^q(X_2|Y).$$

Proof. Assertion (3) follows from 5.1(b). Assertion (1) follows from 5.1(a) and the canonical isomorphism

$$P^* \Omega_{X|Y} \xrightarrow{\sim} \Omega_{X'|Y'}, \quad P : X \times_Y Y' \longrightarrow X.$$

Likewise, (2) follows from 5.1(a) and

$$\mathcal{O}_{X,x} \otimes_{\mathcal{O}_X} \Omega_{X|Y} \xrightarrow{\sim} \Omega_{X_x|Y}.$$

Finally, (5) follows from 5.1(d) and

$$\pi_1^* \Omega_{X_1|Y} \oplus \pi_2^* \Omega_{X_2|Y} \xrightarrow{\sim} \Omega_{X_1 \times_Y X_2|Y}.$$

For (4), first suppose that there is a diagram

$$\begin{array}{ccc} U & \xrightarrow{i} & U' \\ f \downarrow & \swarrow & \\ Y & & \end{array}$$

where U' is smooth over Y at $x' = i(x)$ and $\dim_{x'} U'_{f(x)} = p$. Then

$$\dim_{\kappa(x')} \Omega_{U'|Y}(x') = p,$$

and the sequence

$$i^* \Omega_{U'|Y} \longrightarrow \Omega_{U|Y} \longrightarrow 0$$

is exact. Hence $\dim_{\kappa(x)} \Omega_{U|Y}(x) \leq p$, so $x \notin J_U^p(U|Y)$ by (3). Since

$$J_U^p(U|Y) = J_X^p(X|Y)|_U,$$

we have $x \notin J_X^p(X|Y)$.

Conversely, suppose that $x \notin J_X^p(X|Y)$. The question is local, so write $X = \operatorname{Spec}(B)$ and $Y = \operatorname{Spec}(A)$. Choose an A -presentation

$$0 \longrightarrow J \longrightarrow P \longrightarrow B \longrightarrow 0, \quad P = A[x_1, \dots, x_n].$$

There is an exact sequence

$$J/J^2 \xrightarrow{d} B \otimes_P \Omega_{P|A} \longrightarrow \Omega_{B|A} \longrightarrow 0,$$

and $B \otimes_P \Omega_{P|A}$ is free of rank n over B . Thus

$$I_B^p(\Omega_{B|A}) = \operatorname{Im} \left(\bigwedge^{n-p} (J/J^2) \otimes_B (B \otimes_P \bigwedge^{n-p} \Omega_{P|A})^* \longrightarrow B \right).$$

Consequently, $x \notin J_X^p(X|Y)$ precisely when there are $f_1, \dots, f_{n-p} \in J$ and a choice of $n-p$ among the variables $x_1, \dots, x_n$ for which the corresponding Jacobian minor

$$\det \left(\frac{\partial f_i}{\partial x_j} \right) (x)$$

is nonzero. Equivalently,

$$\operatorname{Spec}(P/(f_1, \dots, f_{n-p})) \longrightarrow \operatorname{Spec}(A)$$

is smooth at the image x' of x . Taking

$$U' = \operatorname{Spec}(P/(f_1, \dots, f_{n-p}))$$

gives the required closed immersion locally. $\square$

Theorem 5.4. Let $X \xrightarrow{f} Y$ be a morphism essentially of finite presentation. Suppose that there is a fixed integer $n \geq 0$ for which one of the following conditions holds:

- (1) f is flat and $\dim_x X_{f(x)} = n$ for every $x \in X$;
- (2) f is a relative complete intersection of virtual dimension n (SGA 6, VIII, 1.9).

Then $x \notin J_X^n(X|Y)$ if and only if f is smooth at x .

Proof. By 5.3(4), $x \notin J_X^n(X|Y)$ if and only if there is a diagram

$$\begin{array}{ccc} U & \xrightarrow{i} & U' \\ f \downarrow & \swarrow f' & \\ Y & & \end{array}$$

such that f' is smooth at x' , with $\dim_{x'} U'_{f(x)} = n$, while i is a closed immersion and $x' = i(x)$. It therefore suffices to show that under either (1) or (2), i is an isomorphism at x .

Set

$$A = \mathcal{O}_{Y, f(x)}, \quad B' = \mathcal{O}_{U', x'}, \quad B = \mathcal{O}_{U, x},$$

and let

$$0 \longrightarrow I \longrightarrow B' \longrightarrow B \longrightarrow 0$$

be exact. We must show that $I = 0$.

Assume (1). Since B is flat over A , the sequence

$$0 \longrightarrow I/\mathfrak{m}I \longrightarrow B'/\mathfrak{m}B' \longrightarrow B/\mathfrak{m}B \longrightarrow 0$$

is exact, where $\mathfrak{m}$ is the maximal ideal of A . By hypothesis,

$$\dim_x U_{f(x)} = \dim_{x'} U'_{f(x)},$$

and hence $\dim(B'/\mathfrak{m}B') = \dim(B/\mathfrak{m}B)$. Since $B'/\mathfrak{m}B'$ is a regular local ring, it follows that $I/\mathfrak{m}I = 0$, and therefore $I = 0$.

Now assume (2), and consider the exact sequence

$$I/I^2 \xrightarrow{d} B \otimes_{B'} \Omega_{B'|A} \longrightarrow \Omega_{B|A} \longrightarrow 0.$$

Since X is a relative complete intersection over Y and $U' \longrightarrow Y$ is smooth at x' , the map

$$d : I/I^2 \longrightarrow B \otimes_{B'} \Omega_{B'|A}$$

is injective at x , and both I/I^2 and $B \otimes_{B'} \Omega_{B'|A}$ are free at x . By the definition of the virtual dimension of a relative complete intersection, we must therefore have

$$n = \dim_{\kappa(x)} \Omega_{B'|A}(x') - \dim_{\kappa(x)} (I/I^2)(x) = n - \dim_{\kappa(x)} (I/I^2)(x).$$

Thus $I/I^2 = 0$, and hence $I = 0$. $\square$ **Definition 5.5.** Let $X \xrightarrow{f} Y$ be a morphism essentially of finite presentation. If there is a fixed integer n satisfying (1) or (2) of 5.4, set

$$\underline{I}_X(X|Y) = \underline{I}_X^n(X|Y), \quad J_X(X|Y) = J_X^n(X|Y).$$

This notion extends naturally to formal schemes. Let $\mathfrak{X}$ be a noetherian formal scheme and $\mathcal{E}$ a coherent $\mathfrak{X}$ -module. For every affine open $U \subset \mathfrak{X}$, define

$$\underline{I}_{\mathfrak{X}}^p(\mathcal{E})(U) = I_{\mathfrak{X}(U)}^p(\mathcal{E}(U)).$$

If $\mathfrak{X} = \varinjlim_{\nu} X_{\nu}$, then $\mathcal{E} = \varprojlim_{\nu} \mathcal{E}_{\nu}$, where

$$\mathcal{E}_{\nu} = \mathcal{E} \otimes_{\mathcal{O}_{\mathfrak{X}}} \mathcal{O}_{X_{\nu}},$$

and noetherianness gives

$$\underline{I}_{\mathfrak{X}}^p(\mathcal{E}) = \varprojlim_{\nu} \underline{I}_{X_{\nu}}^p(\mathcal{E}_{\nu}).$$

Now let S be a noetherian formal scheme and $\mathfrak{X}$ an S -adic formal scheme essentially of finite type. Let $\underline{I}_{\mathfrak{X}}$ be the ideal sheaf defining the closed diagonal immersion

$$\Delta : \mathfrak{X} \longrightarrow \mathfrak{X} \times_S \mathfrak{X}$$

(where the fibre product is taken in the category of formal schemes), so that

$$0 \longrightarrow \underline{I}_{\mathfrak{X}} \longrightarrow \mathcal{O}_{\mathfrak{X} \times_S \mathfrak{X}} \longrightarrow \mathcal{O}_{\mathfrak{X}} \longrightarrow 0$$

is exact. Define

$$\Omega_{\mathfrak{X}|S} = \underline{I}_{\mathfrak{X}} / \underline{I}_{\mathfrak{X}}^2.$$

Writing

$$\mathfrak{X} = \varinjlim_{\nu} X_{\nu}, \quad X_{\nu} = \mathfrak{X} \times_S S_{\nu},$$

we have

$$\Omega_{\mathfrak{X}|S} \simeq \varprojlim_{\nu} \Omega_{X_{\nu}|S_{\nu}},$$

so $\Omega_{\mathfrak{X}|S}$ is a coherent $\mathfrak{X}$ -module. For every formal S -scheme S' of finite type there is a canonical isomorphism

$$\Omega_{\mathfrak{X} \times_S S'|S'} \simeq \mathcal{O}_{S'} \otimes_{\mathcal{O}_S} \Omega_{\mathfrak{X}|S}.$$

Definition 5.6. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S -adic formal scheme essentially of finite type. Define

$$\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S) = \underline{I}_{\mathfrak{X}}^p(\Omega_{\mathfrak{X}|S}),$$

and let $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ be the closed formal subscheme of $\mathfrak{X}$ defined by this ideal sheaf.

Since $\Omega_{\mathfrak{X}|S} = \varprojlim_{\nu} \Omega_{X_{\nu}|S_{\nu}}$, we have

$$\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S) = \varprojlim_{\nu} \underline{I}_{X_{\nu}}^p(X_{\nu}|S_{\nu})$$

and

$$J_{\mathfrak{X}}^p(\mathfrak{X}|S) = \varinjlim_{\nu} J_{X_{\nu}}^p(X_{\nu}|S_{\nu}).$$

The formal scheme $\mathfrak{X}_{(x)}$ is again S -adic and essentially of finite type, and there is a commutative diagram

$$\begin{array}{ccc} \mathfrak{X}_{(\widehat{x})} & \xrightarrow{i_{(\widehat{x})}} & \mathfrak{X} \\ & \searrow & \nearrow i_{(x)} \\ & \mathfrak{X}_{(x)} & \end{array}$$

If x is an isolated point of $\mathfrak{X}$, then $i_{(x)}$ and $i_{(\widehat{x})}$ are both isomorphisms onto the connected component of $\mathfrak{X}$ containing x .

In particular, if

$$|\mathfrak{X}| = \{x_1, \dots, x_m\}$$

is a finite discrete space, then there is a canonical identification

$$\mathfrak{X} = \coprod_{1 \leq i \leq m} \mathfrak{X}_{(x_i)} \quad (\text{disjoint union}).$$

Corollary 5.7. Let S be a noetherian formal scheme and $\mathfrak{X}$ an S -adic formal scheme essentially of finite type. Then:

(1) Formation of $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ commutes with base change. More precisely, if

$$\mathfrak{X}' = \mathfrak{X} \times_S S',$$

then

$$J_{\mathfrak{X}'}^p(\mathfrak{X}'|S') = J_{\mathfrak{X}}^p(\mathfrak{X}|S) \times_S S'.$$

(2) The connected components of $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ are in bijective correspondence with those of $J_{X_0}^p(X_0|S_0)$. For every

$$x \in |J_{X_0}^p(X_0|S_0)|$$

there is a canonical isomorphism

$$J_{\mathfrak{X}}^p(\mathfrak{X}|S)_{(x)} \simeq J_{\mathfrak{X}_{(x)}}^p(\mathfrak{X}_{(x)}|S).$$

In particular, if

$$|J_{X_0}^p(X_0|S_0)| = \{x_1, \dots, x_m\}$$

is discrete, then

$$J_{\mathfrak{X}}^p(\mathfrak{X}|S) = \coprod_{1 \leq i \leq m} J_{\mathfrak{X}_{(x_i)}}^p(\mathfrak{X}_{(x_i)}|S) \quad (\text{disjoint union}).$$

(3) Let S be an ordinary noetherian scheme and X an S -scheme essentially of finite type. Given a closed subscheme $S_0 \subset S$, let $\widehat{S}$ be the formal completion of S along S_0 . If

$$\widehat{X} = X \times_S \widehat{S}$$

is the formal completion of X along $X_0 = X \times_S S_0$, then

$$J_{\widehat{X}}^p(\widehat{X}|\widehat{S}) = J_X^p(X|S) \times_S \widehat{S}.$$

Equivalently, it is the formal completion of $J_X^p(X|S)$ along $J_{X_0}^p(X_0|S_0)$.

(4) Suppose that a fixed integer $n \geq 0$ satisfies condition 5.4(1) or 5.4(2) for every $X_\nu \rightarrow S_\nu$. Then

$$x \notin J_{\mathfrak{X}}^n(\mathfrak{X}|S)$$

if and only if $\mathfrak{X}$ is formally smooth over S at x .

Proof. Assertion (1) follows from

$$\Omega_{\mathfrak{X}'|S'} \simeq \mathcal{O}_{S'} \widehat{\otimes}_{\mathcal{O}_S} \Omega_{\mathfrak{X}|S},$$

whereas (2) follows from

$$\mathcal{O}_{\mathfrak{X}_{(x)}} \widehat{\otimes}_{\mathcal{O}_{\mathfrak{X}}} \Omega_{\mathfrak{X}|S} \simeq \Omega_{\mathfrak{X}_{(x)}|S}.$$

For (3),

$$\begin{aligned} J_{\widehat{X}}^p(\widehat{X}|\widehat{S}) &= \varinjlim_{\nu} J_{X \times_S S_\nu}^p(X \times_S S_\nu|S_\nu) \\ &= \varinjlim_{\nu} (J_X^p(X|S) \times_S S_\nu) = J_X^p(X|S) \times_S \widehat{S}. \end{aligned}$$

Finally, since $\mathfrak{X}$ is S -adic,

$$\begin{aligned} \mathfrak{X} \rightarrow S \text{ is formally smooth at } x &\iff X_\nu \rightarrow S_\nu \text{ is formally smooth at } x \text{ for every } \nu \\ &\iff x \notin J_{X_\nu}^n(X_\nu|S_\nu) \text{ for every } \nu \\ &\iff x \notin \varinjlim_{\nu} J_{X_\nu}^n(X_\nu|S_\nu) = J_{\mathfrak{X}}^n(\mathfrak{X}|S). \end{aligned}$$

□ **Remark 5.8.** Let $\mathfrak{X}$ be an S -adic formal scheme essentially of finite type. For a point $x \in \mathfrak{X}$, consider also the S -adic formal scheme

$$\mathfrak{X}_{(\widehat{x})} = \varinjlim_{\nu} \text{Spec}(\widehat{\mathcal{O}_{X_\nu, x}}).$$

There are natural morphisms

$$\mathfrak{X}_{(\widehat{x})} \longrightarrow \mathfrak{X}_{(x)} \longrightarrow \mathfrak{X}.$$

In general, $\widehat{\mathcal{O}_{X_0, x}}$ is not essentially of finite type over $S_{(y)}$, where y is the image of x under $\mathfrak{X} \rightarrow S$.

For simplicity, put

$$A_\nu = \mathcal{O}_{X_\nu, x}, \quad R_\nu = \mathcal{O}_{S_\nu, y}.$$

The natural map

$$\Omega_{A_\nu|R_\nu} \longrightarrow \widehat{\Omega}_{A_\nu|R_\nu}$$

induces an isomorphism

$$\widehat{\Omega_{A_\nu|R_\nu}} \xrightarrow{\sim} \widehat{\widehat{\Omega}_{A_\nu|R_\nu}},$$

and therefore

$$\varprojlim_\nu \widehat{\Omega_{A_\nu|R_\nu}} \xrightarrow{\sim} \varprojlim_\nu \widehat{\widehat{\Omega}_{A_\nu|R_\nu}}.$$

Here the outer hat denotes the separated completion of a module for the topology induced by the ring. Consequently, if we define

$$\widehat{\Omega}_{\mathfrak{X}(\widehat{x})|S} = \varprojlim_\nu \widehat{\Omega_{\mathcal{O}_{X_\nu, x}|\mathcal{O}_{S_\nu, y}}},$$

then $\widehat{\Omega}_{\mathfrak{X}(\widehat{x})|S}$ is a coherent $\mathfrak{X}(\widehat{x})$ -module.

Define $J_{\mathfrak{X}(\widehat{x})}^p(\mathfrak{X}(\widehat{x})|S)$ to be the closed formal subscheme of $\mathfrak{X}(\widehat{x})$ defined by

$$\underline{I}_{\mathfrak{X}(\widehat{x})}^p(\mathfrak{X}(\widehat{x})|S) = \underline{I}_{\mathfrak{X}(\widehat{x})}^p(\widehat{\Omega}_{\mathfrak{X}(\widehat{x})|S}).$$

Then

$$J_{\mathfrak{X}(x)}^p(\mathfrak{X}(x)|S)_{(\widehat{x})} \simeq J_{\mathfrak{X}(\widehat{x})}^p(\mathfrak{X}(\widehat{x})|S).$$

If x is an isolated point of $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$, the natural morphism

$$J_{\mathfrak{X}(\widehat{x})}^p(\mathfrak{X}(\widehat{x})|S) \longrightarrow J_{\mathfrak{X}(x)}^p(\mathfrak{X}(x)|S)$$

is an isomorphism.

To form the “image” of $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ under $f : \mathfrak{X} \longrightarrow S$, one could simply take the formal subscheme of S defined by the ideal sheaf

$$\ker(\mathcal{O}_S \longrightarrow \mathcal{O}_{\mathfrak{X}}/\underline{I}_{\mathfrak{X}}^p(\mathfrak{X}|S)).$$

For our purposes, however, it is more convenient to use the Fitting ideal once again. **Definition**

5.9. Let S be a noetherian formal scheme and $f : \mathfrak{X} \longrightarrow S$ an S -adic formal scheme essentially of finite type. Suppose that $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ is proper over S , so that

$$f_* \mathcal{O}_{J_{\mathfrak{X}}^p(\mathfrak{X}|S)}$$

is a coherent $\mathcal{O}_S$ -module. Define

$$\underline{I}_S^p(\mathfrak{X}|S) = \underline{I}_S^0(f_* \mathcal{O}_{J_{\mathfrak{X}}^p(\mathfrak{X}|S)}),$$

and let $J_S^p(\mathfrak{X}|S)$ be the closed formal subscheme of S defined by $\underline{I}_S^p(\mathfrak{X}|S)$.

If there is a fixed integer n satisfying condition 5.4(1) or 5.4(2) for every $X_\nu \longrightarrow S_\nu$, set

$$\begin{aligned} \underline{I}_{\mathfrak{X}}(\mathfrak{X}|S) &= \underline{I}_{\mathfrak{X}}^n(\mathfrak{X}|S), & J_{\mathfrak{X}}(\mathfrak{X}|S) &= J_{\mathfrak{X}}^n(\mathfrak{X}|S), \\ \underline{I}_S(\mathfrak{X}|S) &= \underline{I}_S^n(\mathfrak{X}|S), & J_S(\mathfrak{X}|S) &= J_S^n(\mathfrak{X}|S). \end{aligned}$$

Corollary 5.10. Let S be a noetherian formal scheme and $f : \mathfrak{X} \longrightarrow S$ an S -adic formal scheme essentially of finite type such that $J_{\mathfrak{X}}^p(\mathfrak{X}|S)$ is proper over S . Then:

(1)

$$|J_S^p(\mathfrak{X}|S)| = f(|J_{\mathfrak{X}}^p(\mathfrak{X}|S)|).$$

(2) Formation of $J_S^p(\mathfrak{X}|S)$ commutes with base change: if $\mathfrak{X}' = \mathfrak{X} \times_S S'$, then

$$J_{S'}^p(\mathfrak{X}'|S') = J_S^p(\mathfrak{X}|S) \times_S S'.$$

(3) If

$$J_{\mathfrak{X}}^p(\mathfrak{X}|S) = \coprod_{1 \leq i \leq m} \mathfrak{Y}^{(i)}$$

is a disjoint union, then

$$\underline{I}_S^p(\mathfrak{X}|S) = \prod_{1 \leq i \leq m} \underline{I}_S^0(f_* \mathcal{O}_{\mathfrak{Y}^{(i)}}).$$

In particular, if

$$|J_{X_0}^p(X_0|S_0)| = \{x_1, \dots, x_m\}$$

is discrete, then

$$\underline{I}_S^p(\mathfrak{X}|S) = \prod_{1 \leq i \leq m} \underline{I}_S^p(\mathfrak{X}_{(x_i)}|S) = \prod_{1 \leq i \leq m} \underline{I}_S^p(\mathfrak{X}_{(\hat{x}_i)}|S).$$

(4) Let S be an ordinary noetherian scheme, X an S -scheme essentially of finite type, and $S_0 \hookrightarrow S$ a closed subscheme. Let $\hat{S}$ be the formal completion of S along S_0 . If

$$\hat{X} = X \times_S \hat{S}$$

is the formal completion of X along $X_0 = X \times_S S_0$, then

$$J_{\hat{S}}^p(\hat{X}|\hat{S}) = J_S^p(X|S) \times_S \hat{S}.$$

Thus it is the formal completion of $J_S^p(X|S)$ along $J_{S_0}^p(X_0|S_0)$.

Proof. By 5.1(c),

$$\text{Supp } f_* \mathcal{O}_{J_{\mathfrak{X}}^p(\mathfrak{X}|S)} = |J_S^p(\mathfrak{X}|S)|,$$

which proves (1). Assertions (2) and (4) follow from 5.1(a), while (3) follows from 5.1(d) and, for an isolated point $x \in J_{\mathfrak{X}}^p(\mathfrak{X}|S)$, the isomorphism

$$J_{\mathfrak{X}_{(\hat{x})}}^p(\mathfrak{X}_{(\hat{x})}|S) \xrightarrow{\sim} J_{\mathfrak{X}_{(x)}}^p(\mathfrak{X}_{(x)}|S).$$

□

Remark 5.11. Our definition of the Jacobian ideal, adopted from the classical definition over a base field, commutes with base change and, under the mild hypothesis of 5.4, gives a smoothness criterion. In a general setting, however, one can probably define in many ways a canonical subscheme whose support is the nonsmooth locus. We outline one such definition.

For an A -algebra B , put

$$I_{B|A} = \bigcap_E \text{Ann } D^1(B|A, E),$$

where E ranges over all finite-type B -modules. By 3.13(c), the ideal $I_{B|A}$ commutes with localization. Thus a morphism $f : X \rightarrow Y$ determines an ideal sheaf $\underline{I}_{X|Y}$. If f is essentially of finite type, then by 3.13(a), f is smooth at x if and only if

$$x \notin \text{Supp}(\mathcal{O}_X / \underline{I}_{X|Y}).$$

This construction carries over naturally to adic formal schemes essentially of finite type, and is therefore equally suitable for deformation theory.

6. Non-degenerate quadratic singularities

The purpose of this section is to study deformations of a k -scheme whose only singularities are non-degenerate quadratic singularities. In view of 4.13, this becomes, in a suitable situation, a purely local problem.

We return to the notation of §4. Thus Λ is a fixed complete noetherian local ring with residue field k , and $\underline{C}_\Lambda$ denotes the category of local artinian Λ -algebras having residue field k .

Let X be a k -scheme of finite type, and let x be a closed point of X . Recall that x is called a *non-degenerate quadratic singularity rational over k* if

$$\hat{\mathcal{O}}_{X,x} \simeq k[[x_1, \dots, x_n]]/(f)$$

and

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) = (x_1, \dots, x_n).$$

One may always take f to be a quadratic form. Indeed:

Lemma 6.1. Let $f \in k[[x_1, \dots, x_n]]$, and put $M = (x_1, \dots, x_n)$. If

$$\left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) = M,$$

then there are elements $x'_1, \dots, x'_n$ satisfying $x'_i \equiv x_i \pmod{M^2}$ such that

$$f = q(x'_1, \dots, x'_n)$$

for a quadratic form q .

Proof. We use successive approximation. Write

$$f = f_2 + h_3,$$

where f_2 is quadratic and $h_3 \in M^3$. Since the partial derivatives of f generate M , there are $\varepsilon_1^{(1)}, \dots, \varepsilon_n^{(1)} \in M^2$ such that

$$\sum_i \varepsilon_i^{(1)} \frac{\partial f}{\partial x_i} \equiv h_3 \pmod{M^4}.$$

Put $x_i^{(1)} = x_i - \varepsilon_i^{(1)}$. Then

$$\begin{aligned} f(x_1^{(1)}, \dots, x_n^{(1)}) &= f(x_1, \dots, x_n) - \sum_i \varepsilon_i^{(1)} \frac{\partial f}{\partial x_i} + \dots \\ &\equiv f_2(x_1, \dots, x_n) \pmod{M^4}. \end{aligned}$$

Thus

$$f(x_1^{(1)}, \dots, x_n^{(1)}) = f_2(x_1, \dots, x_n) + h_4, \quad h_4 \in M^4.$$

Moreover, the derivatives with respect to the new variables still generate M . Hence one can choose $\varepsilon_1^{(2)}, \dots, \varepsilon_n^{(2)} \in M^3$ such that

$$\sum_i \varepsilon_i^{(2)} \frac{\partial f}{\partial x_i^{(1)}} \equiv h_4 \pmod{M^5}.$$

It follows that

$$\begin{aligned} &f(x_1^{(1)} - \varepsilon_1^{(2)}, \dots, x_n^{(1)} - \varepsilon_n^{(2)}) \\ &= f(x_1^{(1)}, \dots, x_n^{(1)}) - \sum_i \varepsilon_i^{(2)} \frac{\partial f}{\partial x_i^{(1)}} + \dots \\ &\equiv f_2(x_1, \dots, x_n) \pmod{M^5}. \end{aligned}$$

Continuing in this way, one finds elements $\varepsilon_i^{(\nu)} \in M^{\nu+1}$ such that, on setting

$$x_i^{(\nu)} = x_i - (\varepsilon_i^{(1)} + \cdots + \varepsilon_i^{(\nu)}),$$

one has

$$f(x_1^{(\nu)}, \dots, x_n^{(\nu)}) \equiv f_2(x_1, \dots, x_n) \pmod{M^{\nu+3}}.$$

Let $\varepsilon_i = \sum_{\nu} \varepsilon_i^{(\nu)}$. Completeness then gives

$$f(x_1 - \varepsilon_1, \dots, x_n - \varepsilon_n) = f_2(x_1, \dots, x_n),$$

or, equivalently,

$$f(x_1, \dots, x_n) = f_2(x_1 + \varepsilon_1, \dots, x_n + \varepsilon_n).$$

Taking $x'_i = x_i + \varepsilon_i$ proves the assertion. $\square$

Lemma 6.2. Let A_0 be a local k -algebra essentially of finite type such that

$$\widehat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f),$$

where f is a non-degenerate quadratic form, and let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \text{Spec}(A_0)$. Then:

(1) The canonical maps

$$J_{\mathfrak{X}}(\mathfrak{X}|R) \longrightarrow J_R(\mathfrak{X}|R) \longrightarrow \text{Spec}(\Lambda)$$

are both isomorphisms.

(2) The ideal $I_R(\mathfrak{X}|R)$ is principal and generated by a free formal generator of R over Λ ; in other words,

$$I_R(\mathfrak{X}|R) = (t), \quad R = \Lambda[[t]].$$

(3) The morphism $\mathfrak{X} \longrightarrow \text{Spec}(\Lambda)$ is formally smooth.

Proof. Write $\mathfrak{X} = \text{Spf}(A)$. By 4.10(b), $(R, \text{Spf}(\widehat{A}))$ is a formal versal deformation of $\widehat{X}_0 = \text{Spec}(\widehat{A}_0)$, where

$$\widehat{A} = \varprojlim_{\nu} A \otimes_R R_{\nu}.$$

Now

$$\widehat{A}_0 \simeq k[[x_1, \dots, x_n]]/(f), \quad \left(\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n} \right) = (x_1, \dots, x_n).$$

It follows immediately that

$$[\underline{M}_{\widehat{X}_0}(k[\varepsilon])] \simeq k.$$

Consequently, by 4.14,

$$R \simeq \Lambda[[t]], \quad \widehat{A} \simeq R[[x_1, \dots, x_n]]/(F - t),$$

where F is the quadratic polynomial in $\Lambda[x_1, \dots, x_n] \subset R[[x_1, \dots, x_n]]$ obtained from f by lifting its coefficients to Λ .

The sequence

$$0 \longrightarrow \widehat{A} \xrightarrow{dF} \widehat{A} dx_1 \oplus \cdots \oplus \widehat{A} dx_n \longrightarrow \Omega_{\widehat{A}|R} \longrightarrow 0$$

is exact, where

$$1 \longmapsto \frac{\partial F}{\partial x_1} dx_1 + \cdots + \frac{\partial F}{\partial x_n} dx_n.$$

Therefore

$$I_{\widehat{A}}(\widehat{A}|R) = \left(\frac{\partial F}{\partial x_1}, \dots, \frac{\partial F}{\partial x_n} \right) = (x_1, \dots, x_n),$$

and hence

$$\begin{aligned}\widehat{A}/I_{\widehat{A}}(\widehat{A}|R) &= \widehat{A}/(x_1, \dots, x_n) \\ &= R[[x_1, \dots, x_n]]/(x_1, \dots, x_n, F - t) \simeq \Lambda.\end{aligned}$$

Thus the canonical map $\Lambda \longrightarrow \widehat{A}/I_{\widehat{A}}(\widehat{A}|R)$ is an isomorphism. The completion comparison identifies this quotient with $A/I_A(A|R)$, and consequently

$$\Lambda \xrightarrow{\sim} R/I_R(A|R).$$

This proves (1) and (2). Finally, (3) follows from the isomorphisms

$$\begin{aligned}\widehat{A} &\simeq R[[x_1, \dots, x_n]]/(F - t) \\ &\simeq \Lambda[[t, x_1, \dots, x_n]]/(F - t) \simeq \Lambda[[x_1, \dots, x_n]]\end{aligned}$$

of Λ -algebras. $\square$

Corollary 6.3. Let X_0 be a k -scheme of finite type, and let $x \in X_0$ be a non-degenerate quadratic singularity rational over k .

(1) Let $(R, \mathfrak{X})$ and $(R, \mathfrak{Y})$ be two deformations of X_0 , with $R \in \text{ob}(\widehat{\mathcal{C}}_\Lambda)$. Assume that $k = k^2$. Then

$$\mathfrak{X}_{(\widehat{x})} \simeq \mathfrak{Y}_{(\widehat{x})}$$

as R -adic formal schemes if and only if

$$I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R).$$

(2) Let $R \in \text{ob}(\widehat{\mathcal{C}}_\Lambda)$ be a regular local ring, and let $(R, \mathfrak{X})$ be a deformation of X_0 . Then $\mathcal{O}_{\mathfrak{X}, x}$ is regular if and only if

$$I_R(\mathfrak{X}_{(x)}|R) \not\subset \mathfrak{m}_R^2;$$

equivalently, $I_R(\mathfrak{X}_{(x)}|R)$ is a regular divisor in $\text{Spf}(R)$.

Proof. Let (R_0, Z) be a formal versal deformation of $\text{Spec}(\widehat{\mathcal{O}}_{X_0, x})$. Since x is a non-degenerate quadratic singularity rational over k , Lemma 6.2 allows us to take

$$R_0 \simeq \Lambda[[t]], \quad Z \simeq \text{Spf}(R_0[[X_1, \dots, X_n]]/(F - t)).$$

For (1), the deformations $(R, \mathfrak{X}_{(\widehat{x})})$ and $(R, \mathfrak{Y}_{(\widehat{x})})$ are induced from (R_0, Z) by continuous local Λ -algebra homomorphisms

$$\varphi, \psi : R_0 \longrightarrow R.$$

Put $a = \varphi(t)$ and $b = \psi(t)$. Then

$$\begin{aligned}\mathfrak{X}_{(\widehat{x})} &\simeq \text{Spf}(R[[X_1, \dots, X_n]]/(F - a)), \\ \mathfrak{Y}_{(\widehat{x})} &\simeq \text{Spf}(R[[X_1, \dots, X_n]]/(F - b)).\end{aligned}$$

Suppose that $I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{Y}_{(x)}|R)$; equivalently, $(a) = (b)$ in R . Then $a = ub$ for some unit $u \in R$. Since $k = k^2$ and R is complete, $u = v^2$ for some unit $v \in R$, and hence $a = v^2b$. Thus

$$(F - a) \longmapsto v^2(F - b)$$

under $X_i \mapsto vX_i$, and this substitution induces an R -algebra isomorphism

$$R[[X_1, \dots, X_n]]/(F - a) \xrightarrow{\sim} R[[X_1, \dots, X_n]]/(F - b),$$

because F is quadratic. The converse is immediate.

For (2), there is a continuous local Λ -algebra homomorphism

$$\varphi : R_0 = \Lambda[[t]] \longrightarrow R$$

such that

$$\mathfrak{X}_{(\hat{x})} \simeq \operatorname{Spf}(R[[X_1, \dots, X_n]]/(F - \varphi(t))).$$

Moreover,

$$I_R(\mathfrak{X}_{(x)}|R) = I_R(\mathfrak{X}_{(\hat{x})}|R) = (\varphi(t)),$$

and

$$R[[X_1, \dots, X_n]]/(F - \varphi(t)) = \widehat{\mathcal{O}}_{\mathfrak{X},x}$$

is regular if and only if $\mathcal{O}_{\mathfrak{X},x}$ is regular. Since R is regular, $R[[X_1, \dots, X_n]]$ is a regular local ring. If M denotes its maximal ideal, then its quotient by $F - \varphi(t)$ is regular if and only if

$$F - \varphi(t) \notin M^2,$$

which is equivalent to $\varphi(t) \notin \mathfrak{m}_R^2$, and hence to

$$I_R(\mathfrak{X}_{(x)}|R) \not\subset \mathfrak{m}_R^2.$$

□

Theorem 6.4. Let X_0 be a k -scheme of finite type whose nonsmooth points are isolated, and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . Denote the set of nonsmooth points of X_0 by $\{z_1, \dots, z_m\}$. Assume that

$$H^2(X_0, \underline{D}_{X_0}^0) = 0$$

and that every z_i is a non-degenerate quadratic singularity rational over k . Then:

- (1) R is formally smooth over Λ ; that is, R is a formal power-series ring over Λ .
- (2) There is a disjoint-union decomposition

$$J_{\mathfrak{X}}(\mathfrak{X}|R) = \coprod_{1 \leq i \leq m} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R),$$

and the canonical morphism

$$J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) \xrightarrow{\sim} J_R(\mathfrak{X}_{(z_i)}|R)$$

is an isomorphism.

- (3) Each ideal $I_R(\mathfrak{X}_{(z_i)}|R)$ is principal, as is

$$I_R(\mathfrak{X}|R) = \prod_{1 \leq i \leq m} I_R(\mathfrak{X}_{(z_i)}|R).$$

If $I_R(\mathfrak{X}_{(z_i)}|R) = (t_i)$, then $t_1, \dots, t_m$ can be extended to a system of free formal generators of R over Λ .

- (4) The morphism $\mathfrak{X} \rightarrow \operatorname{Spec}(\Lambda)$ is formally smooth; equivalently,

$$J_{\mathfrak{X}}(\mathfrak{X}|\Lambda) = \emptyset.$$

- (5) If X_0 is a complete irreducible curve, then

$$\dim \underline{M}_{X_0} = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \quad a = \dim_k H^0(X_0, \underline{D}_{X_0}^0).$$

Proof. Assertion (1) follows from (2) and (3). The first assertion in (2) follows from 5.7(2). For each i , let $(R_i, \mathfrak{Y}_i)$ be a formal versal deformation of $\operatorname{Spec}(\widehat{\mathcal{O}}_{X_0, z_i})$. Since $(R, \mathfrak{X}_{(\hat{z}_i)})$ is such a deformation, there is a continuous local Λ -algebra homomorphism

$$\varphi_i : R_i \rightarrow R$$

and a cartesian diagram

$$\begin{array}{ccc} \mathfrak{X}_{(\widehat{z}_i)} & \longrightarrow & \mathfrak{Y}_i \\ \downarrow & & \downarrow \\ \mathrm{Spf}(R) & \xrightarrow{\mathrm{Spf}(\varphi_i)} & \mathrm{Spf}(R_i). \end{array}$$

Consequently,

$$\begin{aligned} J_{\mathfrak{X}_{(z_i)}}(\mathfrak{X}_{(z_i)}|R) &= J_{\mathfrak{X}_{(\widehat{z}_i)}}(\mathfrak{X}_{(\widehat{z}_i)}|R) \\ &\simeq J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R), \end{aligned}$$

and

$$\begin{aligned} J_R(\mathfrak{X}_{(z_i)}|R) &= J_R(\mathfrak{X}_{(\widehat{z}_i)}|R) \\ &\simeq J_{R_i}(\mathfrak{Y}_i|R_i) \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R). \end{aligned}$$

By Lemma 6.2(1),

$$J_{\mathfrak{Y}_i}(\mathfrak{Y}_i|R_i) \longrightarrow J_{R_i}(\mathfrak{Y}_i|R_i)$$

is an isomorphism, proving the remaining assertion in (2).

By Lemma 6.2(2),

$$I_{R_i}(\mathfrak{Y}_i|R_i) = (u_i), \quad R_i = \Lambda[[u_i]].$$

It follows that

$$I_R(\mathfrak{X}_{(z_i)}|R) = I_R(\mathfrak{X}_{(\widehat{z}_i)}|R) = (t_i), \quad t_i = \varphi_i(u_i),$$

and therefore

$$I_R(\mathfrak{X}|R) = \prod_{1 \leq i \leq m} I_R(\mathfrak{X}_{(z_i)}|R) = \left(\prod_{i=1}^m t_i \right).$$

By 4.13, the natural morphism

$$\underline{M}_{X_0} \longrightarrow \underline{M}_{X_0, (\widehat{z}_1)} \times \cdots \times \underline{M}_{X_0, (\widehat{z}_m)}$$

is smooth. Hence

$$R_1 \widehat{\otimes}_{\Lambda} \cdots \widehat{\otimes}_{\Lambda} R_m \longrightarrow R$$

is formally smooth. Thus $t_1, \dots, t_m$ extend to a system of free formal generators of R over Λ , proving (1) and (3).

For (4), the projection

$$\mathfrak{X}_{(\widehat{z}_i)} \xrightarrow{\sim} \mathfrak{Y}_i \times_{\mathrm{Spf}(R_i)} \mathrm{Spf}(R) \longrightarrow \mathfrak{Y}_i$$

is formally smooth because $\mathrm{Spf}(R) \longrightarrow \mathrm{Spf}(R_i)$ is formally smooth. Lemma 6.2(3) shows that $\mathfrak{Y}_i \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth. Hence every $\mathfrak{X}_{(\widehat{z}_i)} \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth. Since the z_i are precisely the nonsmooth points of X_0 , it follows that $\mathfrak{X} \longrightarrow \mathrm{Spec}(\Lambda)$ is formally smooth. For (5), R is formally smooth over Λ , and by 4.4 and 4.11 the sequence

$$0 \longrightarrow H^1(X_0, \underline{D}_{X_0}^0) \longrightarrow [\underline{M}_{X_0}(k[\varepsilon])] \longrightarrow H^0(X_0, \underline{D}_{X_0}^1) \longrightarrow 0$$

is exact. Thus

$$\begin{aligned} \dim \underline{M}_{X_0} &= \dim_k H^1(X_0, \underline{D}_{X_0}^0) + \dim_k H^0(X_0, \underline{D}_{X_0}^1) \\ &= \dim_k H^1(X_0, \underline{D}_{X_0}^0) + m. \end{aligned}$$

It is therefore enough to show that

$$\dim_k H^1(X_0, \underline{D}_{X_0}^0) + m = 3g - 3 + \dim_k H^0(X_0, \underline{D}_{X_0}^0),$$

or equivalently

$$\chi(\underline{D}_{X_0}^0) = 3 - 3g + m,$$

where, for a coherent $\mathcal{O}_{X_0}$ -module $\mathcal{E}$, we put

$$\chi(\mathcal{E}) = \dim_k H^0(X_0, \mathcal{E}) - \dim_k H^1(X_0, \mathcal{E}).$$

The Euler characteristic of $\underline{D}_{X_0}^0$ is unchanged by extension of the base field to its algebraic closure, so we may assume that k is algebraically closed. Let z be a nonsmooth point of X_0 , put

$$A = \widehat{\mathcal{O}}_{X_0, z},$$

and let A' be the normalization of A . By hypothesis,

$$A \simeq k[[x, y]]/(f),$$

where f is a non-degenerate quadratic form. Over the algebraically closed field k we may take $f = xy$, so that

$$A \simeq k[[x, y]]/(xy).$$

It follows at once that

$$\ell(A'/A) = 1, \quad \mathrm{Der}_k(A, \mathfrak{m}_A) = \mathrm{Der}_k(A, A).$$

Let C be the conductor of A in A' . Since A is a complete intersection,

$$\ell(A/C) = \ell(A'/A) = 1,$$

and hence $C = \mathfrak{m}_A$. Therefore

$$A' = \{a' \in K \mid a' \mathfrak{m}_A \subset \mathfrak{m}_A\},$$

where K is the total fraction ring of A .

We claim that

$$\mathrm{Der}_k(A, A) \subset \mathrm{Der}_k(A', A')$$

inside $\mathrm{Der}_k(K, K)$. Indeed, let $D \in \mathrm{Der}_k(A, A)$. For $a' \in A'$ and $x \in \mathfrak{m}_A$, one has $xa' \in \mathfrak{m}_A$, so $D(xa') \in \mathfrak{m}_A$ and $a'D(x) \in \mathfrak{m}_A$. The identity

$$D(xa') = a'D(x) + xD(a')$$

therefore gives $xD(a') \in \mathfrak{m}_A$ for every $x \in \mathfrak{m}_A$, whence $D(a') \in A'$.

Moreover, the exact sequence

$$0 \longrightarrow \mathrm{Der}_k(A, A) \longrightarrow \mathrm{Der}_k(A', A') \longrightarrow A'/\mathfrak{m}_A A' \longrightarrow 0$$

may be written

$$0 \longrightarrow \mathrm{Der}_k(A, A) \longrightarrow \mathrm{Der}_k(A', A') \longrightarrow A'/C \longrightarrow 0.$$

If X'_0 is the normalization of X_0 , these local sequences give an exact sequence

$$0 \longrightarrow \underline{D}_{X_0}^0 \longrightarrow \underline{D}_{X'_0}^0 \longrightarrow \mathcal{O}_{X'_0}/C \longrightarrow 0.$$

Since X_0 is locally a complete intersection,

$$\ell(\mathcal{O}_{X'_0}/C) = 2\ell(\mathcal{O}_{X'_0}/\mathcal{O}_{X_0}) = 2m,$$

whereas the Riemann–Roch theorem gives

$$\chi(\underline{D}_{X'_0}^0) = 3 - 3(g - m).$$

Consequently,

$$\begin{aligned}\chi(\underline{D}_{X_0}^0) &= \chi(\underline{D}_{X'_0}^0) - \ell(\mathcal{O}_{X'_0}/C) \\ &= 3 - 3(g - m) - 2m = 3 - 3g + m.\end{aligned}$$

This completes the proof. $\square$

Remark 6.5. In the formula of Theorem 6.4(5), the number $a = \dim_k H^0(X_0, \underline{D}_{X_0}^0)$ is easily computed by the Riemann–Roch theorem:

$$a = \begin{cases} 0, & g \geq 2, \\ 1, & g = 1, \\ 3, & g = 0. \end{cases}$$

We finish with a remark on characteristic 2. If $\text{char } k = 2$, there is no non-degenerate quadratic form in an odd number of variables. Equivalently, there is no non-degenerate quadratic singularity of even dimension in characteristic 2. In this case one may instead consider a quadratic form of maximal non-degeneracy.

Definition 6.6. Let X be a k -scheme of finite type. A closed point $z \in X$ is called an *ordinary quadratic singularity rational over k* if the following conditions hold.

- (i) If either $\text{char } k \neq 2$, or $\text{char } k = 2$ and $\dim \mathcal{O}_{X,z}$ is odd, then z is a non-degenerate quadratic singularity rational over k .
- (ii) If $\text{char } k = 2$ and $\dim \mathcal{O}_{X,z}$ is even, then

$$\bar{k} \otimes_k \widehat{\mathcal{O}}_{X,z} \simeq \bar{k}[[X_0, X_1, \dots, X_{2n}]]/(f),$$

where $\bar{k}$ is an algebraic closure of k and

$$f = X_0^2 + X_1X_2 + X_3X_4 + \dots + X_{2n-1}X_{2n}.$$

Lemma 6.7. Let k be a field of characteristic 2, and put

$$A_0 = k[[X_0, X_1, \dots, X_{2n}]]/(f), \quad f = X_0^2 + X_1X_2 + X_3X_4 + \dots + X_{2n-1}X_{2n}.$$

Let $(R, \mathfrak{X})$ be a formal versal deformation of $X_0 = \text{Spec}(A_0)$. Then

$$R = \Lambda[[t, t_1]],$$

where t, t_1 are free formal generators of R over Λ , and

$$I_R(\mathfrak{X}|R) = (t^2).$$

Moreover, $\mathfrak{X} \rightarrow \text{Spec}(\Lambda)$ is formally smooth.

Proof. Write $\mathfrak{X} = \text{Spf}(A)$. Since

$$\left(f, \frac{\partial f}{\partial X_0}, \dots, \frac{\partial f}{\partial X_{2n}}\right) = (X_0^2, X_1, \dots, X_{2n}),$$

we have

$$D^1(A_0|k, A_0) \simeq k[X_0]/(X_0^2).$$

Thus $[\underline{M}_{X_0}(k[\varepsilon])]$ is a two-dimensional k -vector space. By 4.14,

$$R \simeq \Lambda[[t, t_1]], \quad A \simeq R[[X_0, X_1, \dots, X_{2n}]]/(F),$$

where

$$F = f + t_1 + tX_0.$$

The exact sequence

$$0 \longrightarrow A \xrightarrow{dF} A dX_0 \oplus \cdots \oplus A dX_{2n} \longrightarrow \widehat{\Omega}_{A|R} \longrightarrow 0,$$

in which

$$dF : 1 \mapsto \sum_{0 \leq i \leq 2n} \frac{\partial f}{\partial X_i} dX_i + t dX_0,$$

gives

$$I_A(A|R) = (t, X_1, \dots, X_{2n}) = (X_0^2 + t_1, t, X_1, \dots, X_{2n}).$$

Hence

$$\begin{aligned} A/I_A(A|R) &\simeq A/(X_0^2 + t_1, t, X_1, \dots, X_{2n}) \\ &\simeq \Lambda[[t_1, X_0]]/(X_0^2 + t_1). \end{aligned}$$

As an R -module this last module is isomorphic to $R/(t) \oplus R/(t)$. It follows that

$$I_R(A|R) = (t^2),$$

which proves the first assertion. On the other hand,

$$\begin{aligned} A &\simeq \Lambda[[t, t_1, X_0, X_1, \dots, X_{2n}]]/(X_0^2 + X_1 X_2 + \cdots + X_{2n-1} X_{2n} + t X_0 + t_1) \\ &\simeq \Lambda[[t, X_0, X_1, \dots, X_{2n}]], \end{aligned}$$

so A is formally smooth over Λ . □

Lemma 6.7 gives the following analogue of Theorem 6.4; we leave its proof to the reader.

Corollary 6.8. Let X_0 be a k -scheme of finite type whose nonsmooth points are isolated, assume that

$$H^2(X_0, \underline{D}_{X_0}^0) = 0,$$

and let $(R, \mathfrak{X})$ be a formal versal deformation of X_0 . Denote the set of nonsmooth points by $\{z_1, \dots, z_m\}$. Suppose that $\text{char } k = 2$, that $\dim X_0$ is even, and that all the z_i are ordinary quadratic singularities. Then:

- (1) R is formally smooth over Λ .
- (2) Each ideal $I_R(\mathfrak{X}_{(z_i)}|R)$ is principal, as is

$$I_R(\mathfrak{X}|R) = \prod_{1 \leq i \leq m} I_R(\mathfrak{X}_{(z_i)}|R).$$

More precisely,

$$I_R(\mathfrak{X}_{(z_i)}|R) = (t_i^2),$$

and $t_1, \dots, t_m$ can be extended to a system of free formal generators of R over Λ .

- (3) The morphism $\mathfrak{X} \longrightarrow \text{Spec}(\Lambda)$ is formally smooth.
- (4) If k is algebraically closed and X_0 is a complete irreducible curve, then

$$\dim \underline{M}_{X_0} = 3g - 3 + a,$$

where

$$g = \dim_k H^1(X_0, \mathcal{O}_{X_0}), \quad a = \dim_k H^0(X_0, \underline{D}_{X_0}^0).$$

BIEXTENSIONS OF SHEAVES OF GROUPSby A. Grothendieck0. Introduction

The notion of a biextension was introduced by D. Mumford [6], in the context of formal groups, as a convenient way of expressing the relations between the formal groups associated with an abelian scheme and the dual abelian scheme. In the present exposé we undertake a systematic study of this notion in the general context of sheaves of groups on a given site. In particular, we show that it fits remarkably simply into the classical cohomological formalism of $\otimes^L$ and $\mathbf{R}\mathbf{Hom}$ (3.6.4 and 3.6.5). In the following exposé we shall make explicit the most interesting special cases, notably that of biextensions of group schemes by the multiplicative group $\underline{G}_m$ (working with such sites as the Zariski site, the fpqc site, or an intermediate site). We shall see, in particular, that if A and B are two abelian schemes over a scheme S , biextensions of (A, B) by $\underline{G}_m$ admit a remarkable interpretation as, essentially, the classical divisorial correspondences on A, B . This gives a particularly convenient explicit cohomological meaning to the notion of divisorial correspondence (already implicit in Tate duality for abelian varieties over p -adic fields). This interpretation of divisorial correspondences is also convenient for following the fate of such a correspondence when A and B (assumed to be defined over the fraction field of a discrete valuation ring) are made to degenerate by means of Néron theory.

The results of Exposé VIII will show, for example, that one obtains a biextension of the pair of identity components of the Néron models, hence a biextension of their connected special fibres, which plays the role of a specialization of the original divisorial correspondence. These results will be used in Exposé IX, in conjunction with the monodromy theorem (I 3), to prove certain results about the Néron models of abelian varieties, the most important being the “semistable reduction theorem” for the Néron model.

Throughout what follows, unless expressly stated otherwise, we work in a fixed $\underline{U}$ -topos denoted by $\underline{T}$; that is (SGA 4, Exposé IV), $\underline{T}$ is a category equivalent to the category of sheaves on a suitable $\underline{U}$ -site (which may, if desired, be taken to be $\underline{T}$ itself, endowed with its canonical topology). In our terminology we shall often identify the objects of $\underline{T}$ with the sheaves of sets on $\underline{T}$. All groups, extensions of groups, and so forth considered below are relative to this topos $\underline{T}$.

1. Complements on extensions of groups

1.1. Let P and G be two groups of $\underline{T}$, and let

$$(1.1.1) \quad 1 \longrightarrow G \xrightarrow{i} E \xrightarrow{j} P \longrightarrow 1$$

be an extension of P by G . Thus E is a group, $j : E \rightarrow P$ is an epimorphism of groups, and i induces an isomorphism of G with $\ker j$. We shall reinterpret the datum of such an extension in different terms, which will be convenient for the definition and study of biextensions.

1.1.2. First, the homomorphism i makes G act on E in two ways, on the left and on the right, by setting

$$gxg' = i(g)xi(g')$$

for $g, g' \in G(S)$ and $x \in E(S)$, where S is any object of $\underline{T}$. If, by means of j , we regard E as an object “over P ”, that is, as an object of the induced topos $\underline{T}/_P$, this amounts to saying that the group $G_P = G \times P$ induced by G in $\underline{T}/_P$ acts on E on the left and on the right. Moreover, it is well known that each of these actions makes E a torsor (a left torsor and a right torsor, respectively) over P , with group G_P ; more precisely still, it makes E a bitorsor under (G_P, G_P) . Recall [2]

that this means that E is a right torsor E_d under the right action of G_P , and that the left action induces an isomorphism of G_P with the group of automorphisms of E_d (the sheaf of groups of automorphisms of E that commute with the right action of G_P on E). This condition is equivalent to the symmetric condition: the left action of G_P on E makes E a left torsor E_s , and the right action induces an isomorphism of the opposite group G_P^o with the group of automorphisms of E_s .

1.1.2.1. When $\underline{T}$ is the “point topos”, that is, the category of sets (equivalent to the category of sheaves on the one-point topological space), so that P and G are ordinary groups, the datum of a bitorsor E under (G_P, G_P) is plainly equivalent to the datum, for each $p \in P$, of a bitorsor E_p under (G, G) , namely the fibre of E at p . An analogous interpretation is available in the general case. Indeed, giving a bitorsor E under (G_P, G_P) is equivalent to giving, for every “point” $p \in P(S)$, where S is any object of $\underline{T}$, a bitorsor E_p under (G_S, G_S) over S , in such a way that the formation of E_p is “compatible with base change along every morphism $S' \rightarrow S$ ”. Thus, if $p' : S' \rightarrow P$ is the composite morphism, one is also given an isomorphism of bitorsors from $E_{p'}$ to the inverse image of E_p under $S' \rightarrow S$, and these isomorphisms satisfy the usual transitivity condition for a composite $S'' \rightarrow S' \rightarrow S$.

When stated in canonical language (SGA 1, Exposé VI), this reduces to the following tautology: giving a bitorsor under (G_P, G_P) over P is equivalent to giving a cartesian section, over the category $\underline{T}/_P$, of the fibred category of bitorsors under (G_S, G_S) over a variable base $S \in \text{ob } \underline{T}/_P$. More precisely, the “value at P ” functor induces an equivalence between the category of the cartesian sections in question and the category of bitorsors under (G_P, G_P) . This follows trivially from the fact that P is a final object of the base category $\underline{T}/_P$ of our fibred category: for every category fibred over $\underline{T}/_P$, the “value at P ” functor is an equivalence between the category of cartesian sections of that fibred category and its fibre category over P . Despite its tautological nature, however, this interpretation will be useful for expressing certain compatibilities in language close to the set-theoretic case, using points with values in an object S in place of ordinary points.

1.1.3. In terms of the extension structure (1.1.1), the bitorsor structure of 1.1.2 on E under (G_P, G_P) used only part of the multiplicative structure of E : namely, multiplication of pairs of “points” of which at least one comes from a point of G (via i). We shall now interpret the multiplicative structure of E as an additional structure on the bitorsor under consideration.

For this purpose, first consider two sections p, p' of P . With the notation of 1.1.2.1, these give two bitorsors $E_p, E_{p'}$ under (G, G) , namely the inverse images in E of the sections $p, p' : e \rightarrow P$ (where e denotes the final object of $\underline{T}$). It is then clear that multiplication in E induces a morphism of objects of $\underline{T}$

$$\varphi : E_p \times E_{p'} \rightarrow E_{pp'},$$

which, in terms of the left and right structures of the three bitorsors involved (and, as usual, using points with values in an arbitrary object $S \in \text{ob } \underline{T}$), satisfies

$$(1.1.3.0) \quad \begin{cases} \varphi(gx, x') = g\varphi(x, x'), & \varphi(x, x'g') = \varphi(x, x')g', \\ \varphi(xg, x') = \varphi(x, gx'). \end{cases}$$

Here $g, g' \in G(S)$, $x \in E_p(S)$, and $x' \in E_{p'}(S)$. In terms of the contracted product [2]

$$E_p \overset{G}{\times} E_{p'},$$

the third relation says that φ factors as

$$E_p \times E_{p'} \rightarrow E_p \overset{G}{\times} E_{p'} \xrightarrow{\varphi_{p,p'}} E_{pp'},$$

where the first arrow is the canonical quotient morphism. The first two relations, on the other hand, say that the morphism thus obtained,

$$(1.1.3.1) \quad \varphi_{p,p'} : E_p \overset{G}{\times} E_{p'} \rightarrow E_{pp'},$$

is a homomorphism of bitorsors, where the bitorsor structure on the source is the one for which the left (respectively, right) operations of G come from the left (respectively, right) operations of G on the first factor E_p (respectively, on the second factor $E_{p'}$).

Notice that, like every homomorphism of bitorsors under a fixed pair of groups, $\varphi_{p,p'}$ is necessarily an isomorphism.

The definition of the preceding isomorphism (1.1.3.1) immediately extends to the case in which p, p' are points of P with values in an arbitrary object S of $\underline{T}$, rather than sections of P : one simply applies the preceding discussion to the induced topos $\underline{T}/_S$ and to the extension of groups over S obtained from (1.1.1) by base change. The formation of the isomorphisms $\varphi_{p,p'}$ is then compatible with every base change $S' \rightarrow S$. In other words, it corresponds to a homomorphism of cartesian sections of the fibred category over the category $\underline{T}/_{P \times P}$ of pairs of arrows $p, p' : S \rightrightarrows P$ with target P .

Giving a system of $\varphi_{p,p'}$ satisfying this condition is, of course, equivalent to giving $\varphi_{p,p'}$ in the “universal” case $S = P \times P$, where p, p' are the two projections pr_i ($i = 1, 2$) from $P \times P$ to P . It is then an isomorphism of bitorsors under $(G_{P \times P}, G_{P \times P})$:

$$(1.1.3.2) \quad \varphi : \text{pr}_1^*(E) \text{pr}_2^*(E) \xrightarrow{\sim} \pi^*(E),$$

where

$$\pi : P \times P \rightarrow P$$

is the morphism of objects of $\underline{T}$ defining the multiplicative structure of the group P . The compatibility properties below are most conveniently stated, however, in terms of the data (1.1.3.1), avoiding diagrams that are less close to intuition.

1.1.4. In writing the relations (1.1.3.0), which enabled us to express the given composition law on E in terms of isomorphisms of bitorsors (1.1.3.1), we used associativity of this composition law only for a product of three “points” of E of which at least one comes from a point of G . To express full associativity in terms of the $\varphi_{p,p'}$, one must require, for three points p, p', p'' of P with values in an object S of $\underline{T}$, that the following diagram be commutative:

$$(1.1.4.1) \quad \begin{array}{ccc} & E_{pp'}E_{p''} & \\ \varphi_{p,p'} \wedge \text{id} \nearrow & & \searrow \varphi_{pp',p''} \\ E_p E_{p'} E_{p''} & & E_{pp'p''} \\ \text{id} \wedge \varphi_{p',p''} \searrow & & \nearrow \varphi_{p,p'p''} \\ & E_p E_{p'p''} & \end{array}$$

In this diagram we have written the contracted product of bitorsors without the signs $\overset{G_S}{\times}$. It is understood that the contracted product $Q.R$ of two bitorsors is formed using, in passing to the quotient in $Q \times R$, the action of the structural group G on the right of Q and on the left of R . This contracted-product operation is associative up to a canonical isomorphism [2], and we make the identification

$$(E_p E_{p'}) E_{p''} \simeq E_p (E_{p'} E_{p''}),$$

writing both sides simply as $E_p E_{p'} E_{p''}$. Obvious conventions of this kind will be used throughout the rest of this exposé without being expressly noted each time.

1.1.4.2. Just as the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) are determined by the “universal” case $S = P \times P$, $p = \text{pr}_1$, $p' = \text{pr}_2$ considered in (1.1.3.2), so the associativity condition (1.1.4.1) may likewise be checked only in the universal case $S = P \times P \times P$, $p = \text{pr}_1$, $p' = \text{pr}_2$, $p'' = \text{pr}_3$. We leave it to the reader to write the diagram in whatever notation is preferred.

1.1.5. Conversely, suppose that the groups P and G of $\underline{T}$ are given, together with a bitorsor

$$(1.1.5.1) \quad j : E \longrightarrow P$$

under (G_P, G_P) (the two actions of G_P on E , or equivalently of G on E , being understood in the notation), and a system of isomorphisms of bitorsors over a variable base S ,

$$(1.1.5.2) \quad \varphi_{p,p'} : E_p E_{p'} \xrightarrow{\sim} E_{pp'} \quad (p, p' \in P(S)),$$

compatible with base change $S' \longrightarrow S$; in other words, an isomorphism of bitorsors (1.1.3.2). Suppose that the corresponding diagrams (1.1.4.1) commute. I claim that there is a unique extension structure on E of P by G such that the preceding data are obtained from it by the constructions of 1.1.2 and 1.1.3.

First observe that, on taking p, p' in (1.1.5.2) to be the identity section 1 of P , one obtains an isomorphism

$$(1.1.5.3) \quad \varphi_{1,1} : E_1 \xleftarrow{\sim} E_1 \overset{G}{\times} E_1.$$

Taking the contracted product, on the left say, of both sides with the inverse bitorsor E_1^{-1} of E_1 [2] therefore gives a canonical isomorphism

$$(1.1.5.4) \quad E_1 \simeq \text{the unit bitorsor } {}_s G_d$$

(that is, G acting on itself by left and right translations), in view of the canonical isomorphism

$$E_1^{-1} \cdot E_1 \simeq \text{the unit bitorsor } {}_s G_d$$

of loc. cit., valid for every bitorsor. Equivalently, one has a section 1_E of E_1 , hence of E ,

$$(1.1.5.5) \quad 1_E \in \Gamma(E_1) = \text{Hom}(e, E_1) \subset \Gamma(E),$$

namely the image under (1.1.5.4) of the distinguished section of ${}_s G_d$, satisfying

$$g \cdot 1_E = 1_E \cdot g \quad (g \in G(S)).$$

By construction this section is characterized by

$$(1.1.5.6) \quad \varphi_{1,1}(1_E, 1_E) = 1_E.$$

On the other hand, giving the morphism of torsors φ of (1.1.3.2) is plainly equivalent to giving a morphism

$$(1.1.5.7) \quad \pi_E : E \times E \longrightarrow E, \quad \text{also denoted } (x, y) \longmapsto xy,$$

satisfying

$$(1.1.5.8) \quad j(xy) = j(x)j(y) \quad (x, y \in E(S)),$$

so that j is compatible with the multiplicative structures π_E on E and π on P , together with the three relations (1.1.3.0) (preferably beginning with the last of these). Finally define a morphism of objects of $\underline{T}$,

$$i : G \longrightarrow E,$$

by the formula

$$(1.1.5.9) \quad i(g) = g \cdot 1_E = 1_E \cdot g.$$

The equality of the last two terms was already observed in constructing 1_E . I now claim that the multiplication law π_E on E makes E a group with identity section 1_E , and that $j : E \rightarrow P$ and $i : G \rightarrow E$ are homomorphisms of groups making E an extension of P by G .

Indeed, associativity of the composition law on E is precisely the commutativity of the diagrams (1.1.4.1). The construction of π_E by means of the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1) also shows that, for every $x \in E(S)$, left and right translation by x in $E(S)$ are injective. Since $1_E 1_E = 1_E$ by (1.1.5.6), it follows at once that 1_E is a two-sided identity of E . We leave to the reader the verification that an $x \in E(S)$ lying over $p \in P(S)$ has an inverse; one uses the canonical isomorphism deduced from $\varphi_{p,p^{-1}}$,

$$(1.1.5.10) \quad E_{p^{-1}} \simeq (E_p)^o.$$

Thus E is indeed a group. We have already observed that j is compatible with the composition laws (1.1.5.8), and the same is true of i , since, for $g, g' \in G(S)$,

$$\begin{aligned} i(g)i(g') &= (g \cdot 1_E)(g' \cdot 1_E) \\ &= (g \cdot 1_E \cdot g') 1_E && \text{(the last relation of (1.1.3.0))} \\ &= g \cdot 1_E \cdot g' && \text{(since } 1_E \text{ is an identity)} \\ &= g \cdot (1_E \cdot g') = g \cdot (g' \cdot 1_E) \\ &= (gg') \cdot 1_E && \text{by (1.1.5.9)} \\ &= i(gg'). \end{aligned}$$

Since j is the structural projection of a torsor, it is an epimorphism of objects of $\underline{T}$. It remains only to see that i induces an isomorphism of G with $\ker j = E_1$, which is clear from the definition (1.1.5.9) and the construction of 1_E .

1.1.6. We shall accept that the preceding arguments prove the following. For a fixed topos $\underline{T}$, the preceding constructions give an equivalence between the category of extensions of groups of $\underline{T}$ as in (1.1.1), where P, G, E are all variable, and the category of systems (P, G, E, φ) in which P and G are groups of $\underline{T}$, E is a bitorsor under (G_P, G_P) over P , and φ is an isomorphism of bitorsors (1.1.3.2) (or, equivalently, a system of base-change-compatible isomorphisms of bitorsors (1.1.3.1)), satisfying the associativity condition expressed by commutativity of diagram (1.1.4.2) (or, equivalently, of the diagrams (1.1.4.1)).

We leave it to the reader to spell out, using the preceding dictionary, the evident notion of inverse image (respectively, direct image) of an extension (1.1.1) under a homomorphism of groups

$$P' \rightarrow P \quad (\text{respectively, } G \rightarrow G'),$$

or under both operations simultaneously. We likewise leave to the reader the compatibility of the preceding equivalence of categories with the inverse-image functor f^* associated with a morphism of topoi

$$f : \underline{T}' \rightarrow \underline{T}.$$

For this one should use the datum φ in the form (1.1.3.2), in preference to the $\varphi_{p,p'}$ of (1.1.3.1); it then defines the corresponding datum $f^*(\varphi)$ on the topos $\underline{T}'$.

1.2. The commutative case. From 1.4 onward we shall be concerned only with commutative extensions of groups. We therefore assume that P and G are commutative, and spell out here, in terms of the bitorsor description (P, G, E, φ) of an extension, the condition that E be a commutative group.

An obviously necessary first condition—which in fact expresses that $i(G)$ be a central subgroup of E —is that the bitorsors E_p , for $p \in P(S)$, be ordinary torsors; that is, the left and right actions of G on E_p must coincide. Equivalently, E over P is an ordinary torsor under G_P . (Thus there will henceforth be no need to speak of bitorsors, and we shall consider only ordinary torsors.)

Assuming this preliminary condition, commutativity of E is expressed by commutativity of the following diagrams, for $p, p' \in P(S)$:

$$(1.2.1) \quad \begin{array}{ccc} E_p E_{p'} & \xrightarrow{\varphi_{p,p'}} & E_{pp'} \\ \text{sym} \downarrow & & \downarrow \text{id} \\ E_{p'} E_p & \xrightarrow{\varphi_{p',p}} & E_{p'p}. \end{array}$$

Here the first vertical arrow is the canonical commutativity isomorphism (defined because G is now commutative); moreover $pp' = p'p$ because P is commutative. This condition is plainly equivalent to the corresponding condition in the universal situation $S = P \times P$, $p = \text{pr}_1$, $p' = \text{pr}_2$, which we need not write out again.

1.3. Relations with rigidified or birigidified torsors

1.3.1.0. Let P be an object of $\underline{T}$ equipped with a section e_P , let G be a group of $\underline{T}$, and let E be a torsor under G_P . A rigidification of E , relative to the given section e_P of P , is a trivialization of the torsor $E_1 = e_P^*(E)$ under G ;

that is, a section of this torsor. This terminology is inspired by the case where G is commutative and

$$(1.3.1.1) \quad \text{Hom}_{\text{pt}}(P, G) \simeq (e),$$

where the left-hand side denotes the set of morphisms carrying e_P to the identity section e_G of G . This relation is equivalent to saying that the natural homomorphism

$$(1.3.1.2) \quad \Gamma(G) \xrightarrow{\sim} \text{Hom}(P, G),$$

which associates with a section g of G the constant morphism $P \rightarrow G$ with value g , is an isomorphism. It is immediate that these equivalent conditions are also equivalent to saying that every automorphism of the torsor E preserving the given rigidification is the identity automorphism (such automorphisms correspond precisely to the elements of the left-hand side of (1.3.1.1)). In particular, under condition (1.3.1.1), the category of torsors under G_P rigidified relative to e_P is “rigid”.

1.3.2. Let, in addition, Q be an object of $\underline{T}$ equipped with a section e_Q , and now let E denote a torsor under $G_{P \times Q}$. Since e_Q defines a section $e_{P \times Q/P}$ of $P \times Q$ over P , one may consider rigidifications α of E relative to this section, working in the induced topos $\underline{T}/_P$; explicitly, these are sections of the torsor induced by E on $P \times e_Q$. Interchanging the roles of P and Q , one may likewise consider rigidifications β of E relative to the section $e_{P \times Q/Q}$ of $P \times Q$ over Q defined by e_P . Two such partial rigidifications are said to be compatible if the sections α and β agree on $e_P \times e_Q$, that is, if they define the same section of the torsor $E_{1,1}$ under G , the inverse image

of E under the section (e_P, e_Q) of $P \times Q$. A birigidification of the torsor E relative to the pair of sections (e_P, e_Q) is a pair of compatible partial rigidifications.

If G is commutative, then E admits a birigidification (α, β) if and only if it admits partial rigidifications α and β ; moreover one may find a birigidification in which α (or β) is prescribed in advance. Indeed, it suffices to correct β by the morphism $Q \rightarrow G$ obtained as the composite

$$Q \rightarrow e \xrightarrow{\lambda} G,$$

where λ is defined by $\beta e_Q = (\alpha e_P) \lambda$.

Still in the commutative case, if E, E' are two birigidified torsors under $G_{P \times Q}$, saying that they are isomorphic merely as $G_{P \times Q}$ -torsors is equivalent to saying that they are isomorphic as birigidified $G_{P \times Q}$ -torsors. In other words, if there is an isomorphism $u : E \rightarrow E'$ of $G_{P \times Q}$ -torsors, then there is one $v : E \rightarrow E'$ that preserves the birigidifications. Indeed, measuring the discrepancy of u from the given birigidifications gives morphisms $f : P \rightarrow G$ and $g : Q \rightarrow G$

such that $f(e_P) = g(e_Q) = a \in G(e)$. There is then a morphism $P \times Q \longrightarrow G$ whose restrictions along $e_{P \times Q/P}$ and $e_{P \times Q/Q}$ are respectively f and g , namely

$$h(p, q) = f(p)g(q)a^{-1},$$

and one may take $v = uh$.

1.3.3. The terminology of 1.3.1 extends immediately to the case of a bitorsor E under (G_P, G_P) . A rigidification of the bitorsor E is then required to be an isomorphism from $e_P^*(E)$ to the trivial bitorsor ${}_sG_d$; equivalently, it is defined by a central section e_E of $e_P^*(E)$, one satisfying

$$g \cdot e_E = e_E \cdot g \quad (g \in G(S)).$$

The terminology of birigidifications extends in the same way to bitorsors.

1.3.4. Return to the case in which two groups P and G are given, and choose the identity section e_P in order to define rigidifications of bitorsors over P and birigidifications of bitorsors over $P \times P$. There are natural functors

$$(1.3.4.1) \quad \mathbf{EXT}(P; G) \xrightarrow{i} \mathbf{BITORSRIG}(P, G) \xrightarrow{\delta} \mathbf{BITORSBIRIG}(P, P; G),$$

where the three expressions denote, respectively, the category of extensions of the group P by G , the category of bitorsors under (G_P, G_P) rigidified at e_P , and the category of bitorsors under $(G_{P \times P}, G_{P \times P})$ birigidified at (e_P, e_P) . The functor i associates with an extension E of P by G the corresponding bitorsor over P (1.1.6), rigidified by the identity section of E . The functor δ is defined by

$$(1.3.4.2) \quad \delta(E) = \pi^*(E) \text{pr}_1^*(E)^{-1} \text{pr}_2^*(E)^{-1},$$

where $\pi : P \times P \longrightarrow P$ is the composition law of P , and pr_1, pr_2 are the two projections. There is a canonical birigidification of $\delta(E)$. Indeed, the restriction of $\delta(E)$ to $P \times e_P$ is canonically isomorphic to

$$\text{id}_P^*(E) \text{id}_P^*(E)^{-1} e^*(E)^{-1},$$

where $e = \text{pr}_2 \text{inj}_1$ is the constant morphism $P \longrightarrow P$ with value e_P . The given rigidification of E provides an isomorphism from $e^*(E)$ to the trivial bitorsor, and hence an isomorphism of the restriction of $\delta(E)$ to $P \times e_P$ with the trivial bitorsor. Symmetrically one defines a partial rigidification along $e_P \times P$, and the two partial rigidifications are immediately seen to be compatible.

1.3.4.3. Observe that the composite δi of the functors in (1.3.4.1) is canonically isomorphic to the trivial functor, that is, to the constant functor whose value is the trivial birigidified bitorsor over $P \times P$. Indeed, for every extension E of P by G , the isomorphism φ of (1.1.3.2) plainly defines a trivialization of $\delta i(E)$ compatible with its canonical birigidification.

Proposition 1.3.5. Let P and G be two groups of $\underline{T}$.

a) Suppose that every morphism of sheaves of sets $P \times P \longrightarrow G$ which is trivial on $P \times e_P$ and on $e_P \times P$ is trivial. Then the functor i of (1.3.4.1) is fully faithful. In particular, on a bitorsor E under (G_P, G_P) rigidified relative to e_P , that is, equipped with a central section e_E of $E_1 = e_P^*(E)$, there is at most one extension structure of P by G compatible with its bitorsor structure and having e_E as identity section.

b) Suppose in addition that every morphism of sheaves of sets $P \times P \times P \longrightarrow G$ which is trivial on the partial products $P \times P \times e_P$, $P \times e_P \times P$, and $e_P \times P \times P$ is trivial. Let E be a bitorsor under (G_P, G_P) . In order that E admit an extension structure of P by G compatible with its bitorsor structure, it is necessary and sufficient that the bitorsor under $(G_{P \times P}, G_{P \times P})$ underlying

$$\delta(E) = \pi^*(E) \text{pr}_1^*(E)^{-1} \text{pr}_2^*(E)^{-1}$$

be trivial. When this is so, for every central section e_E of $E_1 = e_P^*(E)$, that is, every rigidification of E relative to e_P , there is a unique extension structure on E compatible with its bitorsor

structure and having e_E as identity section. In other words, E , regarded as a bitorsor rigidified relative to e_P , lies in the essential image of i . Moreover, the birigidified bitorsor $\delta(E)$ has one and only one trivialization.

Thus, in terms of diagram (1.3.4.1), the essential image of the fully faithful functor i is “the kernel” of δ . More picturesquely, diagram (1.3.4.1) is an “exact sequence of pointed categories,” and even of categories endowed with pseudo-group structures (in the terminology introduced in 2.5.1 below).

Let us prove 1.3.5. For a), let E, E' be two extensions of P by G , and let $f : E \rightarrow E'$ be a morphism of the underlying rigidified bitorsors. We show that f is a morphism of extensions, or, what plainly amounts to the same thing, a homomorphism of groups. We must prove $f(xy) = f(x)f(y)$. If $j' : E' \rightarrow P$ denotes the structural morphism of the torsor E' over P , then $j'f(xy) = j'(f(x)f(y))$: the two sides are respectively $j(xy)$ and $j(x)j(y)$, which are equal because j is multiplicative. Hence

$$f(xy) = u(j(x), j(y))f(x)f(y),$$

where

$$u : P \times P \rightarrow G$$

is a uniquely determined morphism. Setting first $x = 1$ and then $y = 1$ in the preceding relation, and using $f(1) = 1$, gives

$$u(p, 1) = u(1, p) = 1.$$

Thus u is trivial on $P \times e_P$ and on $e_P \times P$; by hypothesis it is trivial, and therefore $f(xy) = f(x)f(y)$, as required.

For b), necessity of the stated condition for E to admit an extension structure compatible with its bitorsor structure is immediate from 1.3.4.3. It remains to prove sufficiency for the first assertion. Pulling a trivialization φ of $\delta(E)$ back along the identity section of $P \times P$ gives a trivialization of $E_1 = e_P^*(E)$ (compare 1.5.1), that is, a rigidification of E relative to e_P . Fix this rigidification. A given trivialization φ of $\delta(E)$ need not be compatible with the corresponding birigidification. Relative to the natural partial rigidifications of $\delta(E)$, its restrictions to $P \times e_P$ and $e_P \times P$ are described by central morphisms

$$f : P \rightarrow G, qquad g : P \rightarrow G,$$

whose values at the identity section of P coincide: $f(1) = g(1) = a \in \Gamma G$. The central morphism

$$h : P \times P \rightarrow G, qquad h(p, q) = f(p)g(q)a^{-1},$$

has restrictions f and g to $P \times e_P$ and $e_P \times P$. Correcting φ by h , that is, replacing it by $\varphi(p, q)h(p, q)^{-1}$, therefore gives a trivialization of $\delta(E)$ compatible with its birigidification. Such a trivialization is unique: two of them differ by a central morphism $P \times P \rightarrow G$ trivial on $P \times e_P$ and $e_P \times P$, hence trivial by hypothesis. This proves the last assertion of 1.3.5 b).

It remains to prove that φ , viewed as an isomorphism of the form (1.1.3.2), satisfies the associativity condition (1.1.4.1); by our choice of φ , e_E is already a two-sided identity for the composition law defined by φ . The two composites in (1.1.4.1) are isomorphisms of bitorsors under $(G_{P \times P \times P}, G_{P \times P \times P})$, so they differ by a central morphism

$$u : P \times P \times P \rightarrow G$$

through the formula

$$(xy)z = u(jx, jy, jz) x(yz).$$

Setting successively $z = 1$, $x = 1$, and $y = 1$, and using the fact that 1 is a two-sided identity for the composition law, shows that u is trivial on the three partial products $P \times P \times e_P$, $e_P \times P \times P$, and $P \times e_P \times P$. The hypothesis then implies that u is trivial, so $(xy)z = x(yz)$. This completes the proof.

Corollary 1.3.6. Suppose that every morphism of sheaves of sets $P \times G \longrightarrow G$ which is trivial on $P \times e_G$ and on $e_P \times G$ is trivial. Then, if G is commutative, G is central in every extension of P by G . If P is also commutative and the hypothesis of 1.3.5 a) holds, then every extension E of P by G is commutative.

For the first assertion, we must prove that the action of P on G defined by the extension E is trivial. This action is expressed by a morphism $u : P \times G \longrightarrow G$. The restrictions to $P \times e_G$ and $e_P \times G$ of the corresponding morphism

$$(p, g) \longmapsto u(p, g)g^{-1}$$

are plainly trivial. Hence this last morphism is trivial, so $u(p, g) = g$, as required. Now, if E is an extension of P by G , its opposite group E° is an extension of P° by G° , and the bitorsor structure underlying $i(E^\circ)$ is obtained from $i(E)$ by interchanging the left and right actions. When E is a central extension, one therefore has $i(E) = i(E^\circ)$; if, in addition, P is commutative, so that $P = P^\circ$, the uniqueness assertion 1.3.5 a) gives $E = E^\circ$. Thus E is commutative, proving the corollary.

1.3.7. One interesting special case in which all the hypotheses of 1.3.5 and 1.3.6 hold is

$$(1.3.7.1) \quad \underline{\text{Hom}}_{\text{pt}}(P, G) = e \quad (\text{the final sheaf}),$$

where the left-hand side denotes the sheaf of morphisms of pointed sheaves of sets from P to G . Independently of the marked sections, this condition may also be written in the form, analogous to (1.3.1.2),

$$(1.3.7.2) \quad G \simeq \underline{\text{Hom}}_{\text{sets}}(P, G),$$

where the second member denotes the sheaf of morphisms between the underlying sheaves of sets of P and G . These conditions also mean that, for every object Q of $\underline{T}$, the map $u \longmapsto u \circ \text{pr}_2$

$$(1.3.7.3) \quad \text{Hom}_{\text{sets}}(Q, G) \xrightarrow{\sim} \text{Hom}_{\text{sets}}(P \times Q, G)$$

is bijective. This property of P (with G fixed) is stable under Cartesian products; in particular it implies the same property for $P \times P$, $P \times P \times P$, and so forth. It follows immediately that it implies the hypotheses made in 1.3.5 and 1.3.6. It also implies that all the categories occurring in (1.3.4.1) are rigid, that is, every automorphism of an object of any one of them is the identity.

Moreover, since condition (1.3.7.1) on P, G is stable under every base change $S \longrightarrow e$, descent together with the uniqueness result 1.3.5 a) shows that a rigidified bitorsor E over P comes from an extension of P by G as soon as it does so locally over e . By 1.3.5 b), this may also be expressed by the vanishing of the section

$$\underline{\delta}(E) \in \Gamma R^1 h_*(G_{P \times P})$$

defined by $\delta(E)$, where

$$h : P \times P \longrightarrow e$$

is the structural morphism.

1.3.8. Let $p : P \longrightarrow S$ be a morphism of schemes which is coherent (that is, quasi-compact and quasi-separated) and flat, and suppose that

$$(1.3.8.1) \quad p_*(\mathcal{O}_P) \xleftarrow{\sim} \mathcal{O}_S.$$

Since the formation of $p_*(\mathcal{O}_P)$ commutes with every flat base change (by coherence of p), the stated conditions on the relative scheme P/S are stable under Cartesian products and hence hold, in particular, for the Cartesian powers $P \times P$, $P \times P \times P$, and so forth. They are also stable under every flat base change.

Now let G be an affine scheme over S . Relation (1.3.7.2) follows at once from (1.3.8.1), since

$$\text{Hom}_S(P, G) = \text{Hom}_{\text{Alg}}(\underline{A}, p_*(\mathcal{O}_P)) \simeq \text{Hom}_{\text{Alg}}(\underline{A}, \mathcal{O}_S),$$

where $\underline{A}$ is the quasi-coherent algebra over S such that $G = \text{Spec}(\underline{A})$. By the preceding observation, the same conclusion applies to $P \times P$, $P \times P \times P$, and so on. Likewise, if G is flat over S , the same conclusion applies to the relative scheme P_G over G with respect to the relative scheme G_G over G .

When P and G are group schemes over S , and one works with a topology on $(\text{Sch})/S$ coarser than the canonical topology, so that P and G may be identified with sheaves on this site, every morphism from P , $P \times P$, $P \times P \times P$, and so on to G which is compatible with the marked sections is trivial. Similarly, if G is flat, every S -morphism $P \times G \rightarrow G$ compatible with the marked sections is trivial, as one sees by interpreting it as a G -morphism from P_G to G_G . Thus the hypotheses of 1.3.5 hold, as do those of 1.3.6 when G is flat. Moreover, the three categories occurring in (1.3.4.1) are rigid.

1.3.8.2. The most interesting case in which (1.3.8.1) holds, and continues to hold after every extension of the base, is that in which P is an abelian scheme over S ; then the conditions of 1.3.7 themselves hold. In this case the conclusions of 1.3.5 and 1.3.6 are therefore valid for every affine group G over S . This is the case originally considered by J. P. Serre, and the preceding developments were plainly inspired by it. An interesting case in which (1.3.8.1) holds without holding universally is that in which P is the Néron model of an abelian scheme over the fraction field of a discrete valuation ring with spectrum S .

1.4. Extensions of a free commutative group

Let I be an object of $\underline{T}$, and put

$$(1.4.1) \quad P = \mathbb{Z}[I],$$

the free $\mathbb{Z}$ -module generated by I (SGA 4, IV, 11). With every extension E of P by a given $\mathbb{Z}$ -module G (from now on only commutative group extensions will occur), associate the corresponding torsor under G_P , and then restrict it to I along the canonical morphism

$$I \rightarrow P = \mathbb{Z}[I].$$

This gives a canonical functor

$$(1.4.2) \quad \text{EXT}(P, G) \rightarrow \text{TORS}(I, G_I).$$

I claim that this functor is an equivalence of categories. Equivalently, for every G , the maps

$$(1.4.3) \quad \text{Hom}_{\mathbb{Z}}(P, G) \rightarrow H^0(I, G_I) = G(I)$$

and

$$(1.4.4) \quad \text{Ext}_{\mathbb{Z}}^1(P, G) \rightarrow H^1(I, G_I)$$

are isomorphisms. More generally, if A is a ring (not necessarily commutative) of $\underline{T}$ and $P = A[I]$ is the free A -module generated by I , there are canonical isomorphisms, functorial in the variable A -module G ,

$$(1.4.5) \quad \text{Ext}_A^i(A[I], G) \xrightarrow{\sim} H^i(I, G_I).$$

For $A = \mathbb{Z}$ and $i = 0, 1$, these coincide with (1.4.3) and (1.4.4). To define (1.4.5), recall that its left-hand side is the derived functor of

$$G \mapsto \text{Hom}_A(A[I], G),$$

which, by the definition of $A[I]$, is isomorphic to $G \mapsto G(I) = H^0(I, G_I)$. Since $G \mapsto G_I$ carries injective modules to injective modules of $\underline{T}/I$ (SGA 4, V), the assertion follows. We leave to the reader the verification of the stated compatibility with (1.4.3) and (1.4.4).

Finally, the functor (1.4.2) is compatible with inverse images under a morphism of topoi

$$f : \underline{T}' \longrightarrow \underline{T}.$$

2. The notion of a biextension of abelian sheaves

2.0. Throughout this section, P, Q, G denote commutative groups of the topoi $\underline{T}$. We shall frequently use the Cartesian diagram

$$(2.0.1) \quad \begin{array}{ccc} P & \longleftarrow & P \times Q \\ \downarrow & & \downarrow \\ e & \longleftarrow & Q \end{array}$$

and the groups G_P (respectively $G_Q, G_{P \times Q}$) over P (respectively $Q, P \times Q$), obtained by inverse image from the group G over e . We identify $G_{P \times Q}$ either with the inverse image $(G_P)_{P \times Q}$ under pr_1 , or with the inverse image $(G_Q)_{P \times Q}$ under pr_2 . In this notation the group structures of $P, Q, P \times Q$ play no role; indeed, we shall almost never use the product-group structure on $P \times Q$. We shall instead use on $P \times Q$ the structure of a group over P obtained by inverse image from Q over e ,

$$(2.0.2) \quad P \times Q = Q_P,$$

and, similarly, the structure of a group over Q obtained by inverse image from P over e ,

$$(2.0.3) \quad P \times Q = P_Q.$$

Thus over P we have the two groups G_P, Q_P , and over Q the two groups G_Q, P_Q .

If E is a torsor over $P \times Q$ under $G_{P \times Q}$, it may be viewed either as a torsor over Q_P under $G_{P \times Q} = (G_P)_{Q_P}$, or as a torsor over P_Q under $G_{P \times Q} = (G_Q)_{P_Q}$. It therefore makes sense to equip E with a commutative extension structure in either variable. Applied in the induced topoi $\underline{T}/P$ and $\underline{T}/Q$, 1.1.6 gives a convenient explicit description, using the canonical isomorphisms

$$\underline{T}/P \times Q \simeq (\underline{T}/P)/P \times Q, \quad \underline{T}/P \times Q \simeq (\underline{T}/Q)/P \times Q.$$

An extension structure in the P -direction—that is, of P_Q by G_Q —is equivalent to giving, for every object S of $\underline{T}$ and elements $p, p' \in P(S), q \in Q(S)$, an isomorphism of torsors under G_S

$$(2.0.4) \quad \varphi_{p,p';q} : E_{p,q} E_{p',q} \xrightarrow{\sim} E_{pp',q}.$$

As usual (cf. 1.1.2.1), for $(p, q) \in (P \times Q)(S)$, $E_{p,q}$ denotes the torsor under G_S obtained from E by inverse image along $(p, q) : S \longrightarrow P \times Q$. The data (2.0.4) must be functorial in S , associative, and commutative. The associativity condition is the commutativity of

$$(2.0.5) \quad \begin{array}{ccc} & E_{pp',q} E_{p'',q} & \\ \varphi_{p,p';q} \wedge \text{id} \nearrow & & \searrow \varphi_{pp',p'';q} \\ E_{p,q} E_{p',q} E_{p'',q} & & E_{pp'p'',q} \\ \text{id} \wedge \varphi_{p',p'';q} \searrow & & \nearrow \varphi_{p,p'p'';q} \\ & E_{p,q} E_{p'p'',q} & \end{array}$$

for $p, p', p'' \in P(S)$ and $q \in Q(S)$. Commutativity is expressed by

$$(2.0.6) \quad \begin{array}{ccc} E_{p,q} E_{p',q} & \xrightarrow{\varphi_{p,p';q}} & E_{pp',q} \\ \text{sym} \downarrow & & \downarrow \text{id} \\ E_{p',q} E_{p,q} & \xrightarrow{\varphi_{p',p;q}} & E_{p'p,q}. \end{array}$$

Here $p, p' \in P(S)$ and $q \in Q(S)$.

Symmetrically, an extension structure in the Q -direction—that is, of Q_P by G_P —is described by isomorphisms of torsors under G_S

$$(2.0.7) \quad \psi_{p;q,q'} : E_{p,q} E_{p,q'} \xrightarrow{\sim} E_{p,qq'},$$

for $p \in P(S)$ and $q, q' \in Q(S)$. These are functorial in S and satisfy the symmetric associativity and commutativity conditions

$$(2.0.8) \quad \begin{array}{ccc} & E_{p,qq'} E_{p,q''} & \\ \psi_{p;q,q'} \wedge \text{id} \nearrow & & \searrow \psi_{p;qq',q''} \\ E_{p,q} E_{p,q'} E_{p,q''} & & E_{p,qq'q''} \\ \text{id} \wedge \psi_{p;q',q''} \searrow & & \nearrow \psi_{p;q,q'q''} \\ & E_{p,q} E_{p,q'q''} & \end{array}$$

for $p \in P(S)$ and $q, q', q'' \in Q(S)$, and

$$(2.0.9) \quad \begin{array}{ccc} E_{p,q} E_{p,q'} & \xrightarrow{\psi_{p;q,q'}} & E_{p,qq'} \\ \text{sym} \downarrow & & \downarrow \text{id} \\ E_{p,q'} E_{p,q} & \xrightarrow{\psi_{p;q',q}} & E_{p,q'q}. \end{array}$$

Here $p \in P(S)$ and $q, q' \in Q(S)$.

Definition 2.1. Let P, Q, G be abelian groups of $\underline{T}$, and let E be a torsor under $G_{P \times Q}$ equipped with an extension structure in the P -direction, defined by the isomorphisms (2.0.4), and an extension structure in the Q -direction, defined by the isomorphisms (2.0.7), each functorial in S and subject to the associativity and commutativity conditions above. The two structures are called compatible if, for every object S of $\underline{T}$ and $p, p' \in P(S)$, $q, q' \in Q(S)$, the following diagram commutes:

$$(2.1.1) \quad \begin{array}{ccc} & E_{p,qq'} E_{p',qq'} & \\ \psi_{p;q,q'} \wedge \psi_{p';q,q'} \nearrow & & \searrow \varphi_{p,p';qq'} \\ E_{p,q} E_{p',q'} E_{p',q} E_{p',q'} & & E_{pp',qq'} \\ \downarrow & & \nearrow \psi_{pp';q,q'} \\ E_{p,q} E_{p',q} E_{p,q'} E_{p',q'} & & \\ \searrow \varphi_{p,p';q} \wedge \varphi_{p,p';q'} & & \nearrow \psi_{pp',q,q'} \\ & E_{pp',q} E_{pp',q'} & \end{array}$$

where the vertical arrow is induced by the symmetry isomorphism

$$E_{p,q'}E_{p',q} \xrightarrow{\sim} E_{p',q}E_{p,q'}.$$

A biextension of (P, Q) by G is a torsor E under $G_{P \times Q}$ equipped with compatible extension structures in the P - and Q -directions in the preceding sense.

2.1.1. Equivalently, a biextension structure on the $G_{P \times Q}$ -torsor E is given by two partially defined composition laws $\varphi(x, y)$ and $\psi(x, y)$ on E . The first is defined when $\text{pr}_2 j(x) = \text{pr}_2 j(y)$, and the second when $\text{pr}_1 j(x) = \text{pr}_1 j(y)$, where $j : E \rightarrow P \times Q$ is the structural morphism of the torsor. Besides associativity relations of the type

$$(2.1.1.1) \quad \varphi(gx, y) = \varphi(x, gy) = g\varphi(x, y) \quad (x, y \in E(S), g \in G(S))$$

and the analogous relations for ψ , the axioms say that these partial composition laws are associative and commutative and satisfy

$$(2.1.1.2) \quad \varphi(\psi(x, y), \psi(z, t)) = \psi(\varphi(x, z), \varphi(y, t)),$$

where x, y, z, t lie respectively in $E_{p,q}(S), E_{p,q'}(S), E_{p',q}(S), E_{p',q'}(S)$.

2.2. Identity sections. Consider first the identity section of E regarded as an extension of Q_P by G_P ,

$$(2.2.1) \quad e_{E/P} \in \Gamma(E/P) = \text{Hom}_P(P, E),$$

and, symmetrically, the identity section of E regarded as an extension of P_Q by G_Q ,

$$(2.2.2) \quad e_{E/Q} \in \Gamma(E/Q) = \text{Hom}_Q(Q, E).$$

In particular, the torsor E over $P \times Q$ under $G_{P \times Q}$ is canonically trivialized over $P \times 1_Q$ by $e_{E/P}$, and over $1_P \times Q$ by $e_{E/Q}$. These trivializations coincide over the identity section $1_P \times 1_Q$ of $P \times Q$, and hence define the same section

$$(2.2.3) \quad e_E \in \Gamma(E),$$

which, by an abuse of language, will be called the identity section (or neutral section) of E .

To prove the asserted equality, let e, e' be the two sections of E deduced respectively from $e_{E/P}$ and $e_{E/Q}$. They are characterized as the identity sections in the inverse image $E_{1_P, 1_Q}$ of E along the identity section of $P \times Q$, for the two composition laws φ and ψ , respectively. Applying (2.1.1.2) to the four sections e, e', e', e of $E_{1,1}$ gives e on the left and e' on the right; hence $e = e'$. It follows moreover that the two composition laws φ and ψ on $E_{1,1}$ are identical. Indeed, under transport of structure by the morphisms $g \mapsto g.e$ and $g \mapsto g.e'$, respectively, both correspond to the group law of G , and $e = e'$.

The extension of Q_P by G_P over the base P , restricted over the identity section 1_P of P , gives an extension of Q by G on the inverse image of $1_P \times Q$ in E . The homomorphism $e_{E/Q}$ is immediately seen to be multiplicative for this extension structure, so the extension is canonically split. Symmetrically, the extension of P by G obtained by restricting the extension of P_Q by G_Q over Q to the identity section 1_Q is canonically split by $e_{E/P}$.

2.3. Morphisms of biextensions. Let E be a biextension of (P, Q) by G , and let E' likewise be a biextension of (P', Q') by G' , where P', Q', G' are again commutative groups of $\underline{T}$. A morphism of biextensions from E to E' is a system (u, v, w, f) , where

$$(2.3.1) \quad u : P \rightarrow P', \quad v : Q \rightarrow Q', \quad w : G \rightarrow G'$$

are morphisms of groups, and

$$(2.3.2) \quad f : E \rightarrow E'$$

is a morphism of the sheaves of sets underlying E and E' , such that f is simultaneously a homomorphism of group extensions over P and P' , respectively, associated with the relative group homomorphisms

$$u \times v : Q_P \longrightarrow Q'_{P'}, \quad u \times w : G_P \longrightarrow G'_{P'},$$

and a homomorphism of group extensions over Q and Q' , respectively, associated with the relative group homomorphisms

$$u \times v : P_Q \longrightarrow P'_{Q'}, \quad v \times w : G_Q \longrightarrow G'_{Q'}.$$

Notice moreover that u, v, w are determined by f alone. Indeed, passing to the quotient recovers $u \times v$ from f , and then recovers u and v by restriction to $P \times 1_Q$ and $1_P \times Q$, respectively; restricting f to $G.e_E$ recovers w . Thus no ambiguity arises from identifying the morphism (u, v, w, f) with f .

After taking the usual precaution of adjoining their sources and targets to morphisms (the canonical genuflection), one sees that the biextensions of the topos $\underline{T}$ are the objects of a category, whose objects and morphisms are those defined in 2.1 and 2.3 and whose composition of morphisms is defined in the evident way. Moreover, $(P, Q, G, E) \mapsto (P, Q, G)$ gives a canonical functor

$$(2.3.3) \quad \text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}),$$

where $\text{BIEXT}(\underline{T})$ denotes the category of biextensions over $\underline{T}$, and $\text{GRAB}(\underline{T})$ the category of abelian groups over $\underline{T}$. It is useful to decompose this canonical functor into the two functors $E \mapsto (P, Q)$,

$$(2.3.4) \quad \text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}) \times \text{GRAB}(\underline{T}),$$

and $E \mapsto G$,

$$(2.3.5) \quad \text{BIEXT}(\underline{T}) \longrightarrow \text{GRAB}(\underline{T}).$$

2.4. The category $\text{BIEXT}(P, Q; G)$ for variable P, Q, G .

I claim that the canonical functor (2.3.4) is fibred, and that the canonical functor (2.3.5) is cofibred. The first assertion means, in particular, that if E is a biextension of (P, Q) by G , and if

$$u : P' \longrightarrow P, \quad v : Q' \longrightarrow Q$$

are morphisms of abelian groups, then there is a biextension E' of (P', Q') by G and a morphism of biextensions

$$f : E' \longrightarrow E$$

compatible with u, v, id_G , having the usual universal property for morphisms of biextensions into E compatible with u, v, id_G .

The second assertion is expressed similarly. In particular, if E is a biextension of (P, Q) by G , and if

$$w : G \longrightarrow G'$$

is a morphism of abelian groups, then there is a biextension E' of (P, Q) by G' and a morphism of biextensions

$$g : E \longrightarrow E'$$

compatible with $\text{id}_P, \text{id}_Q, w$, having the usual universal property for morphisms of biextensions from E compatible with $\text{id}_P, \text{id}_Q, w$. We leave it to the reader to spell out the two constructions whose existence has just been asserted; they are obtained easily by paraphrasing the familiar analogous constructions for extensions of ordinary groups.

Thus, for three abelian groups P, Q, G of $\underline{T}$, write

$$(2.4.1) \quad \text{BIEXT}(P, Q; G)$$

for the category of biextensions of (P, Q) by G , defined as the fibre category of (2.3.3) over (P, Q, G) (so its morphisms are the morphisms of biextensions compatible with the identity morphisms of P, Q, G). Then this category depends covariantly on G and contravariantly on P, Q . Moreover, with these variances, the category $\text{BIEXT}(P, Q; G)$ depends additively on each of the three arguments P, Q, G . For example, for the argument G , this means precisely:

- a) $\text{BIEXT}(P, Q; 0)$ is equivalent to the one-point category;
- b) for two abelian groups G, G' of $\underline{T}$, the functor

$$(2.4.2) \quad \text{BIEXT}(P, Q; G \times G') \longrightarrow \text{BIEXT}(P, Q; G) \times \text{BIEXT}(P, Q; G')$$

deduced from the two projections $G \times G' \longrightarrow G$ and $G \times G' \longrightarrow G'$ is an equivalence of categories.

In the language of [4], one also says that, for fixed P, Q , the category of biextensions of (P, Q) by variable groups G is, via the functor $E \mapsto G$ induced by (2.3.5), an additive cofibred category over $\text{GRAB}(\underline{T})$.

Remark 2.4.3. In the terminology of *loc. cit.*, the preceding cofibred category is not merely additive but even left exact [4, 2.2]. We shall not verify this property here, since it will reappear below in 3.6 as a consequence of more precise facts. We only point out that the general theory of *loc. cit.* would then describe the structure of this cofibred category, up to equivalence of fibred categories, in terms of a chain complex $L_\bullet$ in $\text{GRAB}(\underline{T})$ depending only on P and Q . We shall also give below a direct construction of this complex as

$$P \overset{\mathbb{L}}{\otimes} Q,$$

the tensor product taken in a derived category.

2.5. Structure of the category $\text{BIEXT}(P, Q; G)$.

The general formal considerations of [4, no. 1] give several useful consequences of the additive dependence of the categories $\text{BIEXT}(P, Q; G)$ on G .

2.5.1. First, for fixed G , the category $\text{BIEXT}(P, Q; G)$ has a composition law which assigns functorially to two biextensions E, E' of (P, Q) by G a product biextension, denoted $E.E'$. This construction is simply the analogue of the familiar product of two extensions of one commutative group by another. It is compatible with that construction whether E, E' are viewed as group extensions of Q_P by G_P , or as group extensions of P_Q by G_Q . In particular, as a torsor under $G_{P \times Q}$, EE' is the product of the torsors under $G_{P \times Q}$ defined by E and E' .

The formation of $E.E'$ is moreover associative and commutative up to canonical isomorphisms, and this composition law on $\text{BIEXT}(P, Q; G)$ has a two-sided neutral object, called the trivial biextension of (P, Q) by G . It may be constructed, as in *loc. cit.*, from the unique biextension E_0 of (P, Q) by the zero group 0 , up to isomorphism, by means of the homomorphism $0 \longrightarrow G$. From this description one immediately obtains an alternative description of the trivial biextension as the trivial torsor

$$E = P \times Q \times G.$$

Its two group structures over P and over Q are obtained, respectively, from the product group structures on $Q \times G$ and $P \times G$ by the base changes $P \longrightarrow e$ and $Q \longrightarrow e$.

Finally, for every biextension E of (P, Q) by G , one can define the inverse biextension, denoted E^{-1} , together with an isomorphism from EE^{-1} to the trivial biextension. This determines E^{-1} up to unique isomorphism and makes its dependence on E functorial.

In short, the category $\text{BIEXT}(P, Q; G)$ behaves like a group object in the category **Cat** of categories, except that associativity, commutativity, unit, and inverse relations hold only up to specified isomorphisms. One may call it a pseudo-group in the 2-category **Cat**.

2.5.2. For fixed P, Q, G , the preceding properties imply the following facts. For every biextension E of (P, Q) by G , the monoid of endomorphisms of E is canonically isomorphic to the

monoid of endomorphisms of the trivial biextension $\underline{1}$: an endomorphism u of $\underline{1}$ is sent to $u.\text{id}_E$ on $\underline{1}.E \simeq E$. The endomorphism monoid of $\underline{1}$ is in fact a commutative group, denoted

$$(2.5.2.1) \quad \text{Biext}^0(P, Q; G).$$

Its composition law is also given by $(u, v) \mapsto u.v \in \text{End}(\underline{1}.E) \simeq \text{End}(\underline{1})$.

Likewise, the composition law $(E, E') \mapsto EE'$ equips the set

$$(2.5.2.2) \quad \text{Biext}^1(P, Q; G)$$

of isomorphism classes of biextensions of (P, Q) by G with the structure of a commutative group; the inverse of the class of E is the class of E^{-1} .

2.5.3. Given a group homomorphism $G \longrightarrow G'$, the corresponding functor

$$(2.5.3.1) \quad \text{BIEXT}(P, Q; G) \longrightarrow \text{BIEXT}(P, Q; G')$$

is compatible, up to canonical isomorphisms, with the operations $E.E'$ and E^{-1} of 2.5.1. Consequently, the two evident maps

$$(2.5.3.2) \quad \text{Biext}^i(P, Q; G) \longrightarrow \text{Biext}^i(P, Q; G') \quad (i = 0, 1)$$

are group homomorphisms. Moreover, if E is a biextension of (P, Q) by G and E' the corresponding biextension by G' , the natural homomorphism

$$(2.5.3.3) \quad \text{Hom}(E, E) = \text{Aut}(E) \longrightarrow \text{Hom}(E', E') = \text{Aut}(E')$$

is precisely (2.5.3.2) for $i = 0$.

2.5.4. The group $\text{Biext}^0(P, Q; G)$ admits a simple explicit description. If E is a biextension of (P, Q) by G , an automorphism u of E is first of all an automorphism of the underlying torsor under $G_{P \times Q}$, and hence is given by a morphism

$$u_0 : P \times Q \longrightarrow G$$

through the formula

$$u(x) = u_0(j(x))x.$$

Compatibility of u with the first, respectively the second, extension structure on E is equivalent to additivity of u_0 in the second, respectively the first, argument. Thus u is an automorphism of the biextension structure of E if and only if u_0 is bilinear. This gives a canonical bijection

$$(2.5.4.1) \quad \text{Biext}^0(P, Q; G) \simeq \text{Hom}(P \otimes Q, G).$$

It is readily checked that this bijection is independent of E , respects the group structures, and is functorial in G for the functoriality on the left described in (2.5.3.2).

Below we shall determine $\text{Biext}^1(P, Q; G)$ and find that, canonically and functorially in G , it is isomorphic to

$$\text{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G).$$

2.6. In 2.5 only G was varied, while P and Q were held fixed. We now say a few words about variation in P, Q . The essential fact, from which the others follow formally, is that the contravariant dependence of $\text{BIEXT}(P, Q; G)$ on P, Q commutes with its covariant dependence on G . Thus the categories in question may be viewed as the fibre categories of a cofibred category

$$(2.6.1) \quad \text{BIEXT}'(\underline{T}) \longrightarrow \text{GRAB}(\underline{T})^{\text{op}} \times \text{GRAB}(\underline{T})^{\text{op}} \times \text{GRAB}(\underline{T}),$$

where the superscript op denotes the opposite category and the structural functor is $E \mapsto (P, Q, G)$. This cofibred category is additive in each of its three arguments, or, as we shall say, tri-additive.

By paraphrasing standard arguments, which we shall not reproduce, it follows that the composition $E.E'$ in the fibre category $\text{BIEXT}(P, Q; G)$ may be described using the additive structure of any one of the three arguments: the three resulting pseudo-group structures are canonically isomorphic. Consequently, the abelian groups

$$(2.6.2) \quad \text{Biext}^i(P, Q; G), \quad i = 0, 1,$$

may be regarded as two trilinear functors in P, Q, G , respectively in $\text{GRAB}(\underline{T})^{\text{op}}, \text{GRAB}(\underline{T})^{\text{op}}$, and $\text{GRAB}(\underline{T})$. Equivalently, they are trilinear functors of variable P, Q, G in $\text{GRAB}(\underline{T})$, contravariant in P, Q and covariant in G , with values in the category of abelian groups. For this variance, the isomorphism (2.5.4.1) is functorial in all of P, Q, G , and not merely in G . The analogous assertion will hold below for the description of Biext^1 by an Ext^1 .

2.7. The symmetric biextension of a given biextension.

If E is a biextension of (P, Q) by G , define its symmetric biextension sE , which is a biextension of (Q, P) by G . The underlying sheaf of sets of sE is that of E , and the actions of G on E and sE are the same. The structural morphism

$${}^sE \longrightarrow Q \times P$$

is the composite of $E \longrightarrow P \times Q$ with the symmetry $P \times Q \longrightarrow Q \times P$; finally, the left and right extension structures of sE are, respectively, the right and left extension structures of E . Evidently $E \mapsto {}^sE$ is an automorphism of the category $\text{BIEXT}(\underline{T})$ of biextensions, and indeed an involutive automorphism:

$${}^s({}^sE) = E.$$

Every statement about bitorsors $E, \dots$ therefore gives a symmetric statement upon being applied to ${}^sE, \dots$. We may call it the statement obtained by *exchanging left and right*. In general we shall spell out only one of the two symmetric statements, choosing between left and right, for example, according to our political preferences.

2.8. Change of topos. Localization.

Let

$$f : \underline{T}' \longrightarrow \underline{T}$$

be a morphism of topoi. The inverse-image functor f^* transforms a biextension of (P, Q) by G into a biextension $f^*(E)$ of (f^*P, f^*Q) by f^*G , as always for a structure expressible in terms of finite projective limits and arbitrary inductive limits. To prove compatibility of the two extension structures on $f^*(E)$ induced by those on E , one expresses compatibility by commutativity of (2.1.1) in the universal case

$$S = P \times P \times Q \times Q,$$

where p, p', q, q' are the four projections. As expected, $E \mapsto f^*(E)$ gives a canonical functor

$$f^* : \text{BIEXT}(\underline{T}) \longrightarrow \text{BIEXT}(\underline{T}').$$

In what follows we shall use without further mention the fact that every construction made with biextensions is compatible with inverse images under morphisms of topoi. This applies in particular to the product EE' and inverse E^{-1} considered in 2.5.

One may take for $\underline{T}'$ an induced topos $\underline{T}_{/S'}$, and for f the canonical morphism, called the localization morphism (SGA 4 IV 5),

$$f = i_{S'} : \underline{T}_{/S'} \longrightarrow \underline{T}.$$

Then $f^*(E)$ is called the biextension induced by E on S' , and is also denoted, as usual, by $E|_{S'}$ or $E_{S'}$.

Since restriction has the usual transitivity properties, the categories $\text{BIEXT}(\underline{T}_{/S})$, for variable $S \in \underline{T}$, form the fibres of a fibred category over $\underline{T}$ (SGA 1 VI 8). In fact, this category is even

a stack over $\underline{T}$ in the sense of Giraud [2]: morphisms and objects in its fibres can be glued. For example, to give a biextension over S up to canonical isomorphism, it suffices to give a covering family (that is, an epimorphic family) $S_i \rightarrow S$, a biextension E_i over each S_i (an object of $\underline{T}_{/S_i}$), and, for every pair (i, j) , an isomorphism between $E_i|_{S_j}$ and $E_j|_{S_i}$, subject to the usual cocycle condition.

We call this the stack of biextensions over $\underline{T}$. Replacing $\underline{T}$ in (2.3.3), and hence also in (2.3.4) and (2.3.5), by $\underline{T}_{/S}$ for variable S gives morphisms of stacks over $\underline{T}$. The same applies to the operations $E.E'$ and E^{-1} of (2.5.1) in the categories $\text{BIEXT}(P_S, Q_S; G_S)$ for variable S , where P, Q, G are fixed commutative groups over $\underline{T}$.

2.9. Relations with birigidified torsors.

In the terminology of 1.3.2, we saw in 2.2 that, for every biextension E of (P, Q) by G , the underlying torsor structure under $G_{P \times Q}$ is canonically birigidified relative to the pair of identity sections of P and Q . This gives a canonical functor

$$(2.9.1) \quad \text{BIEXT}(P, Q; G) \xrightarrow{i} \text{TORSBIRIG}(P, Q; G),$$

where the category on the right consists of birigidified torsors over $P \times Q$ under $G_{P \times Q}$.

Using the identity sections of P and Q , for every product of n objects each isomorphic to P or Q one obtains a canonical morphism inj_i from the i -th factor into that product. Thus, for a morphism from such a product to G , one may speak of its restriction to the i -th factor. Consider the products

$$(2.9.2) \quad \begin{cases} P \times Q, \\ P \times Q \times Q, & P \times Q \times Q \times Q, \\ P \times P \times Q, & P \times P \times P \times Q, \\ P \times P \times Q \times Q. \end{cases}$$

Proposition 2.9.3. Suppose that every morphism from one of the products (2.9.2) to G which is trivial on each factor of that product (embedded by the identity sections as just explained) is trivial. Then:

a) The canonical functor (2.9.1) is fully faithful, and both sides of (2.9.1) are rigid categories. In particular, if E is a torsor under $G_{P \times Q}$ birigidified relative to (e_P, e_Q) , there is at most one biextension structure on E compatible with its birigidified torsor structure.

b) Such a biextension structure exists on E if and only if E admits an extension structure of Q_P by G_P compatible with its torsor structure under G_P , and an extension structure of P_Q by G_Q compatible with its torsor structure under G_Q (cf. 1.3.5(b)). In that case there is a unique extension structure of Q_P by G_P compatible with the torsor structure under G_P and the given rigidification over $P \times e_Q$, and a unique extension structure of P_Q by G_Q compatible with the torsor structure under G_Q and the given rigidification over $e_P \times Q$; these two extension structures are compatible in the sense of 2.1.

Proof. a) If E is a birigidified torsor over $P \times Q$, its automorphism group identifies with the group of morphisms $P \times Q \rightarrow G$ trivial on the two partial factors. By the hypothesis applied to $P \times Q$, this group contains only the identity automorphism, proving rigidity. Now let E, E' be biextensions of (P, Q) by G . To see that every morphism $f : E \rightarrow E'$ of birigidified torsors is a morphism of biextensions, apply 1.3.5(a) to the two extension structures defined by E, E' over P and over Q , respectively. This uses the hypothesis for $P \times Q \times Q$ and $P \times P \times Q$ in (2.9.2).

b) Necessity is clear. For sufficiency, 1.3.5(b) first shows that the stated condition gives unique extension structures of Q_P by G_P and of P_Q by G_Q , compatible with the torsor structure and the rigidifications. Here the hypothesis is used for $P \times Q \times Q \times Q$ and $P \times P \times P \times Q$. It remains to prove that the two extension structures are compatible, that is, that diagram (2.1.1) commutes. Its two composites are isomorphisms of torsors over $P \times P \times Q \times Q$, and hence differ by a morphism

$$u : P \times P \times Q \times Q \rightarrow G.$$

Because the partial trivializations of E are identity sections for the laws φ and ψ , respectively, u is trivial on each of the four factors. The hypothesis for $P \times P \times Q \times Q$ therefore makes u trivial. This proves the proposition.

Corollary 2.9.4. Let P, Q, G be commutative groups satisfying

$$(2.9.4.1) \quad \underline{\mathrm{Hom}}_{\mathrm{pt}}(P, G) = e, \quad \underline{\mathrm{Hom}}_{\mathrm{pt}}(Q, G) = e$$

(cf. 1.3.7). Then the hypothesis of Proposition 2.9.3 is satisfied, and conclusions (a) and (b) of that proposition hold. Moreover, if E is a torsor under $G_{P \times Q}$ birigidified relative to (e_P, e_Q) , it admits a biextension structure of (P, Q) by G compatible with its given structures if and only if the two morphisms of pointed sheaves which it defines,

$$(2.9.4.2) \quad Q \longrightarrow R^1 p_*(G_P), \quad P \longrightarrow R^1 q_*(G_Q),$$

where $p : P \longrightarrow e$ and $q : Q \longrightarrow e$ are the structural morphisms, are morphisms of groups.

The first assertion was proved in 1.3.7. For the second, necessity is clear. To prove sufficiency, apply Proposition 2.9.3(b) and 1.3.5(b). Multiplicativity of the first morphism in (2.9.4.2) says that the rigidified torsor E under G_P satisfies the condition of 1.3.5(b) locally over P . As observed in 1.3.7, that condition is then true globally over P , so E admits an extension structure of Q_P by G_P compatible with its rigidified torsor structure. Symmetrically, E admits an extension structure of P_Q by G_Q compatible with its rigidified torsor structure under G_Q . The conclusion follows from Proposition 2.9.3(b).

Example 2.9.5. Let S be a scheme, let P and Q be abelian schemes over S , and let $G = \underline{G}_{m,S}$ be the multiplicative group scheme over S . We noted in 1.3.8 that (2.9.4.1) then holds. A birigidified torsor E over $P \times Q$ is equivalently a birigidified invertible sheaf over $P \times Q$. Recall that a divisorial correspondence on P, Q is the isomorphism class of such a birigidified invertible sheaf (see, for example, [1]). In the fppf topology, the morphisms (2.9.4.2) are

$$(2.9.5.1) \quad Q \longrightarrow \underline{\mathrm{Pic}}_{P/S}, \quad P \longrightarrow \underline{\mathrm{Pic}}_{Q/S},$$

defined by the divisorial correspondence. The theorem of the square (*loc. cit.*) says that these morphisms necessarily respect the group structures. Thus 2.9.4 shows that the category of biextensions of (P, Q) by $G = \underline{G}_{m,S}$ is equivalent to the category of birigidified invertible sheaves on $P \times Q$; it is therefore a rigid category, and its set of isomorphism classes identifies with the set of divisorial correspondences on (P, Q) . This identification plainly respects the natural group laws on the two sets, the first being that of (2.5.2.2).

Remark 2.9.6. Additivity of the first morphism in (2.9.4.2) is equivalent to factorization of the second through the subsheaf $\underline{\mathrm{Ext}}^1(Q, G)$; note that, by 1.3.5, the canonical morphism

$$\underline{\mathrm{Ext}}^1(Q, G) \longrightarrow R^1 q_*(G_Q)$$

is injective. Likewise, additivity of the second morphism in (2.9.4.2) means that the first factors through $\underline{\mathrm{Ext}}^1(P, G)$. This gives group isomorphisms

$$(2.9.6.1) \quad \mathrm{Biext}^1(P, Q; G) \simeq \mathrm{Hom}(Q, \underline{\mathrm{Ext}}^1(P, G)) \simeq \mathrm{Hom}(P, \underline{\mathrm{Ext}}^1(Q, G)).$$

In the situation of 2.9.5 these reduce to the familiar isomorphisms

$$(2.9.6.2) \quad \mathrm{Corr}(P, Q) \simeq \mathrm{Hom}(P, Q') \simeq \mathrm{Hom}(Q, P'),$$

where

$$P' = \underline{\mathrm{Ext}}^1(P, \underline{G}_m), \quad Q' = \underline{\mathrm{Ext}}^1(Q, \underline{G}_m)$$

are the dual abelian schemes of P and Q , identified respectively with $\underline{\mathrm{Pic}}_{P/S}^0$ and $\underline{\mathrm{Pic}}_{Q/S}^0$. In both formulas, Hom denotes the group of group homomorphisms.

2.10. Various generalizations of the notion of biextension. We mention them for completeness, although we shall not use them.

2.10.1. One can define a biextension of (P, Q) by G without assuming P or Q commutative, only G being commutative. It is a torsor under $G_{P \times Q}$ equipped with central extension structures of Q_P by G_P and of P_Q by G_Q , each compatible with the torsor structure and compatible with one another, meaning that they make (2.1.1) commute. Most of the preceding developments remain valid, *mutatis mutandis*, for this more general notion. Even when P, Q are commutative, it is more general than the notion of 2.1, which is the case in which the two underlying relative group structures on E are commutative. In the noncommutative case, the category of biextensions of (P, Q) by G still depends additively on G , but not on P and Q .

2.10.2. Paraphrasing 2.1, for a fixed integer $n \geq 1$ one defines an n -extension of a sequence of n commutative groups $(P_1, P_2, \dots, P_n)$ by a commutative group G . It consists of a torsor $\bar{E}$ under $G_{P_1 \times \dots \times P_n}$, equipped with n partial extension structures over the partial products of $n - 1$ factors, with a compatibility relation of type (2.1.1) for every pair of distinct indices between 1 and n .

If $n \geq 3$, the P_i are abelian schemes over a scheme S , and $G = \underline{G}_{m,S}$, every n -extension of $(P_1, \dots, P_n)$ by G is trivial.

2.10.3. The notion of biextension in 2.1 may be viewed as a variant of the notion of a commutative extension of commutative groups, that is, of an extension of $\mathbb{Z}$ -modules. Let A be a ring of $\underline{T}$, assumed commutative for simplicity, and let P, Q, G be three A -modules. More generally one can define an A -module biextension of (P, Q) by G , a variant of an extension of A -modules.

Such a biextension is a biextension in the sense of 2.1 whose underlying partial extension structures are enriched to extension structures of A_P -modules and A_Q -modules, respectively, compatible with the relative module structures already present on Q_P, G_P and P_Q, G_Q . The two relative module structures on E may be interpreted as two actions of the underlying multiplicative monoid of A on the sheaf of sets underlying E , and these two actions are required to commute. This condition is added to the compatibility expressed by (2.1.1), and is itself expressed by an analogous diagram, which we leave to the reader. Finally, the first action of A on E is required to be distributive with respect not only to the first composition law on E but also to the second; the same is required of the second action.

The dictionary of 1.1.6 also has an A -linear variant. An extension of the A -module P by the A -module G amounts to a family of torsors E_p under G_p , parametrized by points $p \in P(S)$ for variable $S \in \text{ob } \underline{T}$, with E_p varying A -linearly, rather than merely additively, in p . Besides the isomorphisms $\varphi_{p,p'}$ of (1.1.3.1), this is expressed by isomorphisms

$$\mu_{a,p} : {}^a E_p \xrightarrow{\sim} E_{ap} \quad (p \in P(S), a \in A(S)),$$

where ${}^a E_p$ denotes the image of the G_S -torsor E_p under extension of the structure group along

$$g \mapsto ag : G_S \longrightarrow G_S.$$

We leave to the reader the conditions on φ and μ which make these data correspond to an extension of A -modules E of P by G (cf. 3.5.4(b)).

3. Supplements on homological algebra

The principal purpose of this section is to define the isomorphism

$$\text{Biext}^1(P, Q; G) \simeq \text{Ext}^1(P \otimes^{\mathbb{L}} Q, G)$$

announced above. For this we shall need certain developments in homological algebra which are of independent interest and will occupy sections 3.1 through 3.4. The developments in 3.1 and 3.3 may be regarded as supplements to [4], especially to section 6.9 therein.

3.1. Construction of an additive cofibred category $\underline{\Psi}_{L_\bullet}$ in terms of a chain complex $L_\bullet$ in an abelian category $\underline{C}$

Let $\underline{C}$ be an abelian category and let $L_\bullet$ be a chain complex in $\underline{C}$,

$$(3.1.1) \quad L_2 \longrightarrow L_1 \longrightarrow L_0 \longrightarrow 0,$$

written with lower-degree conventions, so that the differential has degree -1 . Thus $L_i = 0$ for $i < 0$. We shall define a cofibred category

$$\underline{\Psi} = \underline{\Psi}_{L_\bullet}$$

over $\underline{C}$; up to canonical equivalence of cofibred categories it is the category of the same name in [4, 6.9]. The objects of $\underline{\Psi}_{L_\bullet}$ are pairs

$$(X, \alpha),$$

where X is an extension of L_0 by an object A of $\underline{C}$, depending on X ,

$$(3.1.2) \quad 0 \longrightarrow A \xrightarrow{i} X \xrightarrow{j} L_0 \longrightarrow 0,$$

and α is a trivialization of the inverse image $d_0^*(X)$ of this extension by $d_0 : L_1 \longrightarrow L_0$. It is required that the corresponding trivialization of the inverse image $d_1^*d_0^*(X)$ by $d_1 : L_2 \longrightarrow L_1$ be the evident one, obtained from the transitivity isomorphism

$$d_1^*d_0^*(X) \simeq (d_0d_1)^*(X)$$

and the relation $d_0d_1 = 0$.

Equivalently, α is a lifting of $d_0 : L_1 \longrightarrow L_0$ to a morphism $\alpha : L_1 \longrightarrow X$, subject to $\alpha d_1 = 0$:

$$(3.1.3) \quad \begin{array}{ccccc} & & & 0 & \\ & & & \downarrow & \\ & & & A & \\ & & & \downarrow i & \\ & & & X & \\ & \nearrow \alpha & & \downarrow j & \\ L_2 & \xrightarrow{d_1} & L_1 & \xrightarrow{d_0} & L_0 \\ & & & \downarrow & \\ & & & 0 & \end{array}, \quad \alpha d_1 = 0. \quad (*)$$

Here is a simpler description, pointed out by Deligne: $\underline{\Psi}_{L_\bullet}(A)$ is the category of extensions of the complex $L_\bullet$ by A , regarded as a complex concentrated in degree zero.

In this form one sees that the description of the objects (X, α) depends only on the chain complex $[L_\bullet]$ truncated in degree 1, obtained from $L_\bullet$ by “killing $H_i(L_\bullet)$ for $i \geq 2$ ”: its terms are zero in degrees at least 2, agree with those of $L_\bullet$ in degrees at most 0, and its degree-1 term is the cokernel of $d_1 : L_2 \longrightarrow L_1$.

A homomorphism from (X, α) to (X', α') is a homomorphism of extensions $u : X \longrightarrow X'$, compatible with the identity on L_0 , such that $u\alpha = \alpha'$. Homomorphisms are composed by composing the corresponding homomorphisms of extensions. This gives the announced category $\underline{\Psi}_{L_\bullet}$. Up to canonical isomorphism, it depends only on the degree-1 truncation $[L_\bullet]$ of $L_\bullet$.

There is an evident functor $(X, \alpha) \mapsto A$,

$$(3.1.4) \quad \pi : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{C}.$$

This functor is cofibred. Indeed, let $u : A \longrightarrow B$ be a morphism of $\underline{C}$, and let (X, α) be an object of the fibre $\underline{\Psi}_{L_\bullet}(A)$. Thus X is an extension of L_0 by A . Form the extension $u_*(X)$ of L_0 by B , the direct image of X under u , and equip it with the partial trivialization $\beta = f\alpha$, where $f : X \longrightarrow u_*(X)$ is the canonical morphism. It follows immediately that

$$f : (X, \alpha) \longrightarrow (u_*(X), \beta)$$

is cocartesian for (3.1.4): it has the usual universal property for u -morphisms in $\underline{\Psi}_{L_\bullet}$ with source (X, α) . This is just the familiar and elementary corresponding universal property of $f : X \longrightarrow u_*(X)$.

Finally, the cofibred category $\underline{\Psi}_{L_\bullet}$ is additive; that is, it satisfies

$$(3.1.5) \quad \underline{\Psi}_{L_\bullet}(0) \text{ is equivalent to the one-point category,}$$

and, for arbitrary A, B in $\underline{C}$,

$$(3.1.6) \quad \underline{\Psi}_{L_\bullet}(A \times B) \longrightarrow \underline{\Psi}_{L_\bullet}(A) \times \underline{\Psi}_{L_\bullet}(B) \text{ is an equivalence.}$$

This follows immediately from the corresponding properties of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$. We shall freely use the consequences of the additivity of $\underline{\Psi}_{L_\bullet}$ established in [4, 1] and recalled above in 2.5.

3.1.7. Among these consequences, for every object A of $\underline{C}$ the category $\underline{\Psi}_{L_\bullet}(A)$ has a zero, or trivial, object (2.5.1). It is the trivial extension $X = L_0 + A$ of L_0 by A , equipped with the partial trivialization $\alpha : L_1 \longrightarrow X$ whose components are d_0 and 0.

Moreover, by 2.5.2, for every (X, α) in $\underline{\Psi}_{L_\bullet}(A)$ the automorphism group of this object is canonically isomorphic to the automorphism group of the trivial object. Denote the latter commutative group by $\Psi_{L_\bullet}^0(A)$. The explicit description above gives the canonical isomorphisms

$$(3.1.8) \quad \begin{aligned} \Psi_{L_\bullet}^0(A) &= \text{Ker}(\text{Hom}(L_0, A) \longrightarrow \text{Hom}(L_1, A)) \simeq \text{Hom}(H_0(L_\bullet), A) \\ &\simeq \text{Ext}^0(L_\bullet, A), \end{aligned}$$

induced by the usual identification of the automorphism group of the extension X with $\text{Hom}(L_0, A)$. The isomorphism (3.1.8) is plainly functorial in A .

3.1.9. Finally, recall from 2.5.2 that the natural multiplicative structure on the fibre category $\underline{\Psi}_{L_\bullet}(A)$ endows its set of isomorphism classes with a natural group structure. We denote this group by $\Psi_{L_\bullet}^1(A)$; it is an additive functor of A .

3.2. Determination of $\Psi_{L_\bullet}^1(A)$

Consider an object (X, α) (3.1.3) of $\underline{\Psi}_{L_\bullet}(A)$. Consider the cochain complex of length one

$$(3.2.1) \quad [X \longrightarrow L_0] : \quad 0 \longrightarrow X \longrightarrow L_0 \longrightarrow 0 \longrightarrow \cdots$$

defined by the morphism $j : X \longrightarrow L_0$, with X as its degree-zero component. It is a resolution of A , and hence may be identified with A as an object of the derived category $D(\underline{C})$ [5]. The diagram (3.1.3) then gives a degree-one morphism of complexes

$$(3.2.2) \quad L_\bullet \longrightarrow [X \longrightarrow L_0],$$

made explicit by the following diagram with anticommutative squares:

$$(3.2.3) \quad \begin{array}{ccccccc} \cdots & & L_2 & \xrightarrow{d_1} & L_1 & \xrightarrow{d_0} & L_0 \longrightarrow 0 & \cdots \\ & & & & \downarrow u^{-1}=-\alpha & & \downarrow u^0=\text{id} & \\ & & 0 & \longrightarrow & X & \xrightarrow{j} & L_0 \longrightarrow 0 & \cdots \end{array}$$

(We use the usual convention which associates with the chain complex $L_\bullet$ a complex whose differential has degree 1 and whose degree- i component is L_{-i} .) We thus obtain a degree-one morphism in the derived category

$$(3.2.3) \quad c(X, \alpha) : L_\bullet \longrightarrow A \quad (\text{a degree-one morphism in } D(\underline{C})),$$

which plainly depends only on the isomorphism class of (X, α) . We have thereby defined a canonical map

$$(3.2.4) \quad \Psi_{L_\bullet}^1(A) \longrightarrow \text{Ext}^1(L_\bullet, A) \stackrel{\text{def}}{=} \text{Hom}_{D(\underline{C})}(L_\bullet, A[1]).$$

Theorem 3.2.5. The map (3.2.4) is an isomorphism of groups, functorial in the object A of $\underline{C}$.

We leave to the reader the verification that (3.2.4) is additive and functorial in A , and shall confine ourselves to proving that it is bijective. For injectivity, let us first show that if the morphism of complexes u in (3.2.3) is zero in $D(\underline{C})$, then it is already zero in the homotopy category $K(\underline{C})$; that is, there exists a morphism

$$s : L_0 \longrightarrow X$$

such that

$$(3.2.5.1) \quad js = u^0 = \text{id}_{L_0}, \quad sd_0 = -u^{-1} = \alpha.$$

Indeed, the hypothesis says that the conclusion becomes true after the morphism of complexes under consideration is composed with a suitable quasi-isomorphism v

$$(*) \quad \begin{array}{ccccccc} 0 & \longrightarrow & X & \xrightarrow{j} & L_0 & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & C^0 & \xrightarrow{j'} & C^1 & \longrightarrow & C^2 \longrightarrow \cdots \end{array}$$

(and, after replacing C^i by 0 for $i \geq 2$ and C^1 by $\text{Im}(C^0 \longrightarrow C^1) = \text{Ker}(C^1 \longrightarrow C^2)$ if necessary, we may suppose that $C^i = 0$ for $i \geq 2$). But the morphism $(*)$ induces the identity on A , and therefore identifies X with the fibre product of L_0 and C^0 over C^1 . A morphism $s' : L_0 \longrightarrow C^0$ defining a homotopy $vu \sim 0$, and hence satisfying $j's' = v^1u^0$, consequently factors through a morphism $s : L_0 \longrightarrow X$ satisfying $js = u^0 = \text{id}_{L_0}$. Thus s splits the extension X of L_0 by A , so the complex $X \longrightarrow L_0$ is isomorphic in $K(\underline{C})$ to the complex $K^\bullet$ concentrated in A . For the latter complex it is clear that

$$\text{Hom}_{K(\underline{C})}(L_\bullet, K^\bullet) \longrightarrow \text{Hom}_{D(\underline{C})}(L_\bullet, K^\bullet)$$

is an isomorphism, proving the assertion.

Now the choice of a morphism $s : L_0 \longrightarrow X$ satisfying (3.2.5.1) is precisely equivalent to a trivialization of the extension X of L_0 by A compatible with the partial trivialization α . This proves the injectivity of (3.2.4).

For surjectivity, start with an element of the right-hand side. By definition it may be described by a degree-one morphism of complexes

$$(3.2.5.2) \quad \begin{array}{ccccccc} \dots & & L_2 & \xrightarrow{d_1} & L_1 & \xrightarrow{d_0} & L_0 \longrightarrow 0 \\ & & & & \downarrow u^{-1} & & \downarrow u^0 \\ & & 0 & \longrightarrow & C^0 & \longrightarrow & C^1 \longrightarrow \dots \end{array}$$

with anticommutative squares, where $C^\bullet$ is a resolution of A and, as above, we may suppose that $C^i = 0$ for $i \geq 2$. Thus C^0 is an extension of C^1 by A , and we may form the extension X obtained by pulling it back along $L_0 \rightarrow C^1$. Anticommutativity in (3.2.5.2) shows that $-u^{-1}$ factors through a morphism $\alpha : L_1 \rightarrow X$ which is a partial splitting, that is, $j\alpha = d_0$ (with j as in (3.1.2)). Moreover, the relation $u^{-1}d_1 = 0$ in (3.2.5.2) implies $\alpha d_1 = 0$. Hence (X, α) is indeed an object of $\underline{\Psi}_{L_\bullet}(A)$. By construction, the morphism (3.2.5.2) is the composite of the morphism (3.2.3) associated with (X, α) and the morphism

$$[X \rightarrow L_0] \rightarrow C^\bullet = [C^0 \rightarrow C^1]$$

whose components are (f, u^0) , where $f : X \rightarrow C^0$ is the canonical morphism obtained from the definition of X as the pullback of the extension C^0 .

The preceding morphism is a morphism of resolutions of A and induces the identity on A . It follows, by the definition $\text{Ext}^1 = \text{Hom}_{D(\underline{C})}^1$, that the given element of $\text{Ext}^1(L_\bullet, A)$ described by (3.2.5.2) is precisely $c(X, \alpha)$. This proves the surjectivity of (3.2.4) and completes the proof of 3.2.5.

3.3. Properties of the cofibred category $\underline{\Psi}_{L_\bullet}$.

In this section, which is not essential for understanding what follows, we freely use the notions of [4], our purpose being to compare the preceding results with those of that work.

Theorem 3.3.1. The additive cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{C}$ is left exact ([4, 2.2]), admits a quasi-maximal object ([4, 4.4]), and, for every object $E = (X, \alpha)$ of this category, there exists a typical complex $L_\bullet^E$ for E ([4, 3.1]). Moreover, the typical pro-complex of $\underline{\Psi}_{L_\bullet}$ ([4, 3.7 and 4.4]) is canonically isomorphic in $D(\underline{C})$ to the truncated complex $[L_\bullet]$ considered in 3.1.

We now spell out the meaning of the several assertions in 3.3.1.

3.3.2. To say that the cofibred category $\underline{\Psi}_{L_\bullet}$ is left exact means that it is additive and also satisfies the following condition:

3.3.2.1. For every exact sequence

$$0 \rightarrow A' \xrightarrow{u} A \xrightarrow{v} A'' \rightarrow 0$$

in $\underline{C}$, the natural functor (in the notation of [4, 1.8])

$$\underline{\Psi}_{L_\bullet}(A) \rightarrow \underline{\Psi}_{L_\bullet}([A \rightarrow A''])$$

is an equivalence of categories.

Recall that if $\underline{\Psi}$ is a cofibred category over $\underline{C}$ and $v : A \rightarrow A''$ is a morphism in $\underline{C}$, then $\underline{\Psi}([A \rightarrow A''])$ is the category of pairs (E, ρ) in which E is an object of $\underline{\Psi}(A)$ and ρ is a “trivialization” of the corresponding object $v_*(E)$ of $\underline{\Psi}(A'')$; that is, an isomorphism from it to the zero object (2.5.1) of $\underline{\Psi}(A'')$. The definition of the functor

$$\underline{\Psi}(A') \rightarrow \underline{\Psi}([A \rightarrow A''])$$

in 3.3.2.1 is evident. When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the objects of $\underline{\Psi}_{L_\bullet}([A \rightarrow A''])$ are explicitly the triples (X, α, ρ) , where (X, α) is as in (3.1.3) and $\rho : X \rightarrow A''$ is a morphism in $\underline{C}$ satisfying

$$(*) \quad \rho i = v, \quad \rho \alpha = 0.$$

If $v : A \rightarrow A''$ belongs to a short exact sequence as in 3.3.2.1, then, as is well known, giving a morphism satisfying the first relation (*) is equivalent, by the left exactness of the cofibred category of extensions of L_0 by variable objects of $\underline{C}$, to giving an extension X' of L_0 by $A' \simeq \text{Ker } v$ which is a subextension of X : one takes $X' = \text{Ker } \rho$. The second relation (*) says that α factors through a morphism $\alpha' : L_1 \rightarrow X'$. Up to the compatibilities whose verification is immediate, this proves the left exactness stated in 3.3.2.1.

3.3.3. Recall that an object E_0 of a cofibred category $\underline{\Psi}$ over $\underline{C}$ is called quasi-maximal if, for every object E of $\underline{\Psi}$ lying over an object A of $\underline{C}$, there is a morphism $E \rightarrow E'$ in $\underline{\Psi}$ inducing a monomorphism $A \rightarrow A'$ in $\underline{C}$, such that E is dominated by E_0 , meaning that there is a morphism $E_0 \rightarrow E'$. If $\underline{C}$ has enough injective objects, this is equivalent to saying that every object of $\underline{\Psi}$ over an injective object of $\underline{C}$ is dominated by E_0 .

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, there is the following canonical quasi-maximal object. Put

$$L'_1 = \text{Coker}(d_1 : L_2 \rightarrow L_1)$$

and let $\bar{d}_0 : L'_1 \rightarrow L_0$ be the morphism induced by $d_0 : L_1 \rightarrow L_0$. Let $E_0 = (X_0, \alpha_0)$ be the object of $\underline{\Psi}_{L_\bullet}(L'_1)$ for which $X_0 = L_0 \oplus L'_1$ is the trivial extension of L_0 by L'_1 and α_0 is the composite

$$L_1 \rightarrow L'_1 \xrightarrow{(\bar{d}_0, \text{id}_{L'_1})} L_0 \oplus L'_1.$$

To see that E_0 is quasi-maximal, observe that for every object (X, α) of $\underline{\Psi}_{L_\bullet}$, one can find a morphism from it to an object (X', α') , inducing a monomorphism $A \rightarrow A'$ in $\underline{C}$, such that the extension X' of L_0 by A' is split. Indeed, it suffices to take $A' = X$ and $(X', \alpha') = i_*(X, \alpha)$, with the notation of (3.1.2). On the other hand, it is immediate that any object (X', α') of $\underline{\Psi}$ for which the extension X' is split is dominated by E_0 .

3.3.4. Let E be an object of an additive cofibred category $\underline{\Psi}$ over $\underline{C}$, lying above an object A of $\underline{C}$, and consider the following functor of the variable object B of $\underline{C}$:

$$(3.3.4.1) \quad B \mapsto \text{Hom}(E, \theta_B),$$

where θ_B denotes the trivial object of $\underline{\Psi}(B)$. We say that E admits a typical complex if this functor is representable by an object Ω_E of $\underline{C}$. Since the functor in question maps canonically to the functor $B \mapsto \text{Hom}(A, B)$ represented by A , one obtains a canonical morphism

$$(3.3.4.2) \quad A \rightarrow \Omega_E,$$

and the corresponding length-one chain complex $A \rightarrow \Omega_E$ in $\underline{C}$ is called the typical complex of E .

When $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$ and $E = (X, \alpha)$, one checks immediately that the functor (3.3.4.1) is represented by $\text{Coker } \alpha$, and that the canonical morphism (3.3.4.2) is simply the composite

$$A \xrightarrow{i} X \xrightarrow{\text{can}} \text{Coker } \alpha.$$

3.3.5. If an additive cofibred category $\underline{\Psi}$ over $\underline{C}$ admits a quasi-maximal object (3.3.3), and the typical complexes of its objects exist, then the typical complex $T_\bullet$ of a quasi-maximal object E_0 , regarded as an object of the derived category $D(\underline{C})$, is independent, up to canonical isomorphism, of the choice of E_0 ([4, 4.4]). It is called the typical complex of the additive cofibred category $\underline{\Psi}$. If, in addition, $\underline{\Psi}$ is left exact (3.3.2), then $\underline{\Psi}$ can be reconstructed from the typical complex $T_\bullet$, up to an equivalence of fibred categories over $\underline{C}$ unique up to unique isomorphism ([4, 6.10]). There are also canonical isomorphisms, functorial in $A \in \underline{C}$ ([4, 5.4]),

$$(3.3.5.1) \quad \Psi^0(A) \xrightarrow{\sim} \text{Ext}^0(T_\bullet, A), \quad \Psi^1(A) \xrightarrow{\sim} \text{Ext}^1(T_\bullet, A).$$

To complete the comparison with [4], we shall prove that when $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, its typical complex is the truncation $[L_\bullet]$. This gives another definition of the isomorphisms (3.1.8) and (3.2.4) in terms of (3.3.5.1), and one checks that it is compatible with the definitions in 3.1 and 3.2.

For this assertion we may plainly restrict to the case where $L_\bullet$ is already truncated in degree 1, since 3.1 noted that truncation does not change $\underline{\Psi}_{L_\bullet}$ up to an isomorphism of cofibred categories. The explicit construction in 3.3.4 then shows immediately that the typical complex of the quasi-maximal object described in 3.3.3 is canonically isomorphic to $L_\bullet$.

Remark 3.3.6. Let

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

be an exact sequence in $\underline{\mathcal{C}}$. It gives an exact sequence of Ext^i groups

$$0 \longrightarrow \text{Ext}^0(L_\bullet, A') \longrightarrow \text{Ext}^0(L_\bullet, A) \longrightarrow \cdots \longrightarrow \text{Ext}^1(L_\bullet, A) \longrightarrow \text{Ext}^1(L_\bullet, A''),$$

which could of course be continued to the right by introducing Ext^i for $i \geq 2$. Using the isomorphisms (3.1.8) and (3.2.4), we obtain the exact sequence

$$(3.3.6.1) \quad 0 \longrightarrow \Psi_{L_\bullet}^0(A') \longrightarrow \Psi_{L_\bullet}^0(A) \longrightarrow \Psi_{L_\bullet}^0(A'') \xrightarrow{\partial} \Psi_{L_\bullet}^1(A') \longrightarrow \Psi_{L_\bullet}^1(A) \longrightarrow \Psi_{L_\bullet}^1(A'').$$

All arrows other than ∂ are defined simply by functoriality of $\Psi_{L_\bullet}^0$ and $\Psi_{L_\bullet}^1$. On the other hand, [4, 2.5.2] defines an exact sequence (3.3.6.1), for every left-exact cofibred category $\underline{\mathcal{C}}$ over $\underline{\mathcal{C}}$, through a morphism

$$(3.3.6.2) \quad \partial : \Psi^0(A'') \longrightarrow \Psi^1(A').$$

It associates with an element u of the left-hand side the class, in the right-hand side, of the object of $\underline{\Psi}(A')$, determined up to canonical isomorphism, whose image in $\underline{\Psi}([A \longrightarrow A''])$ (notation of 3.3.2) is the pair (X_0, u) , where X_0 is the trivial object of $\underline{\Psi}(A)$.

For $\underline{\Psi} = \underline{\Psi}_{L_\bullet}$, the morphism (3.3.6.2) defined by this “geometric” construction is the negative of the boundary morphism ∂ in (3.3.6.1). Thus, after the identifications above and a change of sign in ∂ , the latter exact sequence is precisely that of [4, 2.5.2]. This follows from the final assertion of [4, 5.4], in view of the compatibilities noted in 3.3.5.

Remark 3.3.6.3. We take this opportunity to clarify a point in [4, no. 3.11] concerning the fundamental isomorphisms (3.11.3) and (3.11.4), which play an essential role in the proof of [4, 5.4]. The diagram on page 35 of [4] is commutative, not anticommutative, and therefore does not as it stands define a morphism

$$u : [L_1^X \longrightarrow L_0^X] \longrightarrow K^\bullet[1].$$

It must be made anticommutative by replacing u^1 by $-u^1$. One could instead adopt the opposite convention and replace u^0 by $-u^0$; however, the stated convention, in which u^1 becomes $-u^1$, is the one for which (3.11.3) generalizes the isomorphism (3.5.3) of [4], as claimed there. With this clarification, the claim made in [4, 5.4] must also be corrected: the two boundary morphisms considered there differ by a sign.

3.3.7. In [4, 6.13] the variance of the cofibred category $\underline{\Psi}_{L_\bullet}$ over $\underline{\mathcal{C}}$ as $L_\bullet$ varies was also made explicit, in particular by reconstructing the category of cocartesian functors from $\underline{\Psi}_{L_\bullet}$ to $\underline{\Psi}_{L'_\bullet}$ in terms of the chain complexes $L_\bullet$ and $L'_\bullet$. The set of isomorphism classes of such cocartesian functors is canonically in bijection with

$$\text{Hom}_{D(\underline{\mathcal{C}})}(L'_\bullet, L_\bullet).$$

We shall use only a small part of these results, which we develop in the next section using the simple description of $\underline{\Psi}_{L_\bullet}$ given in 3.1.

3.4. Variance of $\underline{\Psi}_{L_\bullet}$ with $L_\bullet$

Consider a morphism of complexes

$$(3.4.1) \quad f : L'_\bullet \longrightarrow L_\bullet$$

given by a commutative diagram

$$(3.4.2) \quad \begin{array}{ccccccc} & \longrightarrow & L'_2 & \longrightarrow & L'_1 & \longrightarrow & L'_0 \longrightarrow 0 \\ & & \downarrow f_2 & & \downarrow f_1 & & \downarrow f_0 \\ & \longrightarrow & L_2 & \longrightarrow & L_1 & \longrightarrow & L_0 \longrightarrow 0. \end{array}$$

We associate with it a canonical functor

$$(3.4.3) \quad f^* : \underline{\Psi}_{L_\bullet} \longrightarrow \underline{\Psi}_{L'_\bullet}.$$

It is in fact a cocartesian functor of cofibred categories over $\underline{\mathcal{C}}$: it is compatible with the structural functors to $\underline{\mathcal{C}}$ and carries cocartesian morphisms to cocartesian morphisms; see SGA 1, Exposé VI.

If (X, α) is an object of $\underline{\Psi}_{L_\bullet}$, define $f^*(X, \alpha) = (X', \alpha')$ as follows. The extension X' of L'_0 by A is the pullback, along f_0 , of the given extension X of L_0 by A , and α' is the partial trivialization deduced from α by transitivity in the second square of (3.4.2). The compatibility of α' with the relation $d'_0 d'_1 = 0$ follows immediately from the analogous compatibility for α . By construction, (3.4.3) commutes with the structural functors to $\underline{\mathcal{C}}$. Its cocartesian character follows at once from the fact that, in the category of extensions of objects of $\underline{\mathcal{C}}$, direct images of extensions commute with inverse images of extensions.

The functor (3.4.3) induces morphisms of functors

$$(3.4.4) \quad f^i : \Psi_{L_\bullet}^i \longrightarrow \Psi_{L'_\bullet}^i, \quad i = 0, 1$$

(compare 2.6). The morphism (3.4.1), on the other hand, induces morphisms

$$(3.4.5) \quad f^i : \text{Ext}^i(L_\bullet, -) \longrightarrow \text{Ext}^i(L'_\bullet, -), \quad i \in \mathbb{Z}.$$

For $i = 0, 1$, the morphisms (3.4.4) and (3.4.5) are compatible with the isomorphisms (3.1.8) and (3.2.4). In other words, for $A \in \text{Ob } \underline{\mathcal{C}}$ and $i = 0, 1$, the following diagram is commutative:

$$(3.4.6) \quad \begin{array}{ccc} \Psi_{L_\bullet}^i(A) & \xrightarrow{\sim} & \text{Ext}^i(L_\bullet, A) \\ \downarrow & & \downarrow \\ \Psi_{L'_\bullet}^i(A) & \xrightarrow{\sim} & \text{Ext}^i(L'_\bullet, A). \end{array}$$

The verification of this compatibility is essentially immediate and is left to the reader.

Proposition 3.4.7. The functor (3.4.3) is faithful (respectively fully faithful, respectively an equivalence of categories) if and only if

$$H_0(f) : H_0(L'_\bullet) \longrightarrow H_0(L_\bullet)$$

is an epimorphism (respectively, if $H_0(f)$ is an isomorphism and $H_1(f)$ is an epimorphism; respectively, if $H_0(f)$ and $H_1(f)$ are isomorphisms).

Indeed, with f^i denoting the functorial morphism (3.4.4), the required condition is that f^0 be a monomorphism (respectively, that f^0 be an isomorphism and f^1 a monomorphism; respectively, that f^0 and f^1 be isomorphisms). By the commutativity of (3.4.6), these conditions may be

expressed by the analogous conditions on the functors Ext^i , $i = 0, 1$. The proposition then reduces to a straightforward statement of homological algebra, which we leave to the reader as an exercise.

3.4.8. Consider now an exact sequence of chain complexes

$$(3.4.8.1) \quad 0 \longrightarrow L_{\bullet} \xrightarrow{f} L'_{\bullet} \xrightarrow{g} L''_{\bullet} \longrightarrow 0.$$

It gives functors of the form (3.4.3),

$$\underline{\Psi}_{L''_{\bullet}} \xrightarrow{g^*} \underline{\Psi}_{L'_{\bullet}} \xrightarrow{f^*} \underline{\Psi}_{L_{\bullet}}.$$

The transitivity property implicit in functors of the form (3.4.3) gives an isomorphism of functors

$$f^* g^* \simeq (gf)^* = (0_{L_{\bullet} \rightarrow L''_{\bullet}})^*.$$

The last member is evidently canonically isomorphic to the cocartesian zero functor from $\underline{\Psi}_{L''_{\bullet}}$ to $\underline{\Psi}_{L_{\bullet}}$. The preceding isomorphism therefore defines a canonical functor

$$(3.4.8.2) \quad \underline{\Psi}_{L''_{\bullet}} \longrightarrow \underline{\Psi}_{[L_{\bullet} \rightarrow L'_{\bullet}]}.$$

Here $\underline{\Psi}_{[L_{\bullet} \rightarrow L'_{\bullet}]}$ denotes the category of pairs (X', t) in which X' is an object of $\underline{\Psi}_{L'_{\bullet}}$ and t is a trivialization of its inverse image $f^*(X')$, that is, an isomorphism

$$f^*(X') \simeq 0_{L_{\bullet}, A},$$

where $0_{L_{\bullet}, A}$ denotes the zero object of $\underline{\Psi}_{L_{\bullet}}(A)$ and A is the object of $\underline{\mathcal{C}}$ underlying X' . We claim that the functor (3.4.8.2) is an equivalence of categories. Together with the left exactness in A established in 3.3.1, this completes the exactness property in $L_{\bullet}$ of the expression $\underline{\Psi}_{L_{\bullet}}(A)$.

Let us sketch the proof. To give an object of the target of (3.4.8.2) is to give an extension

$$0 \longrightarrow A \xrightarrow{i'} X' \xrightarrow{j'} L'_0 \longrightarrow 0$$

of L'_0 by an object A of $\underline{\mathcal{C}}$, together with a morphism $\alpha' : L'_1 \rightarrow X'$ satisfying $j'\alpha' = d'_0$ and $\alpha'd'_1 = 0$, and a morphism $t : L_0 \rightarrow X'$ satisfying $j't = f_0$ and $td_0 = \alpha'f_1$. These data are displayed in the following diagram:

$$\begin{array}{ccccccc}
 & & & & 0 & & \\
 & & & & \swarrow A & & \\
 & & & & i' & & \\
 & & & & \nearrow X' & & \\
 & & & & j' & & \\
 & & & & \alpha' & & \\
 0 & \longrightarrow & L_0 & \xrightarrow{f_0} & L'_0 & \xrightarrow{g_0} & L''_0 \longrightarrow 0 \\
 & & \uparrow d_0 & & \uparrow d'_0 & & \uparrow d''_0 \\
 0 & \longrightarrow & L_1 & \xrightarrow{f_1} & L'_1 & \xrightarrow{g_1} & L''_1 \longrightarrow 0 \\
 & & \uparrow d'_1 & & \uparrow d''_1 & & \\
 0 & \longrightarrow & L_2 & \longrightarrow & L'_2 & \longrightarrow & L''_2 \longrightarrow 0
 \end{array}
 \quad
 \begin{cases}
 j'\alpha' = d'_0, \\
 \alpha'd'_1 = 0, \\
 j't = f_0, \\
 td_0 = \alpha'f_1.
 \end{cases}$$

The familiar left exactness, in the first argument, of extensions of an object L by an object A , applied to the first row of the diagram, shows that a morphism t satisfying $j't = f_0$ allows one to interpret X' as the inverse image of an extension X'' of L''_0 by A . Under this interpretation, t corresponds to the natural trivialization of the inverse image of that extension along $g_0 f_0 = 0$.

The morphism α' satisfying $j'\alpha' = d'_0$ and $\alpha'f_1 = td_0$ is interpreted by essentially the same dictionary, now applied to the second row of the diagram, as arising from a morphism $\alpha'' : L''_1 \rightarrow X''$ satisfying $j''\alpha'' = d''_0$. Here α' and α'' are regarded as trivializations of the inverse images of X' and X'' along d'_0 and d''_0 , respectively. Finally, the relation $\alpha'd'_1 = 0$ is then equivalent to $\alpha''d''_1 = 0$. Thus the data (X', α', t) , an object of $\underline{\Psi}_{[L_\bullet \rightarrow L'_\bullet]}$, are equivalent to the data (X'', α'') , an object of $\underline{\Psi}_{L''_\bullet}$. This proves, in substance, that (3.4.8.2) is indeed an equivalence of categories.

3.4.9. Continue to consider the exact sequence (3.4.8.1), and the exact sequence of Ext^i groups that it defines:

$$(3.4.9.1) \quad 0 \rightarrow \text{Ext}^0(L''_\bullet, A) \xrightarrow{g^0} \text{Ext}^0(L'_\bullet, A) \xrightarrow{f^0} \text{Ext}^0(L_\bullet, A) \xrightarrow{\partial^0} \text{Ext}^1(L''_\bullet, A) \\ \xrightarrow{g^1} \text{Ext}^1(L'_\bullet, A) \xrightarrow{f^1} \text{Ext}^1(L_\bullet, A) \rightarrow \dots$$

The canonical isomorphisms of the forms (3.1.8) and (3.2.4) give a geometric interpretation of the first six terms of this exact sequence in terms of the categories $\underline{\Psi}_{L_\bullet}$, $\underline{\Psi}_{L'_\bullet}$, and $\underline{\Psi}_{L''_\bullet}$. Compatibility (3.4.6) likewise gives a geometric interpretation of the arrows f^0, g^0, f^1, g^1 . We shall now give such an interpretation of ∂^0 , using the result just proved that (3.4.8.2) is an equivalence of categories.

Let

$$u \in \text{Ext}^0(L_\bullet, A), \quad \text{i.e.} \quad u : L_0 \rightarrow A, \quad ud_0 = 0.$$

Consider the object $(0, \bar{u})$ of $\underline{\Psi}_{[L_\bullet \rightarrow L'_\bullet]}$. Here 0 is the zero object of $\underline{\Psi}_{L'_\bullet}$, whose inverse image along f_0 is therefore identified with the zero object of $\underline{\Psi}_{L_\bullet}$, and $\bar{u}$ is the automorphism of the latter defined by u through the canonical isomorphism (3.1.8). Since (3.4.8.2) is an equivalence, $(0, \bar{u})$ comes from an object of $\underline{\Psi}_{L''_\bullet}(A)$, determined up to unique isomorphism. Its isomorphism class is an element of

$$\Psi_{L''_\bullet}^1(A) \simeq \text{Ext}^1(L''_\bullet, A).$$

We claim that this element is precisely $\partial^0(u)$.

We prove this compatibility by reducing it to the analogous one in [4, Cor. 5.5], where the sign must be corrected as indicated in 3.3.6.3. The argument also gives another proof of the equivalence (3.4.8.2), by reducing it to [4, Prop. 2.7]. Consider the opposite category $\underline{C}^\circ$ and interpret the construction of $\underline{\Psi}_{L_\bullet}(A)$, for fixed A , together with the constructions of 3.4.8 and the present section, in terms of $\underline{C}^\circ$ and the category $\underline{E}$ cofibred over $\underline{C}^\circ$ whose objects are extensions of the fixed object A of $\underline{C}^\circ$ by a variable object L of $\underline{C}^\circ$. This is an additive cofibred category over $\underline{C}^\circ$, and is even left exact there (in the sense of [4, 2.2]).

A chain complex $L_\bullet$ of $\underline{C}$ may be interpreted as a cochain complex of $\underline{C}^\circ$. Under this interpretation, the construction of $\underline{\Psi}_{L_\bullet}(A)$ coincides with the opposite of the category $\widehat{\underline{E}}(L_\bullet)$ in the sense of [4, 1.8]. The functor (3.4.8.2) is then identified with the opposite of the functor of the form [4, (2.1.2)], with E replaced by its extension $\widehat{E}$ to the category of cochain complexes. Consequently, the assertion that this functor is an equivalence is the special case [4, 2.7, a) $\Rightarrow$ d)].

On the other hand, the exact sequence (3.4.9.1) is identified with the exact sequence of the groups $\text{Ext}^i(A, L''_\bullet)$, $\text{Ext}^i(A, L'_\bullet)$, and $\text{Ext}^i(A, L_\bullet)$, apart from a factor $(-1)^i$ on ∂^i (Verdier dixit). Similarly, the isomorphisms (3.1.8) and (3.2.4) are identified with the analogous isomorphisms (3.11.5) and (3.11.4). The asserted compatibility is therefore the special case of [4, Cor. 5.5] corrected as above.

Remark 3.4.9.2. The editor, being very afraid of signs, confesses that he does not have complete confidence in the preceding proof: he fears that signs may have been omitted in passing from $\underline{C}$ to $\underline{C}^\circ$. He recommends that a courageous reader verify the asserted compatibility directly (or, respectively, its opposite). Fortunately, these sign questions will play no role in the following exposés.

3.5. A canonical partial flat resolution of an arbitrary $\mathbb{Z}$ -module

We apply the preceding results when $\underline{C}$ is the category of commutative groups of the topos $\underline{T}$, that is, the category of $\mathbb{Z}$ -modules of $\underline{T}$. Let P be an object of $\underline{C}$. We shall define, functorially in

P , a chain complex $L_\bullet(P)$ augmented to P :

$$(3.5.1) \quad \begin{array}{ccccccc} 0 & \rightarrow & \overbrace{\mathbb{Z}[P \times P] \times \mathbb{Z}[P \times P \times P]}^{L_2} & \xrightarrow{d_1} & \overbrace{\mathbb{Z}[P \times P]}^{L_1} & \xrightarrow{d_0} & \overbrace{\mathbb{Z}[P]}^{L_0} \rightarrow 0 \\ & & & & & & \downarrow \varepsilon \\ & & & & & & P \end{array}$$

with $L_i(P) = 0$ for $i \geq 3$. The components are displayed in (3.5.1), using the notation of 1.4. To describe the differentials d_0, d_1 and the augmentation ε , write $[p]$ for the section of $\mathbb{Z}[P](S)$ defined by $p \in P(S)$, and similarly write $[p, q]$ and $[p, q, r]$ for the corresponding sections of $\mathbb{Z}[P \times P](S)$ and $\mathbb{Z}[P \times P \times P](S)$. Then set

$$(3.5.2) \quad \begin{cases} \varepsilon[p] = p, \\ d_0[p, q] = [p + q] - [p] - [q], \\ d_1[p, q] = [p, q] - [q, p], \\ d_1[p, q, r] = [p + q, r] - [p, q + r] + [p, q] - [q, r]. \end{cases}$$

These homomorphisms of commutative groups do define a complex $L_\bullet(P)$ augmented to P :

$$\varepsilon d_0 = 0, \quad d_0 d_1 = 0.$$

The first relation is immediate. The second splits into two relations according to the two factors of $L_2(P)$; they express, respectively, commutativity and associativity of the composition law on P .

The augmented complex depends functorially on P . A morphism $u : P \rightarrow P'$ gives a commutative diagram

$$\begin{array}{ccc} L_\bullet(P) & \xrightarrow{L_\bullet(u)} & L_\bullet(P') \\ \downarrow & & \downarrow \\ P & \xrightarrow{u} & P'. \end{array}$$

Its components are flat, as is every free $\mathbb{Z}$ -module.

Proposition 3.5.3. One has $H_1(L_\bullet(P)) = 0$, and the augmentation ε induces an isomorphism

$$H_0(L_\bullet(P)) \xrightarrow{\sim} P.$$

For the proof one can make the standard reduction to the case in which $\underline{T}$ is the category of sets and P is an ordinary $\mathbb{Z}$ -module, where (3.5.1) is likely familiar. We instead give a direct proof that does not use this reduction, applying 3.4.7 to the augmentation

$$\varepsilon : L_\bullet(P) \rightarrow P,$$

where P is regarded, as usual, as a chain complex concentrated in degree zero. It remains to prove that, for every object G of $\underline{C}$, the functor

$$(*) \quad \varepsilon^* : \underline{\Psi}_P(G) \rightarrow \underline{\Psi}_{L_\bullet(P)}(G)$$

is an equivalence of categories.

The source of $(*)$ is plainly isomorphic to the category of extensions of the $\mathbb{Z}$ -module P by G . To describe its target, use 1.4 to identify the categories of extensions of L_0, L_1, L_2 by G , respectively, with the category of torsors under G_P , the category of torsors under $G_{P \times P}$, and the category of pairs consisting of a torsor under $G_{P \times P}$ and a torsor under $G_{P \times P \times P}$.

Thus an object of the target of $(*)$ is a pair (E, φ) , where E is a torsor under G_P and φ is a trivialization of the torsor E' on $P \times P$ corresponding to the inverse image along d_0 of the extension of $\mathbb{Z}[P]$ by G defined by E . The trivialization is subject to the compatibility with $d_0 d_1 = 0$ described below. Formula (3.5.2) immediately gives

$$E' = \pi^* E \operatorname{pr}_1^*(E)^{-1} \operatorname{pr}_2^*(E)^{-1},$$

and a trivialization of this torsor is the same as an isomorphism, also denoted by φ ,

$$\varphi : \operatorname{pr}_1^* E \operatorname{pr}_2^* E \xrightarrow{\sim} \pi^* E.$$

Expanding the compatibility of φ with $d_0 d_1 = 0$ gives the commutativity condition (1.2.1) and the associativity condition (1.1.4.1), corresponding to the two factors of L_2 . The assertion that $(*)$ is an equivalence is therefore precisely the dictionary of 1.1.6. This proves 3.5.3.

Remark 3.5.4. a) It would be interesting to strengthen 3.5.3 by constructing a resolution $L_\bullet(P)$ of P , functorial in P , whose components are functorially isomorphic to sums of free $\mathbb{Z}$ -modules generated by cartesian powers of P . In view of isomorphisms of the form (1.4.5), this would give a method for partially computing $\operatorname{Ext}^i(P, A)$ as the abutment of a spectral sequence whose initial term is expressed through groups

$$H^j(P \times P \times \cdots \times P, A).$$

Here we obtain only very partial information about Ext^1 , and no information about the higher Ext^i . There is, in any case, a natural way to continue the partial resolution $L_\bullet(P)$ by one degree.¹

b) There is a variant of (3.5.1) and 3.5.3 in the category of A -modules, where A is a fixed ring of $\underline{T}$. The form of $L_\bullet(P)$ is more complicated because the description 1.1.6 must be enlarged (compare 2.10.3). One obtains

$$(3.5.5) \quad \begin{cases} L_0 = A[P], \\ L_1 = A[P \times P] + A[A \times P], \\ L_2 = A[P \times P \times P] + A[P \times P] + A[A \times A \times P] + A[A \times P \times P]. \end{cases}$$

The augmented-complex structure is given by (3.5.2), supplemented by

$$(3.5.6) \quad \begin{cases} d_0[\lambda, p] = [\lambda p] - \lambda[p], \\ d_1[\lambda, \mu, p] = [\lambda, \mu p] - [\lambda \mu, p] + \lambda[\mu, p], \\ d_1[\lambda, p, q] = [\lambda, p + q] - [\lambda, p] - [\lambda, q] + \lambda[p, q] - [\lambda p, \lambda q]. \end{cases}$$

The relation $d_0 d_1 = 0$ follows from the analogous relation on the components appearing in (3.5.2), which, as we saw, is equivalent to associativity and commutativity of addition on P , together with the relations

$$d_0 d_1[\lambda, \mu, p] = 0, \quad d_0 d_1[\lambda, p, q] = 0.$$

The latter express, respectively, associativity of the given action of A on P ,

$$\lambda(\mu p) = (\lambda \mu)p,$$

and its distributivity over addition,

$$\lambda(p + q) = \lambda p + \lambda q.$$

For a torsor E under G_P , a trivialization of $d_0^* E$ amounts to an internal composition law on E (satisfying the compatibility conditions already described for the internal law on E and its torsor structure), together with an external composition law of A on E , compatible with the action of A

1. L. Breen constructed such a resolution (editor's note).

on P and on G . The compatibility of this trivialization with the evident trivialization of $d_1^* d_0^* E$ is equivalent to the conditions already described above, which say that the internal law on E is a commutative-group law extending P by G , together with associativity of the external law and distributivity of the latter with respect to the internal law. These conditions say exactly that E , with its two composition laws and its torsor structure, is an A -module extension of the A -modules P and G .

It follows, by essentially the same method as in 3.5.3, that $L_\bullet$ is a partial resolution of the module P , that is, $H_1(L_\bullet) = 0$. It would again be interesting to extend this partial resolution to an infinite resolution of the same kind, with components of the form

$$A[A \times A \times \cdots \times P \times P \times \cdots].$$

Compare the analogous question in a) for commutative groups.

3.6. Application to the description of the cofibred category of biextensions of a given pair (P, Q) of commutative groups

Let P and Q be two $\mathbb{Z}$ -modules of the topos $\underline{T}$. Choose a flat resolution $L'_\bullet(P)$ of P that agrees in degrees at most 2 with the augmented complex $L_\bullet(P)$ of 3.5. This is possible by 3.5.3 and because every $\mathbb{Z}$ -module on $\underline{T}$ is a quotient of a free, hence flat, $\mathbb{Z}$ -module. Similarly choose a flat resolution $L'_\bullet(Q)$ of Q agreeing with $L_\bullet(Q)$ in degrees at most 2. We have canonical homomorphisms

$$(3.6.1) \quad L_\bullet(P) \longrightarrow L'_\bullet(P), \quad L_\bullet(Q) \longrightarrow L'_\bullet(Q),$$

and hence a canonical homomorphism

$$(3.6.2) \quad \begin{aligned} L_\bullet(P, Q) &\stackrel{\text{def}}{=} L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow L'_\bullet(P, Q) \\ &\stackrel{\text{def}}{=} L'_\bullet(P) \otimes L'_\bullet(Q) \simeq P \overset{\mathbb{L}}{\otimes} Q. \end{aligned}$$

It is an isomorphism on components in degrees at most 2. Consequently, by definition,

$$(3.6.3) \quad \underline{\Psi}_{L_\bullet(P, Q)} = \underline{\Psi}_{L'_\bullet(P, Q)}.$$

Theorem 3.6.4. The additive cofibred category over the category

$$\underline{C} = \text{GRAB}(\underline{T})$$

of commutative groups of $\underline{T}$, consisting of biextensions of (P, Q) by a variable abelian group G (2.4), is canonically equivalent to the additive cofibred category

$$\underline{\Psi}_{L_\bullet(P, Q)} = \underline{\Psi}_{L'_\bullet(P, Q)}$$

defined from $L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q)$ as in 3.1.

This and 3.2.5 give the principal result of the present section, announced above:

Corollary 3.6.5. For fixed P and Q there is a canonical isomorphism, functorial in G ,

$$(3.6.5.1) \quad \text{Biext}^1(P, Q; G) \xrightarrow{\sim} \text{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G).$$

To prove 3.6.5, first make explicit the components of degrees 0, 1, 2 of $L_\bullet(P, Q)$:

$$(3.6.6) \quad \left\{ \begin{array}{l} L_0(P, Q) = L_0(P) \otimes L_0(Q) = \mathbb{Z}[P \times Q], \\ L_1(P, Q) = L_0(P) \otimes L_1(Q) + L_1(P) \otimes L_0(Q) \\ \quad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times P \times Q], \\ L_2(P, Q) = L_0(P) \otimes L_2(Q) + L_1(P) \otimes L_1(Q) + L_2(P) \otimes L_0(Q) \\ \quad = \mathbb{Z}[P \times Q \times Q] + \mathbb{Z}[P \times Q \times Q \times Q] \\ \quad \quad + \mathbb{Z}[P \times P \times Q \times Q] \\ \quad \quad + \mathbb{Z}[P \times P \times Q] + \mathbb{Z}[P \times P \times P \times Q]. \end{array} \right.$$

To describe the cofibred category $\underline{\Psi}_{L_\bullet(P,Q)}$ through its fibres $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, use 1.4 to express extensions of $L_i(P, Q)$, for $i = 0, 1, 2$, by G in terms of torsors. An extension of $L_0(P, Q)$ by G is a torsor under $G_{P \times Q}$. An extension of $L_1(P, Q)$ by G is a pair consisting of a torsor under $G_{P \times Q \times Q}$ and one under $G_{P \times P \times Q}$. An extension of $L_2(P, Q)$ by G is similarly a system of five torsors, under the groups obtained from G by base change to

$$\begin{aligned} P \times Q \times Q, & \quad P \times Q \times Q \times Q, & P \times P \times Q \times Q, \\ P \times P \times Q, & \quad P \times P \times P \times Q. \end{aligned}$$

Under this dictionary, a pair (X, α) consisting of an extension of $L_0(P, Q)$ by G and a trivialization of d_0^*X on $L_1(P, Q)$ becomes a torsor E on $P \times Q$ equipped with two trivializations (φ, ψ) of the torsors (E_1, E_2) on $P \times Q \times Q$ and $P \times P \times Q$, respectively, obtained from E by inverse image along

$$d_0 : L_1(P, Q) \longrightarrow L_0(P, Q).$$

Expanding d_0 , as in the proof of 3.5.3, identifies φ and ψ with isomorphisms

$$\varphi : q_1^*E q_2^*E \xrightarrow{\sim} \pi_Q^*E, \quad \psi : p_1^*E p_2^*E \xrightarrow{\sim} \pi_P^*E.$$

Here $\pi_Q, q_1, q_2 : P \times Q \times Q \longrightarrow P \times Q$ are induced by the composition law and the first and second projections $Q \times Q \longrightarrow Q$; the maps

$$\pi_P, p_1, p_2 : P \times P \times Q \longrightarrow P \times Q$$

are defined symmetrically. Thus φ and ψ are data of the forms (2.0.4) and (2.0.7) appearing in 2.1.

Finally, the condition that (X, α) be an object of $\underline{\Psi}_{L_\bullet(P,Q)}(G)$, namely compatibility of the partial trivialization α with $d_0 d_1 = 0$, becomes five compatibility relations for the structure (E, φ, ψ) on torsors over, respectively,

$$\begin{aligned} P \times Q \times Q, & \quad P \times Q \times Q \times Q, & P \times P \times Q \times Q, \\ P \times P \times Q, & \quad P \times P \times P \times Q. \end{aligned}$$

They correspond to the five terms in the expression (3.6.6) for $L_2(P, Q)$. As in the proof of 3.5.3, the first two are precisely the commutativity relation (2.0.6) and the associativity relation (2.0.5); together they say that φ gives E the structure of a commutative extension of Q_P by G_P . Symmetrically, the last two say that ψ gives E the structure of a commutative extension of P_Q by G_Q . The middle relation, on torsors over $P \times P \times Q \times Q$, is exactly the commutativity of diagram (2.1.1). It says that φ and ψ are compatible, hence, in view of the other conditions, define on E a biextension structure of (P, Q) by G .

The verification of these interpretations is immediate and is left to the reader. The preceding discussion indeed defines an equivalence of categories cofibred over $\underline{C} = \text{GRAB}(\underline{T})$, as asserted in 3.6.4.

Remark 3.6.7. Each of the three generalizations of biextensions mentioned in 2.10 has a corresponding variant of 3.6.4 and hence of 3.6.5. The generalization to any number of factors (2.10.2) is immediate, as is the generalization to biextensions of A -modules (2.10.3), once the variant of $L_\bullet(P)$ described in 3.5.4 b) has been constructed. From the viewpoint of homological algebra, the case in which P and Q are not assumed commutative (2.10.1) is more interesting.

Associate with a not necessarily commutative group P of $\underline{T}$ a complex $K_\bullet(P)$ truncated in degree 2. It is described as $L_\bullet(P)$ was in 3.5, except that

$$K_2(P) = \mathbb{Z}[P \times P \times P],$$

the first factor occurring in $L_2(P)$ being omitted; d_0 and d_1 are defined as in (3.5.2). Then

$$H_0(K_\bullet(P)) = P/[P, P] \simeq \underline{H}_1(P, \mathbb{Z}),$$

and similarly

$$H_1(K_\bullet(P)) \simeq \underline{H}_2(P, \mathbb{Z}).$$

Indeed, $K_\bullet(P)$ is precisely the truncation in degree 2 of the chain complex $C_\bullet(P, \mathbb{Z})$ of P with coefficients in $\mathbb{Z}$.

The argument proving 3.5.3 also shows that the cofibred category $\underline{\Psi}_{K_\bullet(P)}$ over $\underline{C} = \text{GRAB}(\underline{T})$ is canonically equivalent to the category of central extensions of P by variable commutative groups G . Likewise, for two groups P, Q , the cofibred category of biextensions of (P, Q) in the sense of 2.10.1 is equivalent to $\underline{\Psi}_{K_\bullet(P, Q)}$, where

$$K_\bullet(P, Q) = K_\bullet(P) \otimes^{\mathbb{L}} K_\bullet(Q).$$

These considerations suggest that, when P is commutative, the canonical resolution sought in 3.5.4 a) might be obtained by a suitable modification of $C_\bullet(P, \mathbb{Z})$.

3.7. Various compatibilities

In this section we examine the behaviour, for variable P and Q , of the equivalence of cofibred categories in 3.6.4. The variance of the cofibred category of biextensions of (P, Q) by a variable G with respect to variable P, Q is defined in principle in 2.4; it may also be expressed by saying that one has a cofibred functor (2.6.1). Notice that the variance in P, Q is contravariant. The variance of the cofibred category $\underline{\Psi}_{L_\bullet(P, Q)}$ with respect to variable P, Q is also clear, since $L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q)$ depends functorially on the pair (P, Q) , while $\underline{\Psi}_{L_\bullet}$ depends contravariantly on $L_\bullet$ (3.4). With this understood, a first compatibility is expressed, for every triple of morphisms

$$(3.7.1) \quad P' \longrightarrow P, \quad Q' \longrightarrow Q, \quad G \longrightarrow G',$$

by the fact that the diagram of functors

$$(3.7.2) \quad \begin{array}{ccc} \underline{\Psi}_{L_\bullet(P, Q)}(G) & \longrightarrow & \underline{\Psi}_{L_\bullet(P', Q')}(G') \\ \downarrow & & \downarrow \\ \text{BIEXT}(P, Q; G) & \longrightarrow & \text{BIEXT}(P', Q'; G') \end{array}$$

is commutative up to a canonical isomorphism. We shall admit this compatibility: making the commutativity isomorphism explicit could only be as obvious as it is tedious (as, indeed, could the complete explication it presupposes of the equivalence of cofibred categories in 3.6.4).

3.7.3. It follows in particular from the preceding compatibility that the isomorphism (3.6.5.1) is functorial not only in G , but also in P and Q . One may state, similarly and more trivially, the analogous functoriality of the isomorphism defined through (3.1.8), just as (3.6.5.1) was defined through (3.2.4):

$$(3.7.4) \quad \text{Biext}^0(P, Q; G) \xrightarrow{\sim} \text{Ext}^0(P \otimes^{\mathbb{L}} Q, G) \simeq \text{Hom}(P \otimes Q, G).$$

But such an isomorphism was already obtained in (2.5.4.1), and was observed in 2.6 to be functorial in its three arguments. It is immediate to verify that the two isomorphisms (3.7.4) and (2.5.4.1) are equal.

3.7.5. The value of the isomorphism in 3.6.5 is that it expresses the remarkable group of geometric origin $\text{Biext}^1(P, Q; G)$ in terms of the very simple homological-algebra invariant $\text{Ext}^1(P \otimes^{\mathbb{L}} Q, G)$. The latter belongs to the sequence of invariants $\text{Ext}^i(P \otimes^{\mathbb{L}} Q, G)$, to which the usual formalism of homological algebra applies. In particular, the sequence of these Ext^i is a cohomological functor

in the sense of [3] (or a connected sequence of functors in the terminology of [5]) in each of the three arguments P, Q, G . Thus an exact sequence

$$(3.7.5.1) \quad 0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$

(respectively,

$$(3.7.5.2) \quad 0 \longrightarrow P' \longrightarrow P \longrightarrow P'' \longrightarrow 0, \quad \text{or} \quad 0 \longrightarrow Q' \longrightarrow Q \longrightarrow Q'' \longrightarrow 0)$$

gives rise to an infinite exact sequence of the Ext^i . Its first six terms, which involve only Ext^0 and Ext^1 , may be interpreted in terms of biextensions by 3.6.5 and 3.7.4. We have already indicated in 3.3.6 a geometric interpretation of the exact sequence deduced from (3.7.5.1), and especially of the boundary map

$$(3.7.5.3) \quad \partial : \text{Biext}^0(P, Q; G'') \longrightarrow \text{Biext}^1(P, Q; G'),$$

within the framework of a left exact cofibred category. One may seek an analogous interpretation of the six-term exact sequence associated with either short exact sequence in (3.7.5.2), say the first; equivalently, in view of 3.7.3, one may seek a geometric interpretation of the boundary homomorphism

$$(3.7.5.4) \quad \partial : \text{Biext}^0(P', Q; G) \longrightarrow \text{Biext}^1(P'', Q; G).$$

Indeed, we have the following result.

Proposition 3.7.6. The cofibred category over

$$\text{GRAB}(\underline{T})^{\text{op}} \times \text{GRAB}(\underline{T})^{\text{op}} \times \text{GRAB}(\underline{T})$$

of biextensions of (P, Q) by G , with P, Q, G variable (cf. (2.6.1)), is left exact not only in the argument G , but also in each of the arguments P and Q . Moreover, the boundary homomorphism (3.7.5.4) deduced from the exact sequence of the Ext^i associated with (3.7.5.2), by means of the isomorphisms 3.6.5 and 3.7.4, is the boundary homomorphism in the theory of left exact cofibred categories over $\underline{C} = \text{GRAB}(\underline{T})^{\text{op}}$ recalled in 3.3.6, applied to the cofibred category of biextensions of (P, Q) by $\underline{G}$, where Q and G are fixed and P varies (subject to a possible sign error in 3.4.9; cf. 3.4.9.2).

We sketch the proof of 3.7.6. By 3.6.4 and compatibility (3.7.2), we interpret biextensions of (P, Q) by G as objects of $\underline{\Psi}_{L_\bullet(P, Q)}(G)$. Consider the homomorphism

$$L_\bullet(P, Q) = L_\bullet(P) \otimes L_\bullet(Q) \longrightarrow P \otimes L_\bullet(Q)$$

defined by the augmentation $L_\bullet(P) \longrightarrow P$. It follows from 3.5.3 and the flatness of $L_\bullet(Q)$ that this homomorphism induces isomorphisms on H_0 and H_1 , and hence by 3.4.7 an equivalence

$$\underline{\Psi}_{P \otimes L_\bullet(Q)} \longrightarrow \underline{\Psi}_{L_\bullet(P, Q)}.$$

We may therefore interpret biextensions of (P, Q) by G as objects of $\underline{\Psi}_{P \otimes L_\bullet(Q)}(G)$. The exact sequence (3.7.5.2) now gives an exact sequence of chain complexes

$$(3.7.6.1) \quad 0 \longrightarrow P' \otimes L_\bullet(Q) \longrightarrow P \otimes L_\bullet(Q) \longrightarrow P'' \otimes L_\bullet(Q) \longrightarrow 0.$$

The asserted left exactness follows from 3.4.8 applied to this exact sequence of complexes. (A direct verification would in any event be immediate.) Moreover, the homomorphism (3.7.5.4) is precisely the boundary homomorphism $\text{Ext}^0 \longrightarrow \text{Ext}^1$ deduced from this exact sequence by taking the Ext^i with values in G . The last assertion of 3.7.6 is consequently a special case of 3.4.9.

3.8. Biextensions of (P, Q) trivialized over extensions of P and Q

First consider two exact sequences of abelian groups

$$(3.8.1) \quad 0 \longrightarrow L_1 \xrightarrow{i} L_0 \xrightarrow{u} P \longrightarrow 0, \quad 0 \longrightarrow M_1 \xrightarrow{j} M_0 \xrightarrow{v} Q \longrightarrow 0.$$

We shall describe the category of biextensions E of (P, Q) by an abelian group G for which the inverse image $(u, v)^*(E)$ is trivial; that is, those for which there exists a homomorphism s making the following triangle commutative:

$$(3.8.2) \quad \begin{array}{ccc} E & & \\ \downarrow & \searrow s & \\ P \times Q & \xleftarrow{u \times v} & L_0 \times M_0 \end{array}$$

and satisfying the additivity condition required in each of its two arguments. The choice of such an s therefore identifies $(u, v)^*(E)$ with the trivial biextension $E_0 = L_0 \times M_0 \times G$ of (L_0, M_0) by G . The left exactness in P, Q of the cofibred category of biextensions of (P, Q) by fixed G (3.7.6) then shows that to give the biextension E of (P, Q) , together with the trivialization s of its inverse image, is essentially equivalent to giving two partial trivializations of the trivial biextension E_0 of (L_0, M_0) by G , over (L_1, M_0) and (L_0, M_1) respectively, which agree over (L_1, M_1) . Since the biextensions induced by E_0 over (L_1, M_0) , (L_0, M_1) , and (L_1, M_1) identify with the trivial biextensions, a partial trivialization over (L_1, M_0) (respectively over (L_0, M_1)) is equivalent to giving a homomorphism (or pairing)

$$(3.8.3) \quad f : L_1 \otimes M_0 \longrightarrow G \quad (\text{respectively } g : L_0 \otimes M_1 \longrightarrow G).$$

The compatibility in question simply means that the two homomorphisms agree over $L_1 \otimes M_1$, namely

$$(3.8.4) \quad f(x_1 \otimes y_1) = g(x_1 \otimes y_1),$$

for sections x_1, y_1 of L_1, M_1 over an object of $\underline{T}$. In terms of the homomorphism s of (3.8.2), the forms f and g are characterized by

$$(3.8.5) \quad s(x_1, y_0) = f(x_1 \otimes y_0)e_{E/Q}(y_0), \quad s(x_0, y_1) = g(x_0 \otimes y_1)e_{E/P}(x_0),$$

which again makes clear the necessity of condition (3.8.4).

Once the trivialization s has been chosen, the other trivializations are the morphisms $s' : L_0 \times M_0 \longrightarrow E$ of the form

$$(3.8.6) \quad s'(x_0, y_0) = s(x_0, y_0) + h(x_0 \otimes y_0),$$

where

$$h : L_0 \otimes M_0 \longrightarrow G$$

is an arbitrary form. It is then clear from (3.8.5) that the forms f, g defined by s are changed into

$$(3.8.7) \quad \begin{aligned} f'(x_1 \otimes y_0) &= f(x_1 \otimes y_0) + h(x_1 \otimes y_0), \\ g'(x_0 \otimes y_1) &= g(x_0 \otimes y_1) + h(x_0 \otimes y_1). \end{aligned}$$

Thus we obtain:

Proposition 3.8.8. The subgroup of $\text{Biext}^1(P, Q; G)$ formed by the classes of biextensions whose inverse image under (u, v) is a trivial biextension of (L_0, M_0) (that is, the kernel of $\text{Biext}^1(P, Q; G) \longrightarrow \text{Biext}^1(L_0, M_0; G)$) is canonically isomorphic, by the preceding description,

to the quotient of the group of pairs of forms (f, g) in (3.8.3) satisfying compatibility condition (3.8.4), by the pairs of forms which arise by restriction from a form $h : L_0 \otimes M_0 \rightarrow G$.

We now interpret this isomorphism in terms of homological algebra, taking into account the canonical isomorphism

$$\text{Biext}^1(P, Q; G) \simeq \text{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$$

of 3.6.5. To this end, interpret the data (3.8.1) as two complexes resolving P and Q , respectively:

$$(3.8.9) \quad L_\bullet : 0 \rightarrow L_1 \rightarrow L_0 \rightarrow 0, \quad M_\bullet : 0 \rightarrow M_1 \rightarrow M_0 \rightarrow 0,$$

and consider the tensor-product complex $L_\bullet \otimes M_\bullet$

$$0 \rightarrow L_1 \otimes M_1 \xrightarrow{d_2} L_1 \otimes M_0 + L_0 \otimes M_1 \xrightarrow{d_1} L_0 \otimes M_0 \rightarrow 0.$$

There is, moreover, a canonical homomorphism in the derived category

$$P \overset{\mathbb{L}}{\otimes} Q \rightarrow L_\bullet \otimes M_\bullet,$$

and hence an equally canonical homomorphism

$$(3.8.10) \quad H^1(\text{Hom}^\cdot(L_\bullet \otimes M_\bullet, G)) \rightarrow \text{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G).$$

The group of 1-cocycles of $\text{Hom}^\cdot(L_\bullet \otimes M_\bullet, G)$ identifies with the group of pairs (f, g) as in (3.8.3) which satisfy compatibility condition (3.8.4): to such a pair one associates the homomorphism $F : L_1 \otimes M_0 + L_0 \otimes M_1 \rightarrow G$ with components (f, g) ,

$$(3.8.11) \quad F(x_1 \otimes y_0 + x_0 \otimes y_1) = f(x_1 \otimes y_0) + g(x_0 \otimes y_1).$$

Moreover, the coboundary of the 0-cochain $h : L_0 \otimes M_0 \rightarrow G$ is precisely the pair of its restrictions to $L_1 \otimes M_0$ and $L_0 \otimes M_1$. Consequently, the group of pairs (f, g) modulo the pairs (h, h) described in 3.8.8 is canonically isomorphic to the first term of (3.8.10). With this understood, one has the following compatibility.

Proposition 3.8.12. If

$$\xi \in \text{Ker}(\text{Biext}^1(P, Q; G) \rightarrow \text{Biext}^1(L_0, M_0; G))$$

is described by the pair (f, g) as in 3.8.8, then the image of ξ in $\text{Ext}^1(P \overset{\mathbb{L}}{\otimes} Q, G)$ under the isomorphism (3.6.5.1) is the image under (3.8.10) of the element of its first term defined by the cocycle (3.8.11).

The verification of this compatibility is left to the reader.

3.8.13. We shall apply this especially in the case

$$L_0 = \mathbb{Z}[P], \quad M_0 = \mathbb{Z}[Q],$$

where L_1 and M_1 are respectively the kernels of the canonical homomorphisms $\mathbb{Z}[P] \rightarrow P$ and $\mathbb{Z}[Q] \rightarrow Q$. Notice the canonical isomorphisms

$$L_0 \overset{\mathbb{L}}{\otimes} M_0 \simeq L_0 \otimes M_0 \simeq \mathbb{Z}[P \times Q].$$

Since L_0 and M_0 are flat, (1.4.5) gives canonical isomorphisms

$$\text{Ext}^i(L_0 \overset{\mathbb{L}}{\otimes} M_0, G) \simeq H^i(P \times Q, G_{P \times Q}).$$

It follows from this and 3.6.5, or directly from the definitions and 1.4, that the functor which assigns to each biextension of (L_0, M_0) by G the $G_{P \times Q}$ -torsor over $P \times Q$ that it induces is

an equivalence of categories. (This remains valid without using an additive structure on P, Q .) Consequently, if E is a biextension of (P, Q) by G , giving a trivialization of its inverse image over (L_0, M_0) is equivalent to giving a section of E considered only as an object over $P \times Q$, that is, a trivialization of E considered as a torsor under $G_{P \times Q}$.

Biextensions equipped with this additional structure are therefore described by the pairs (f, g) of (3.8.3) satisfying compatibility (3.8.4), and their classification up to isomorphism is the one given in 3.8.8.

With the notation of 3.5, one may also regard L_1 as the cokernel of $L_2(P) \rightarrow L_1(P)$, and similarly for M_1 . This makes it possible to express the data (f, g) in terms of a system of homomorphisms

$$\begin{cases} f' : L_1(P) \otimes L_0(Q) = \mathbb{Z}(P \times P \times Q) \rightarrow G, \\ g' : L_0(P) \otimes L_1(Q) = \mathbb{Z}(P \times Q \times Q) \rightarrow G, \end{cases}$$

that is, homomorphisms of sheaves of sets

$$P \times P \times Q \rightarrow G, \quad P \times Q \times Q \rightarrow G,$$

satisfying five compatibility conditions. These express that f' and g' factor through f and g —which gives 2×2 equations, in view of the expression of $L_2(P)$ and $L_2(Q)$ as sums of two terms—together with condition (3.8.4). One then sees that f' and g' are precisely the partial multiplication data defining a biextension, and that the five conditions in question are exactly the five axioms for biextensions considered in 2.1 (cf. the proof of 3.6.4).

Bibliography to Exposé VII

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